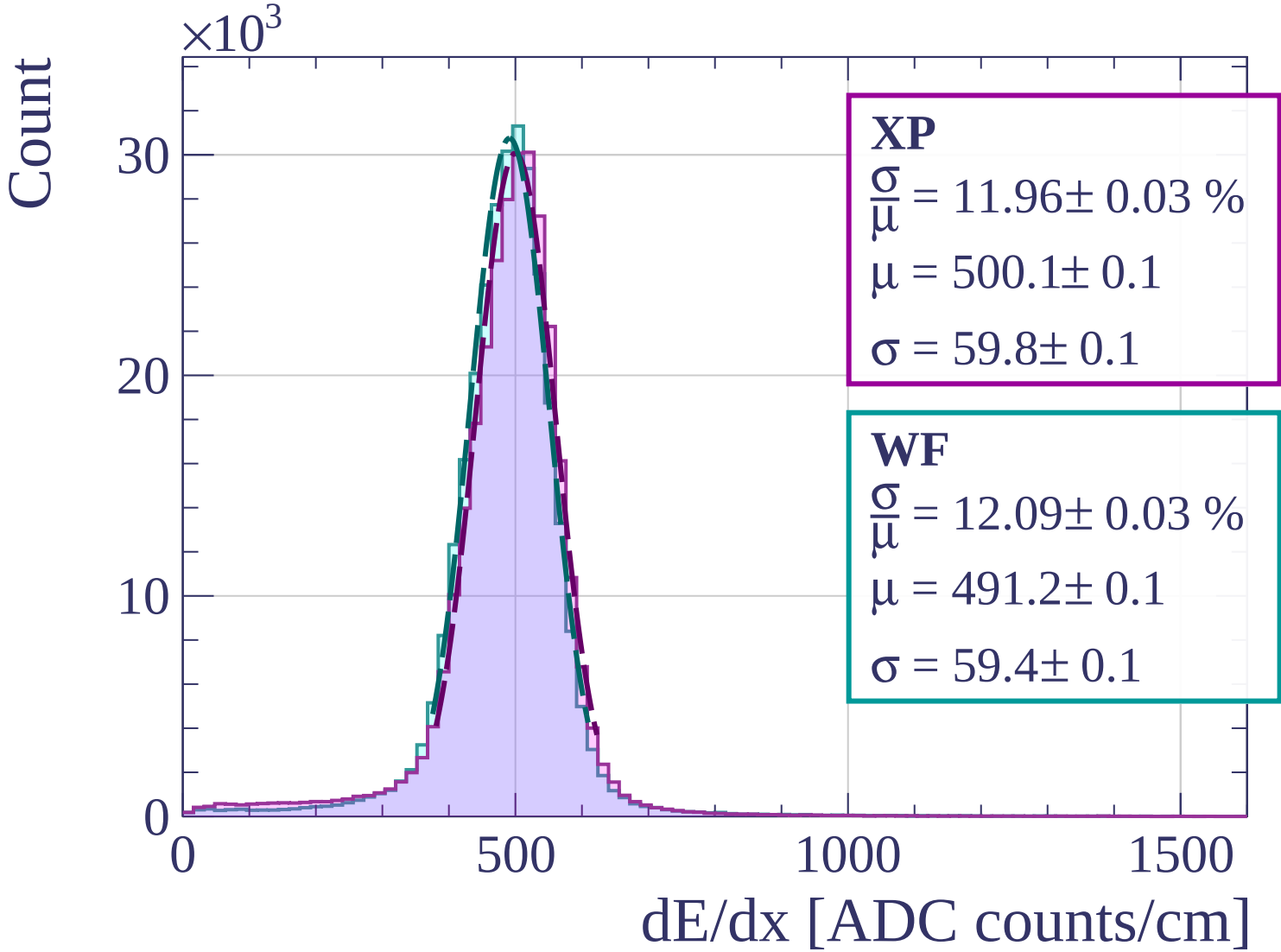
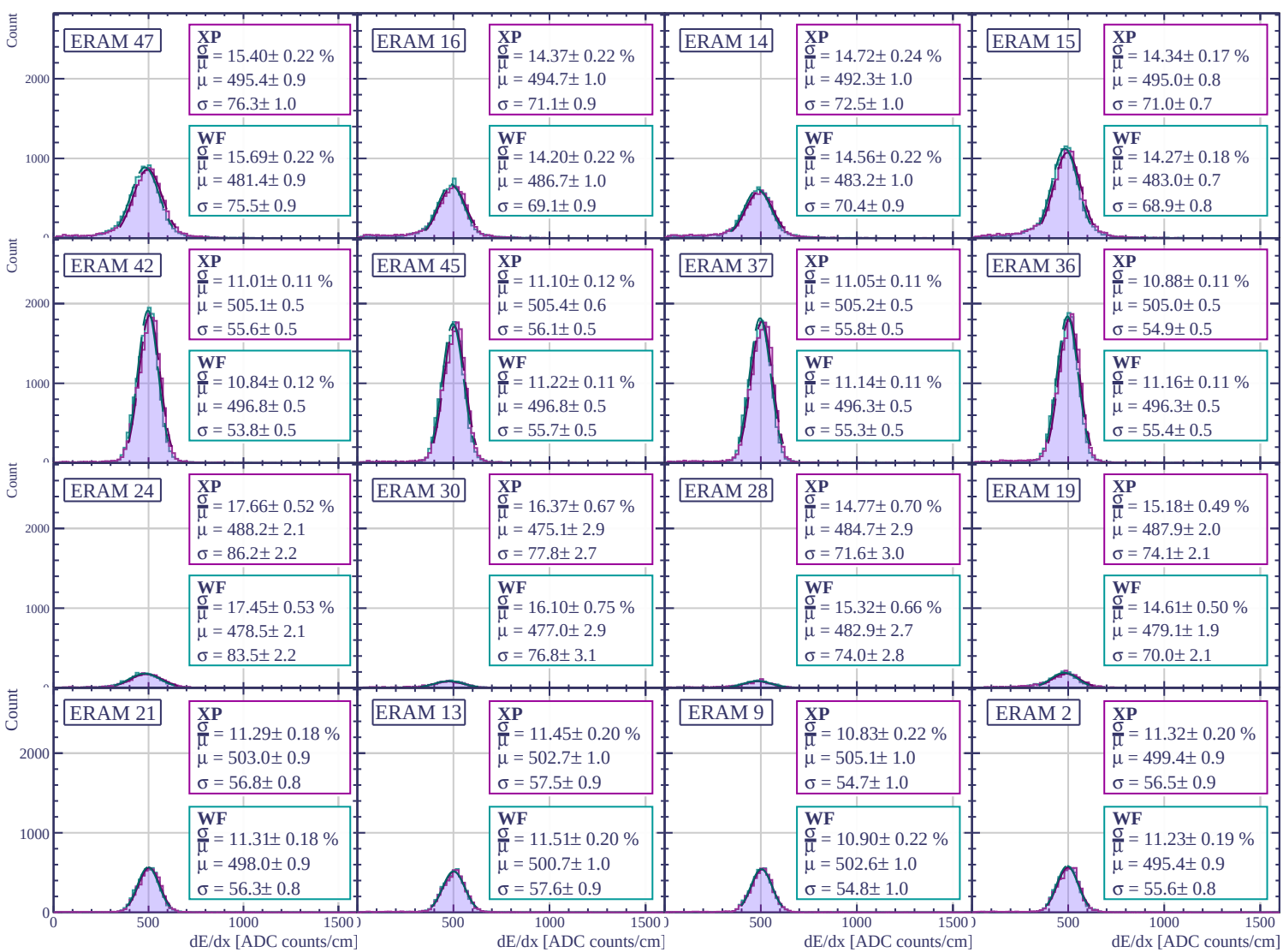
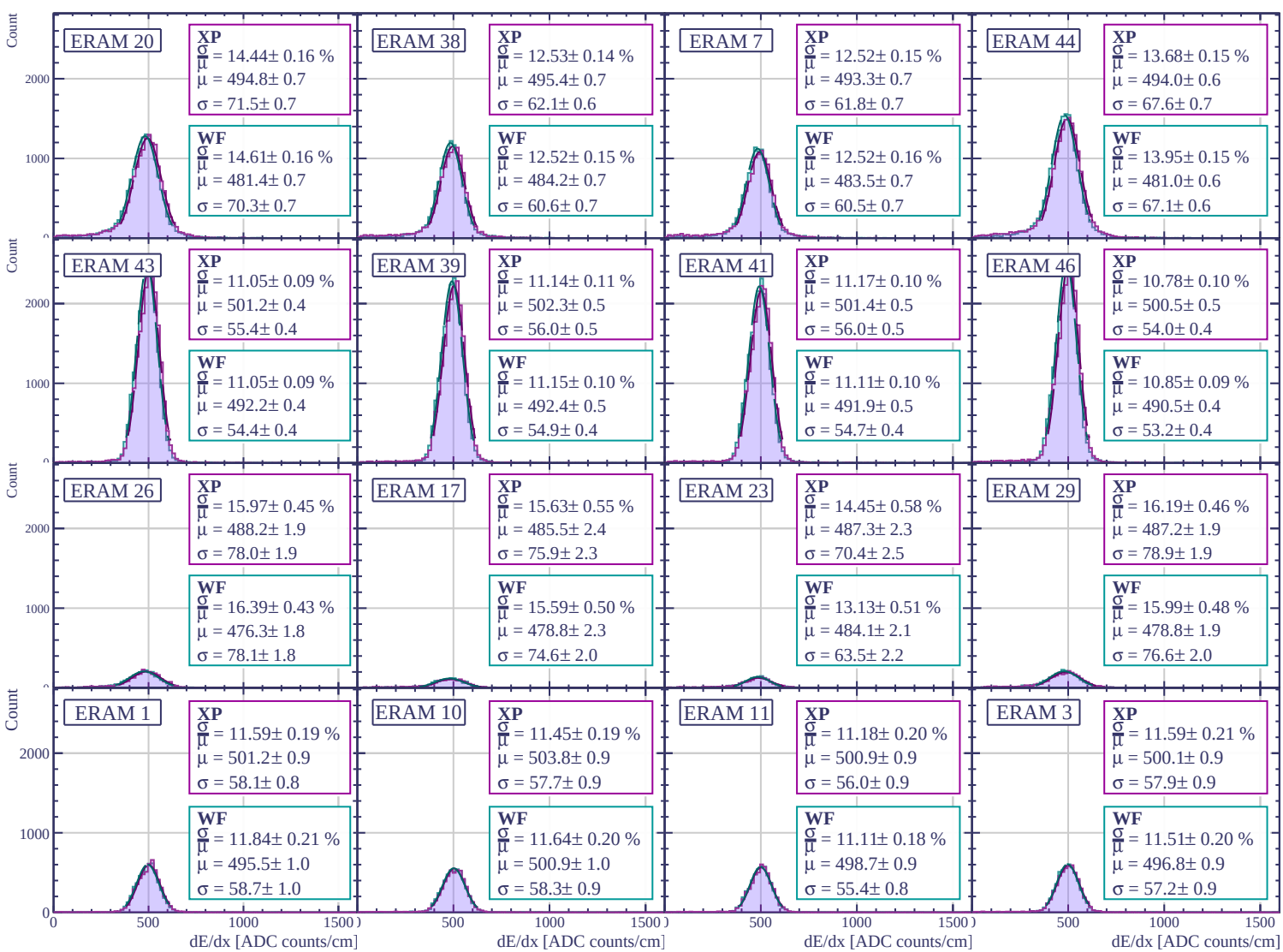
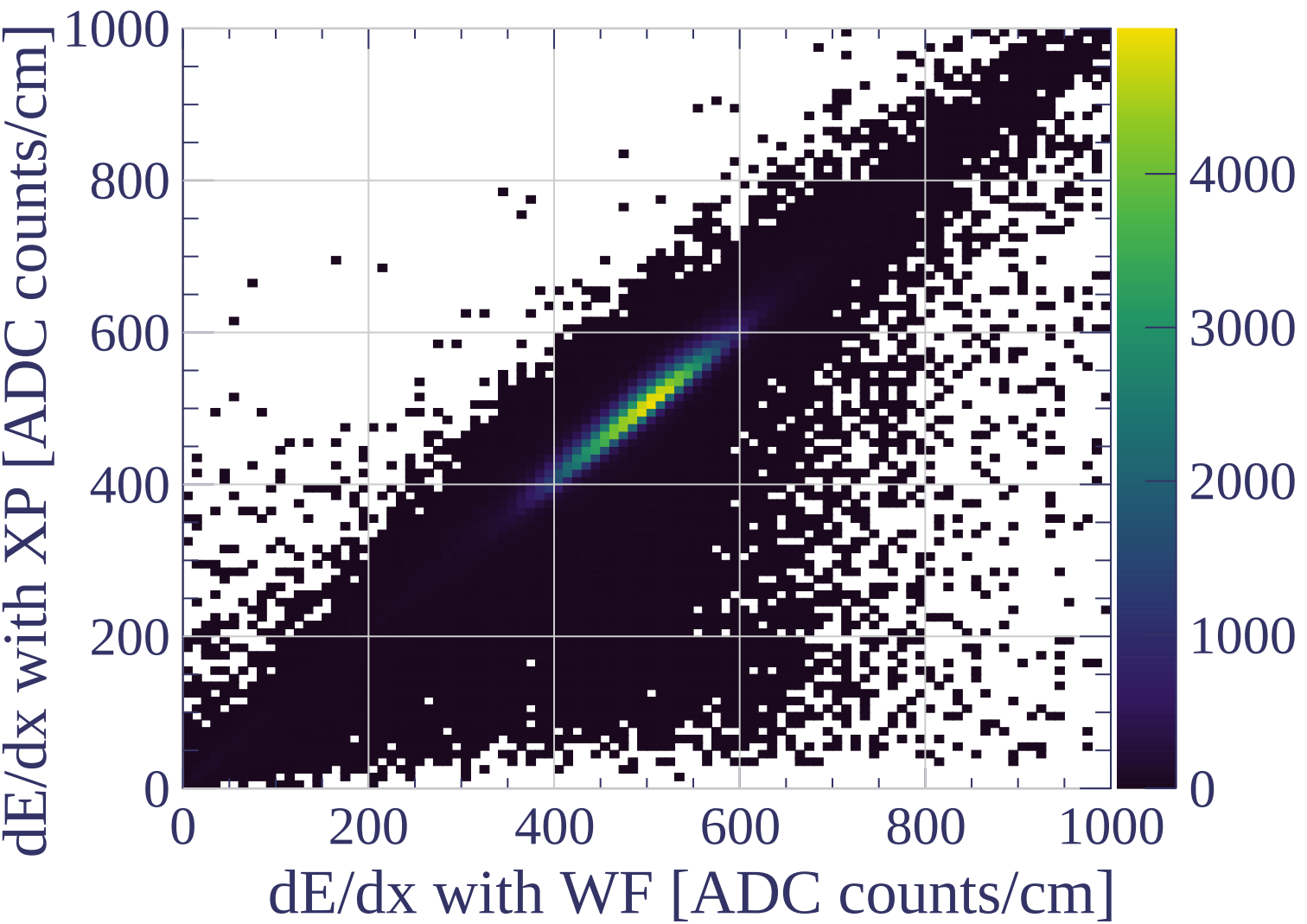
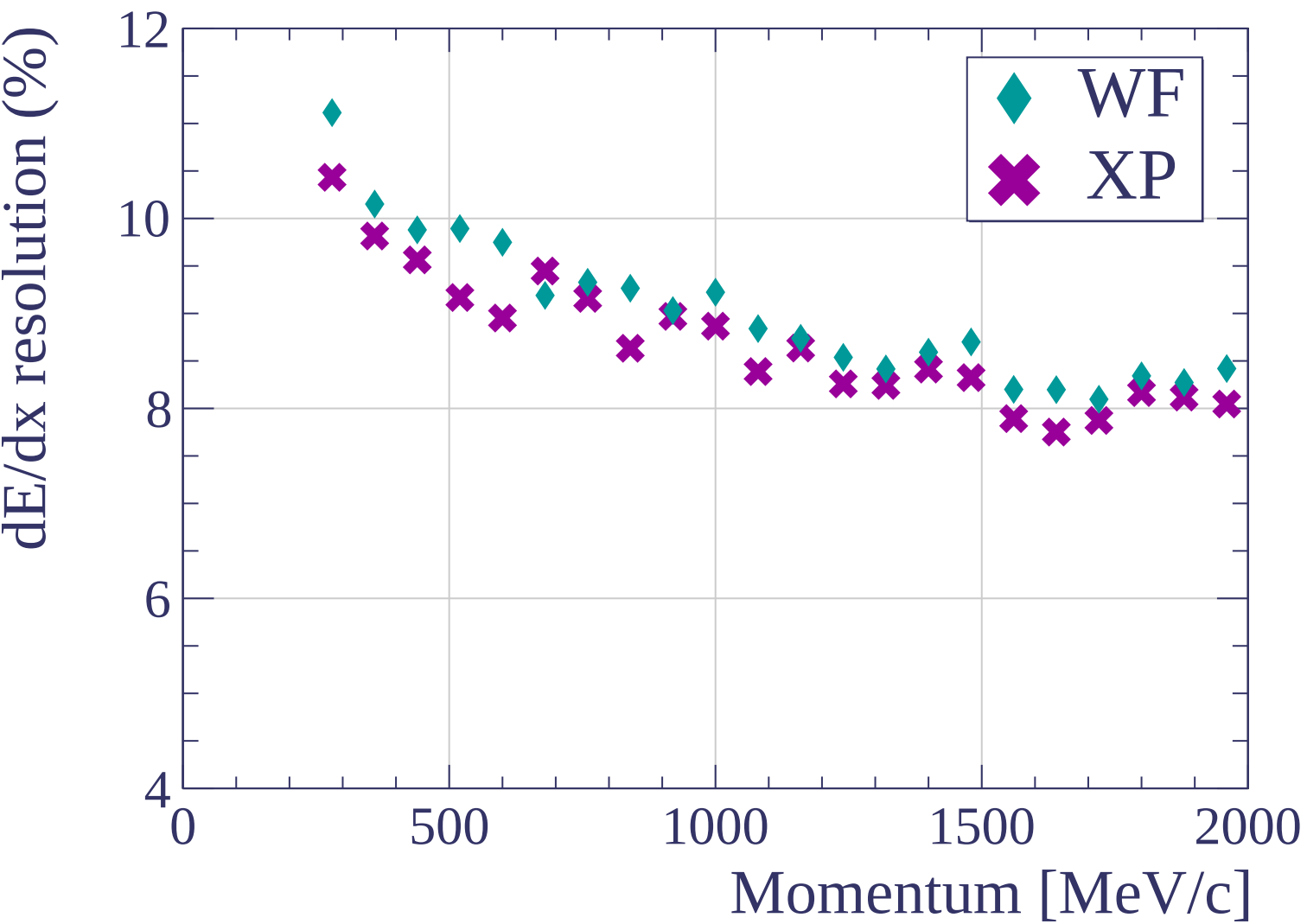


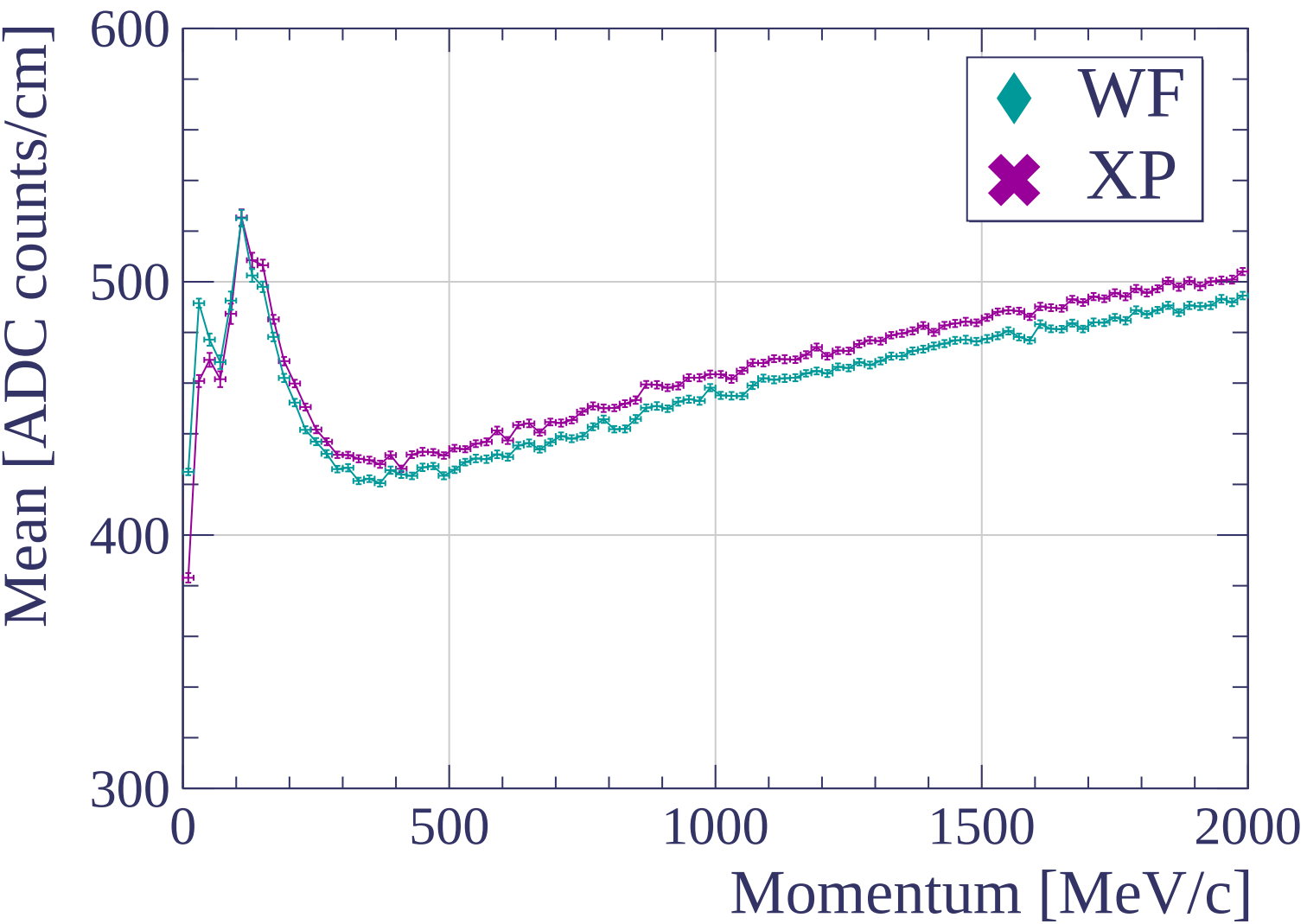


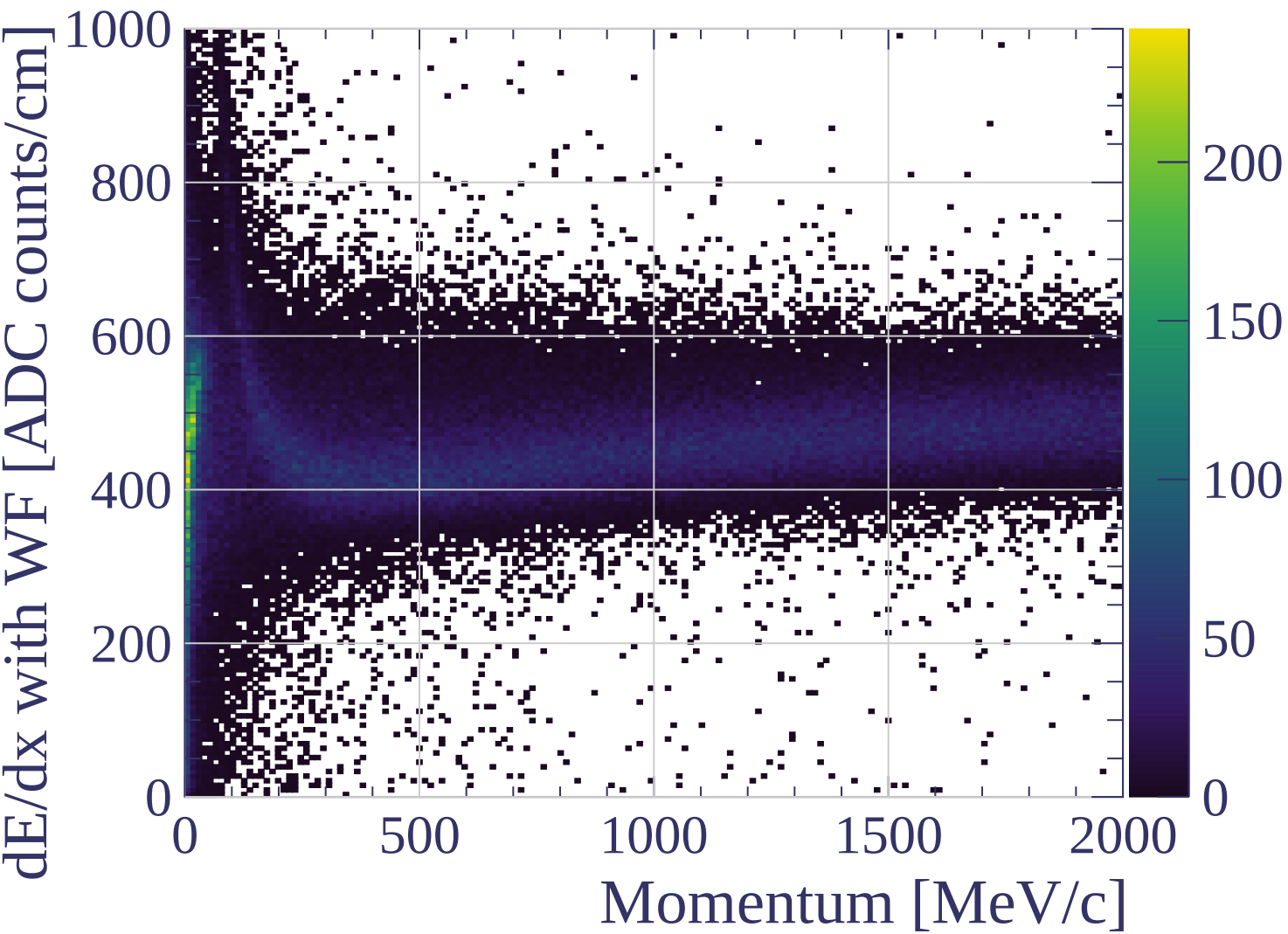


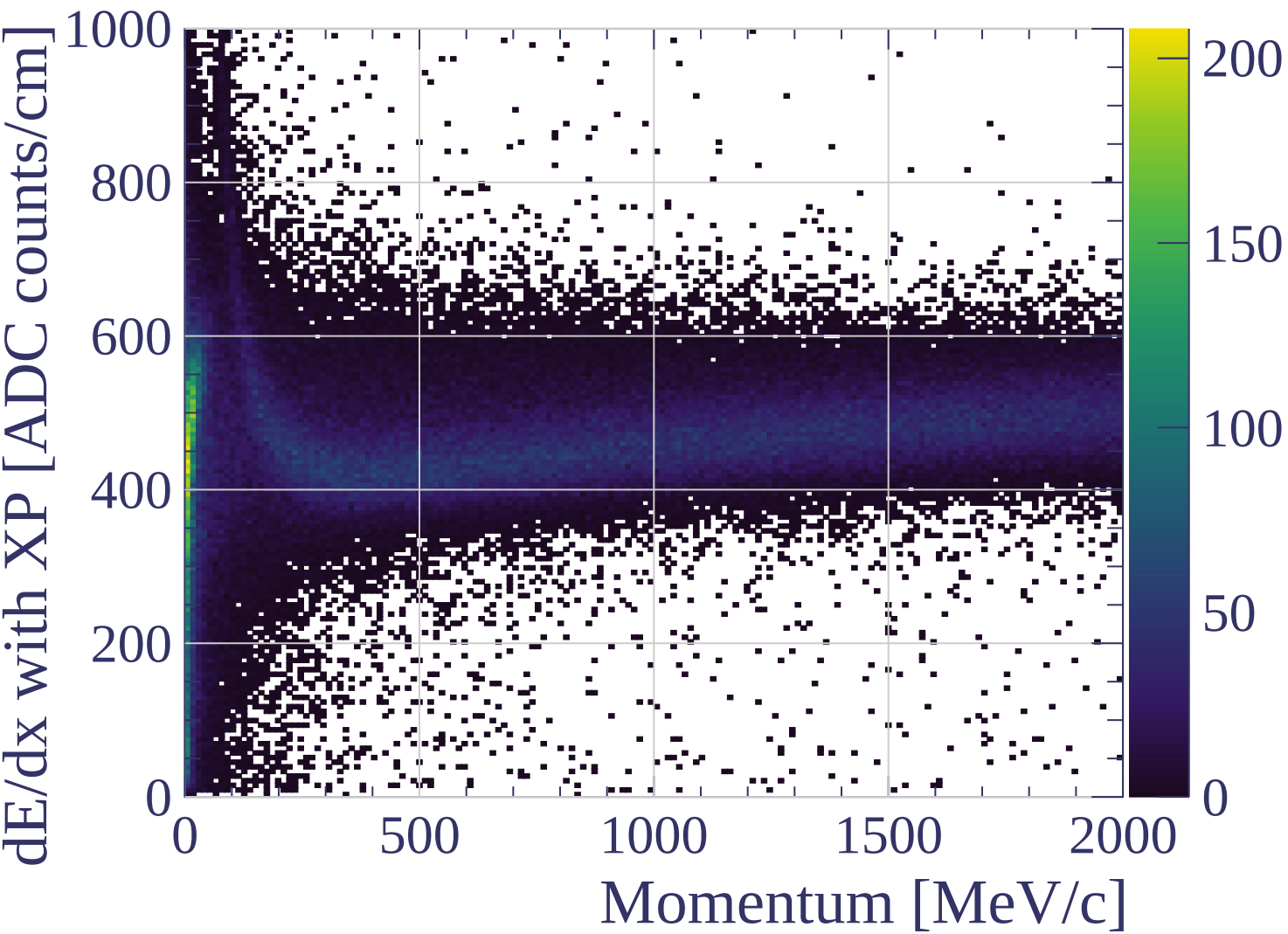




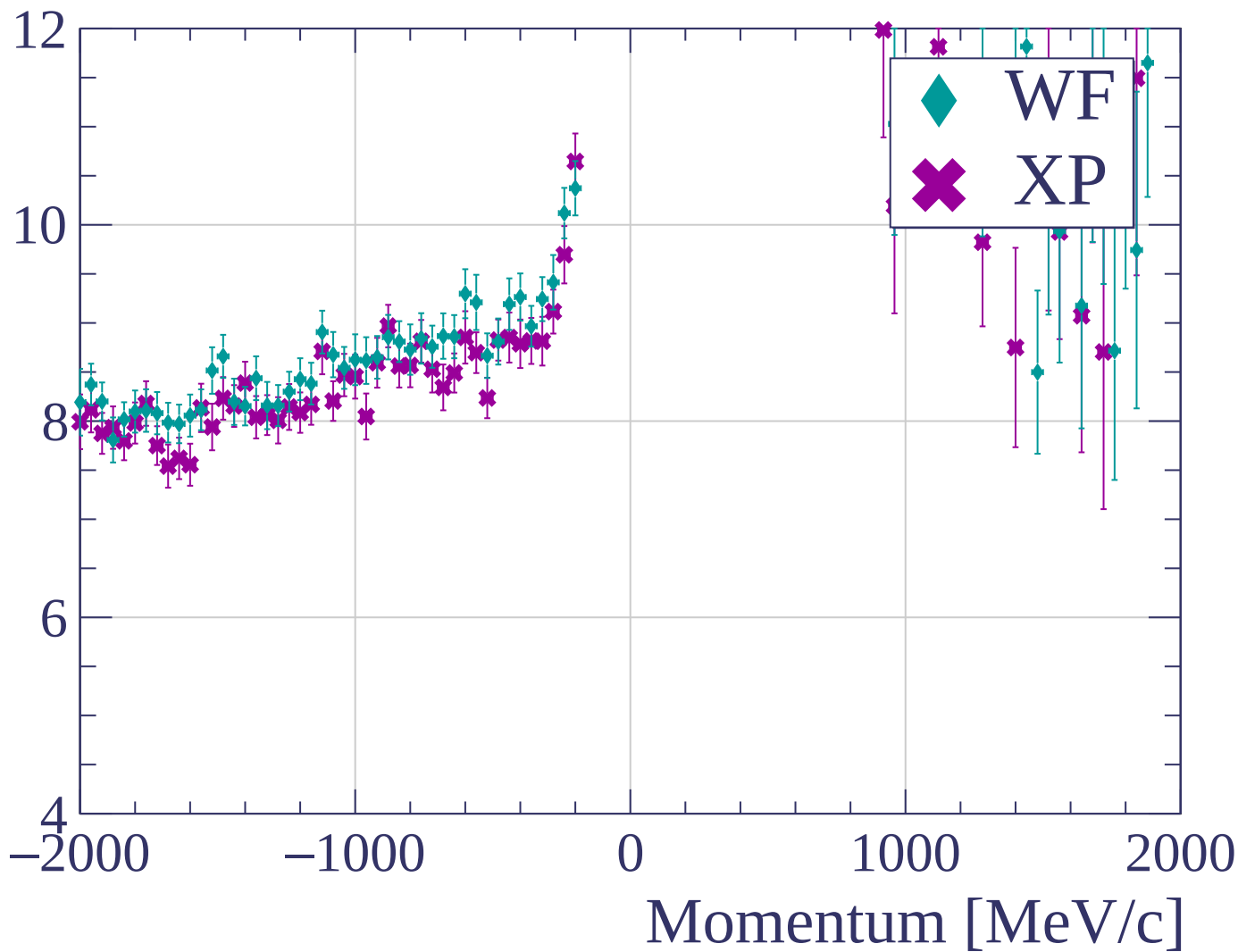


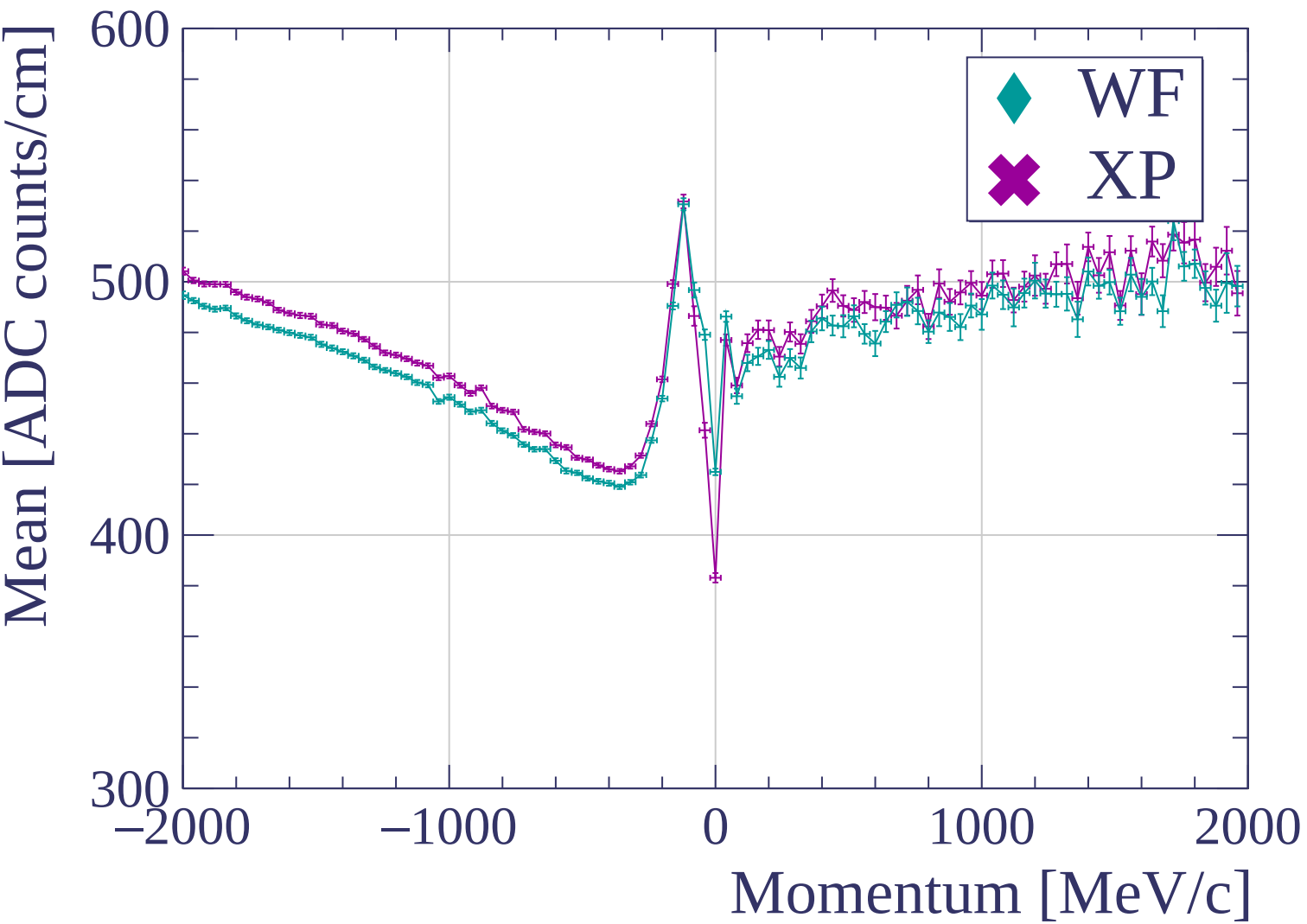


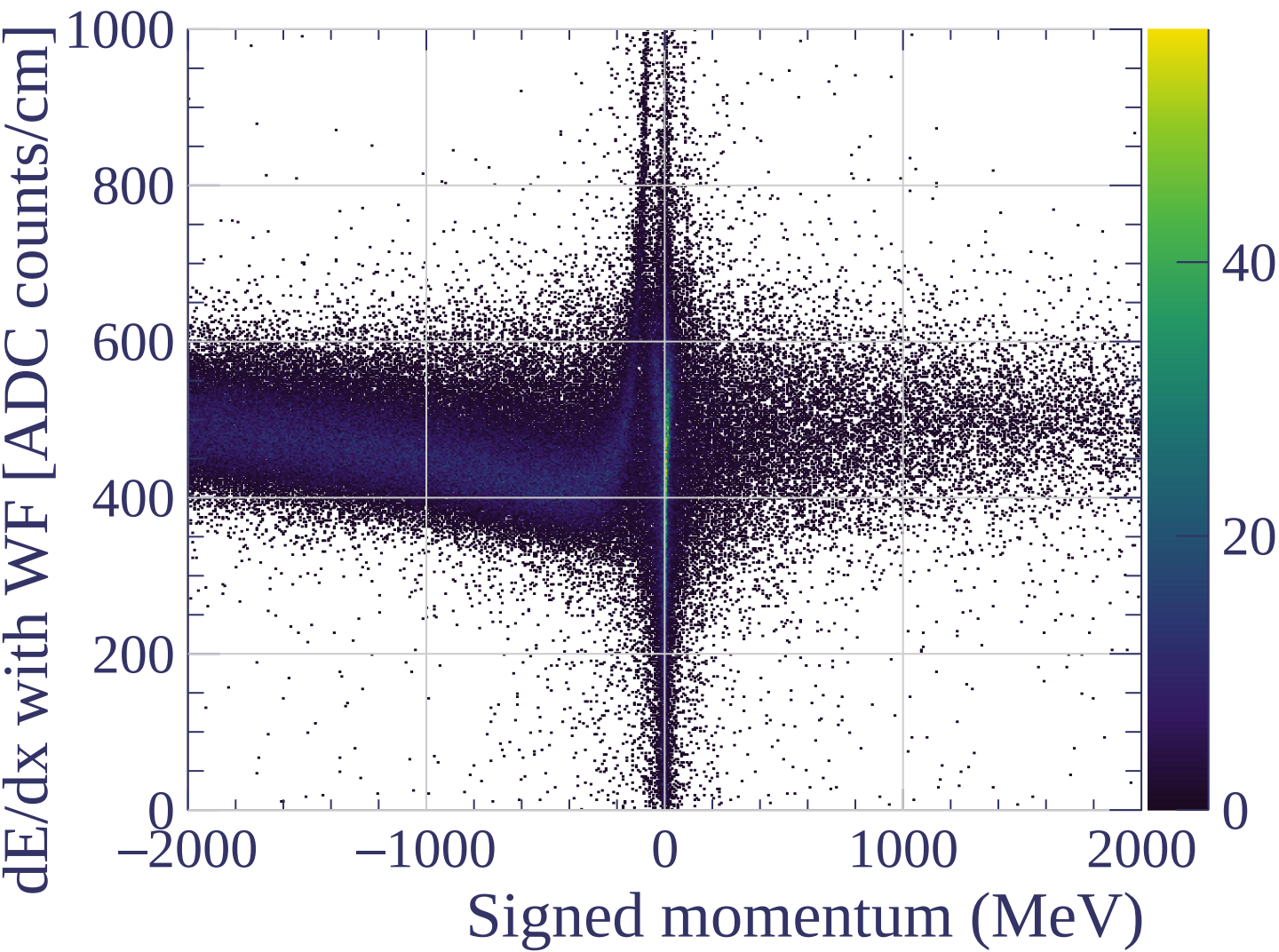


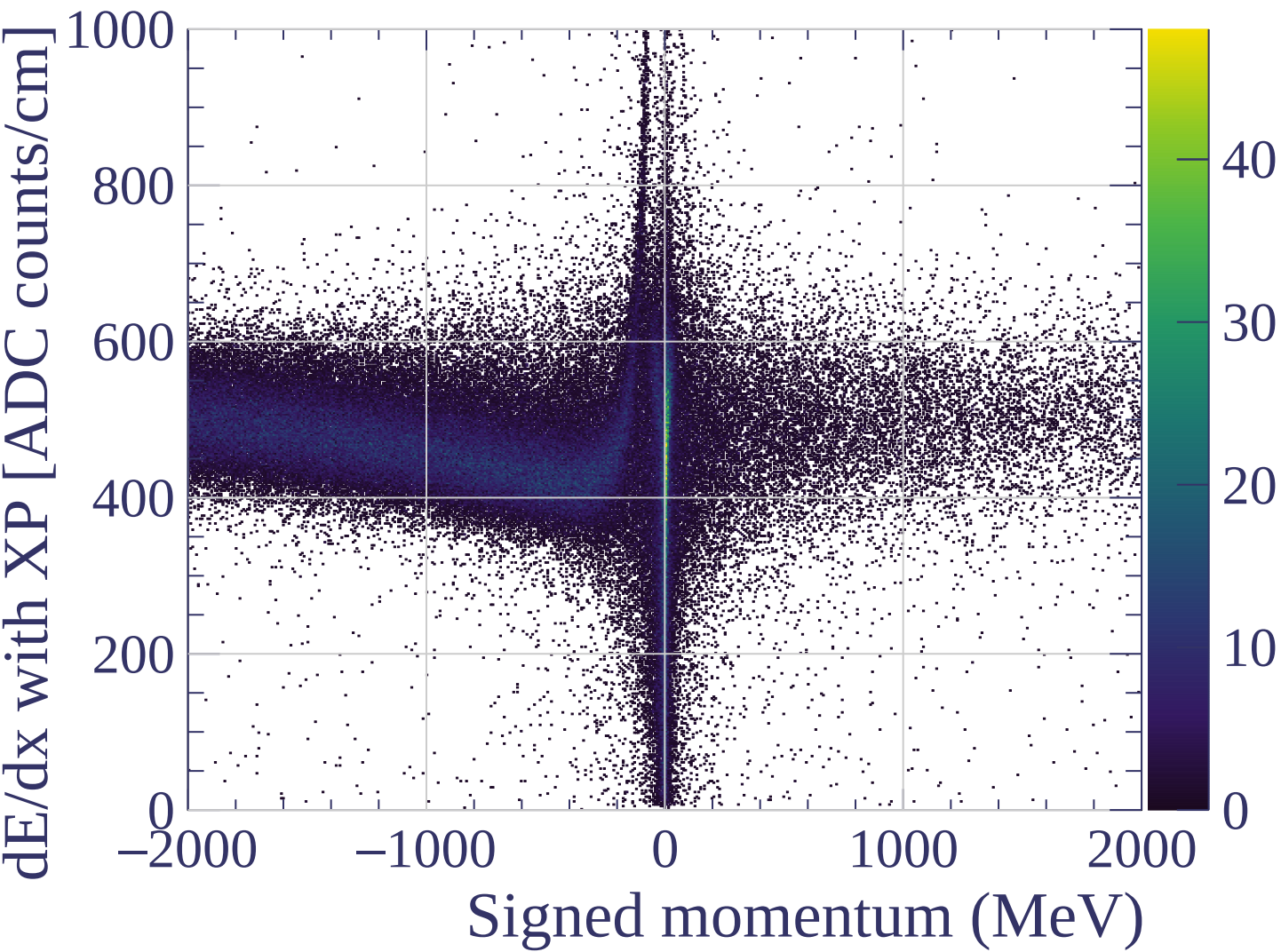


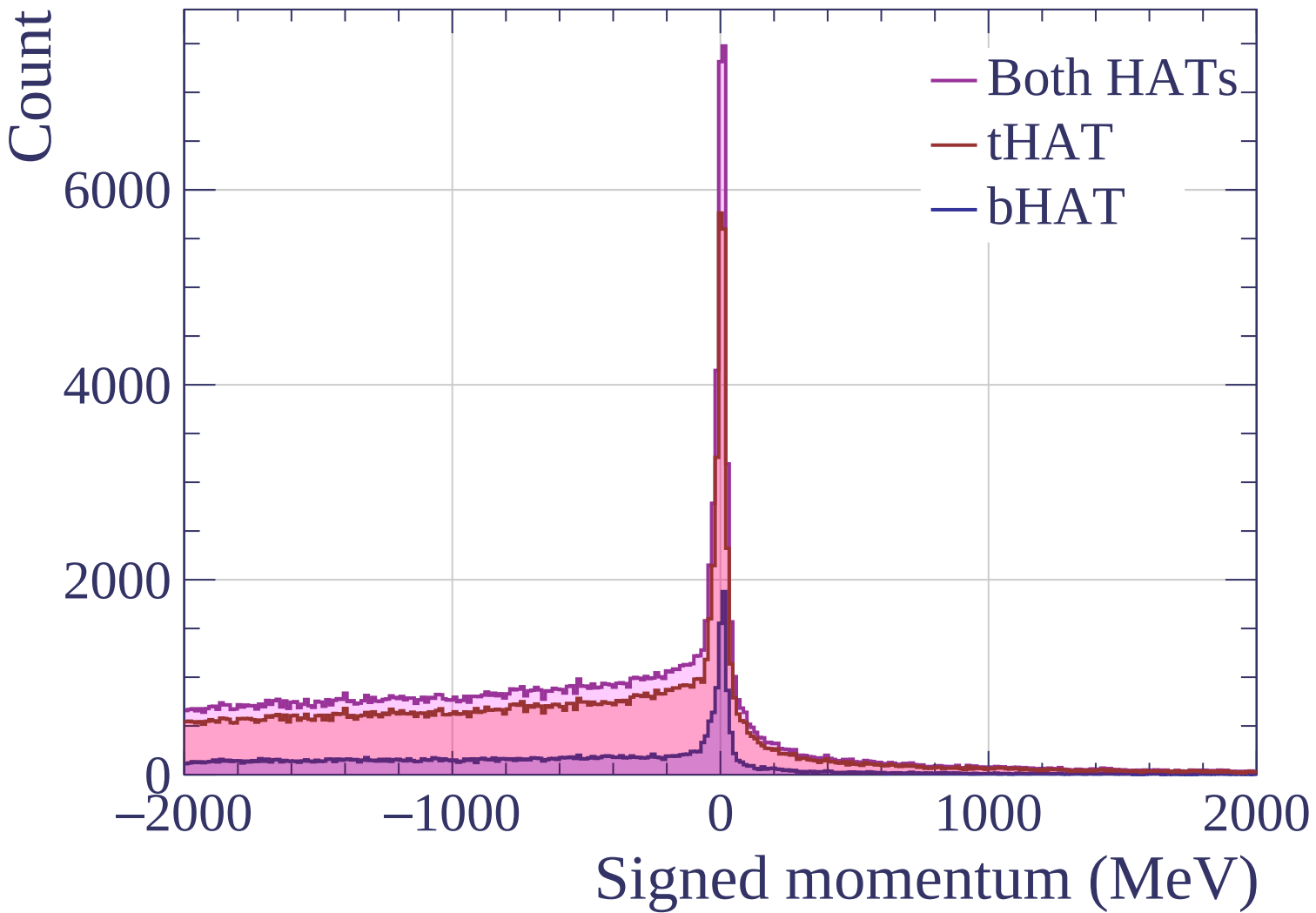
dE/dx resolution (%)

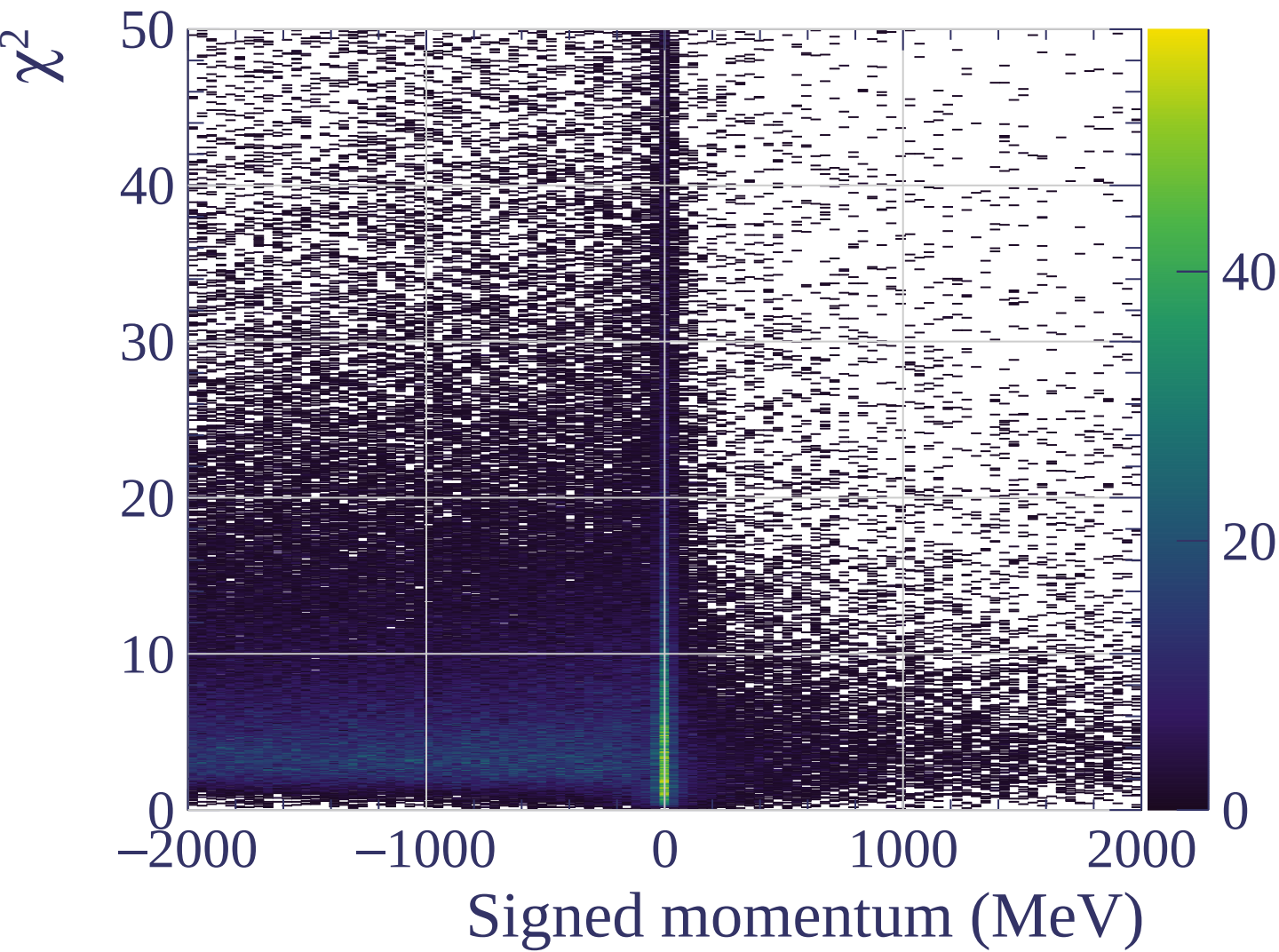


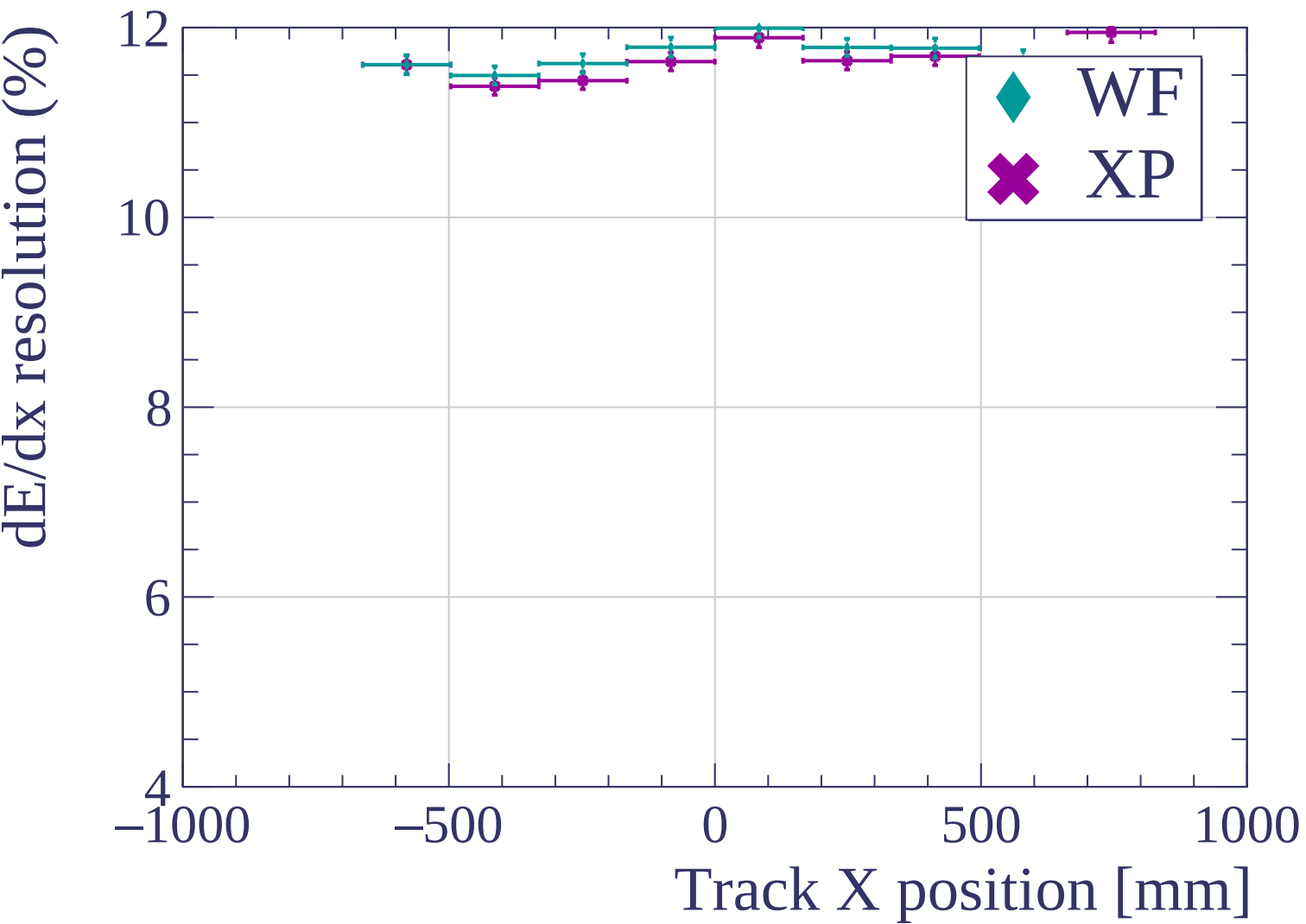


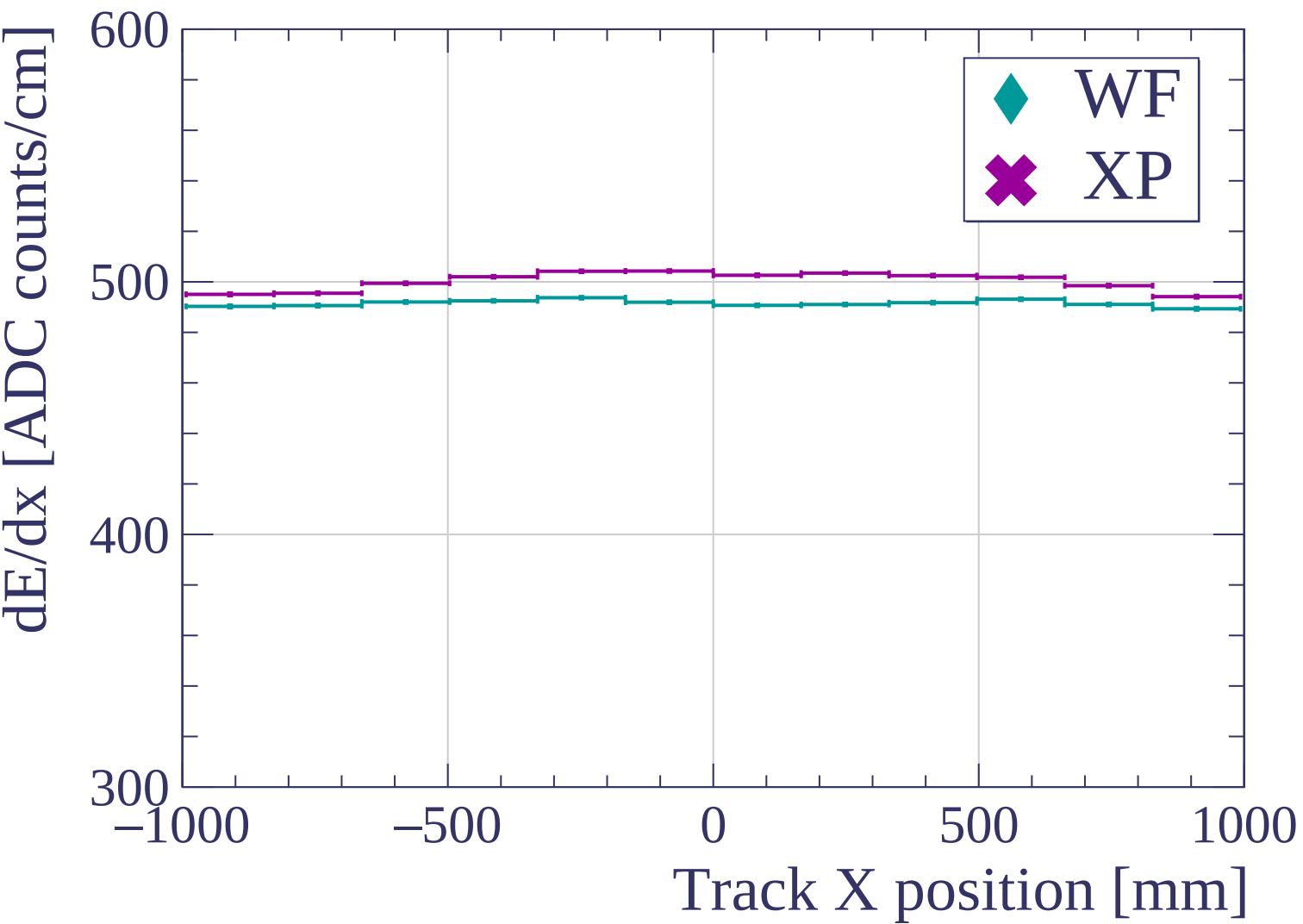


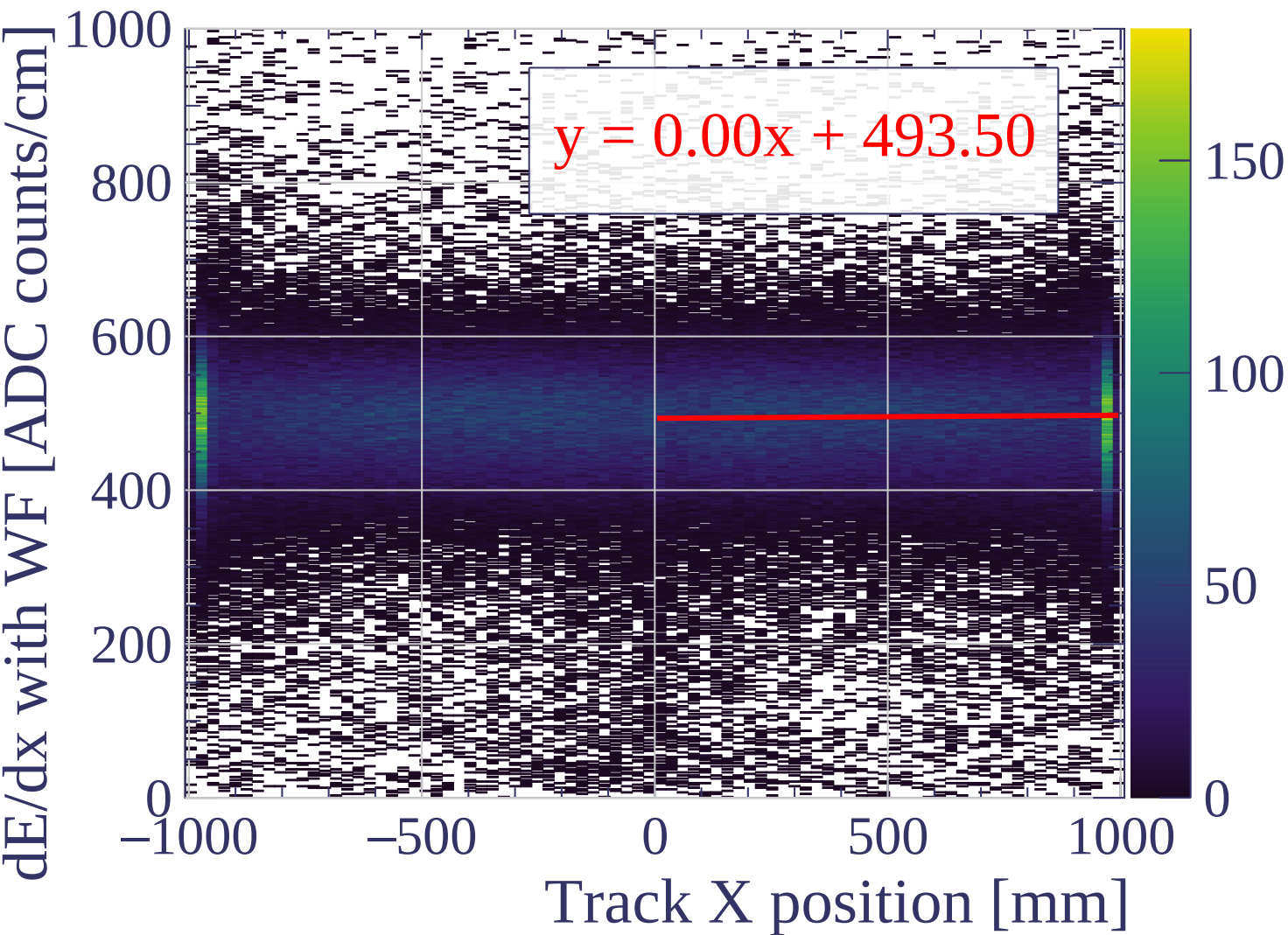


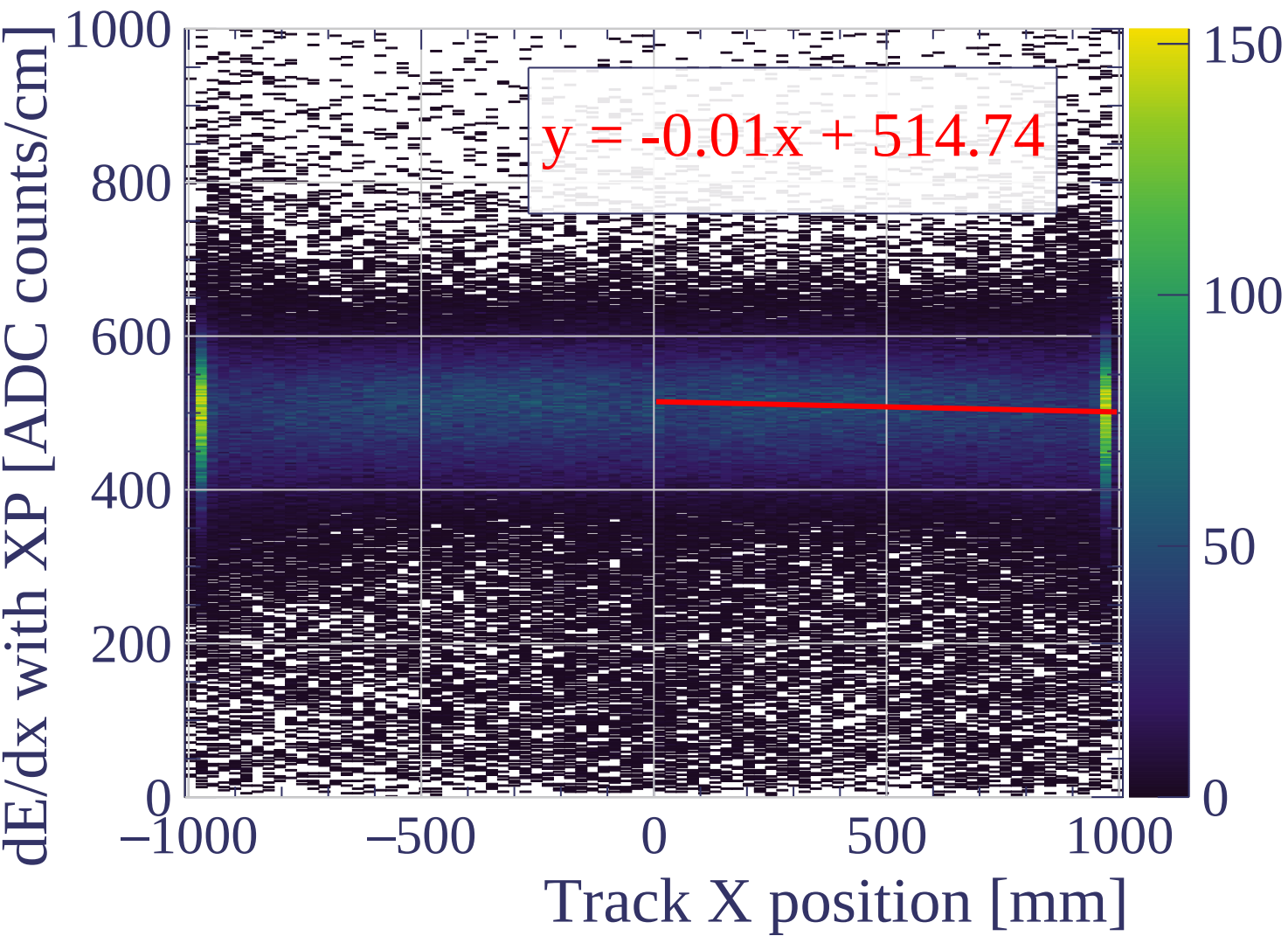


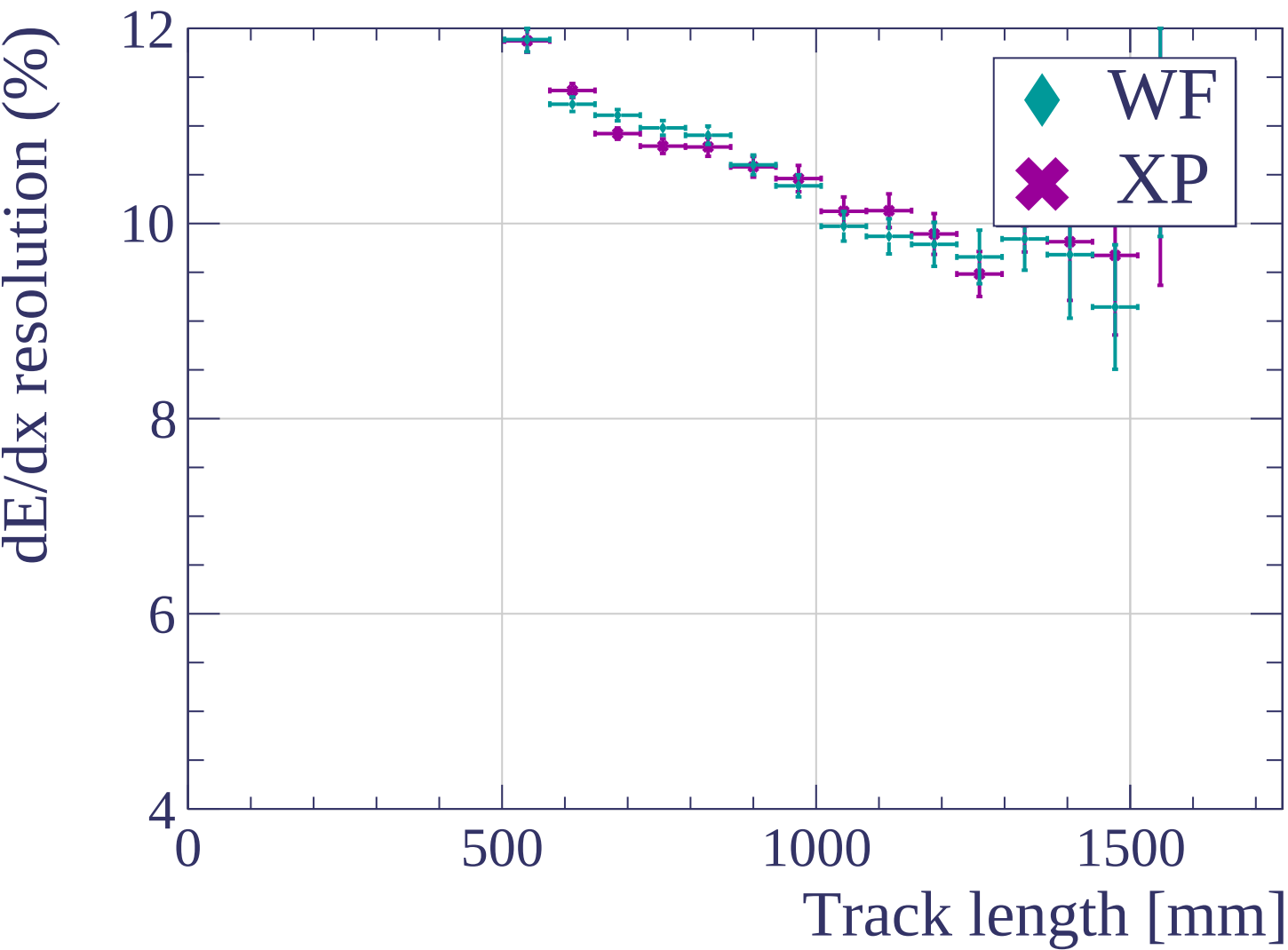


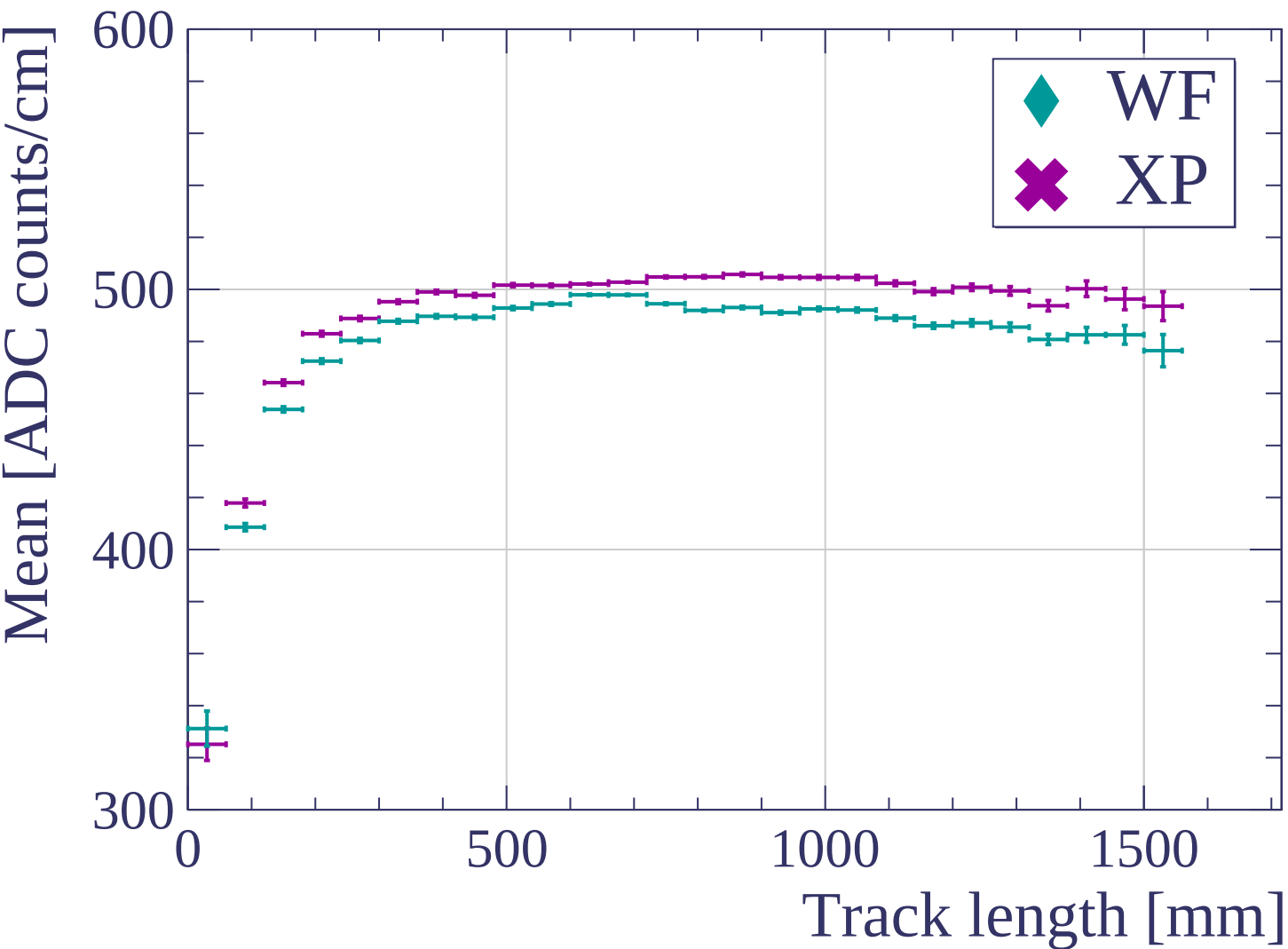


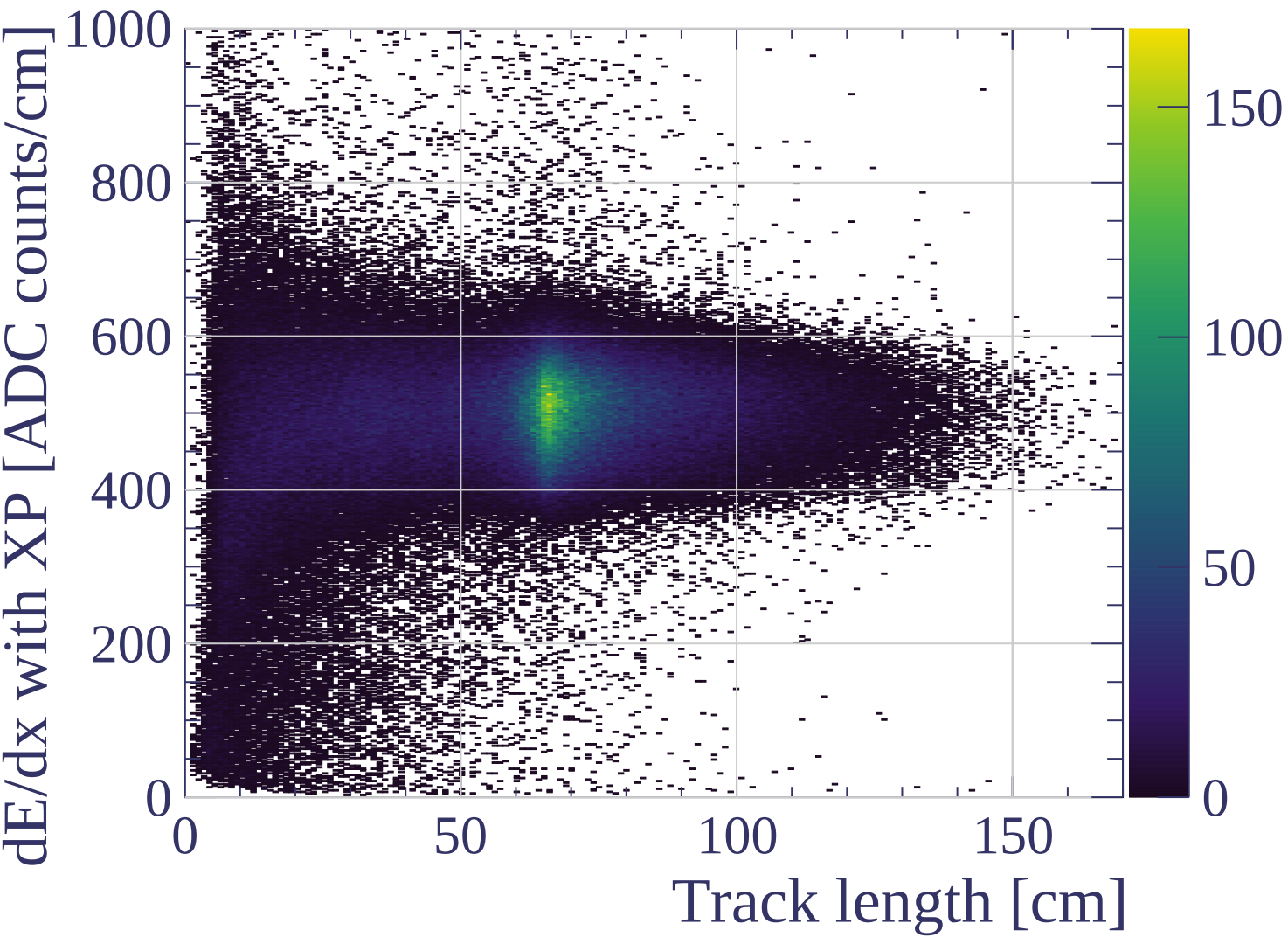


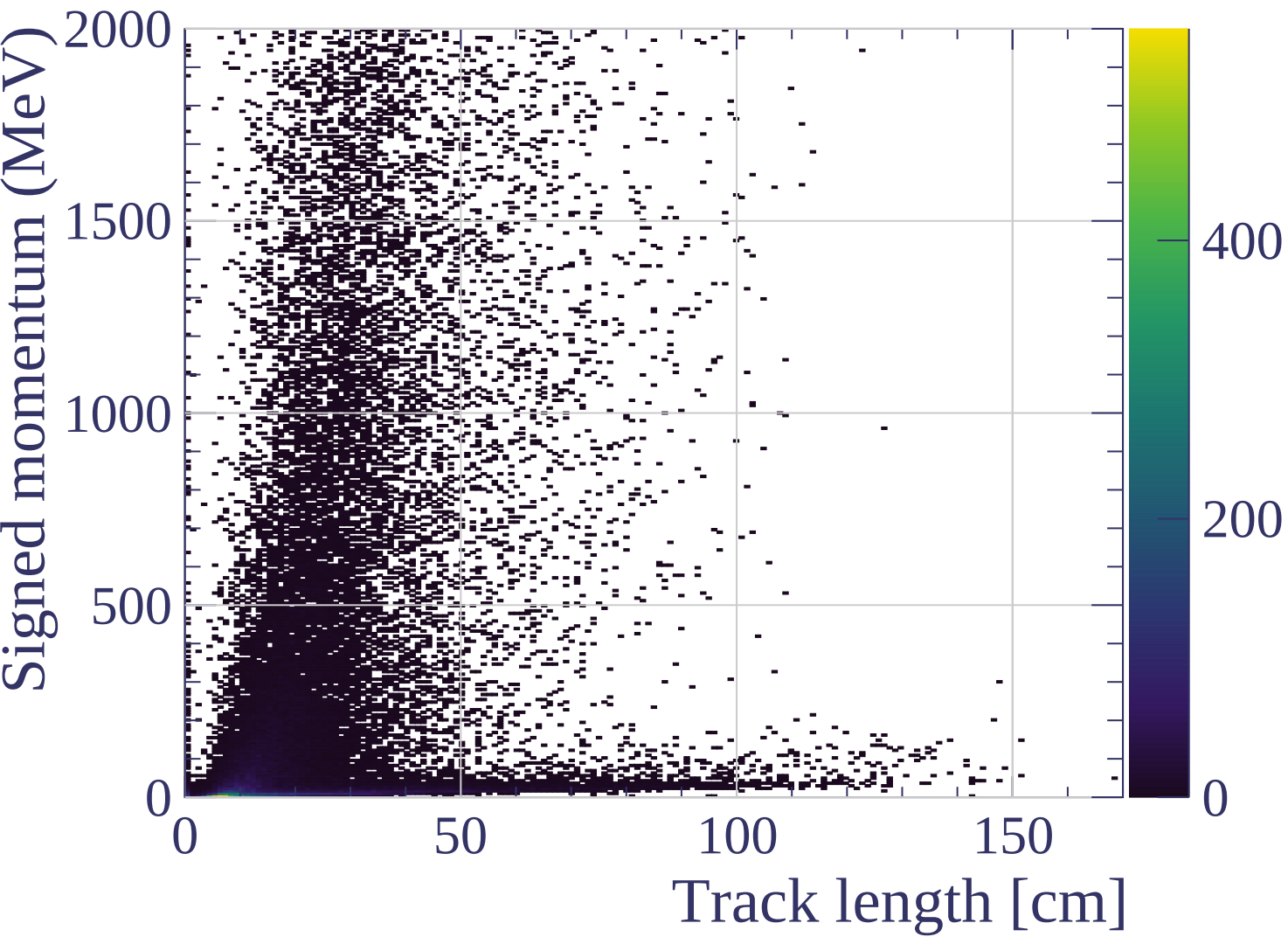


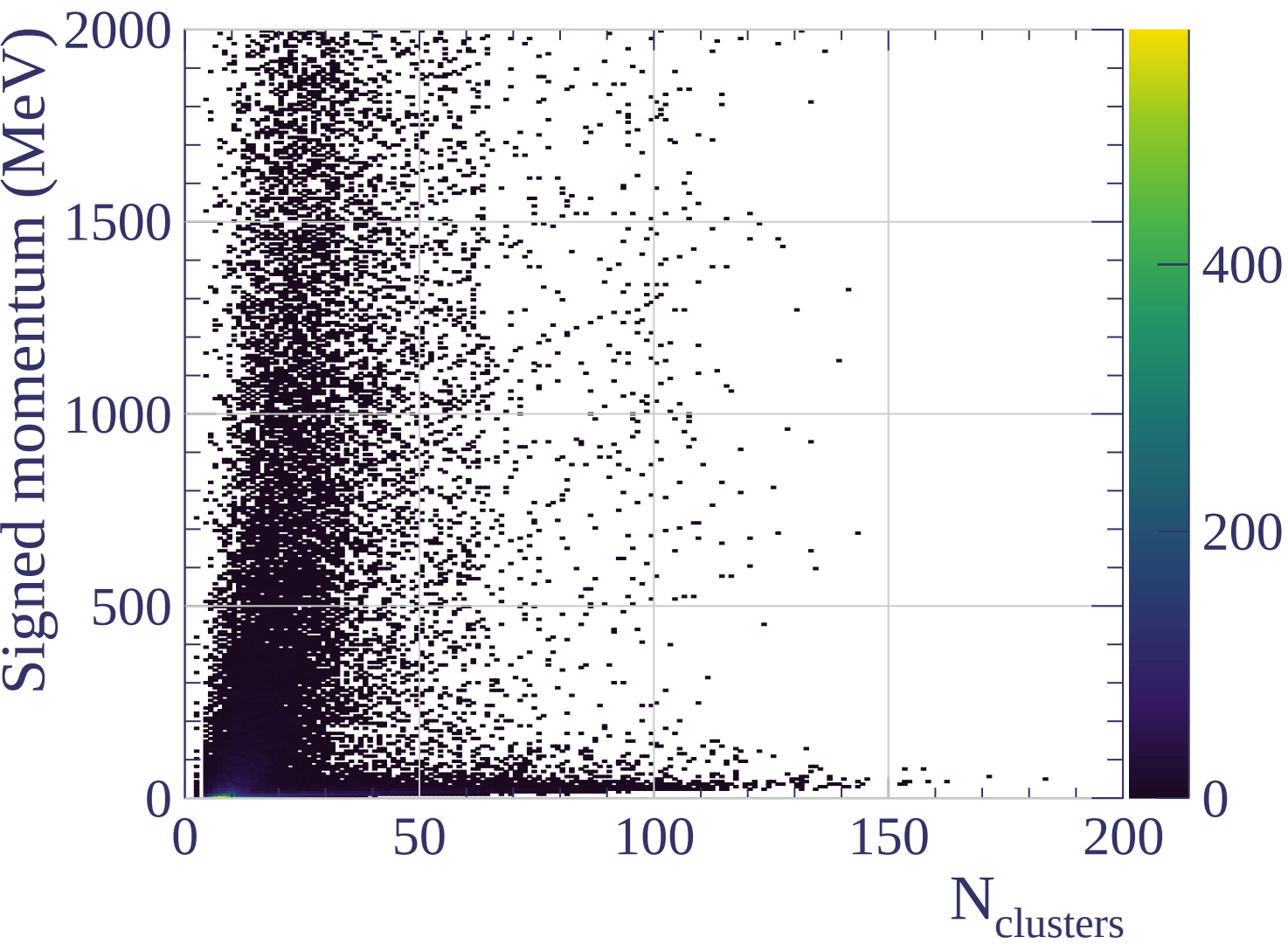


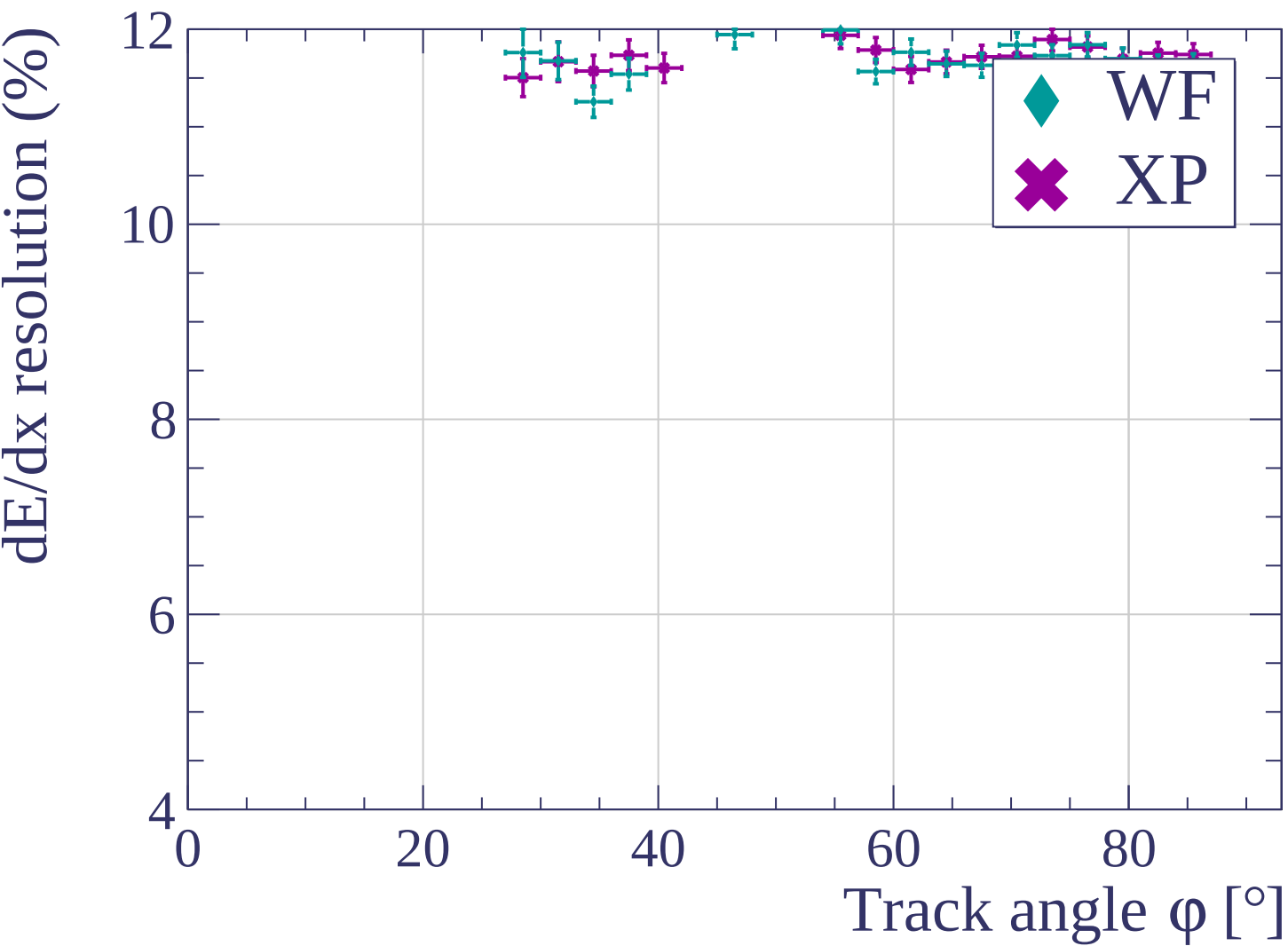


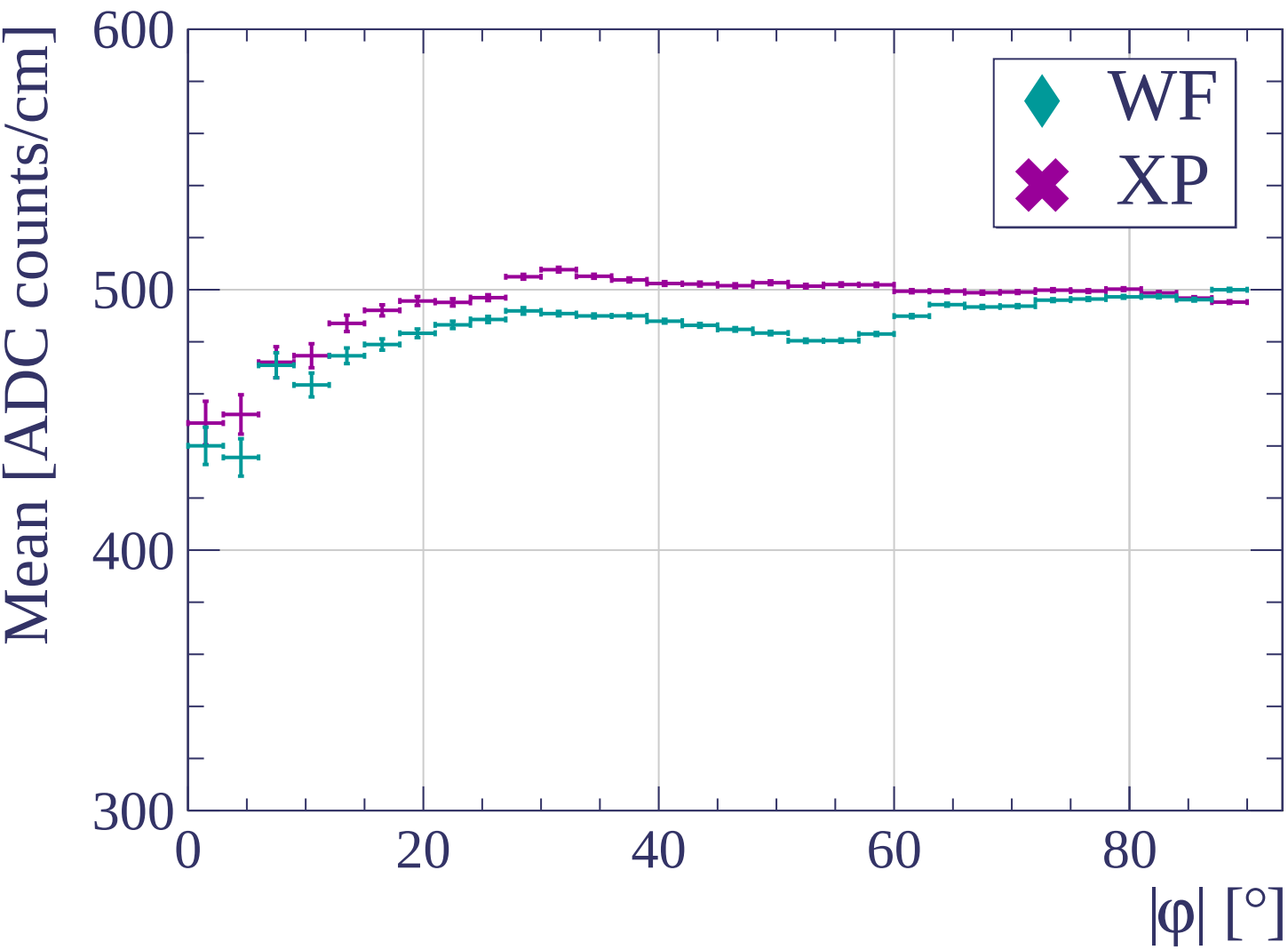




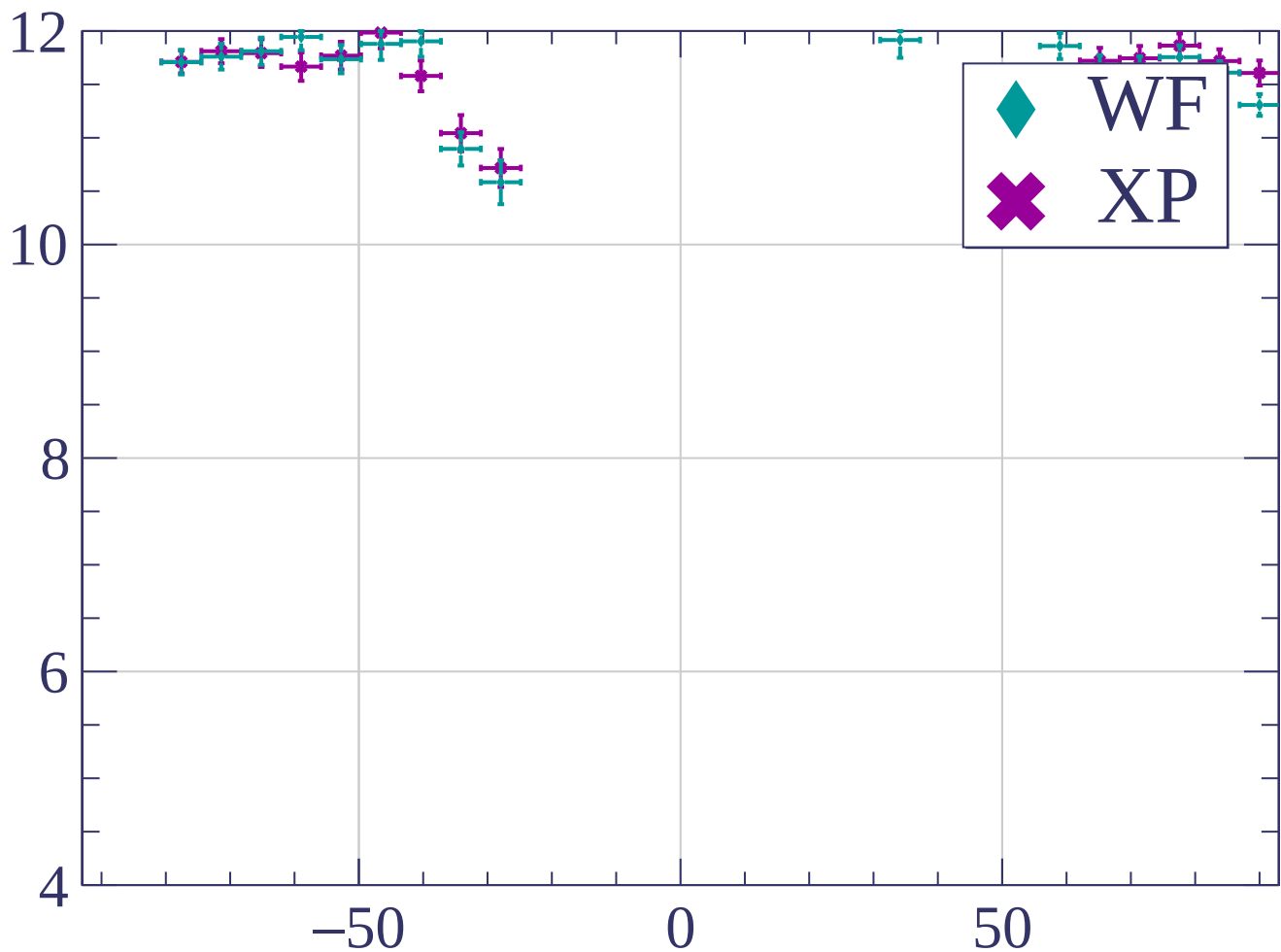




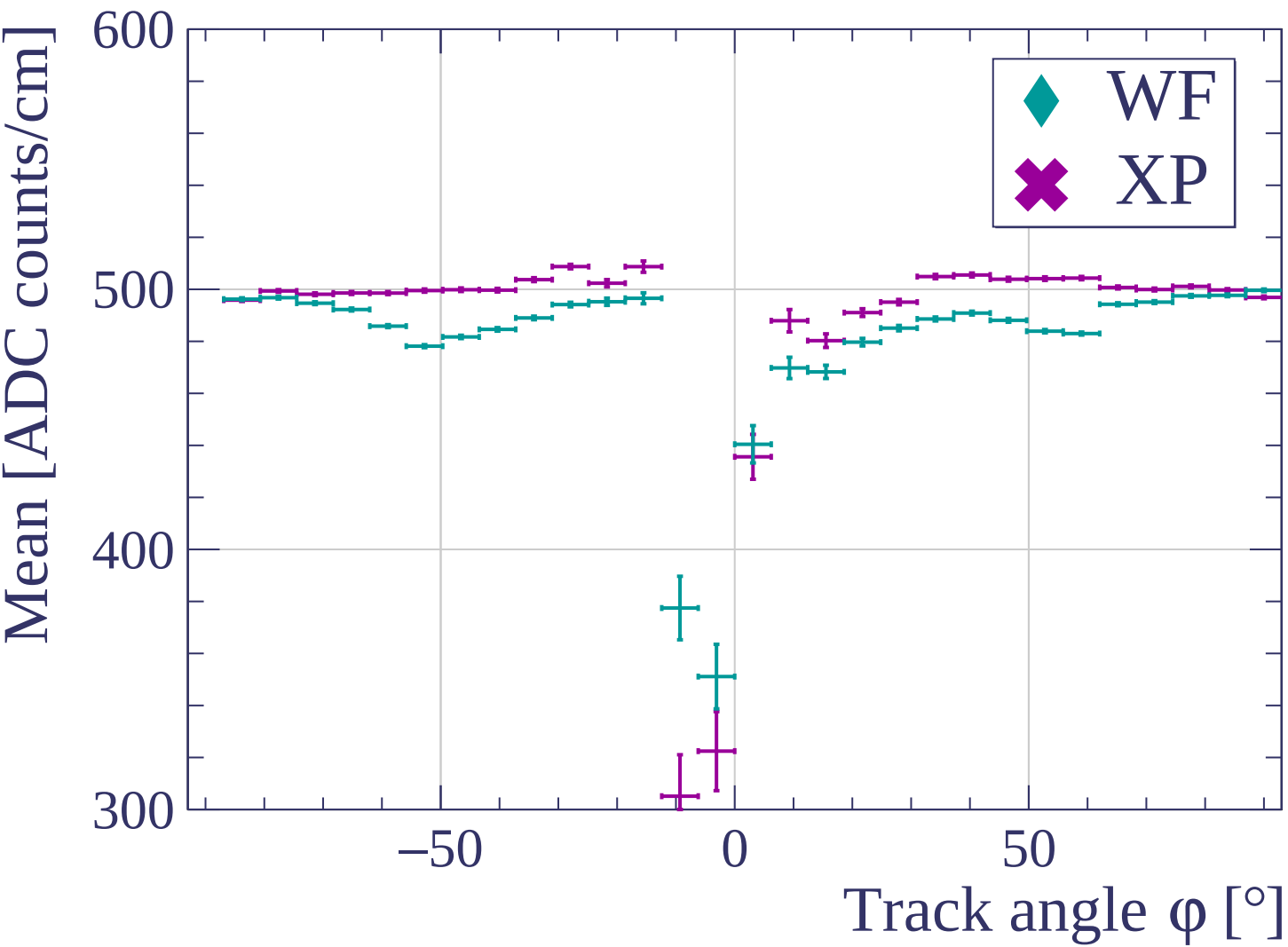


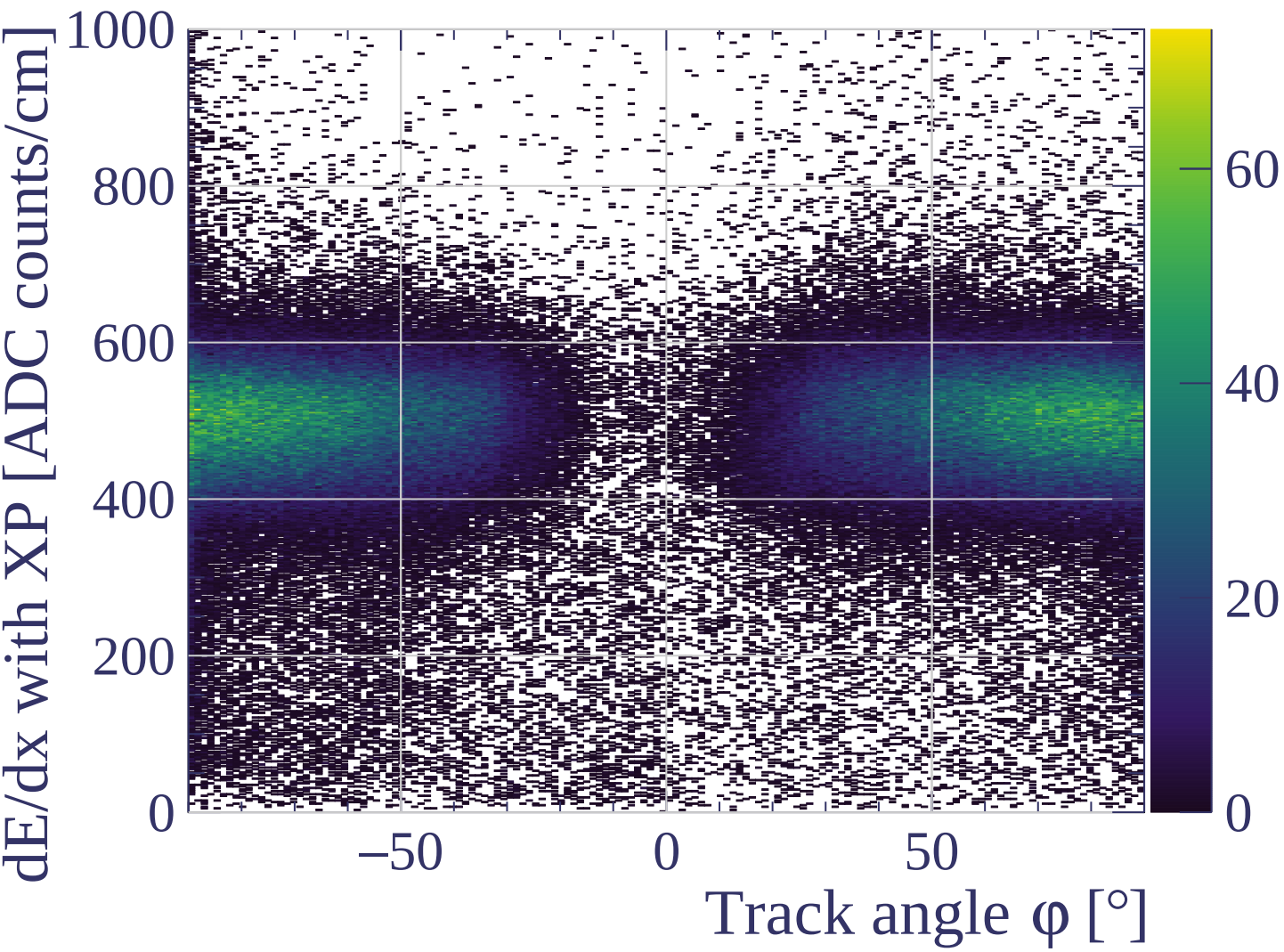


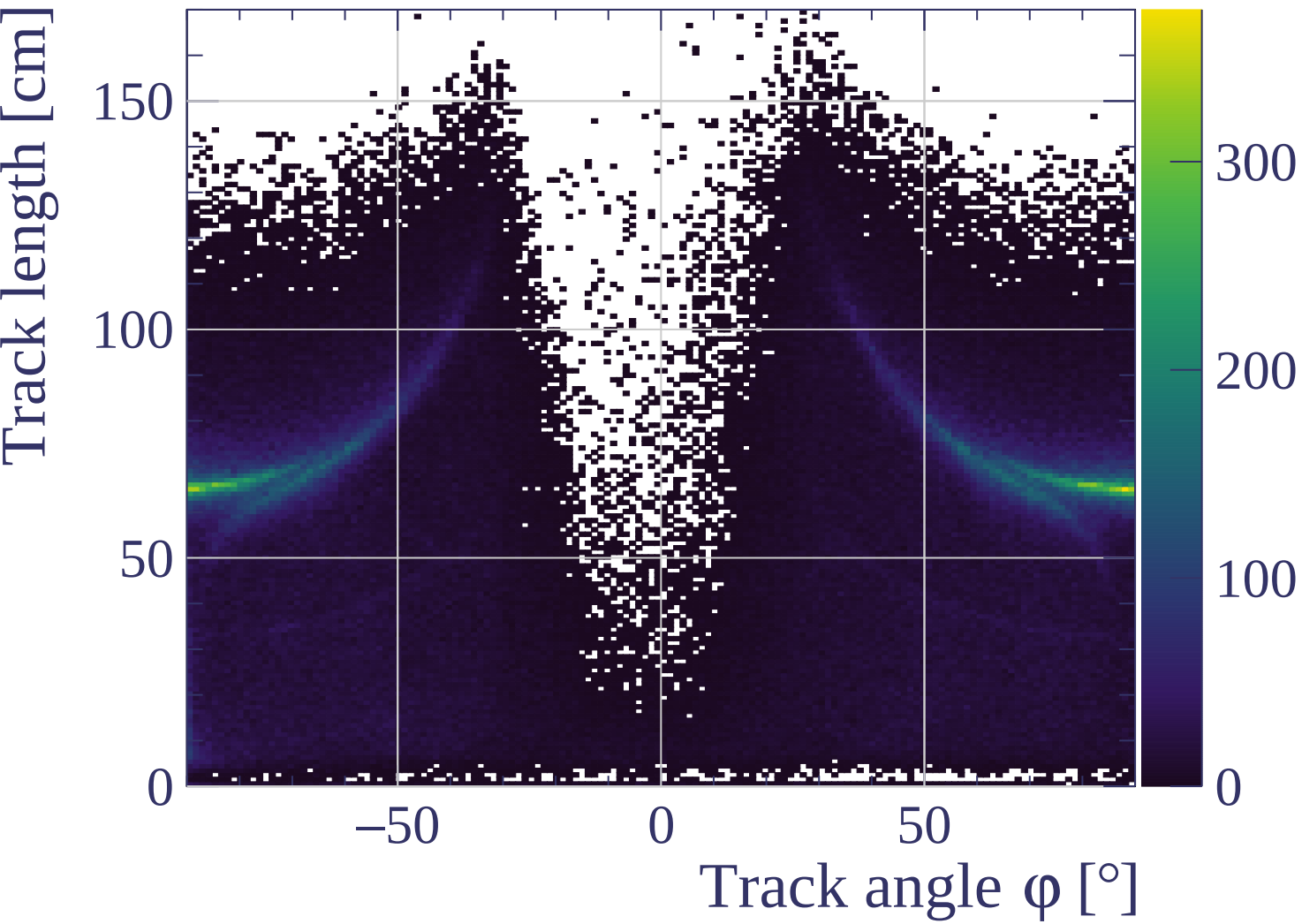
dE/dx resolution (%)

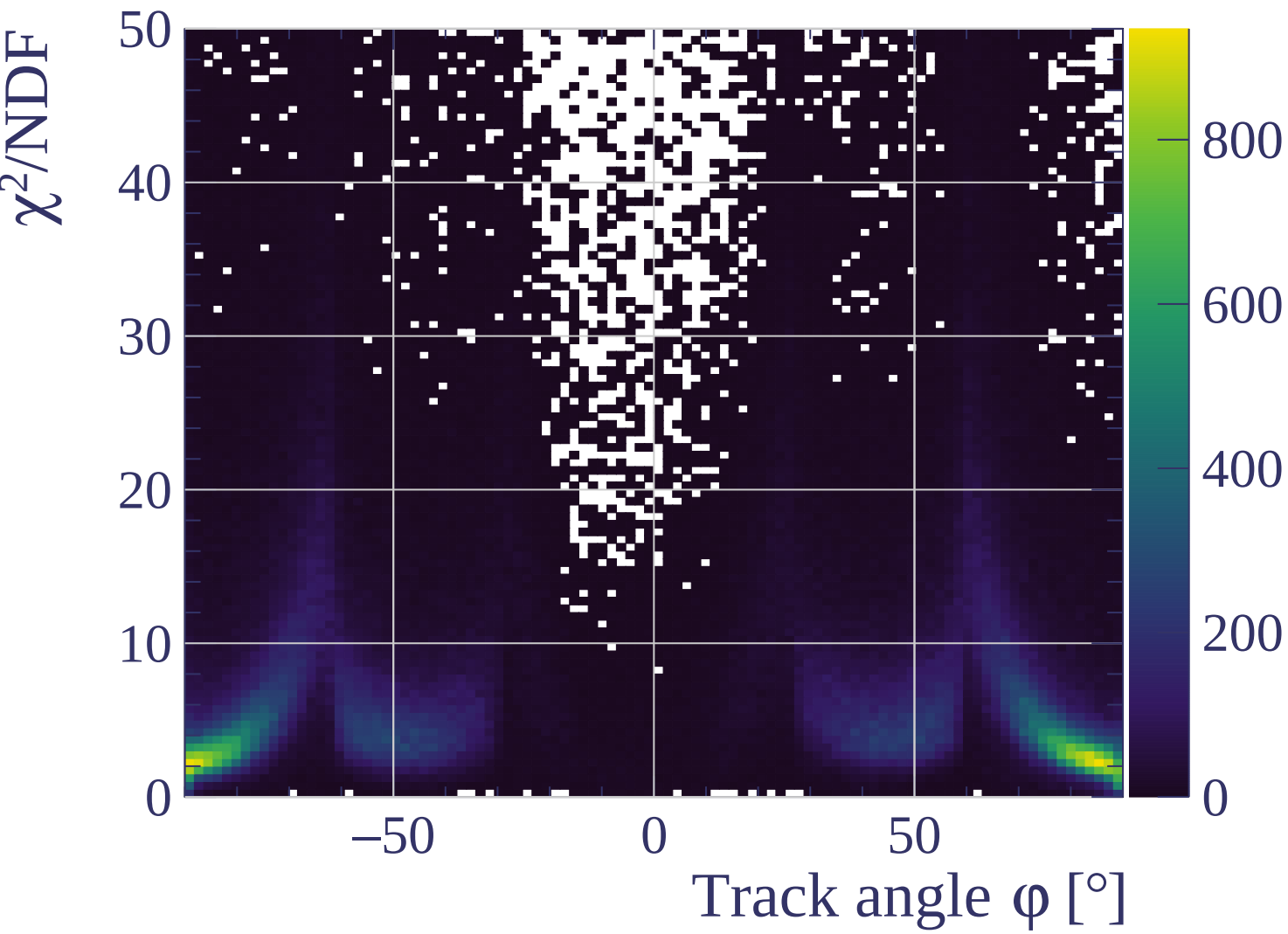


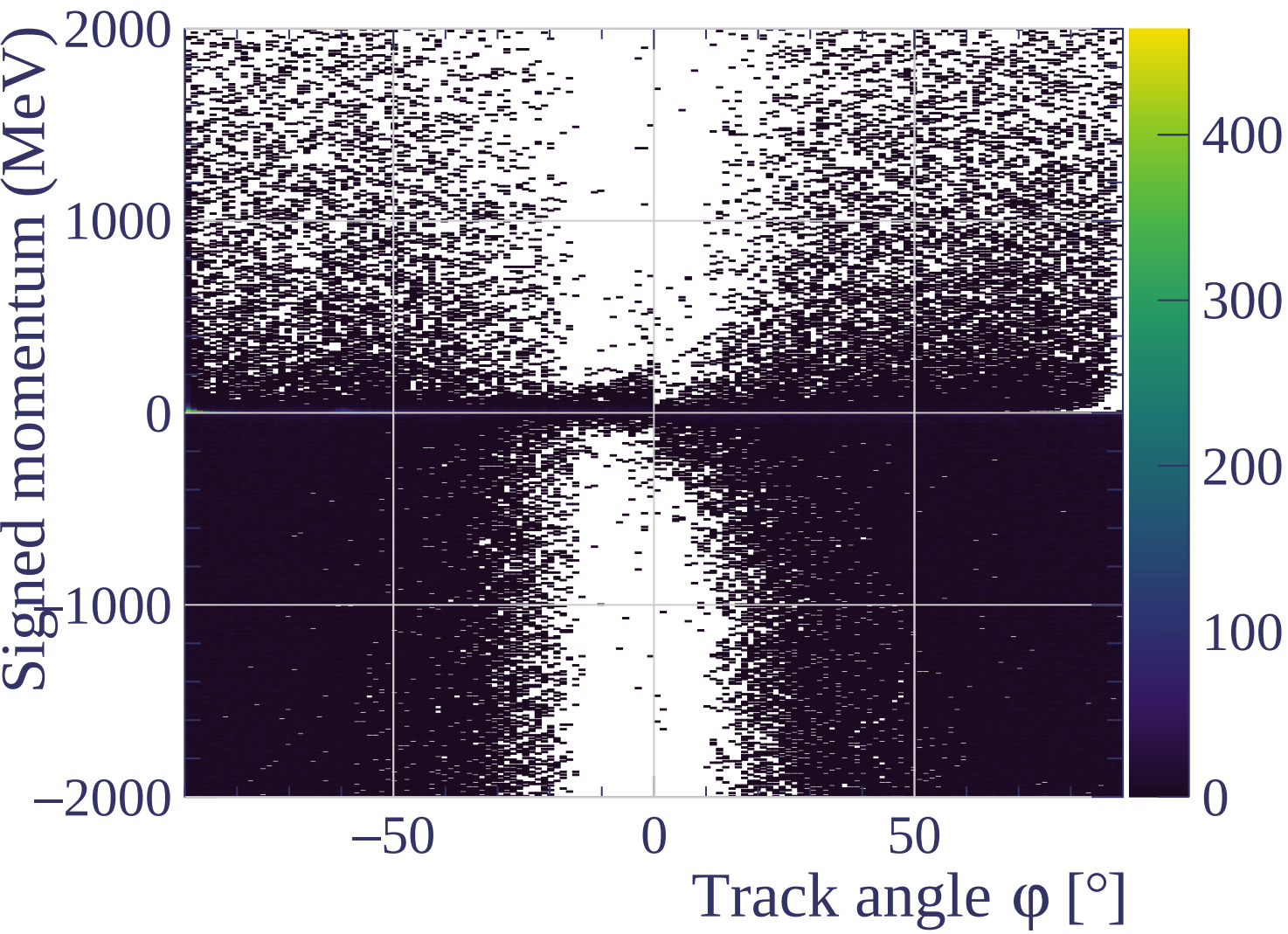
Track angle ϕ [°]

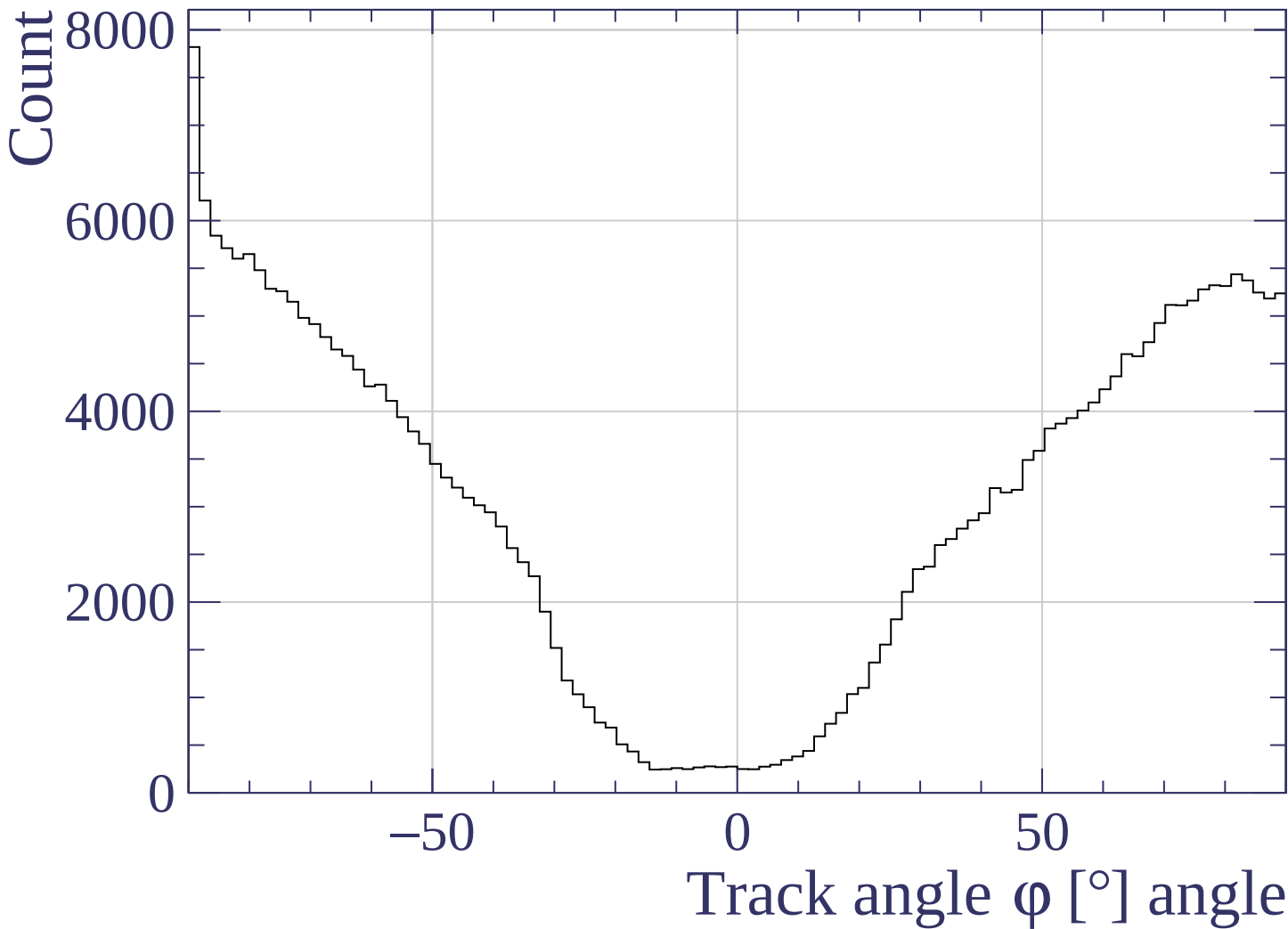


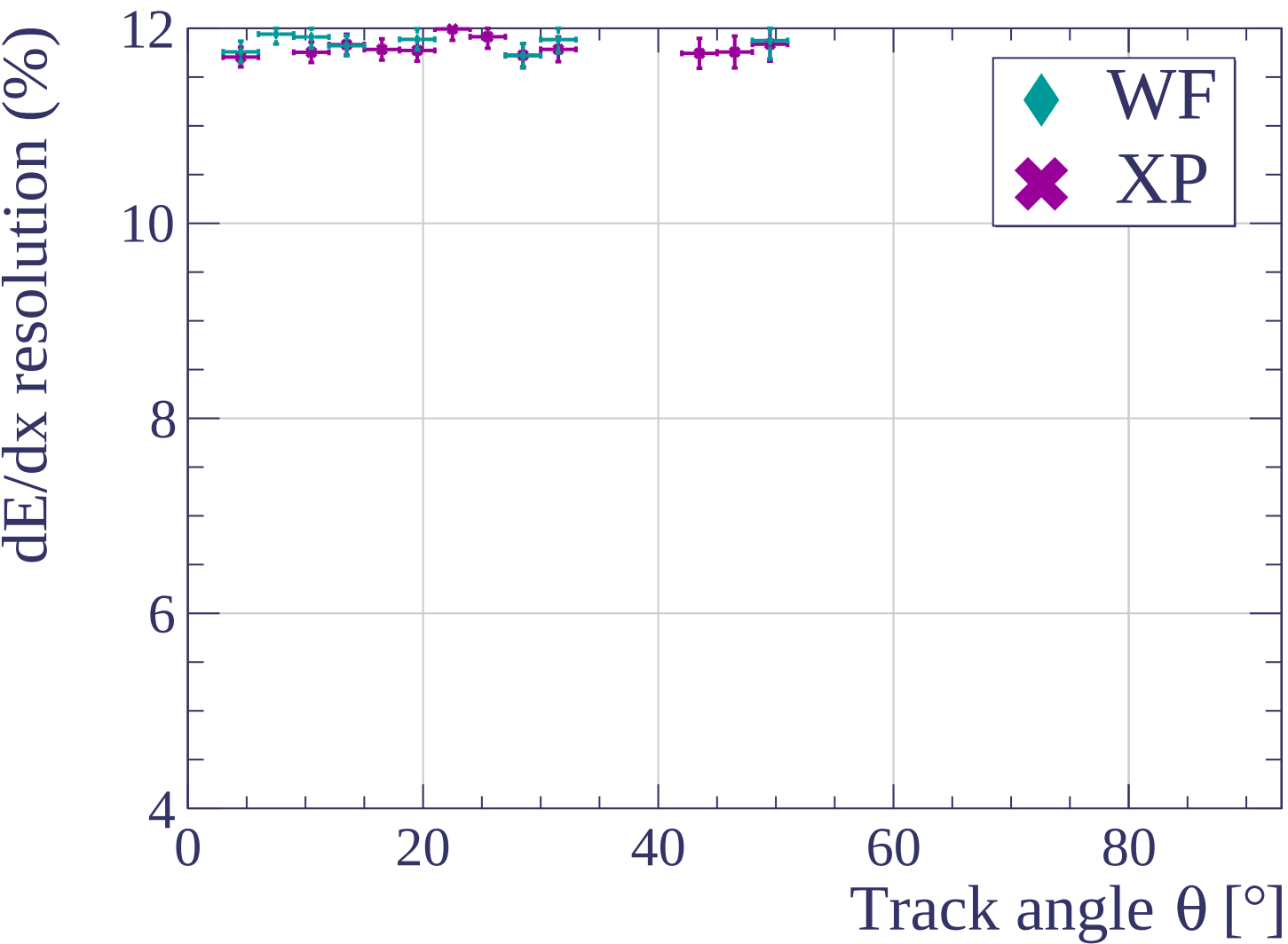


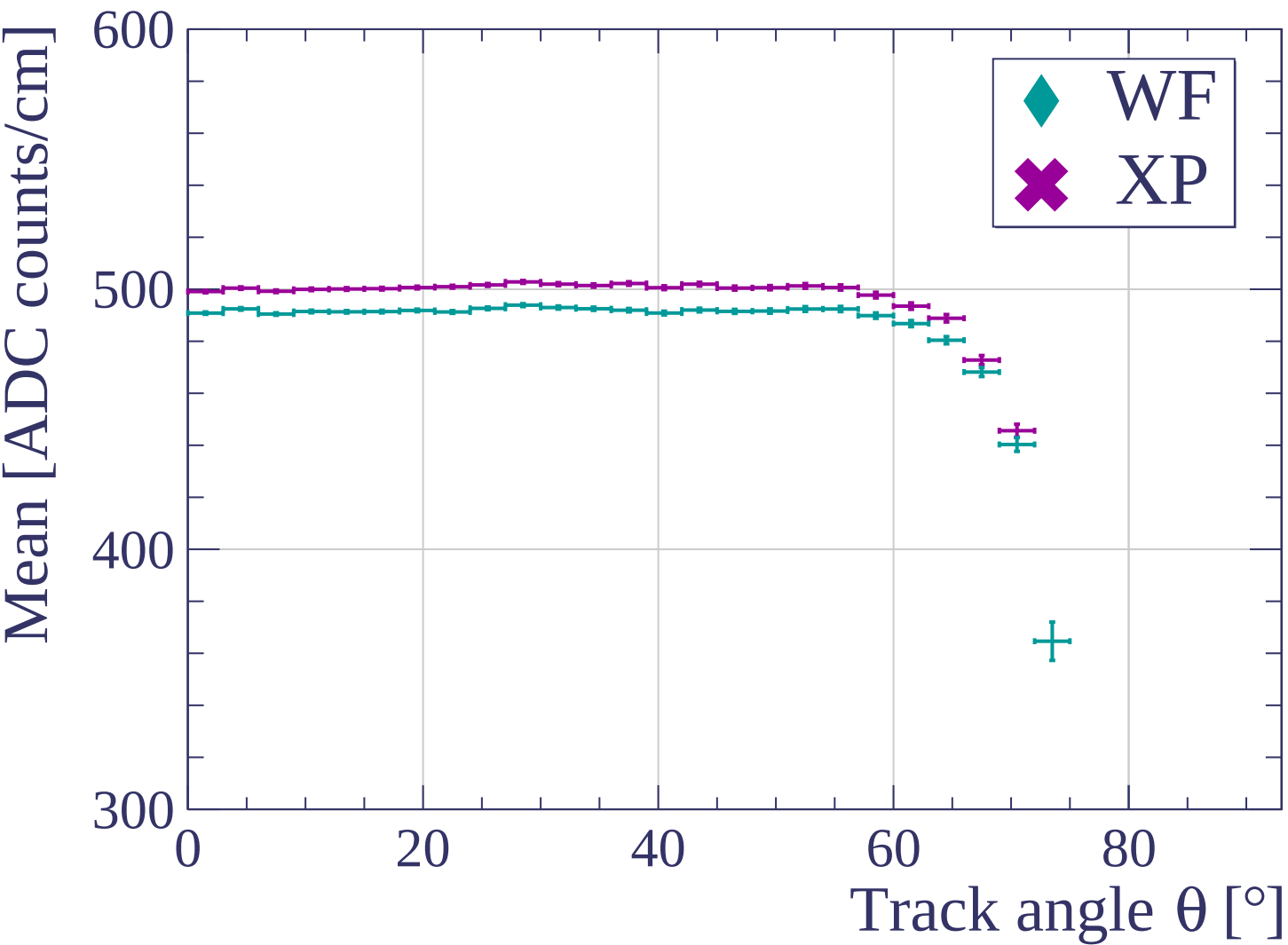


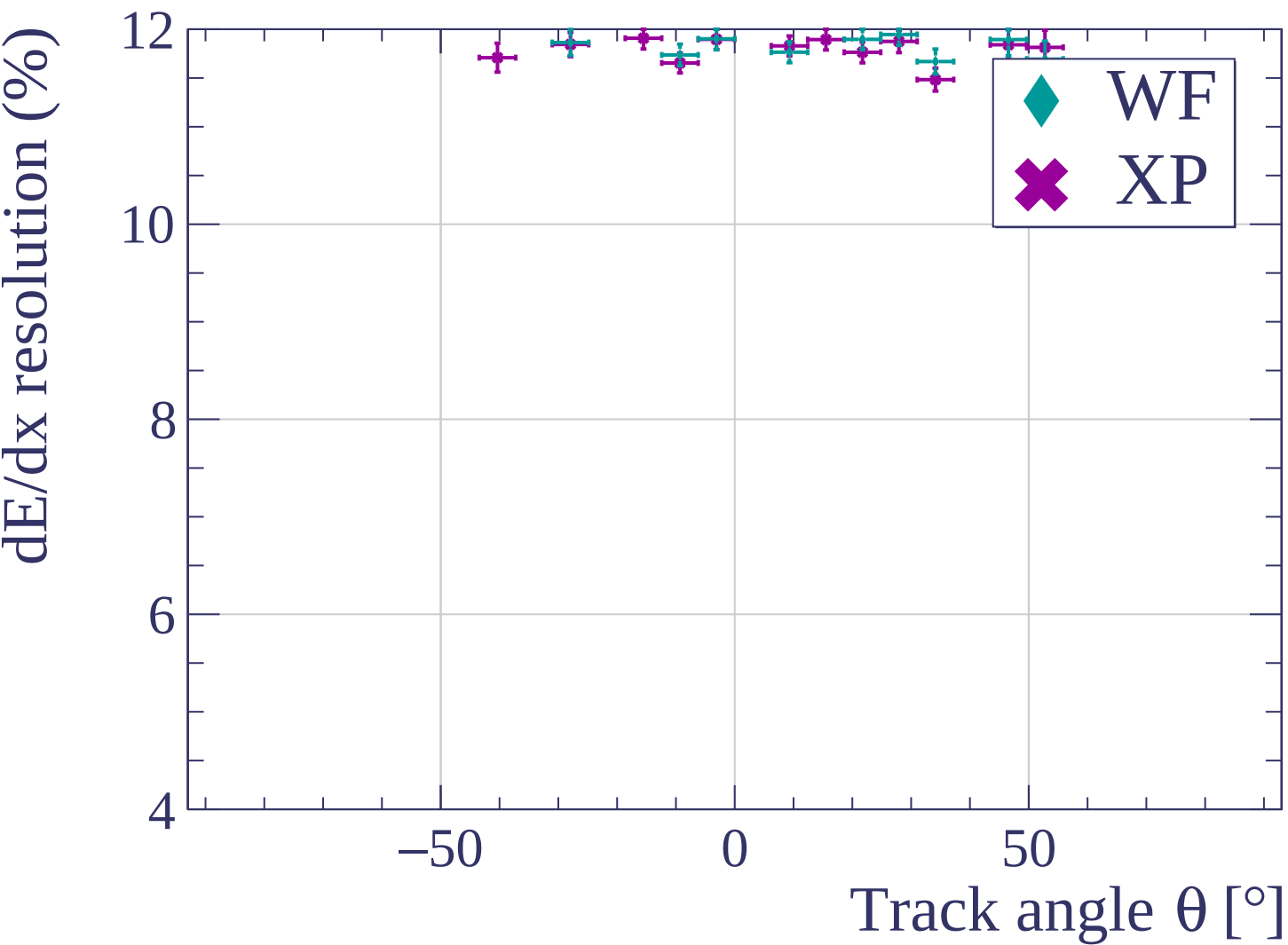


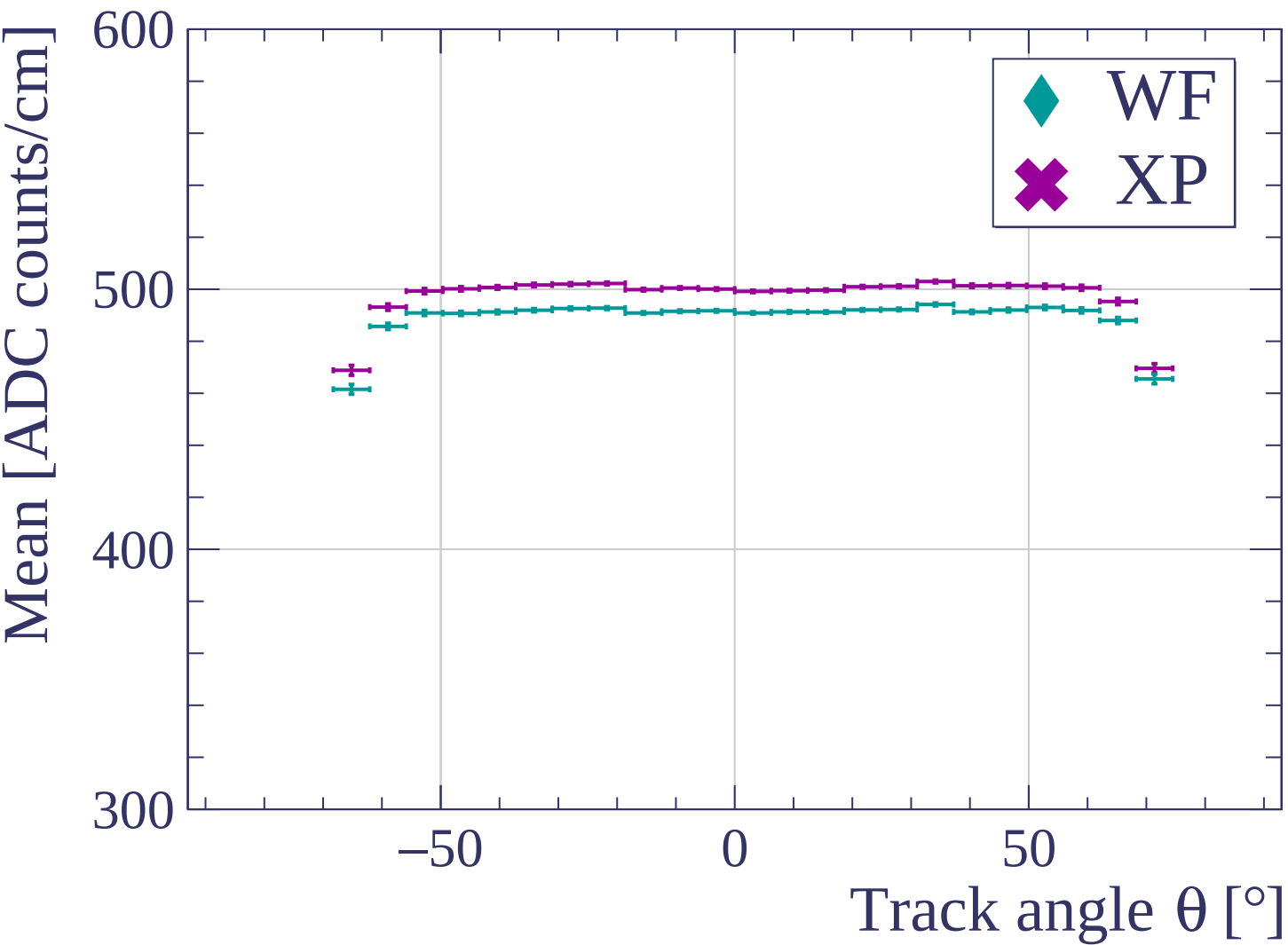


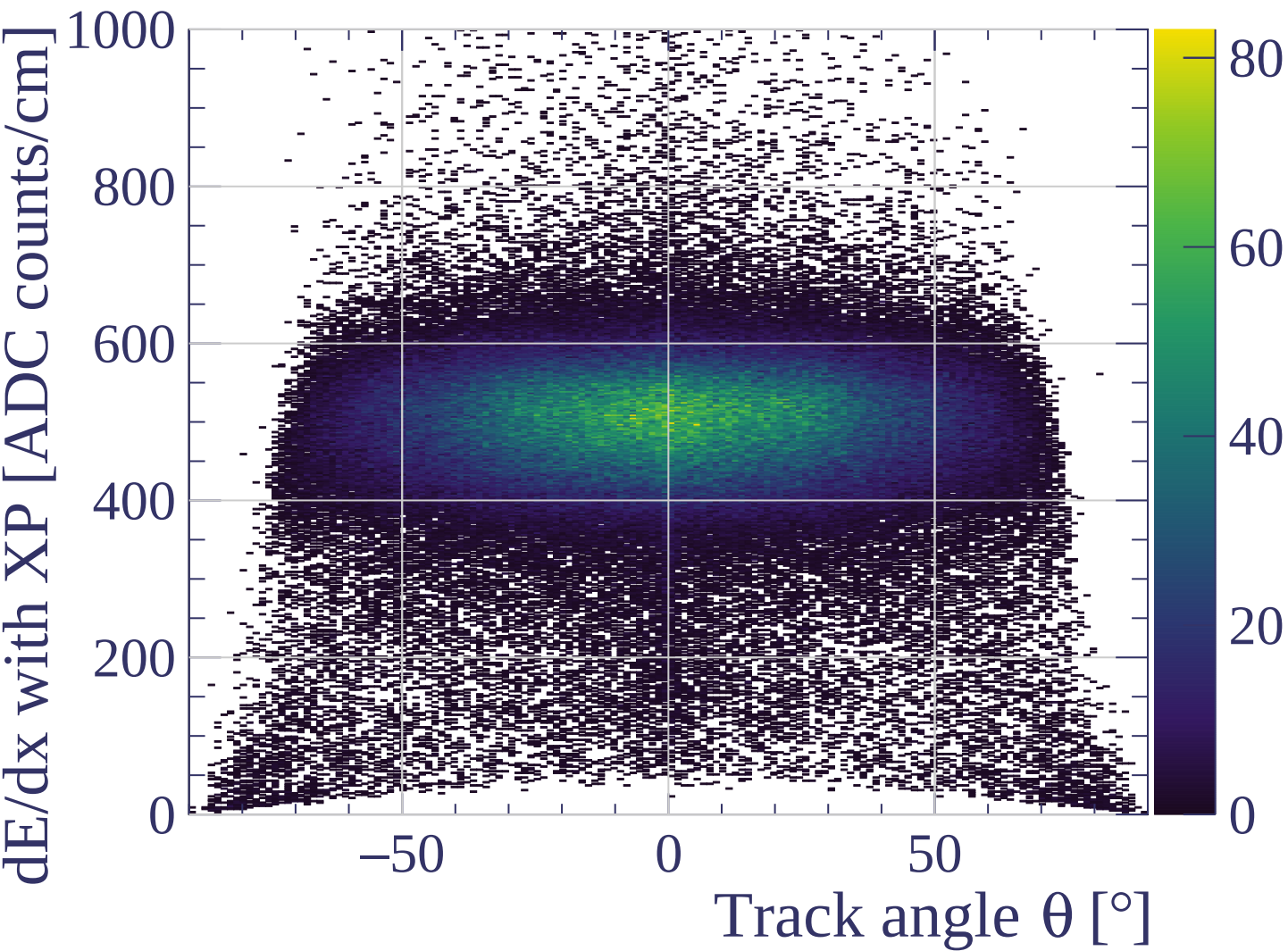




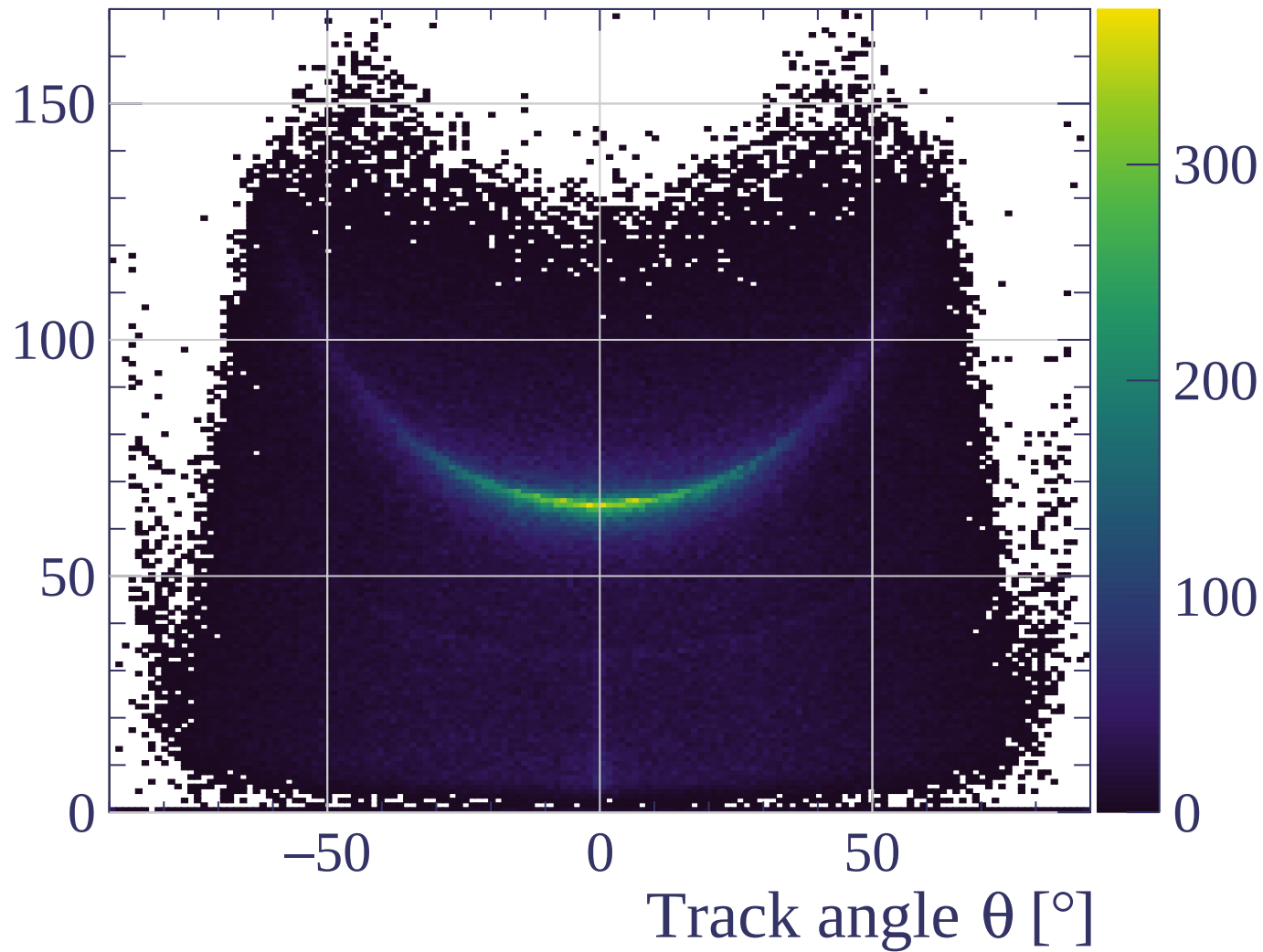


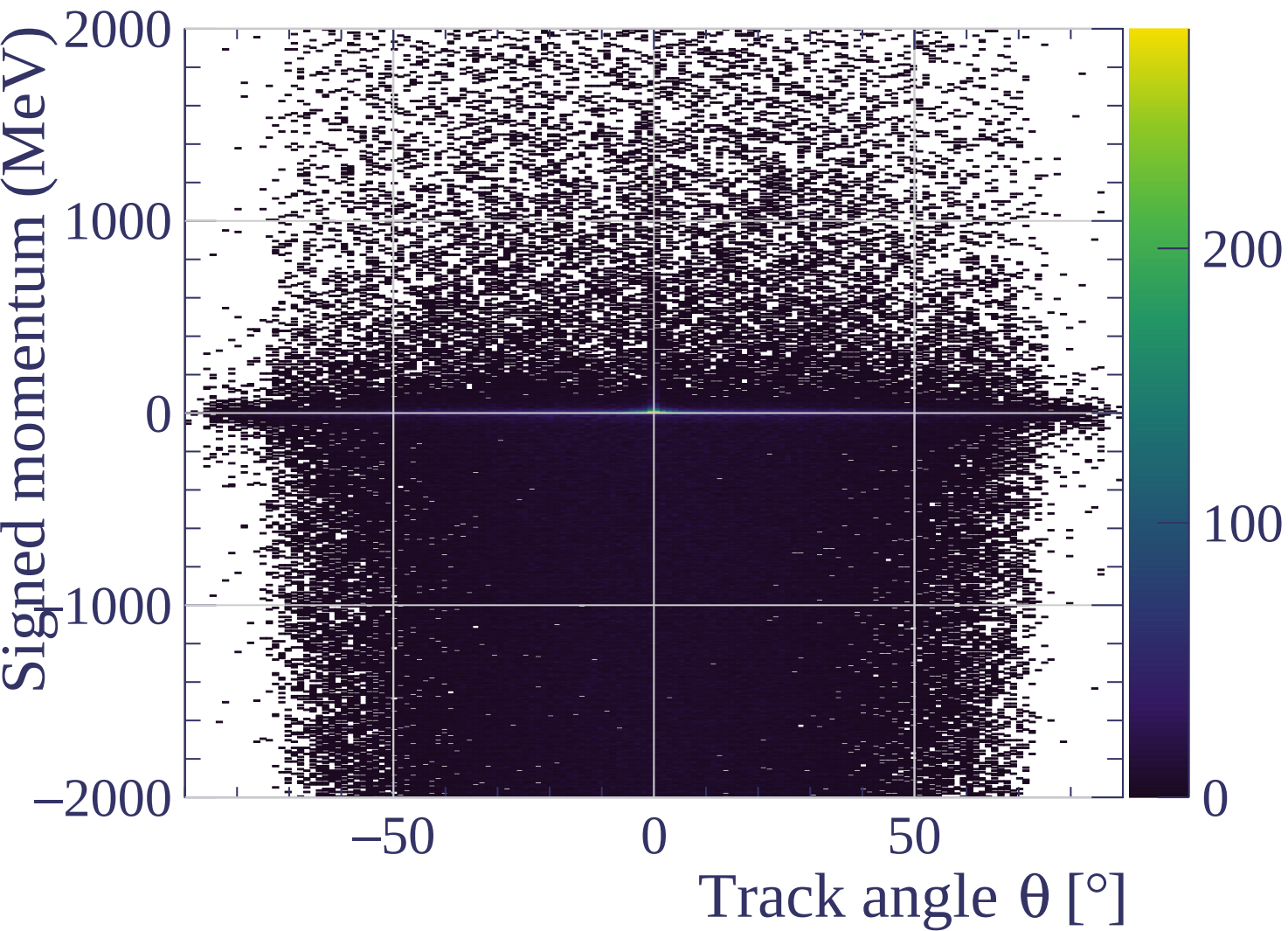




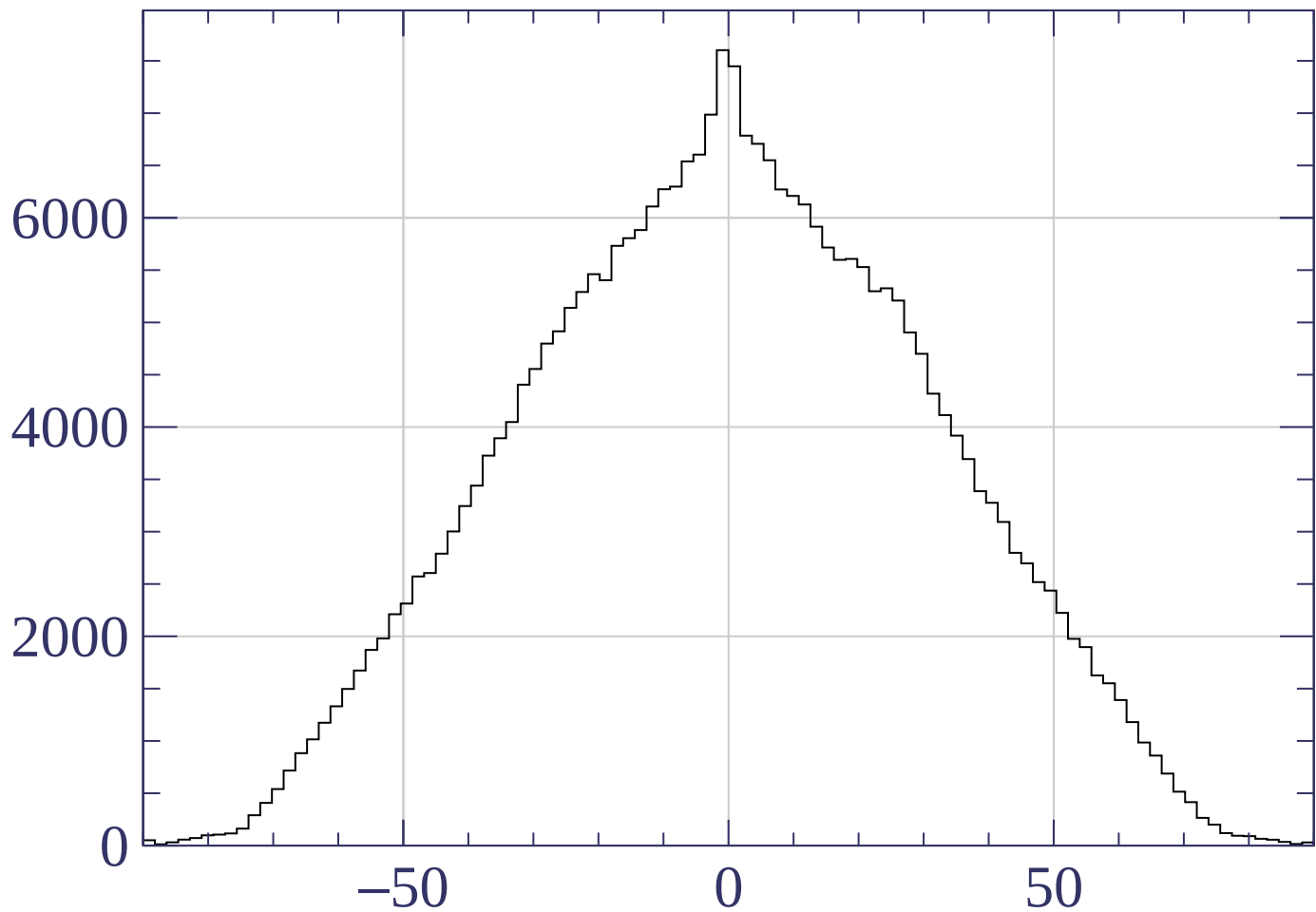


Track length [cm]



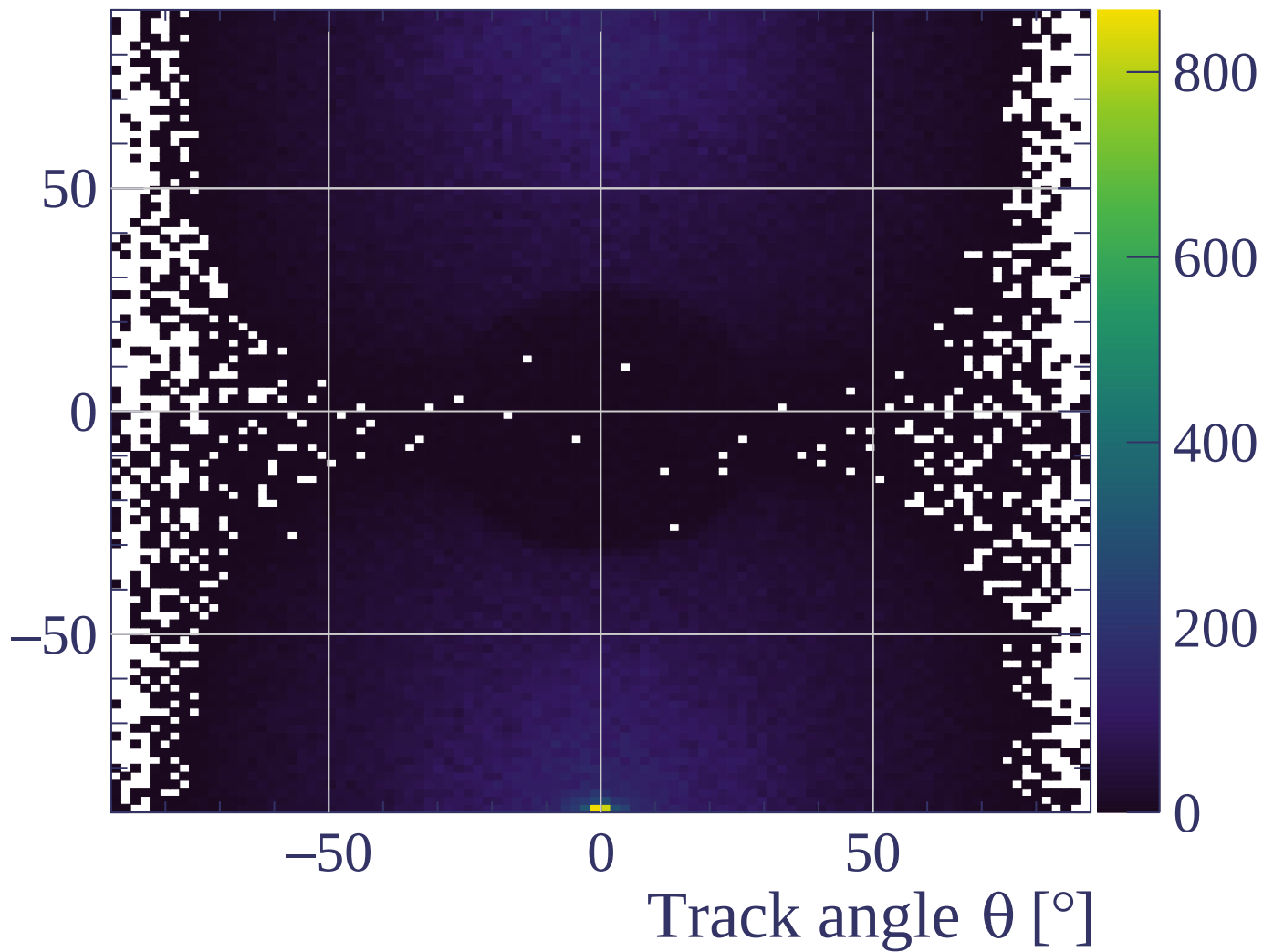


Count

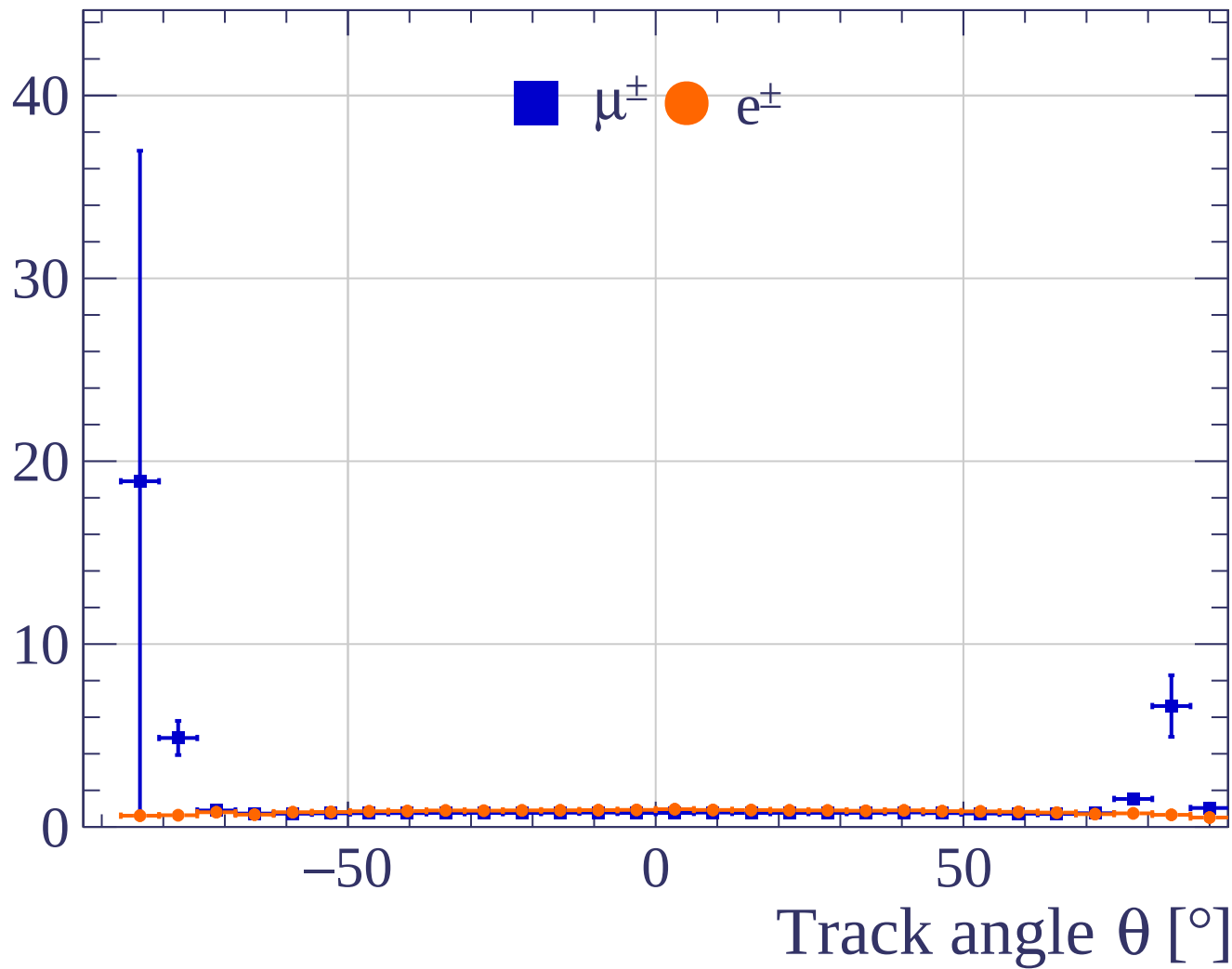


Track angle θ [°]

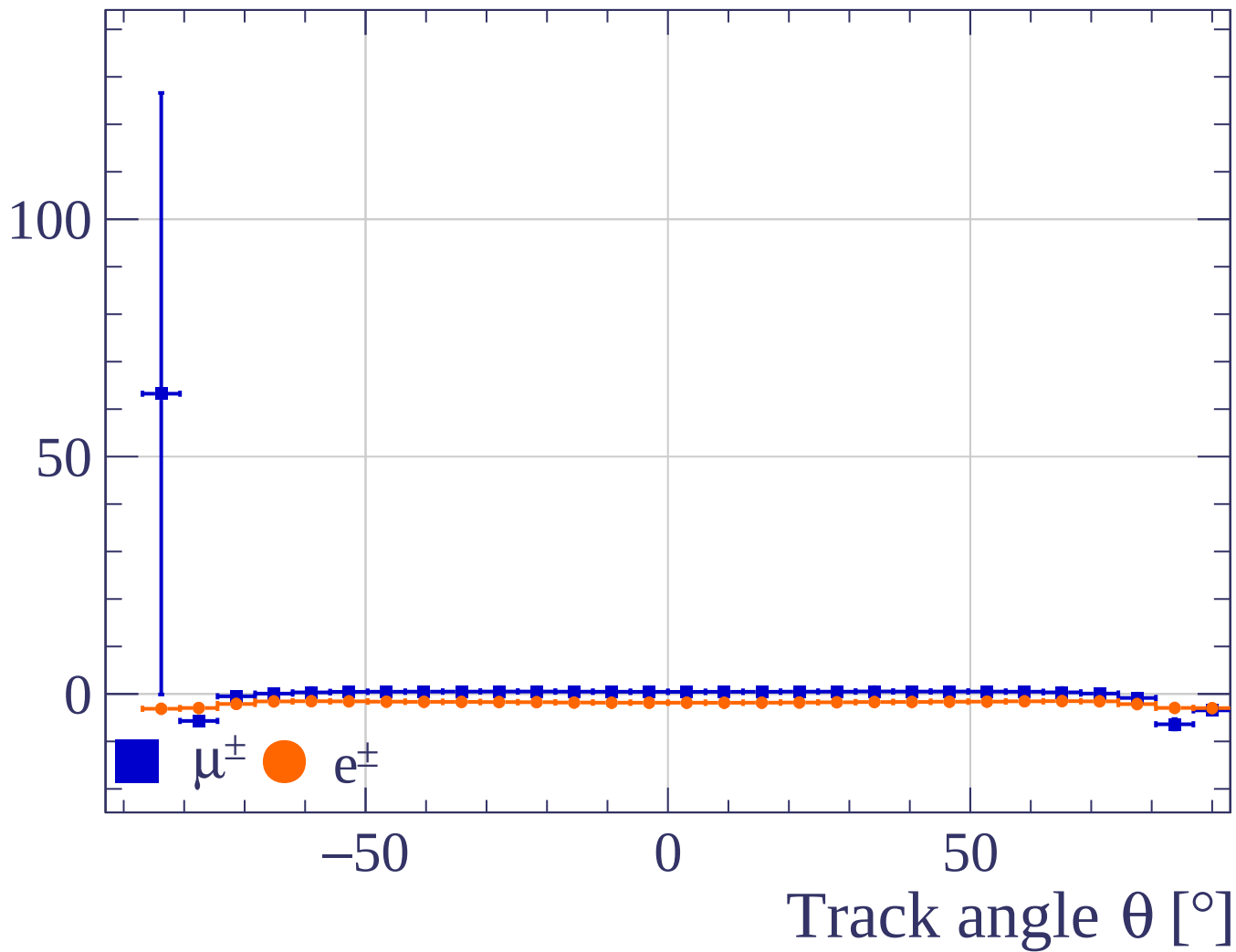
Track angle ϕ [°]



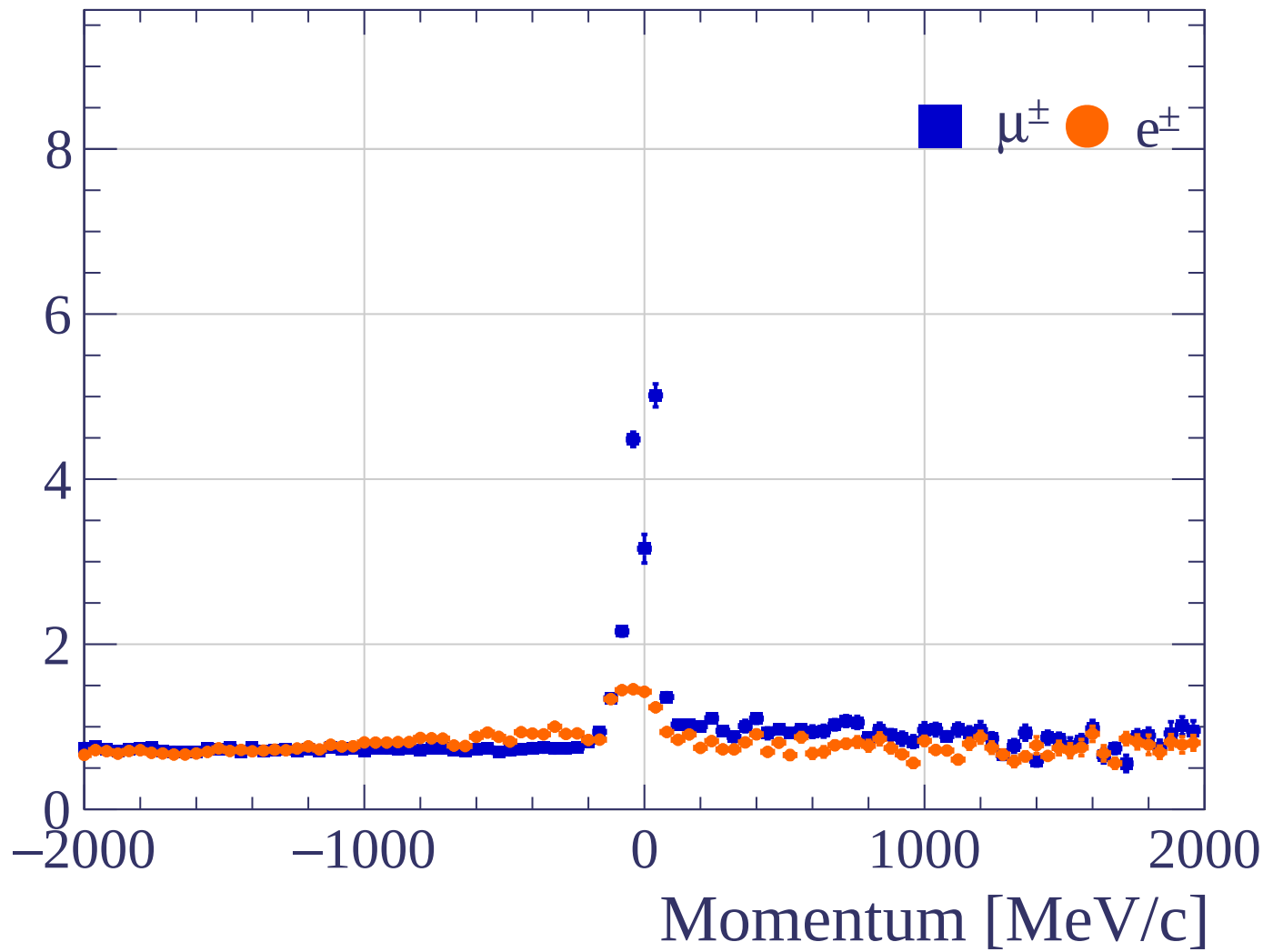
Pulls standard deviation



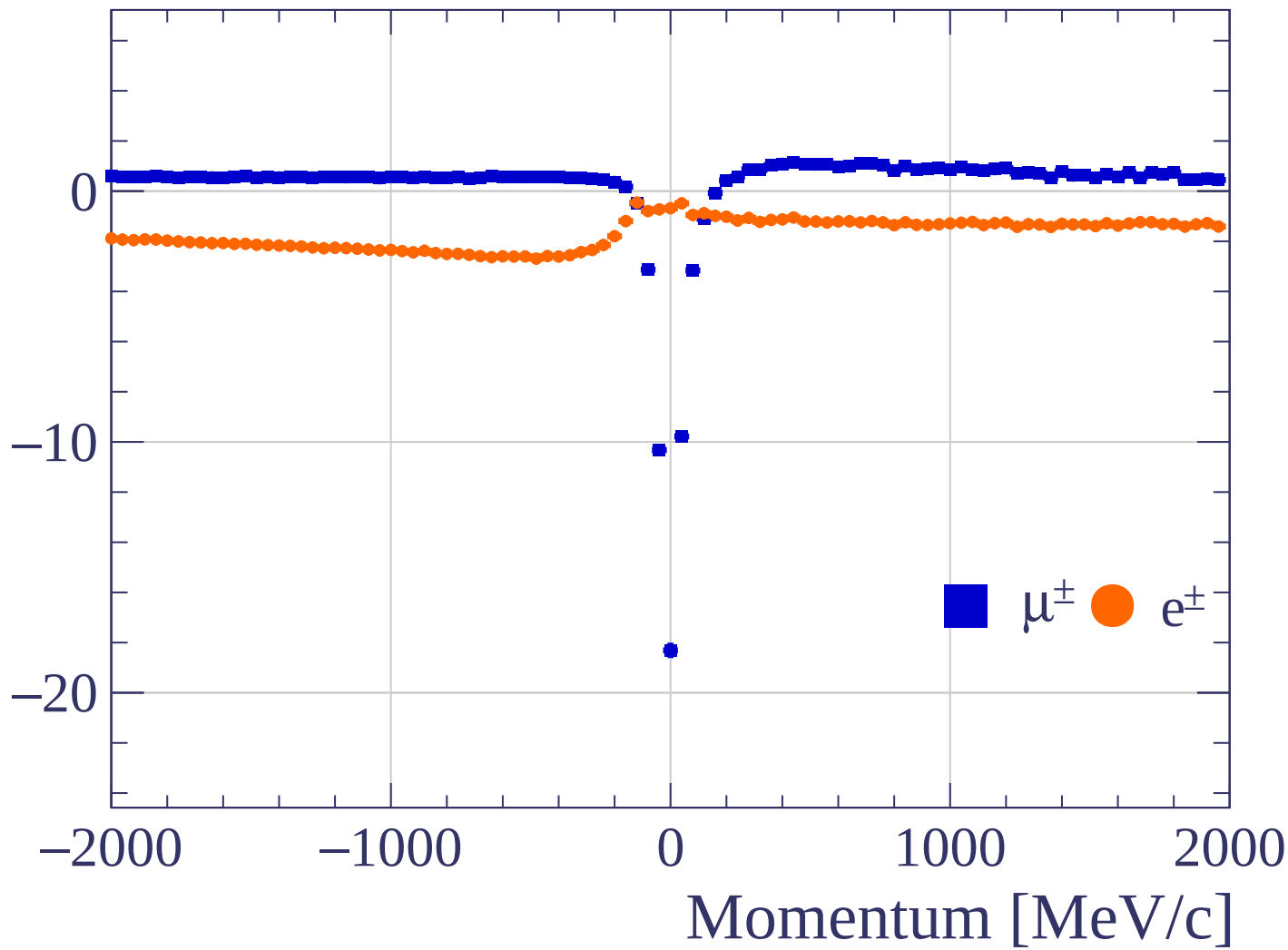
Pulls mean

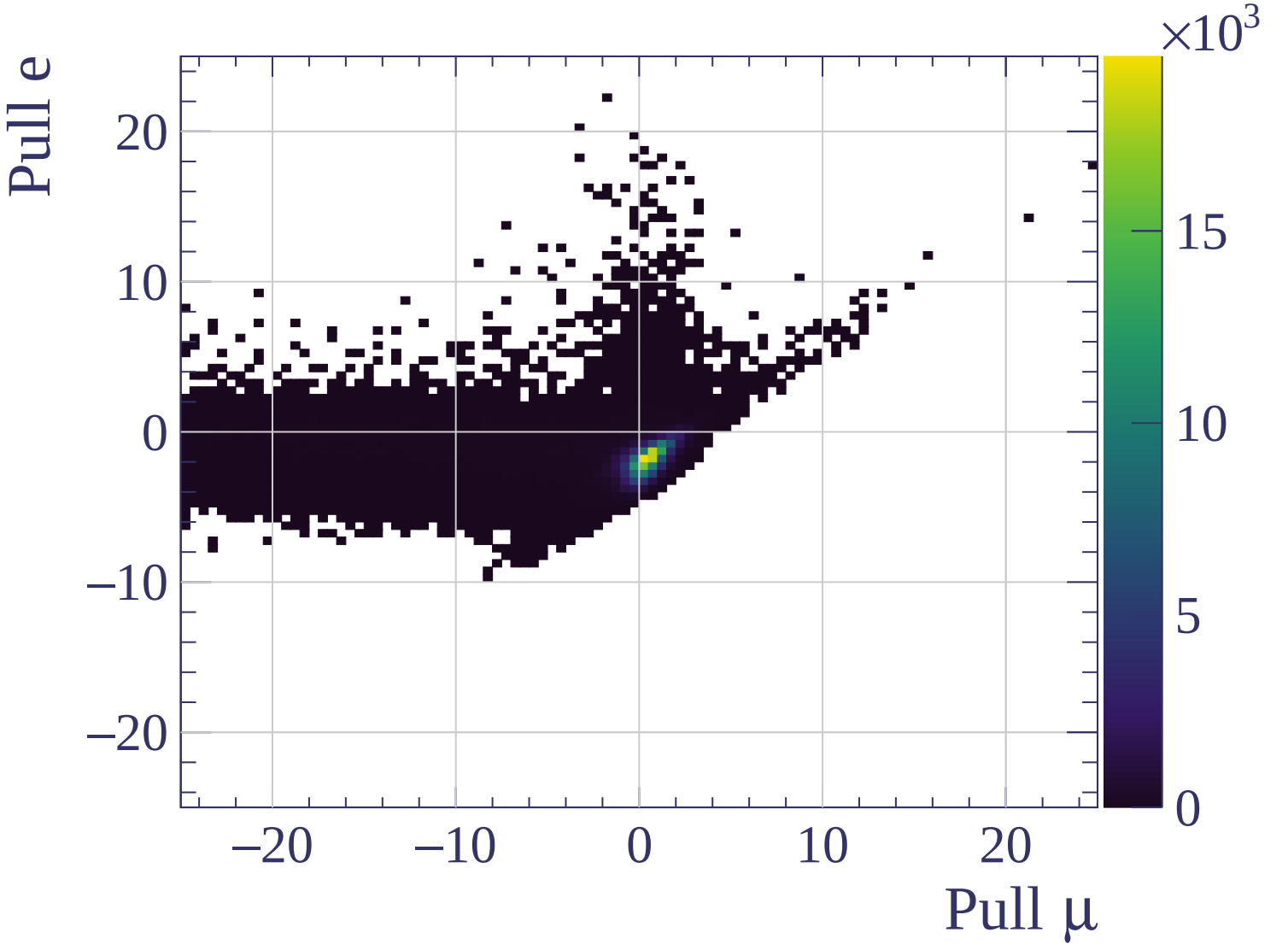


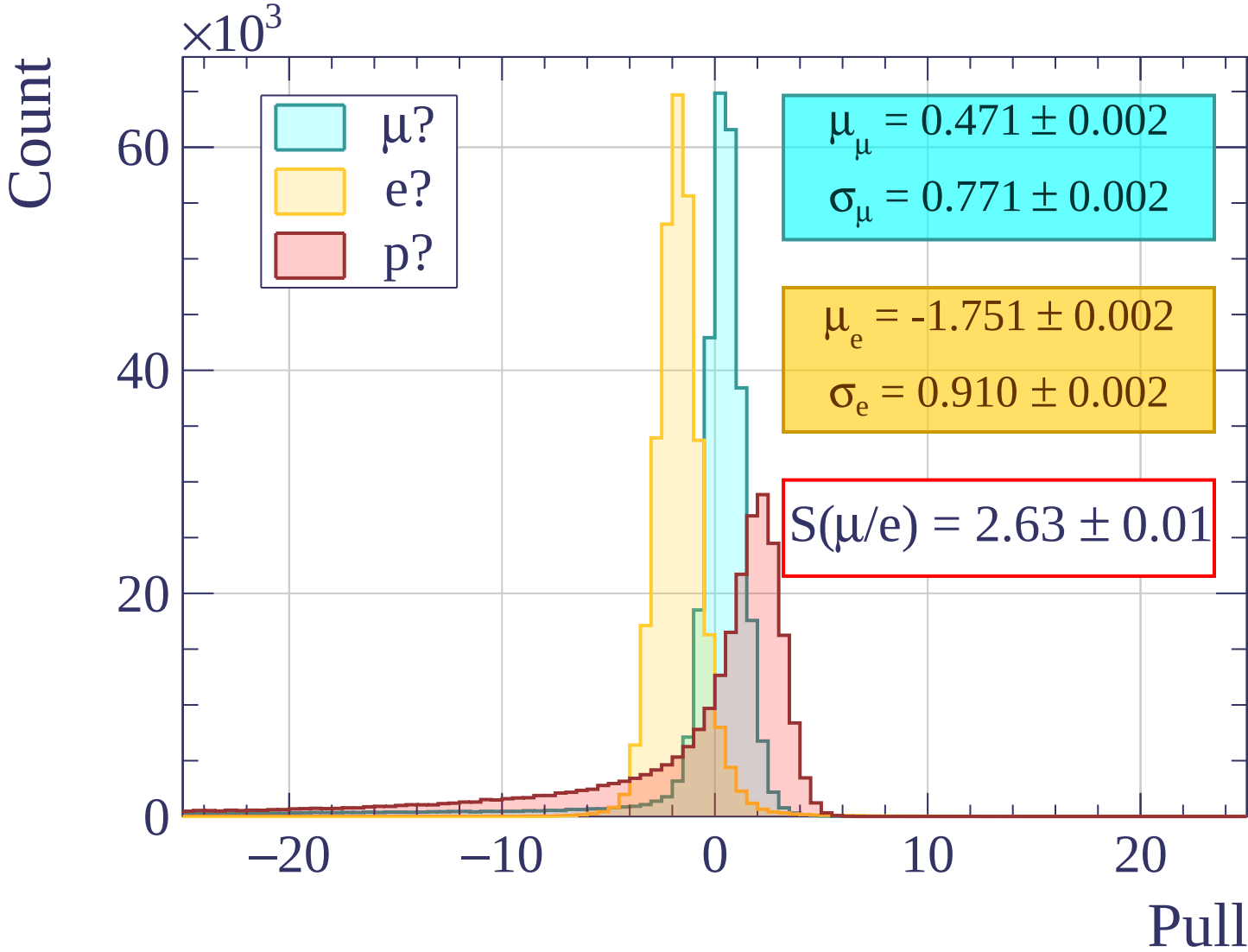
Pulls standard deviation

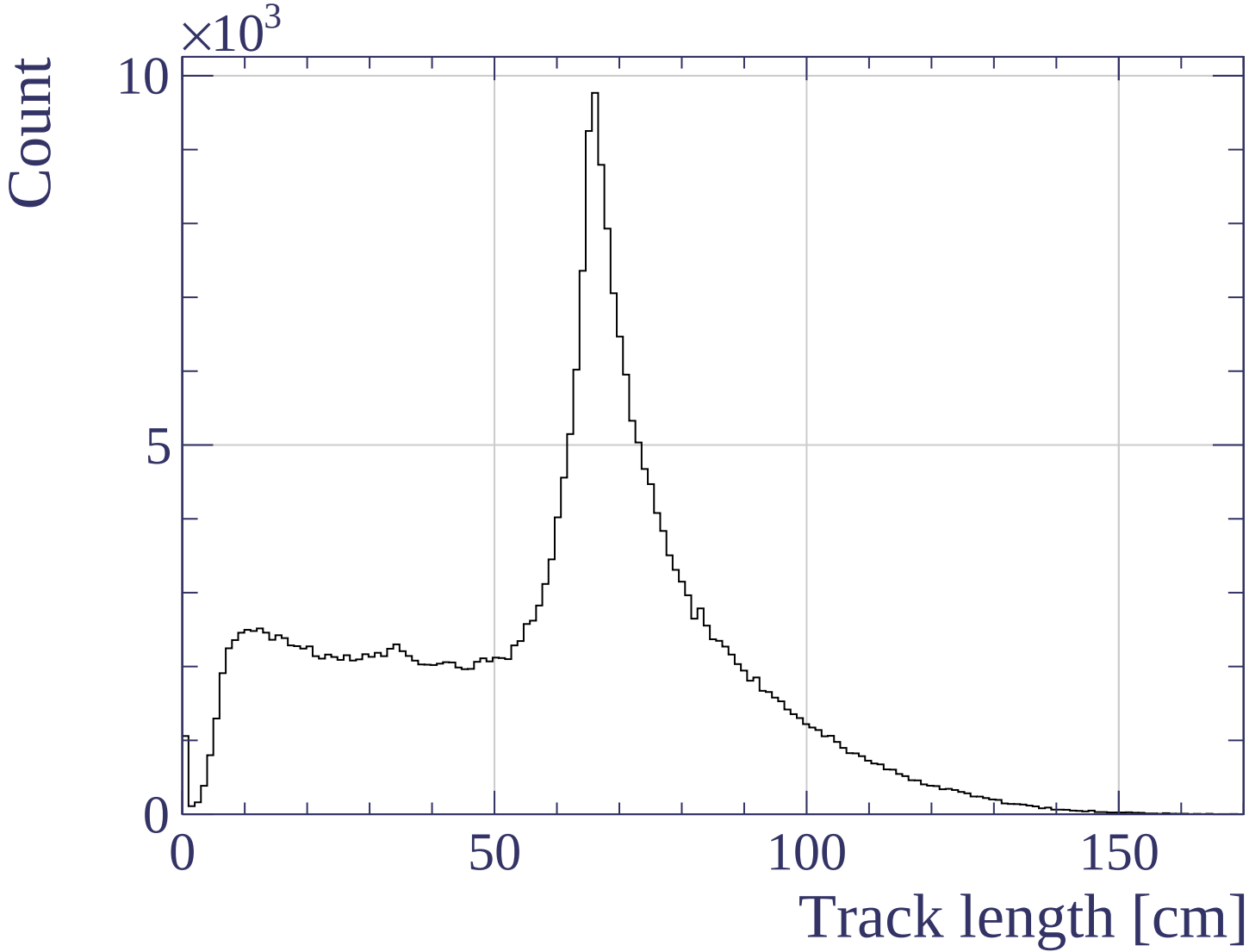


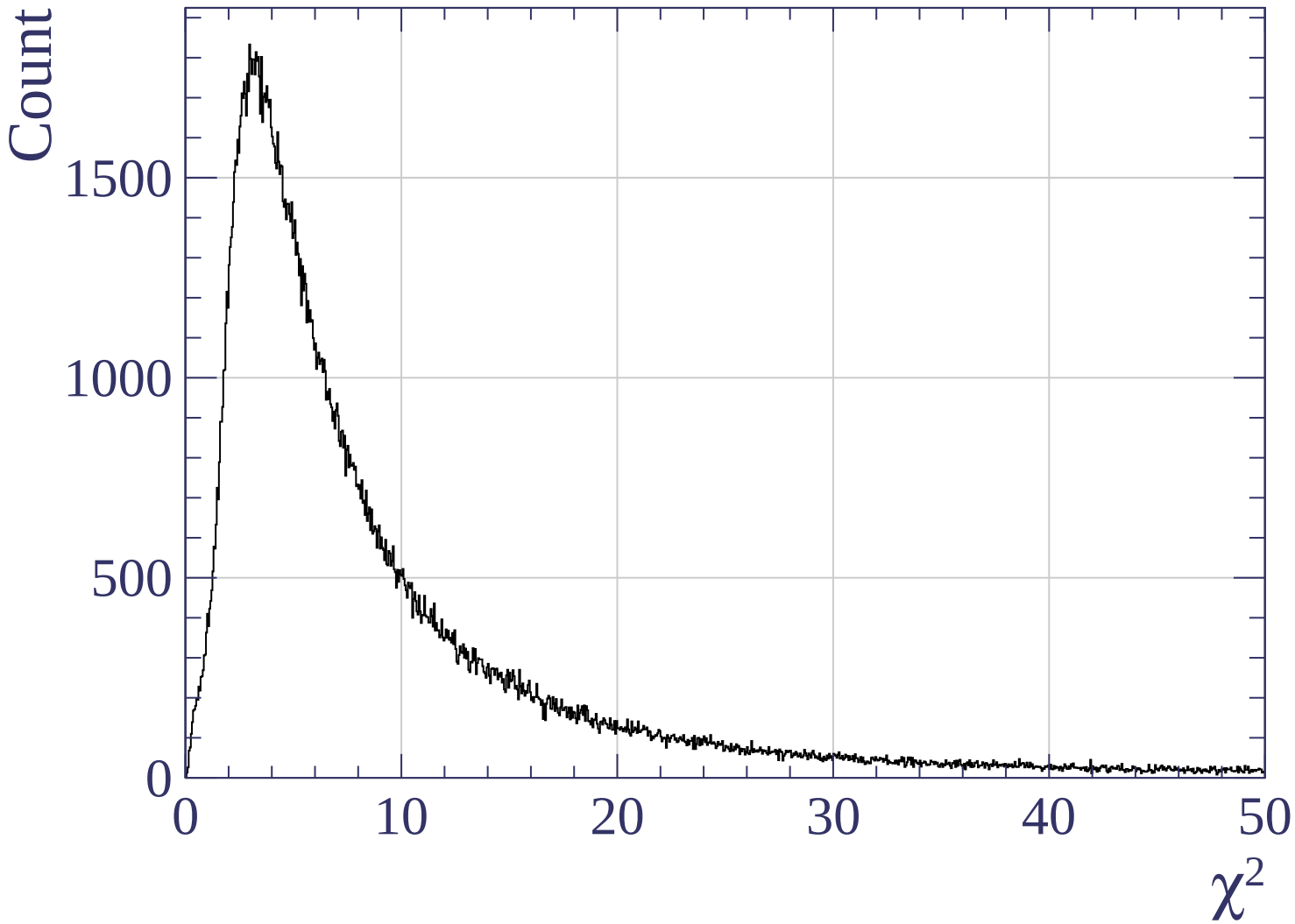
Pulls mean

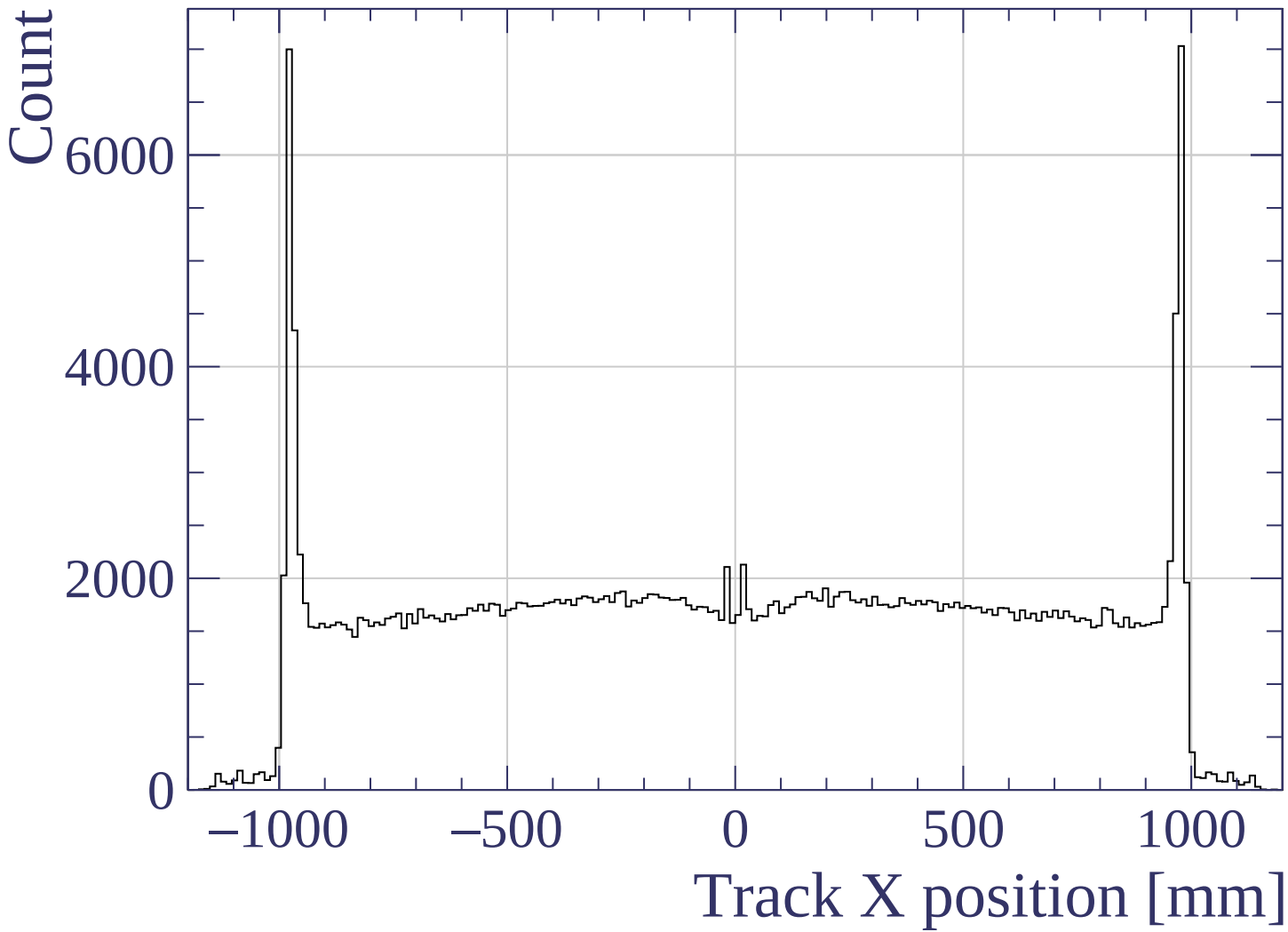


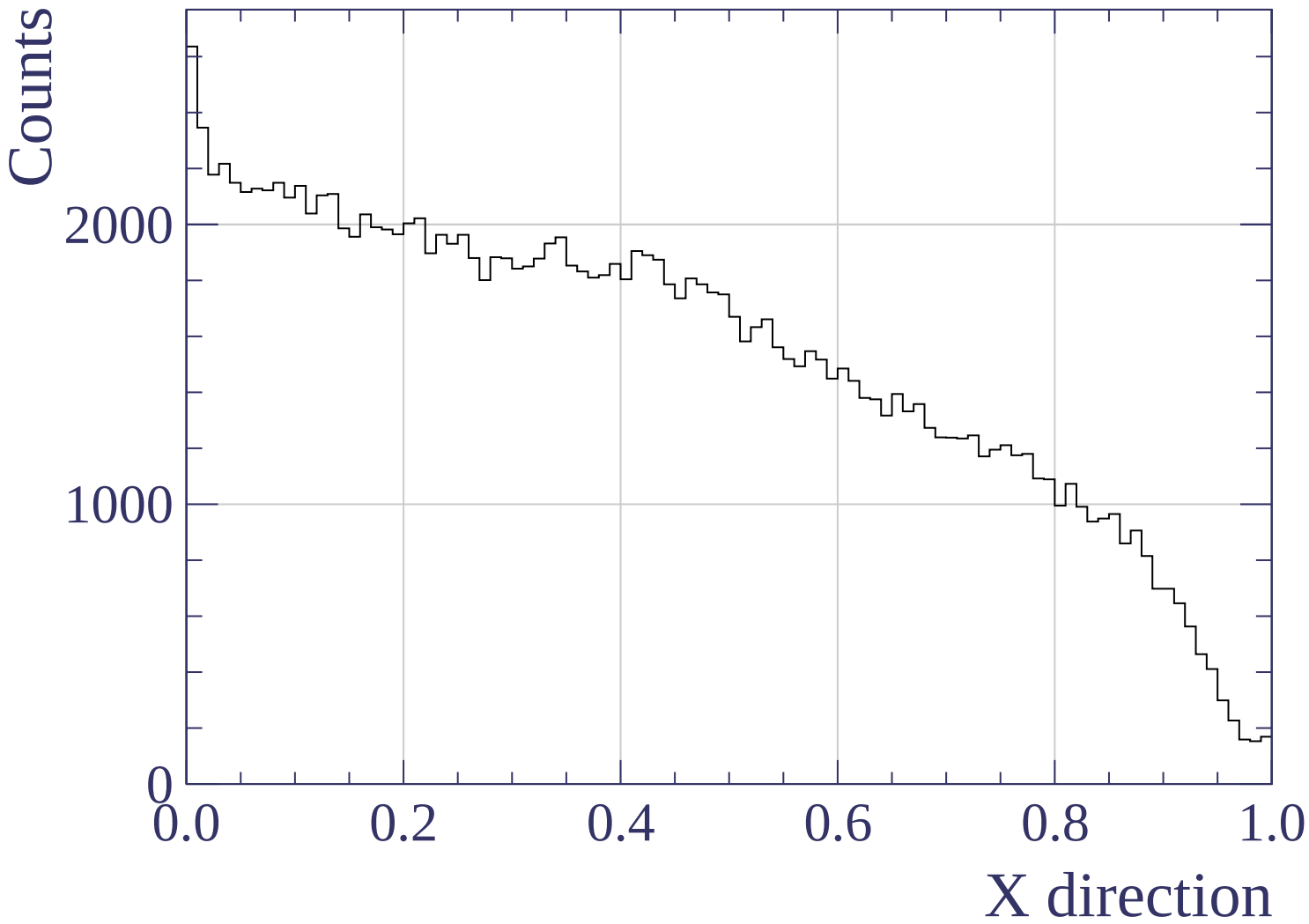


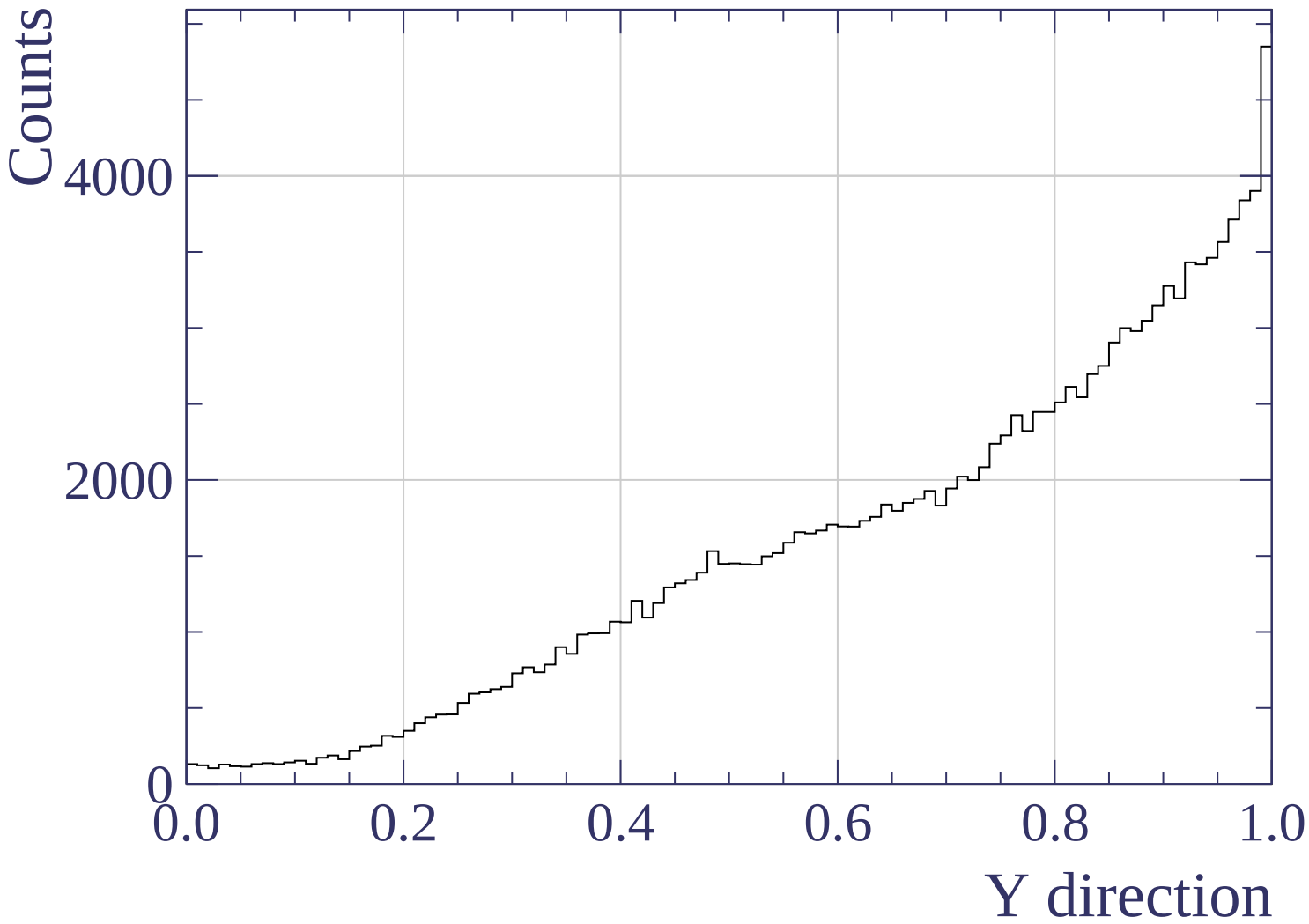


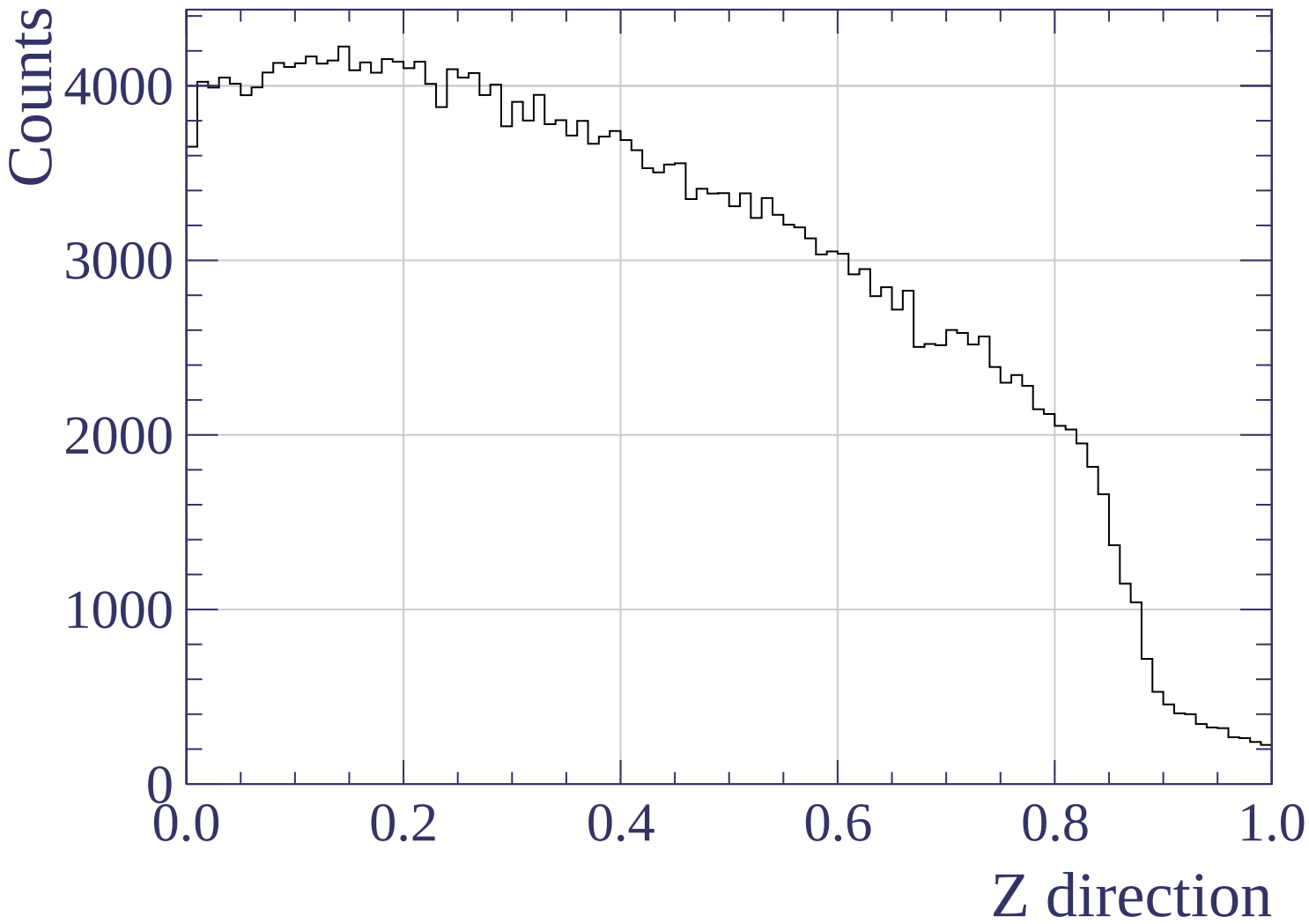


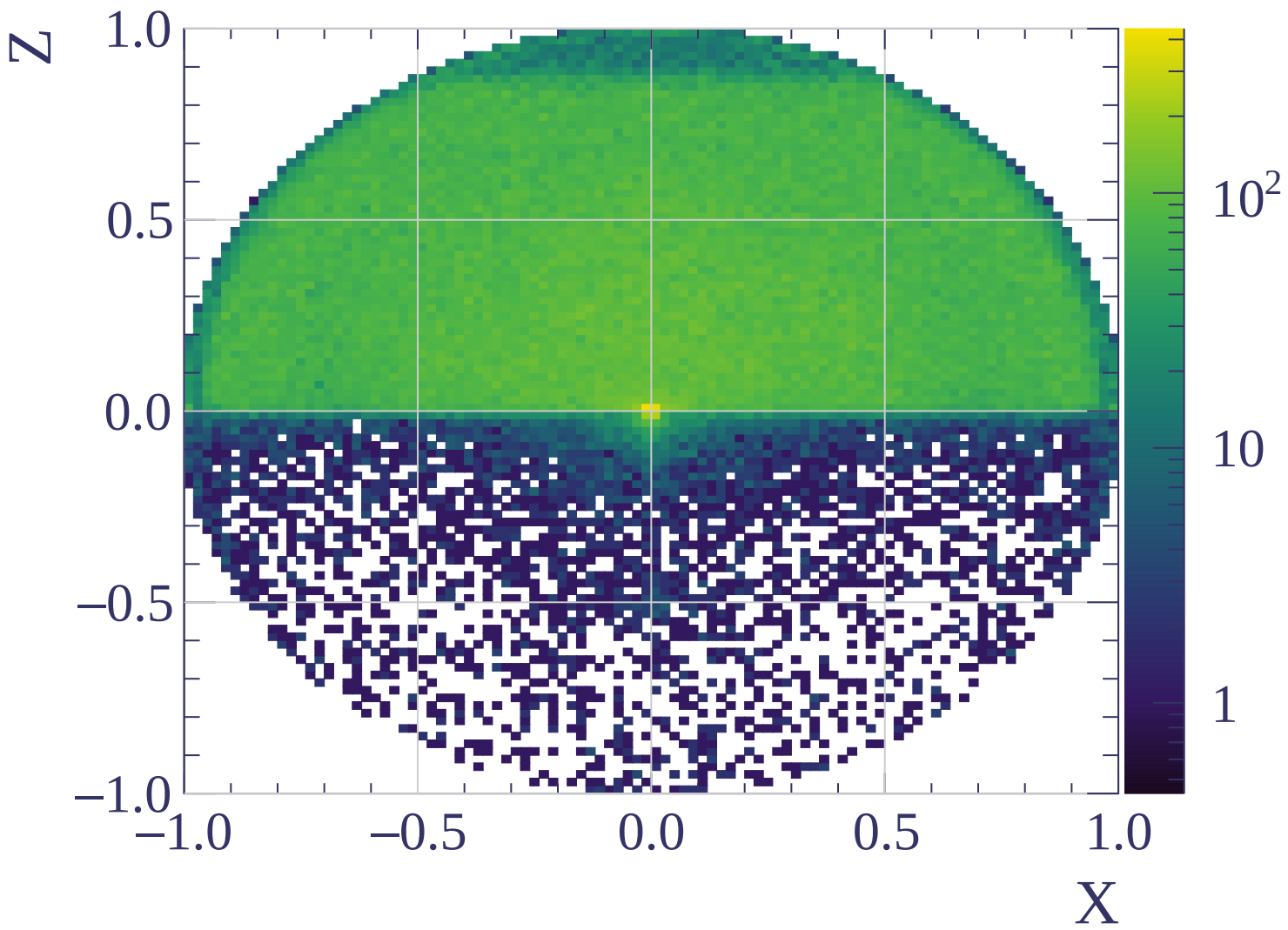


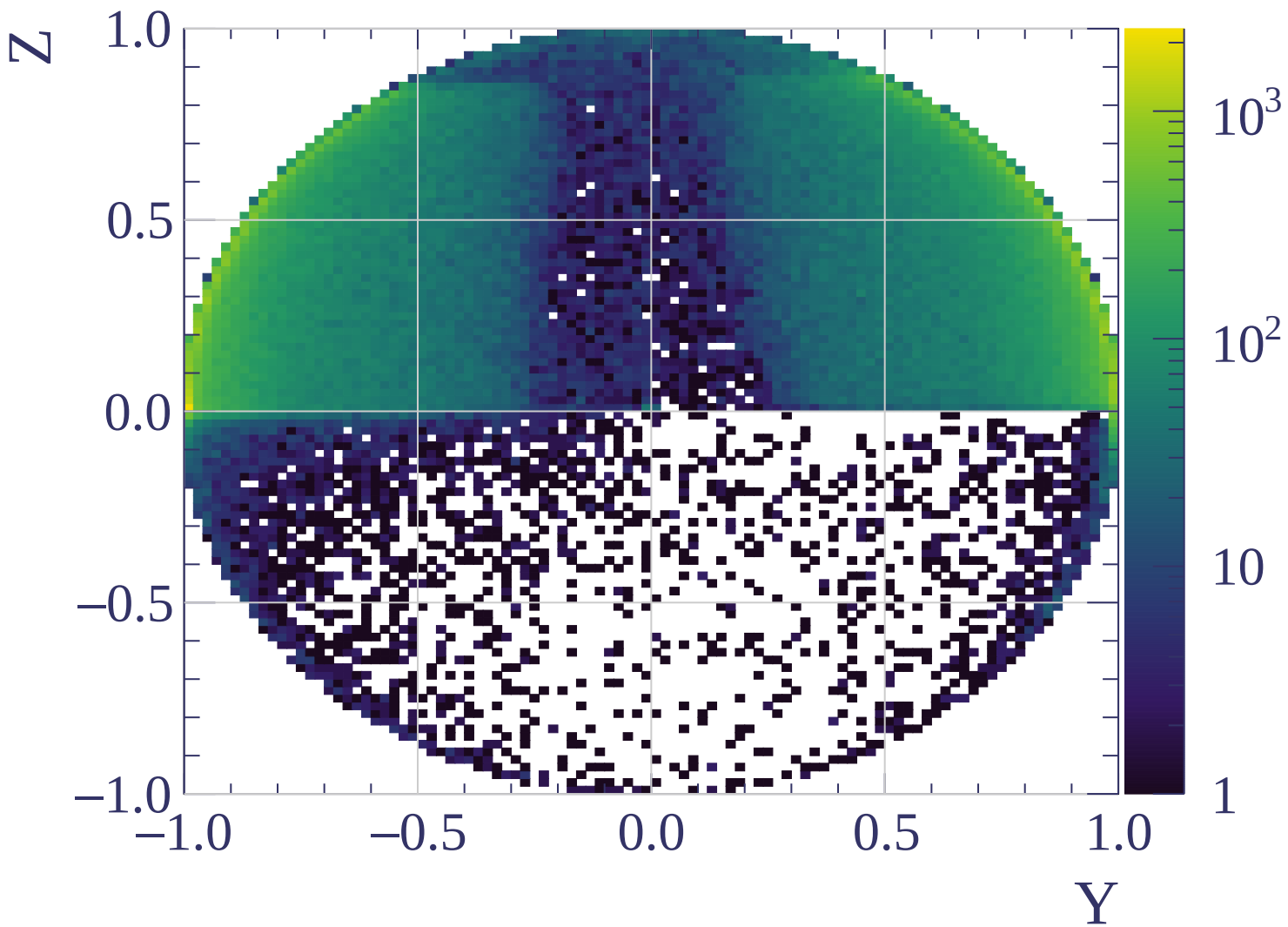


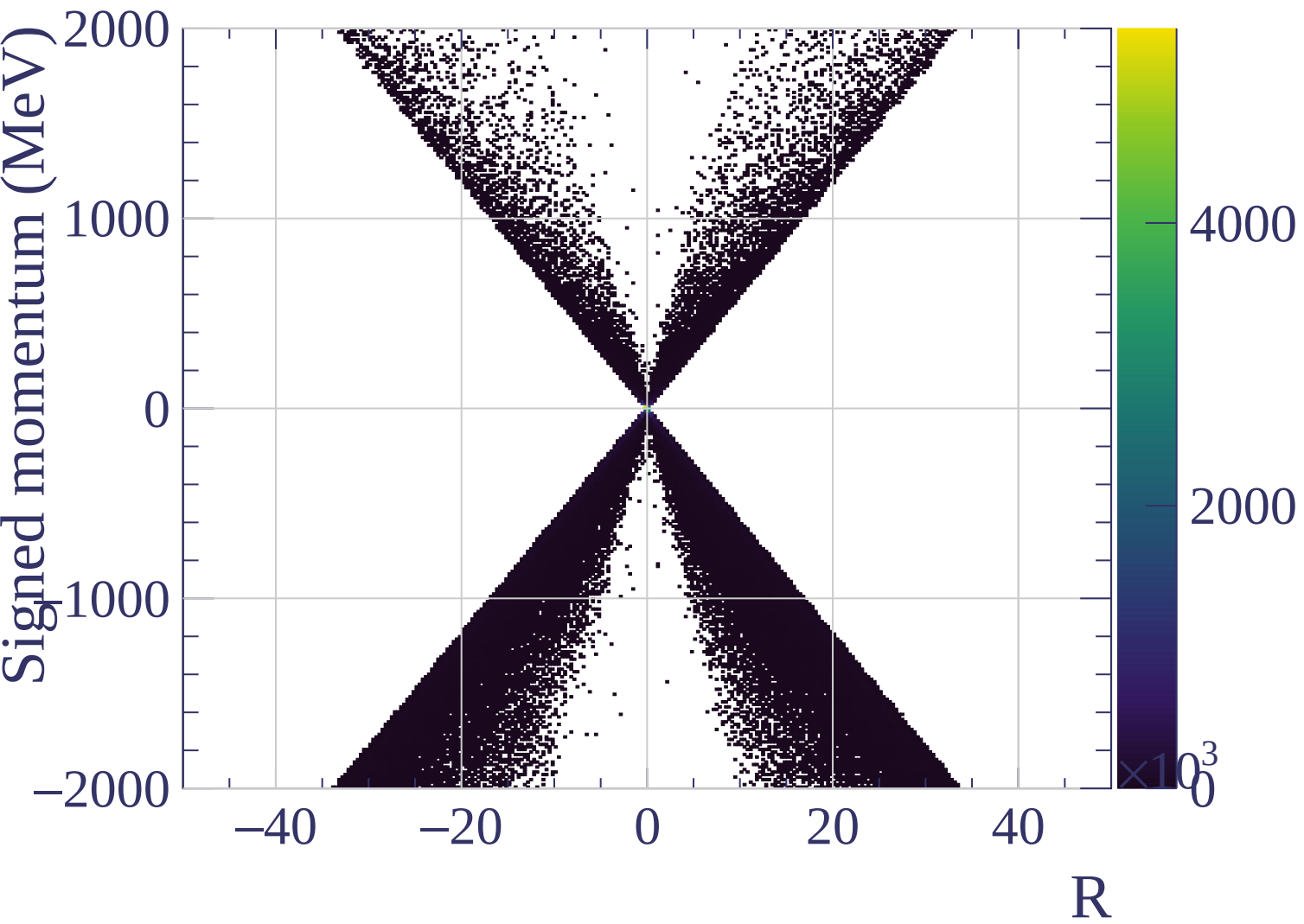


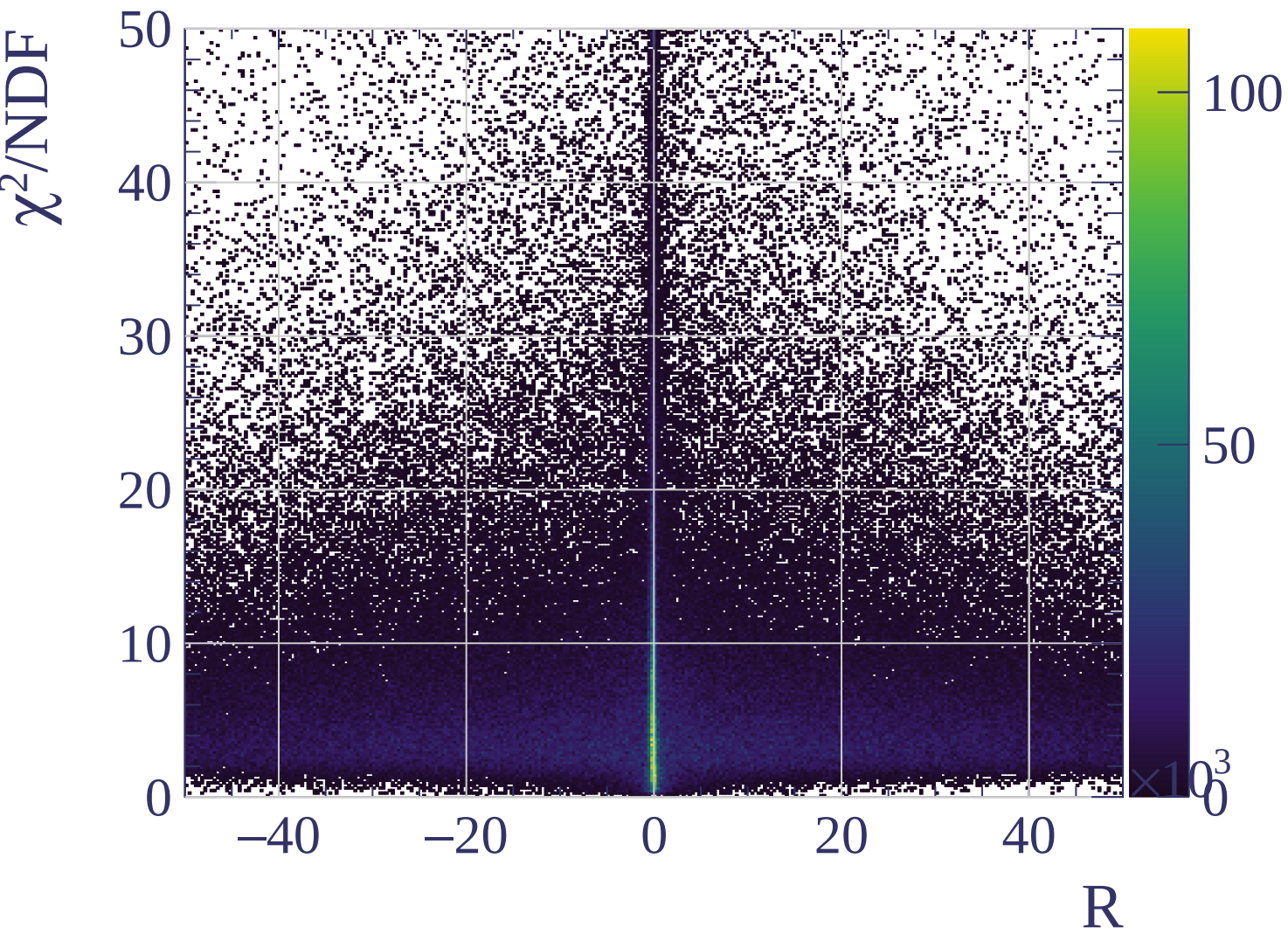












Energy loss | $0 < p < 80$

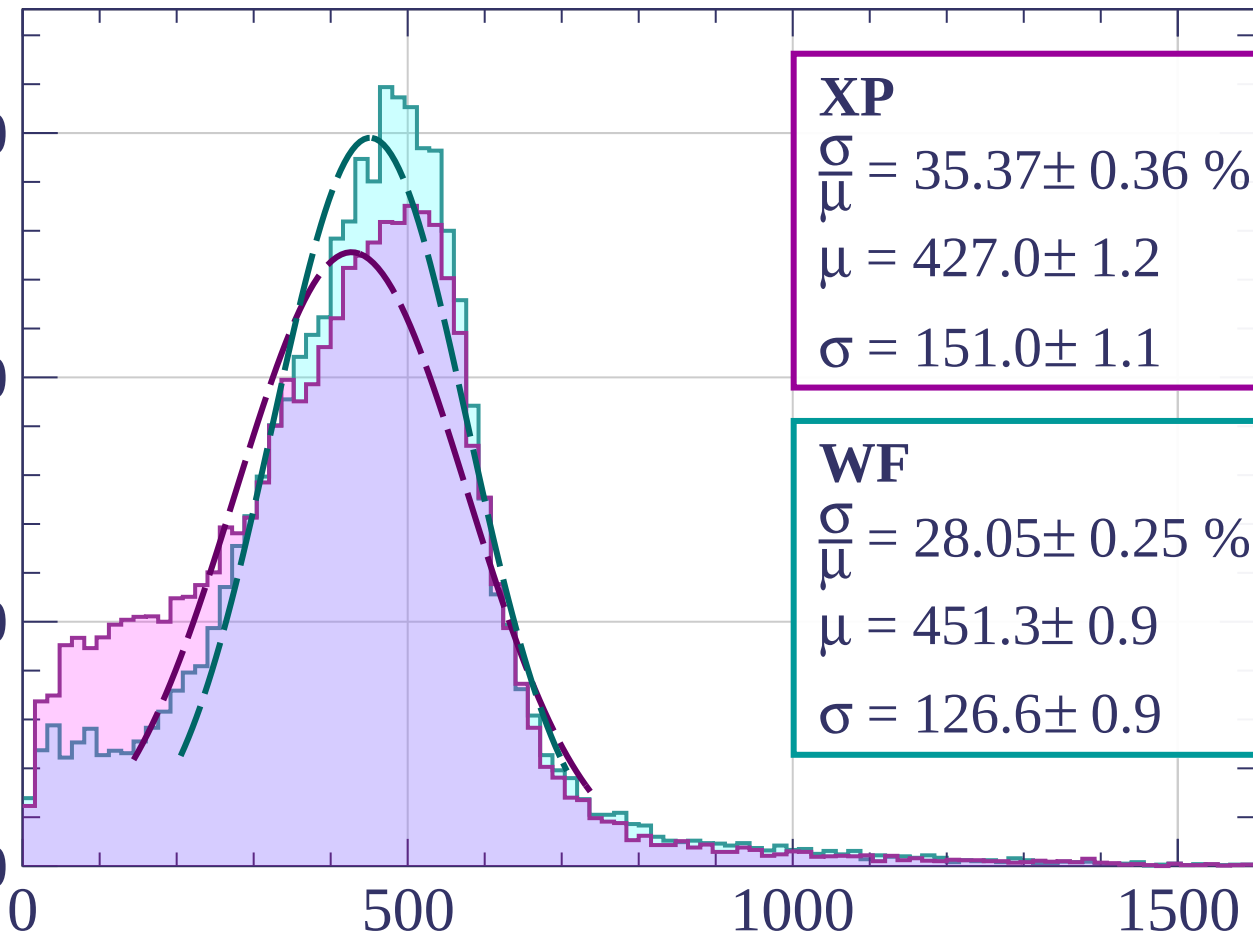
Count

1500

1000

500

0



XP

$$\frac{\sigma}{\mu} = 35.37 \pm 0.36 \%$$

$$\mu = 427.0 \pm 1.2$$

$$\sigma = 151.0 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 28.05 \pm 0.25 \%$$

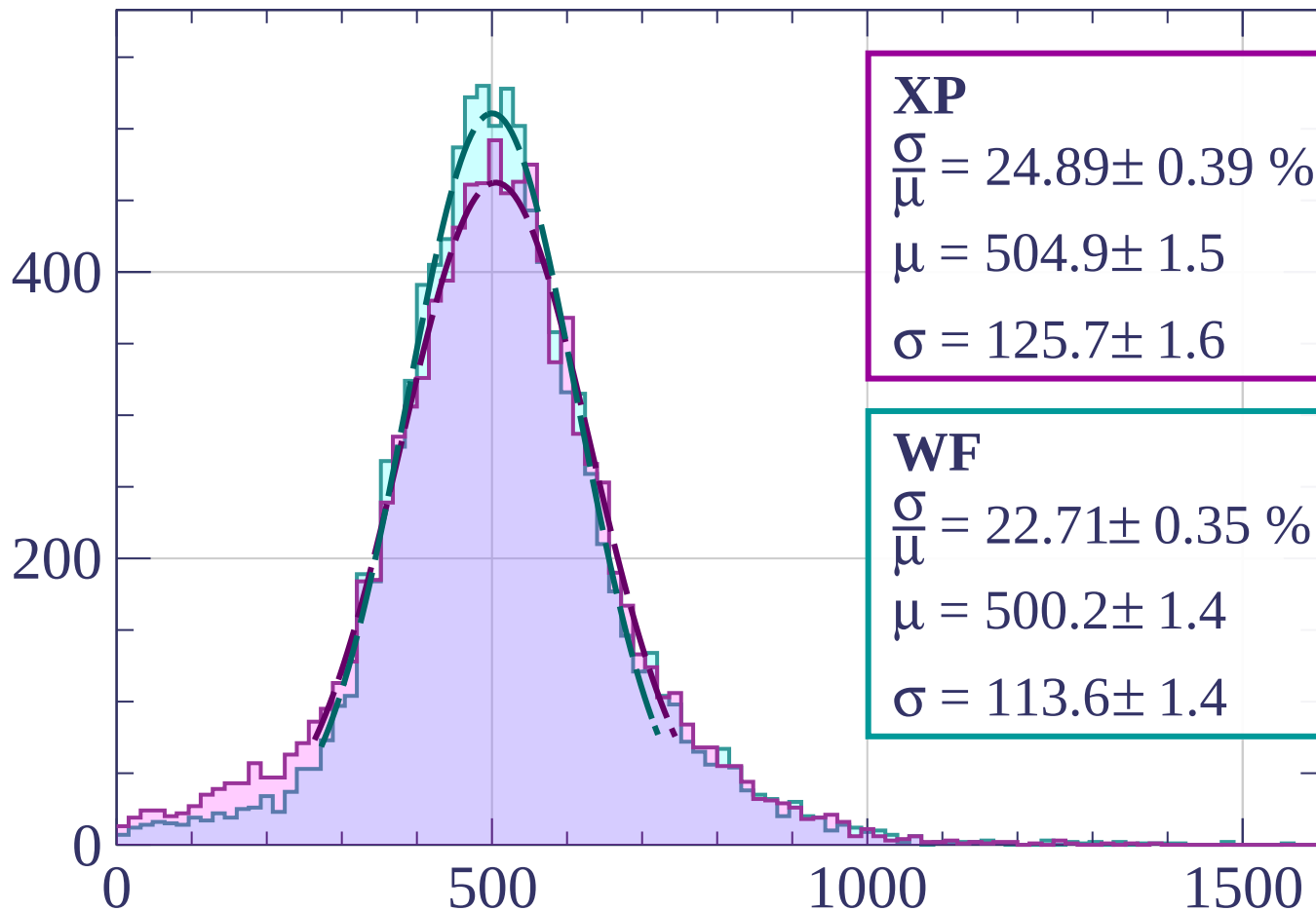
$$\mu = 451.3 \pm 0.9$$

$$\sigma = 126.6 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $80 < p < 160$

Count



XP

$$\frac{\sigma}{\mu} = 24.89 \pm 0.39 \%$$

$$\mu = 504.9 \pm 1.5$$

$$\sigma = 125.7 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 22.71 \pm 0.35 \%$$

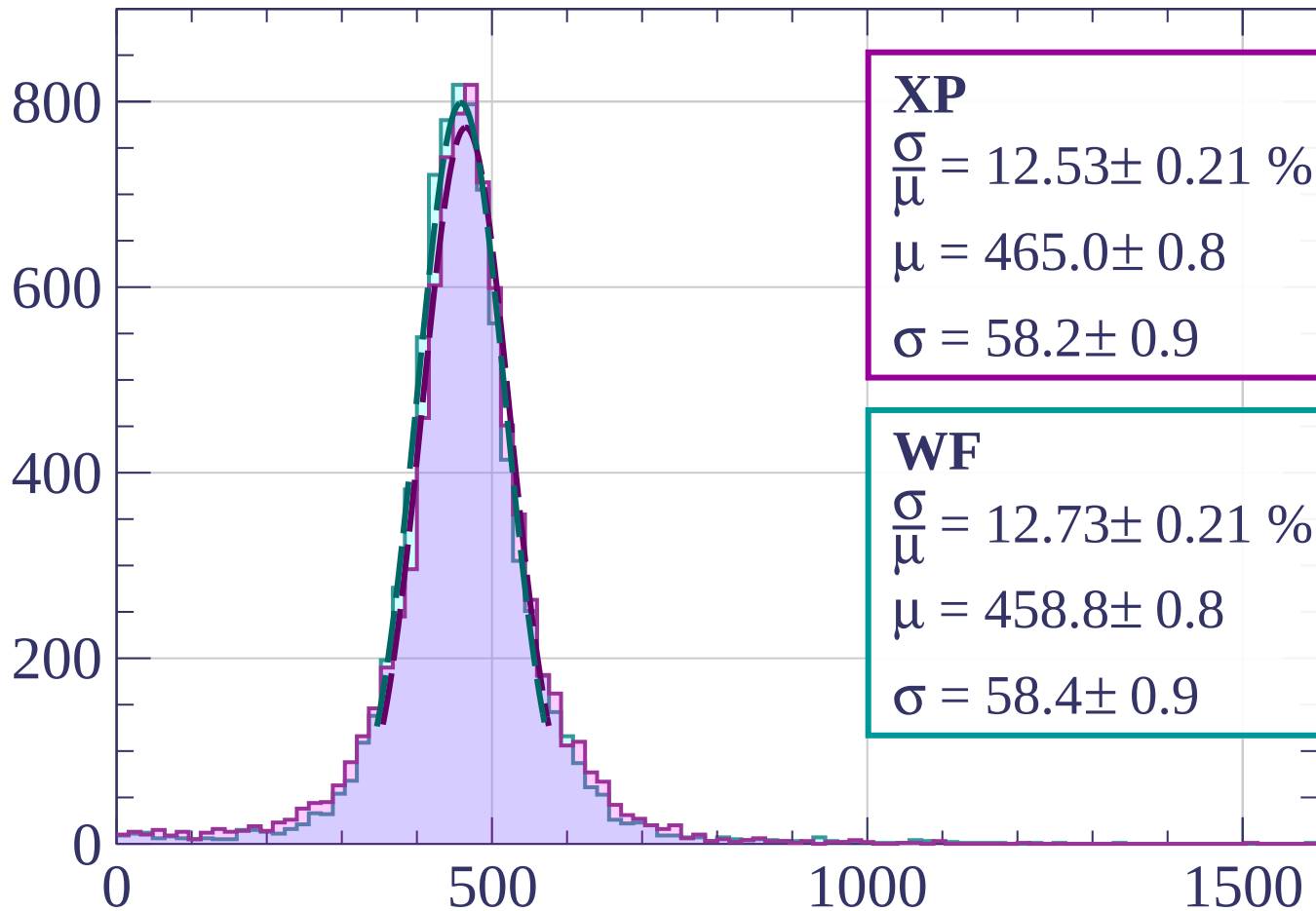
$$\mu = 500.2 \pm 1.4$$

$$\sigma = 113.6 \pm 1.4$$

dE/dx [ADC counts/cm]

Energy loss | $160 < p < 240$

Count



XP

$$\frac{\sigma}{\mu} = 12.53 \pm 0.21 \%$$

$$\mu = 465.0 \pm 0.8$$

$$\sigma = 58.2 \pm 0.9$$

WF

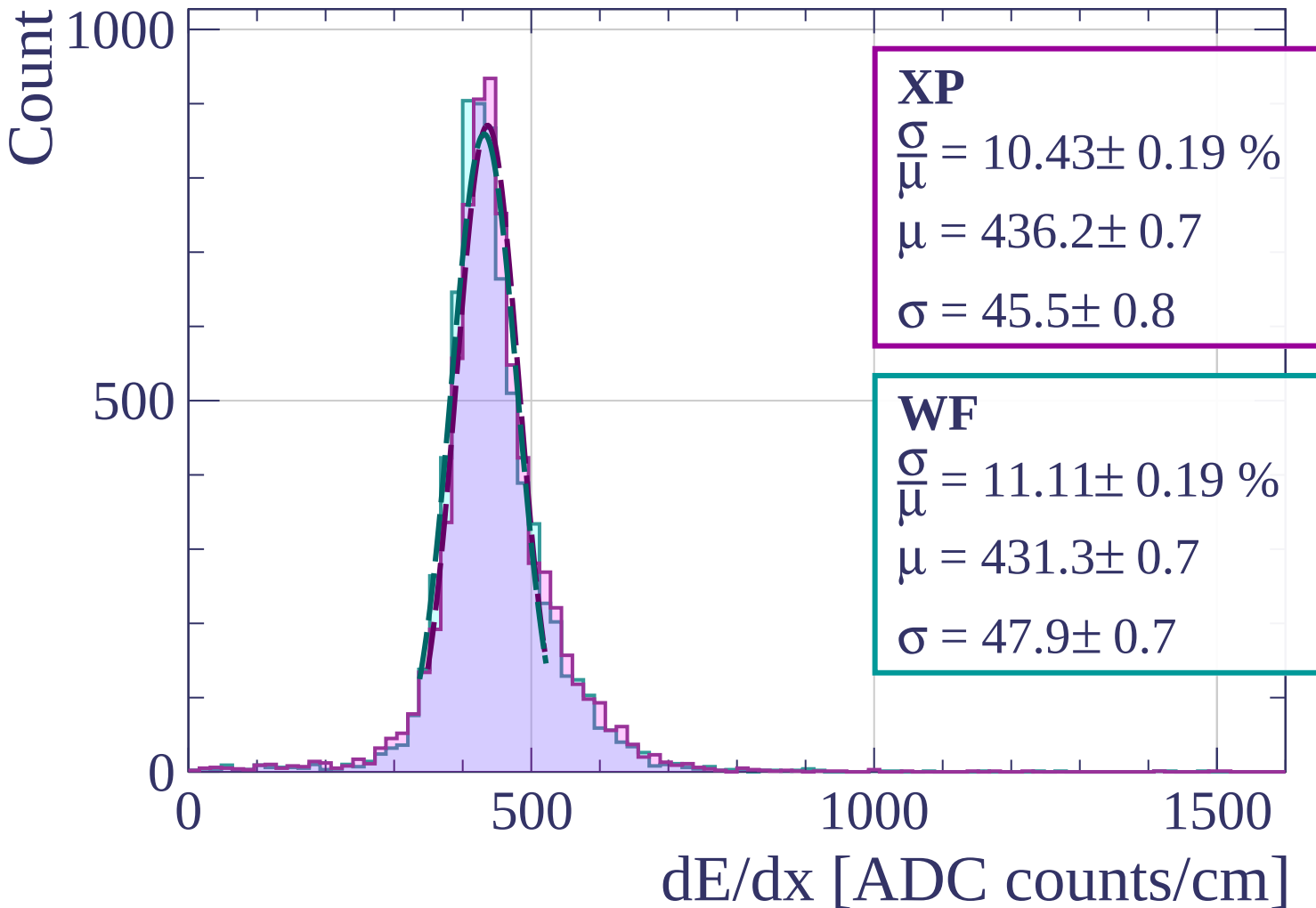
$$\frac{\sigma}{\mu} = 12.73 \pm 0.21 \%$$

$$\mu = 458.8 \pm 0.8$$

$$\sigma = 58.4 \pm 0.9$$

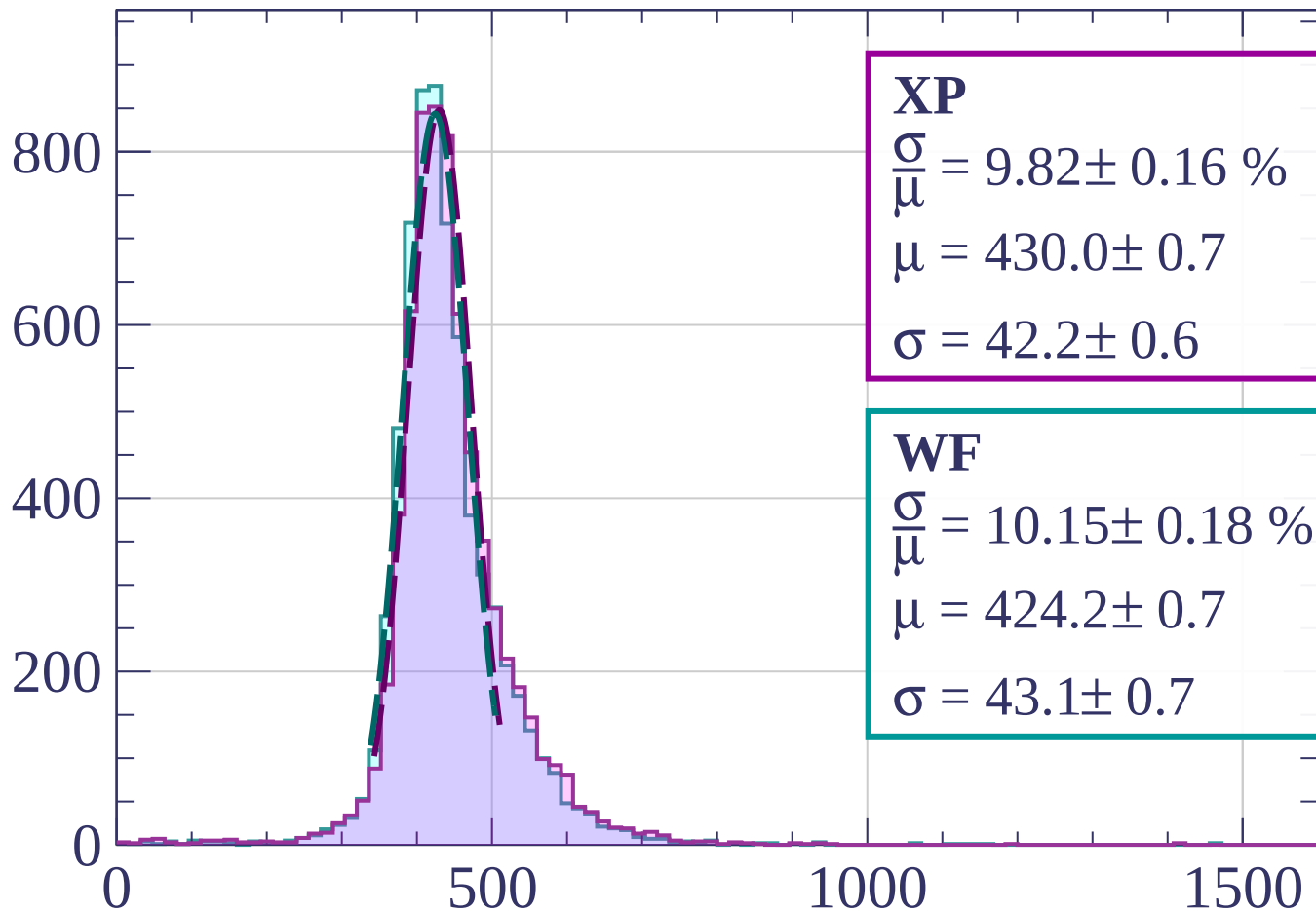
dE/dx [ADC counts/cm]

Energy loss | $240 < p < 320$



Energy loss | $320 < p < 400$

Count



XP

$$\frac{\sigma}{\mu} = 9.82 \pm 0.16 \%$$

$$\mu = 430.0 \pm 0.7$$

$$\sigma = 42.2 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 10.15 \pm 0.18 \%$$

$$\mu = 424.2 \pm 0.7$$

$$\sigma = 43.1 \pm 0.7$$

dE/dx [ADC counts/cm]

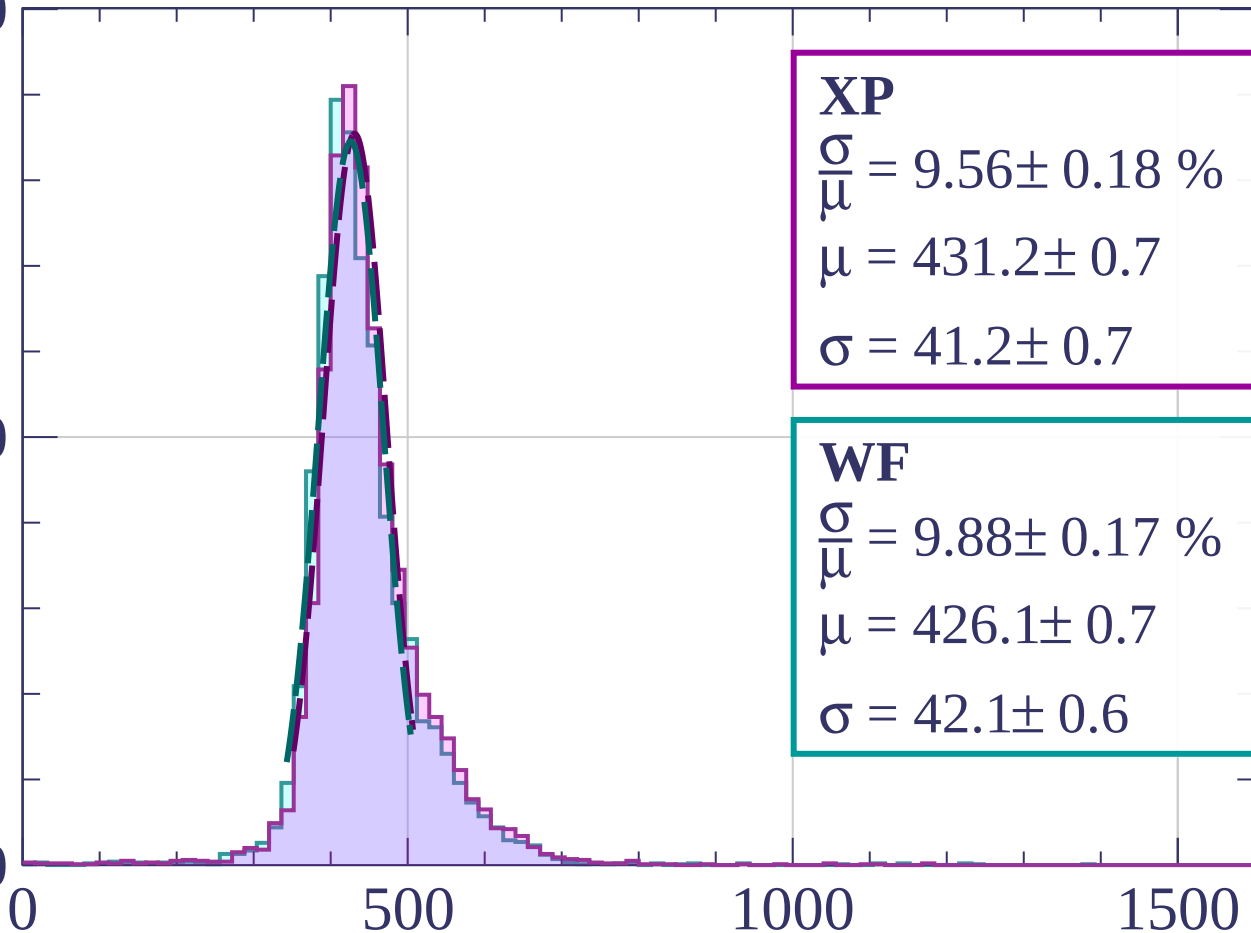
Energy loss | $400 < p < 480$

Count

1000

500

0



XP

$$\frac{\sigma}{\mu} = 9.56 \pm 0.18 \%$$

$$\mu = 431.2 \pm 0.7$$

$$\sigma = 41.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.88 \pm 0.17 \%$$

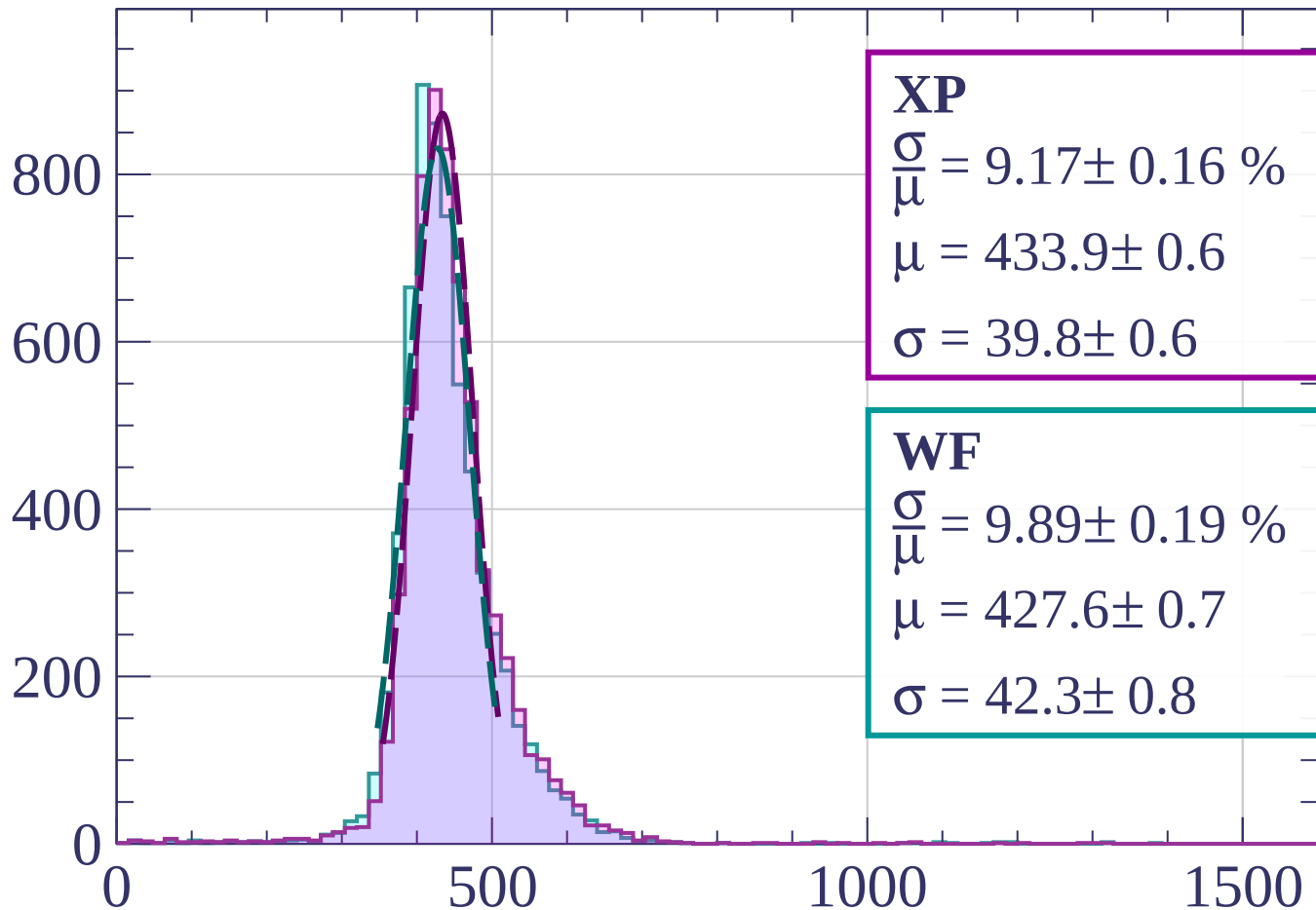
$$\mu = 426.1 \pm 0.7$$

$$\sigma = 42.1 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $480 < p < 560$

Count



XP

$$\frac{\sigma}{\mu} = 9.17 \pm 0.16 \%$$

$$\mu = 433.9 \pm 0.6$$

$$\sigma = 39.8 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.89 \pm 0.19 \%$$

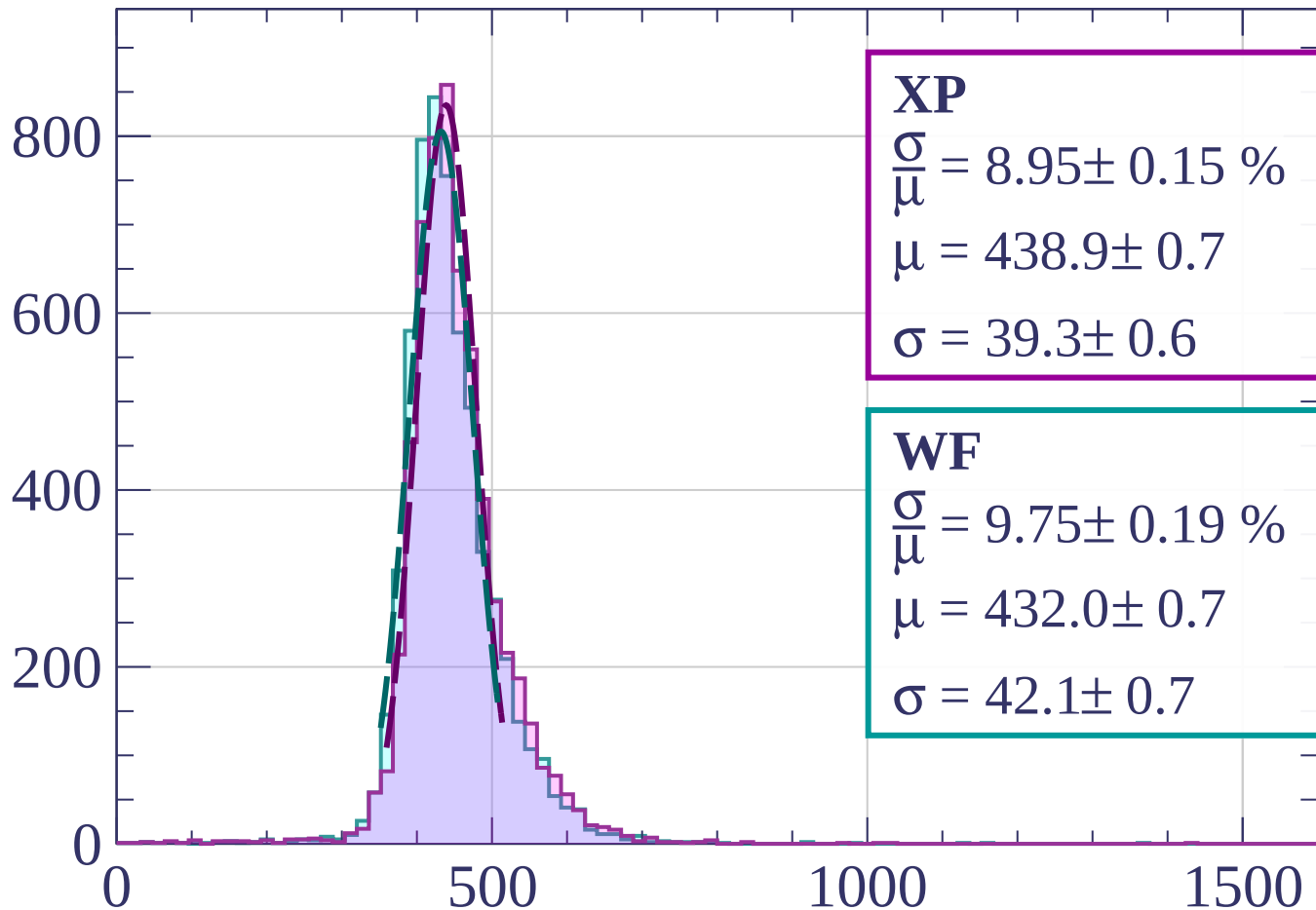
$$\mu = 427.6 \pm 0.7$$

$$\sigma = 42.3 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $560 < p < 640$

Count



XP

$$\frac{\sigma}{\mu} = 8.95 \pm 0.15 \%$$

$$\mu = 438.9 \pm 0.7$$

$$\sigma = 39.3 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.75 \pm 0.19 \%$$

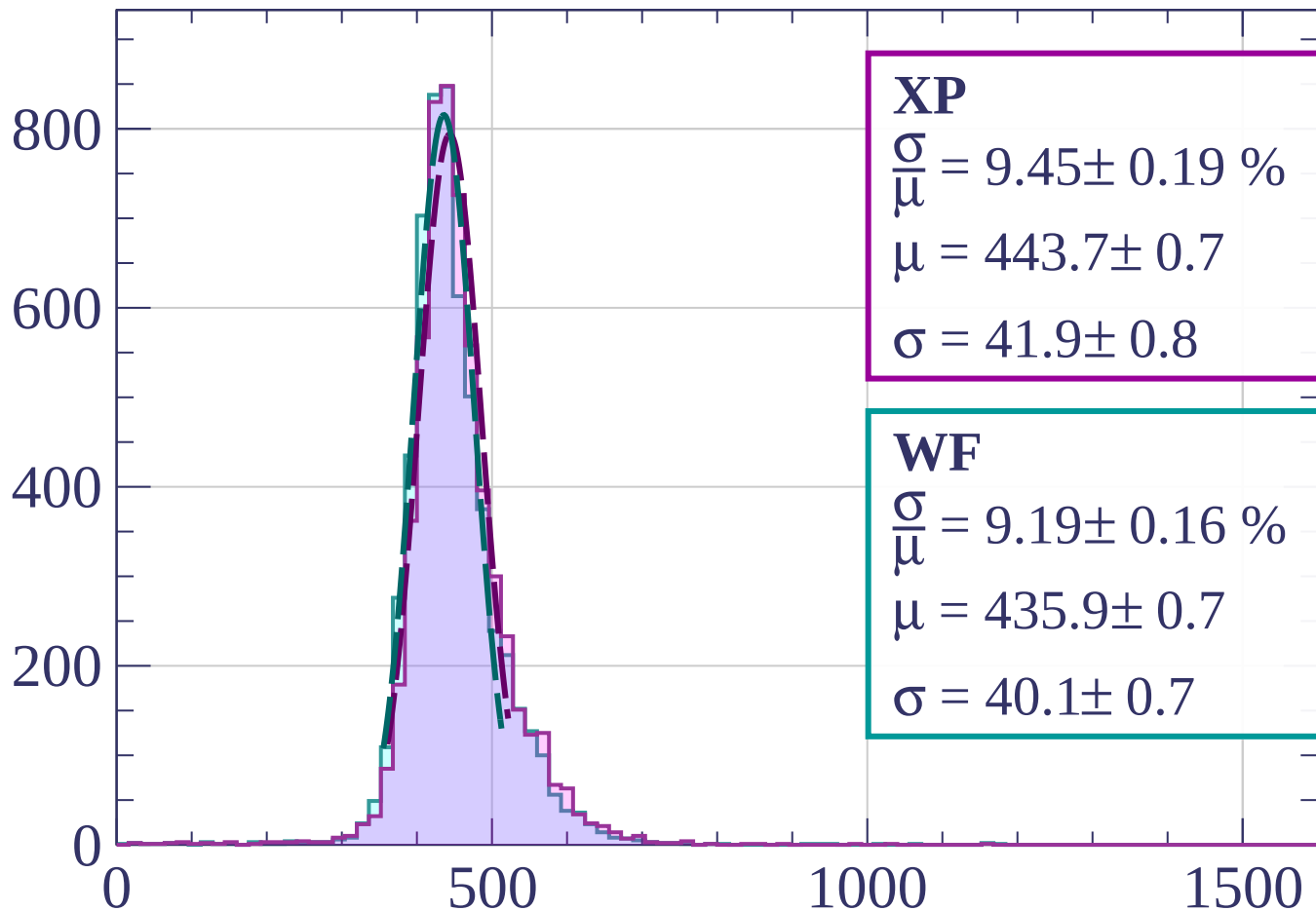
$$\mu = 432.0 \pm 0.7$$

$$\sigma = 42.1 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $640 < p < 720$

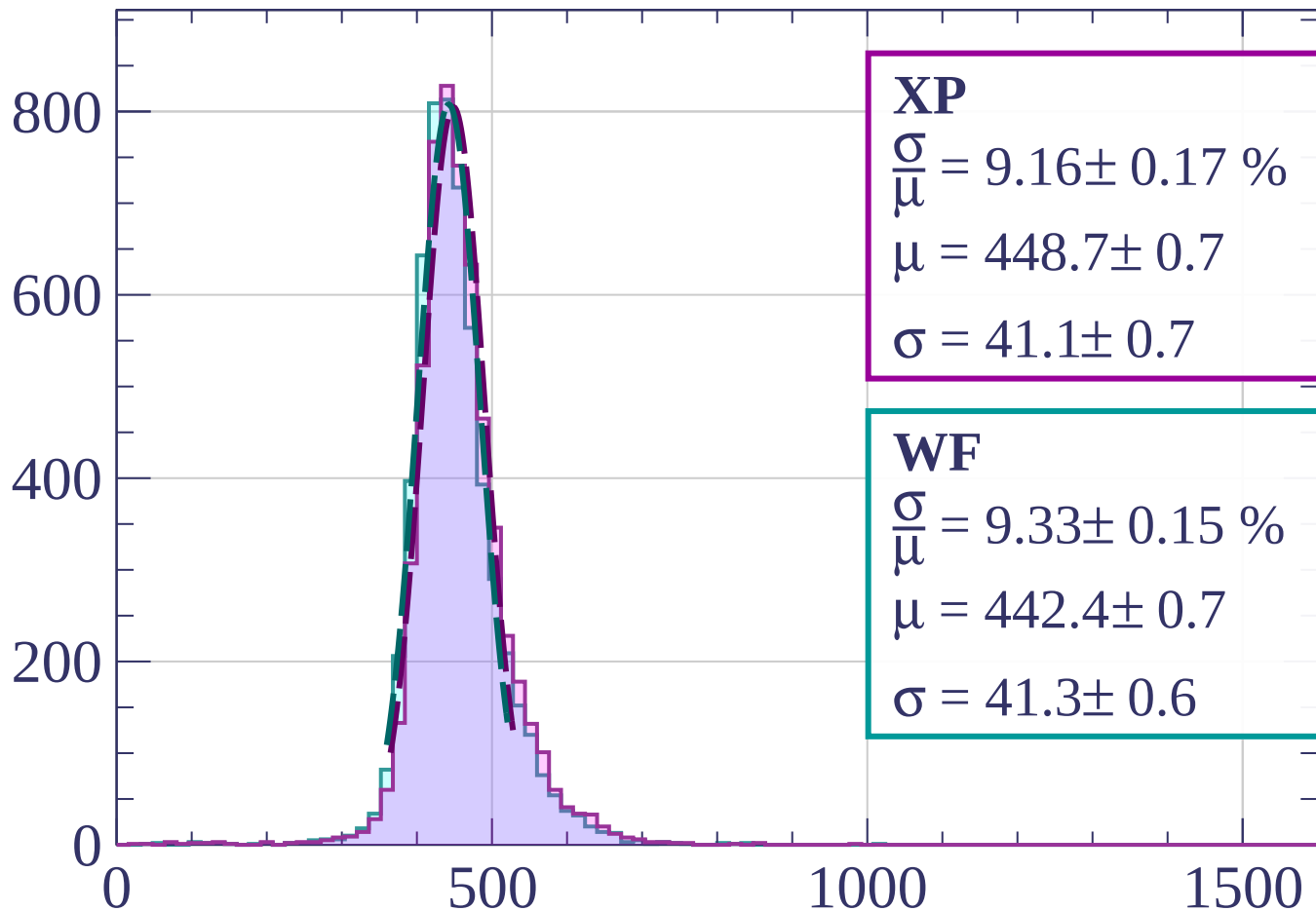
Count



dE/dx [ADC counts/cm]

Energy loss | $720 < p < 800$

Count



XP

$$\frac{\sigma}{\mu} = 9.16 \pm 0.17 \%$$

$$\mu = 448.7 \pm 0.7$$

$$\sigma = 41.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.33 \pm 0.15 \%$$

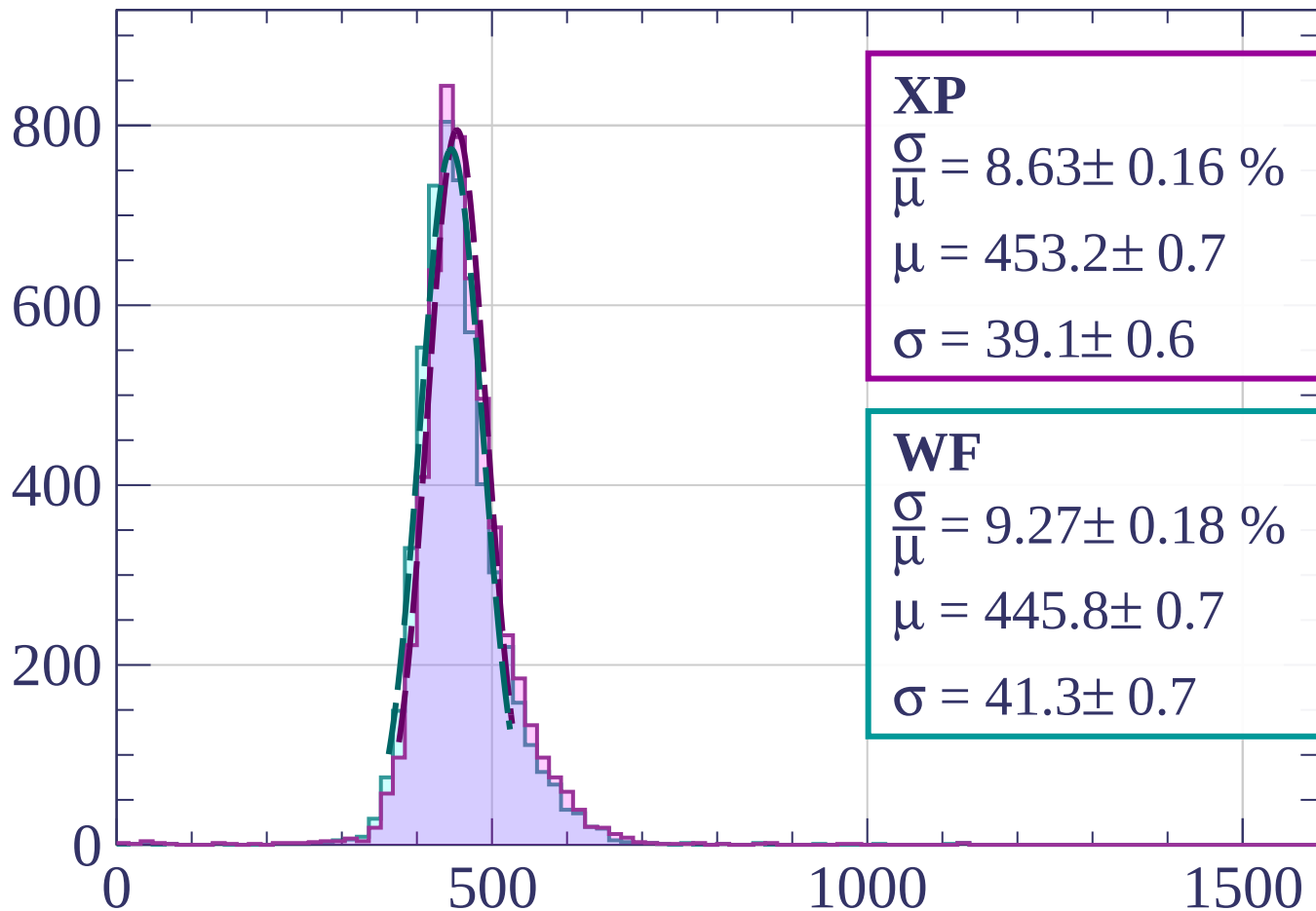
$$\mu = 442.4 \pm 0.7$$

$$\sigma = 41.3 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $800 < p < 880$

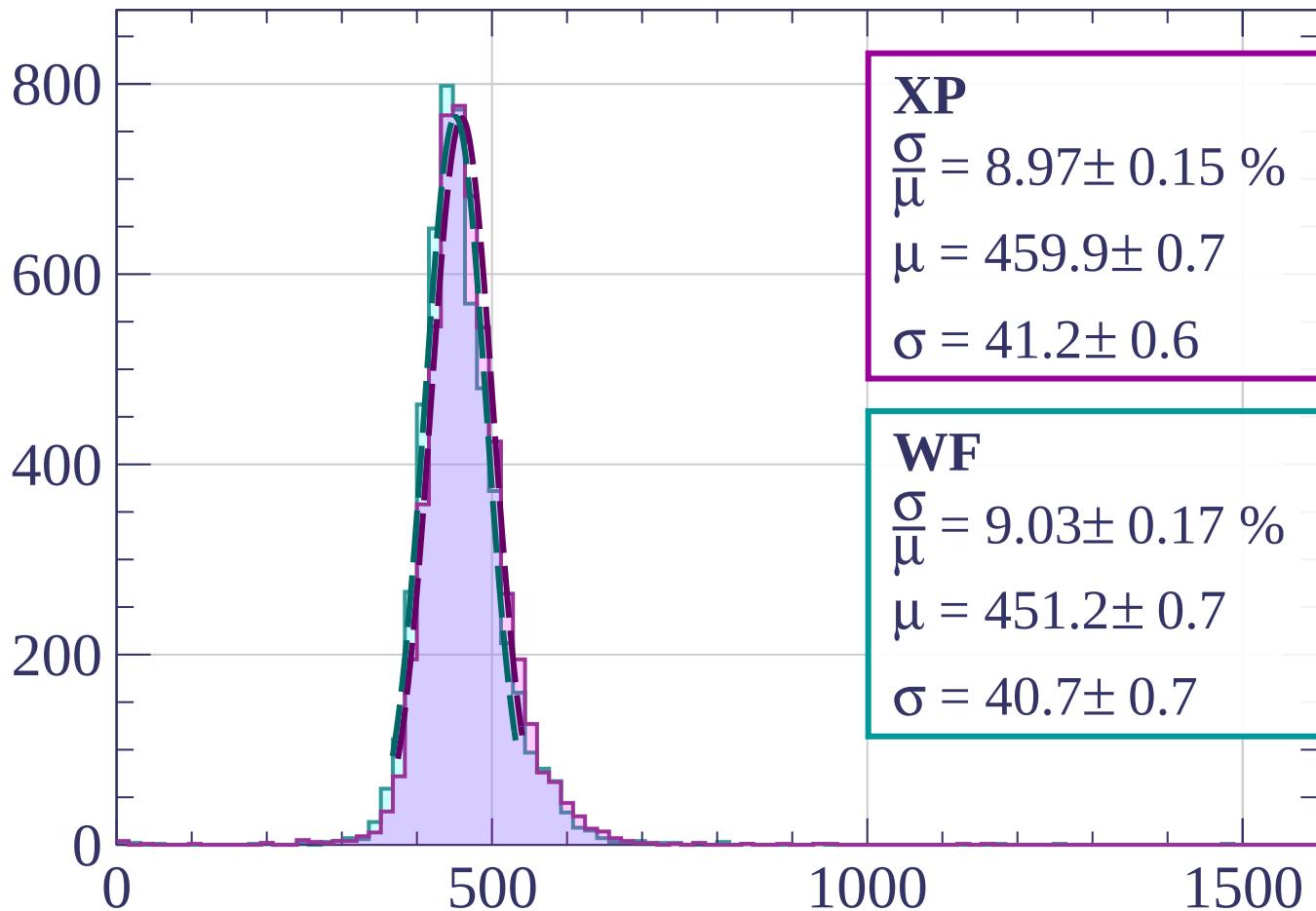
Count



dE/dx [ADC counts/cm]

Energy loss | $880 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 8.97 \pm 0.15 \%$$

$$\mu = 459.9 \pm 0.7$$

$$\sigma = 41.2 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.03 \pm 0.17 \%$$

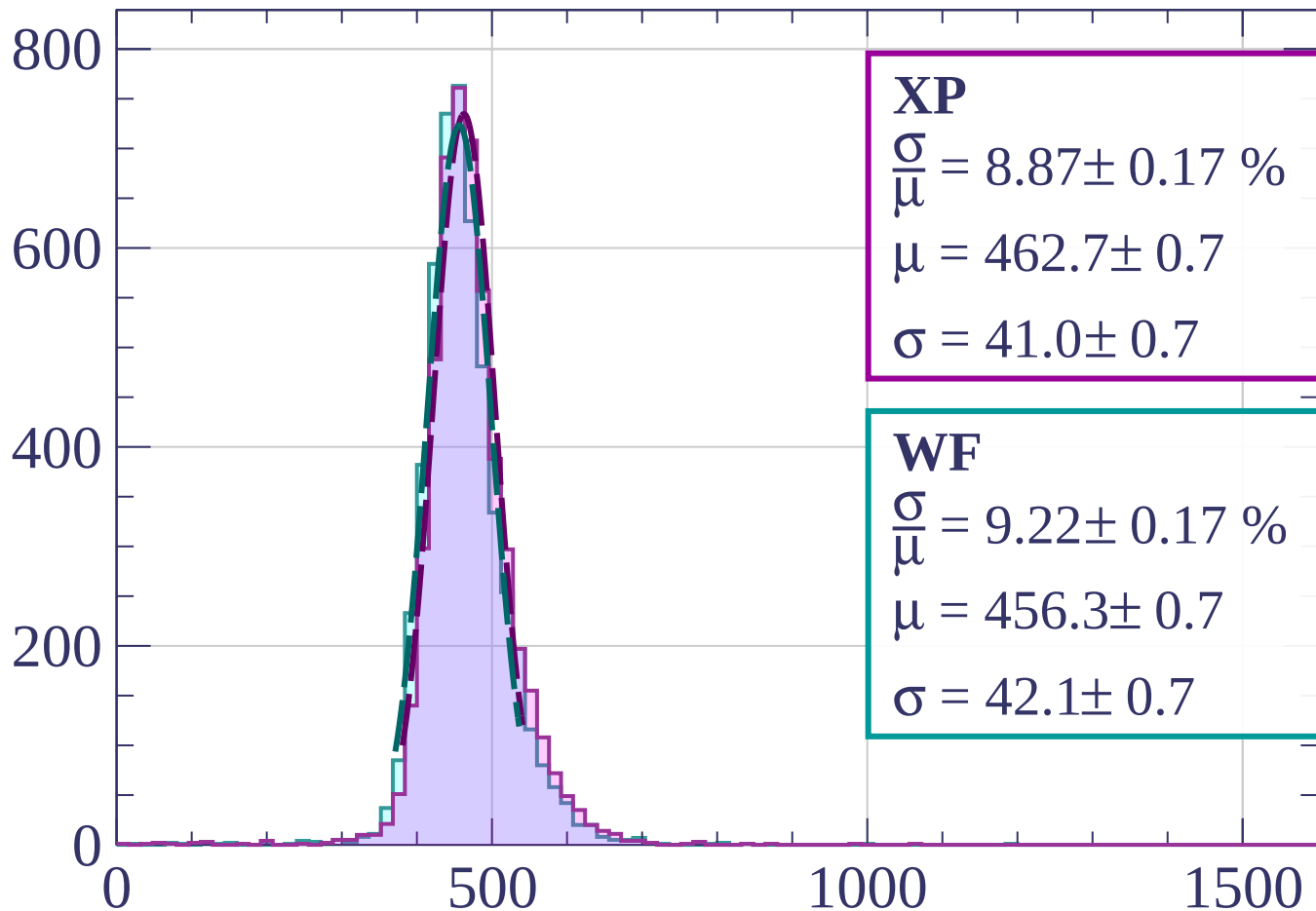
$$\mu = 451.2 \pm 0.7$$

$$\sigma = 40.7 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $960 < p < 1040$

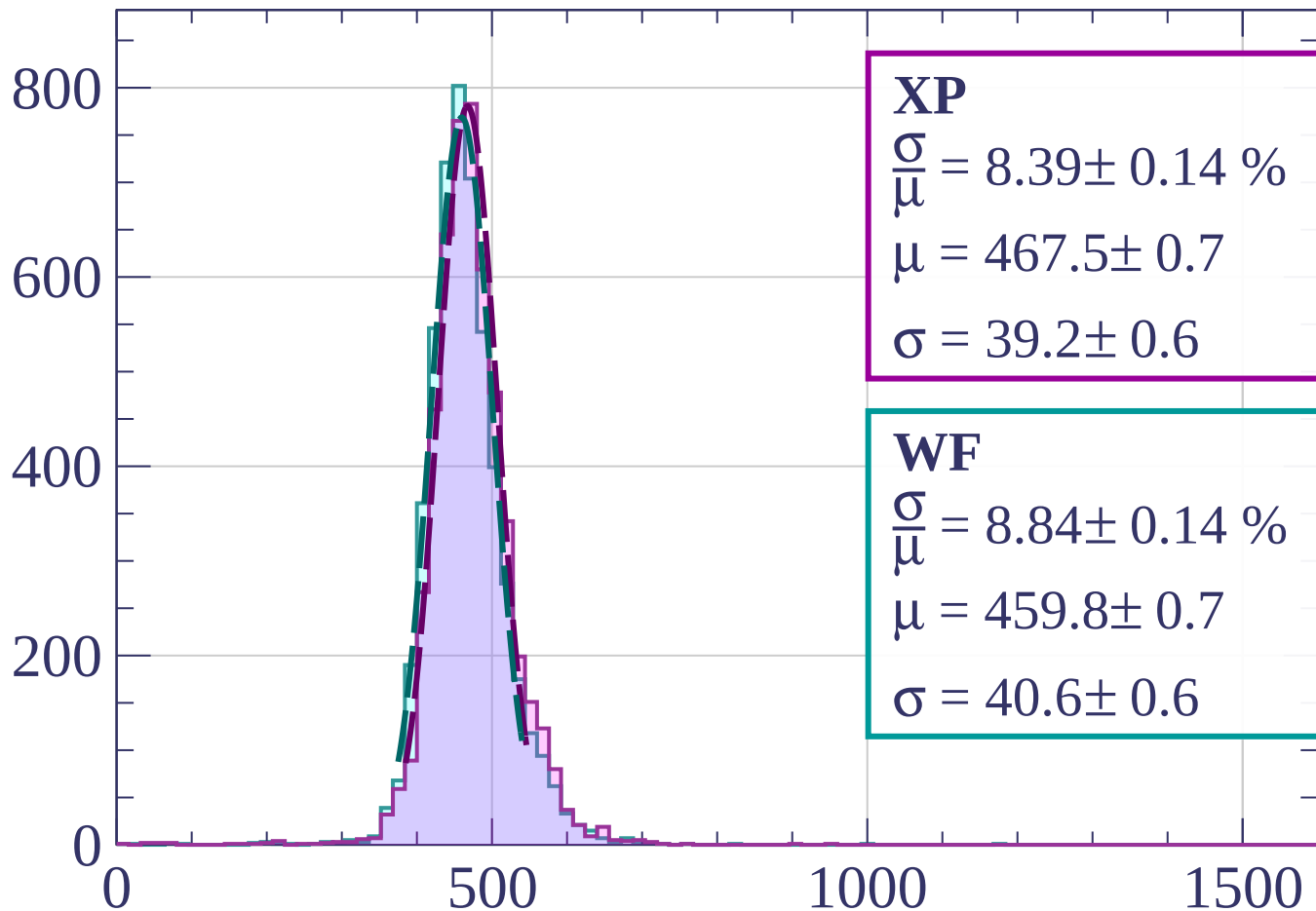
Count



dE/dx [ADC counts/cm]

Energy loss | $1040 < p < 1120$

Count



XP

$$\frac{\sigma}{\mu} = 8.39 \pm 0.14 \%$$

$$\mu = 467.5 \pm 0.7$$

$$\sigma = 39.2 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.84 \pm 0.14 \%$$

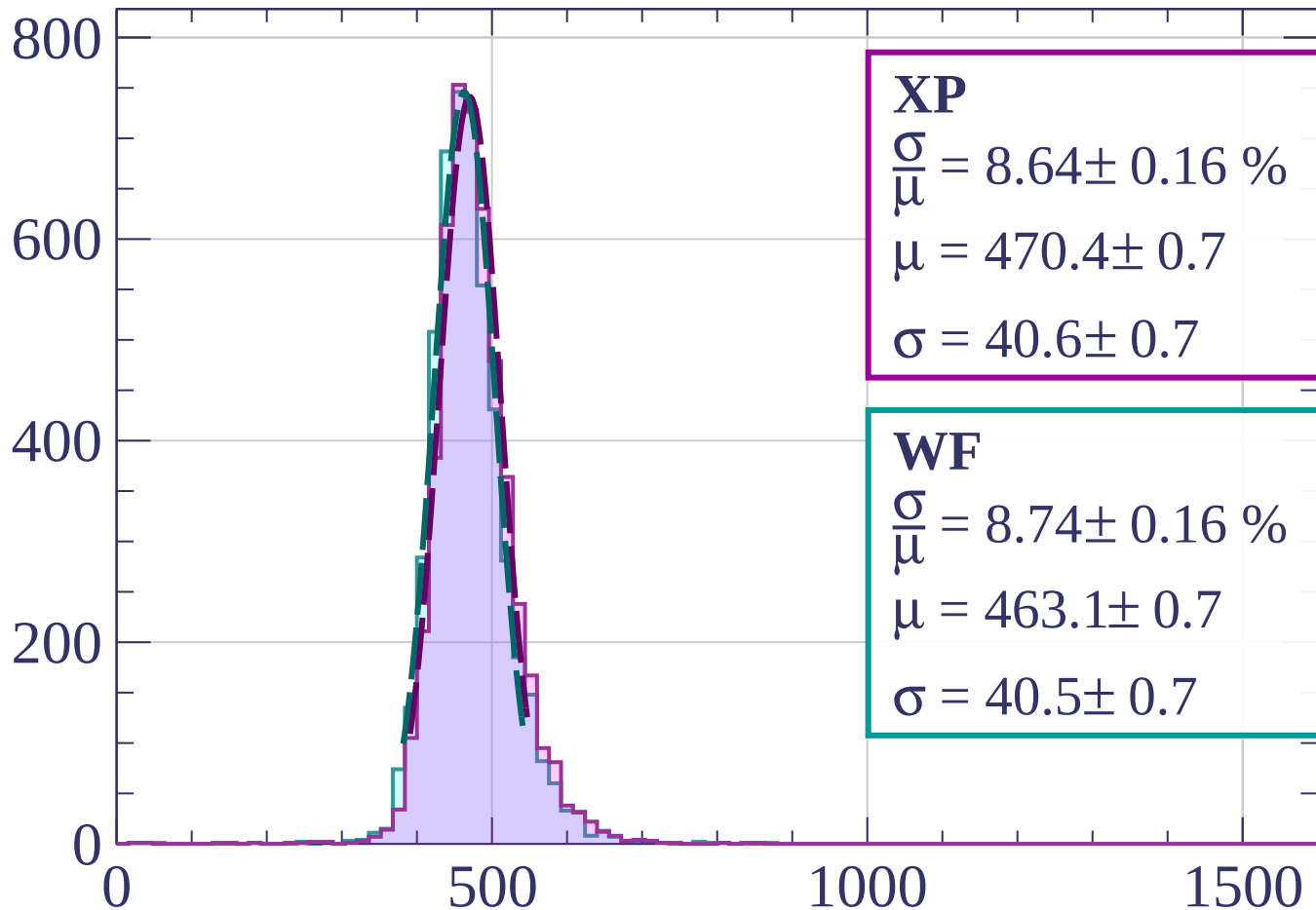
$$\mu = 459.8 \pm 0.7$$

$$\sigma = 40.6 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $1120 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 8.64 \pm 0.16 \%$$

$$\mu = 470.4 \pm 0.7$$

$$\sigma = 40.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.74 \pm 0.16 \%$$

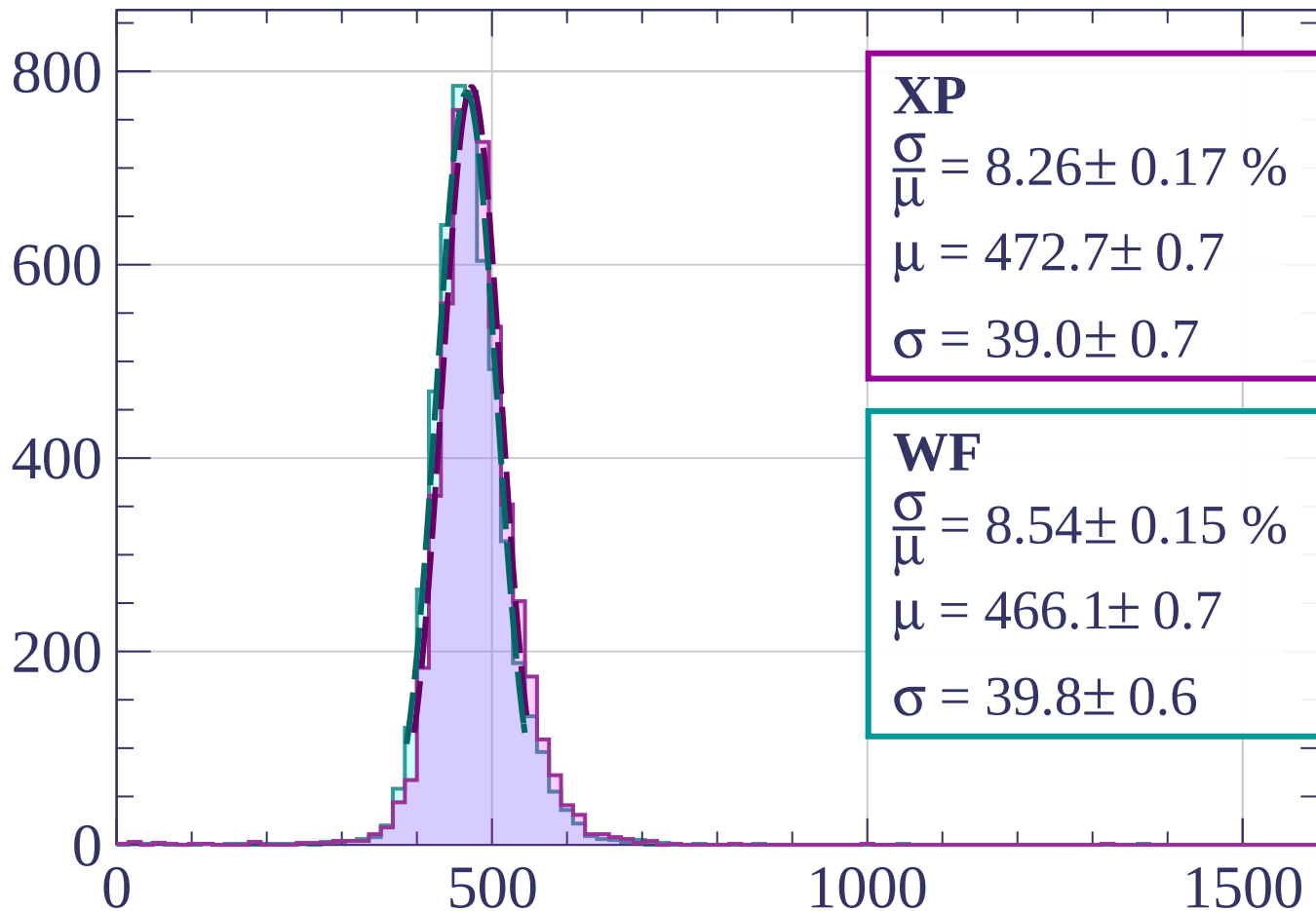
$$\mu = 463.1 \pm 0.7$$

$$\sigma = 40.5 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $1200 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 8.26 \pm 0.17 \%$$

$$\mu = 472.7 \pm 0.7$$

$$\sigma = 39.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.54 \pm 0.15 \%$$

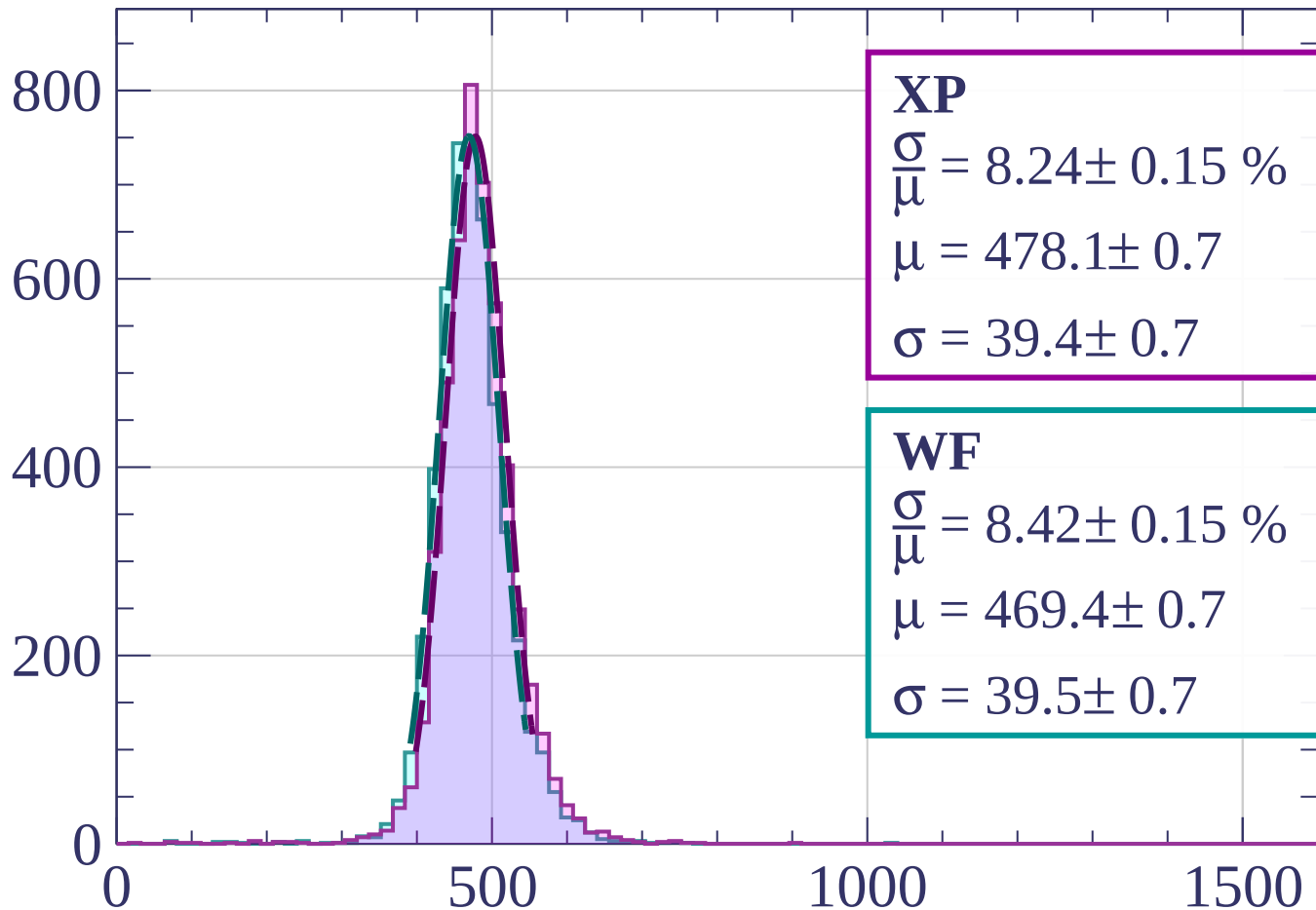
$$\mu = 466.1 \pm 0.7$$

$$\sigma = 39.8 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $1280 < p < 1360$

Count



XP

$$\frac{\sigma}{\mu} = 8.24 \pm 0.15 \%$$

$$\mu = 478.1 \pm 0.7$$

$$\sigma = 39.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.42 \pm 0.15 \%$$

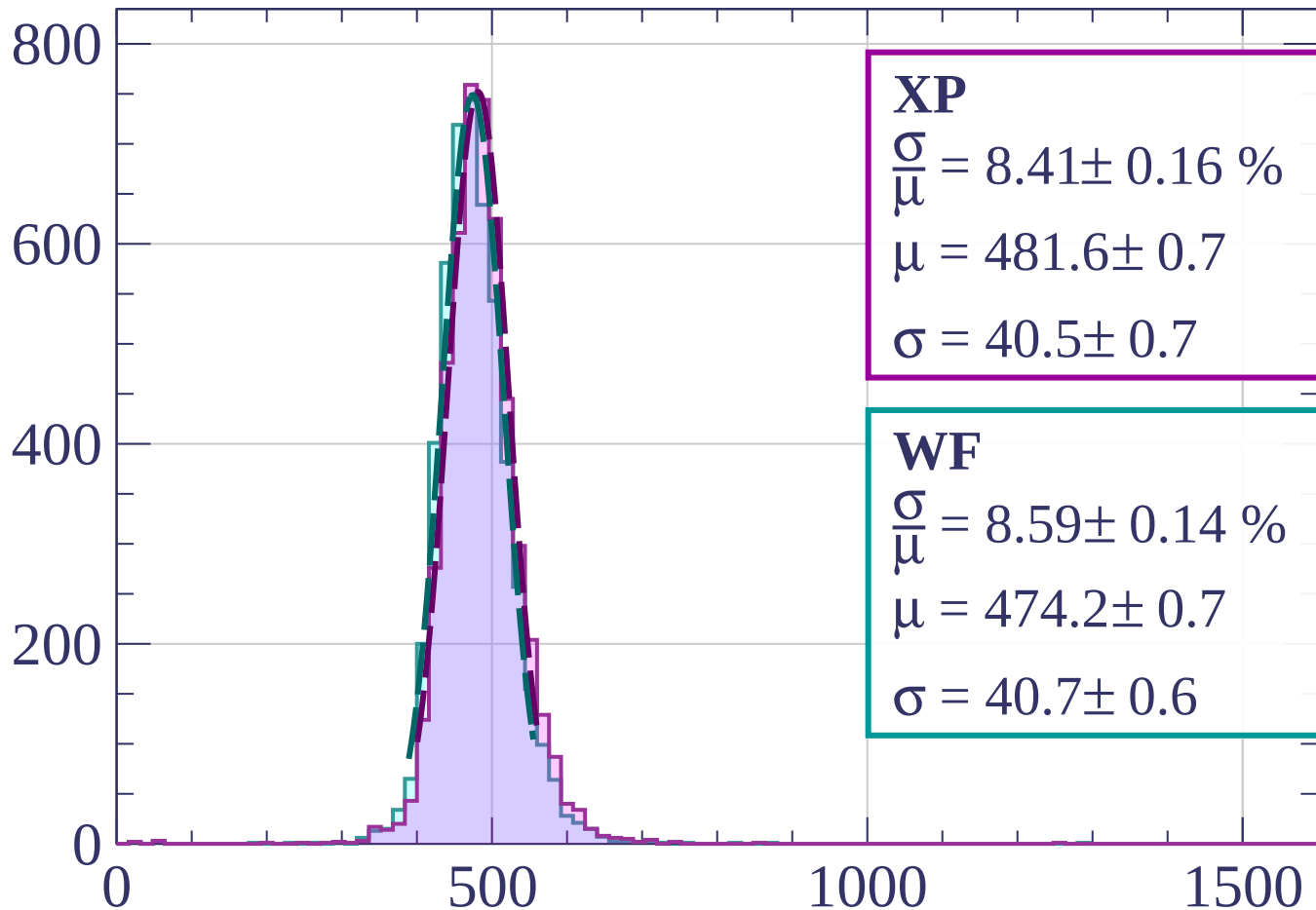
$$\mu = 469.4 \pm 0.7$$

$$\sigma = 39.5 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $1360 < p < 1440$

Count



XP

$$\frac{\sigma}{\mu} = 8.41 \pm 0.16 \%$$

$$\mu = 481.6 \pm 0.7$$

$$\sigma = 40.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.59 \pm 0.14 \%$$

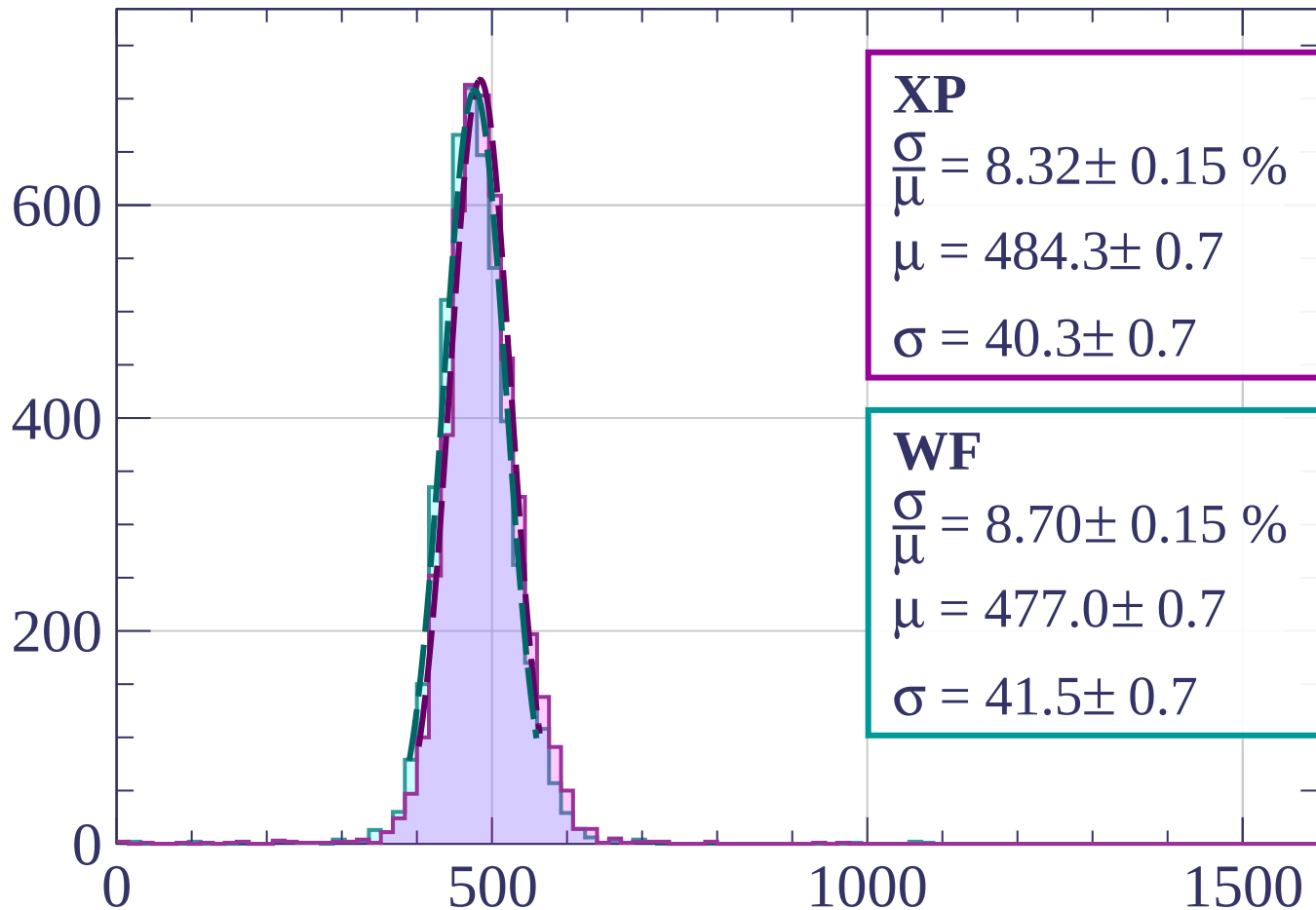
$$\mu = 474.2 \pm 0.7$$

$$\sigma = 40.7 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $1440 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 8.32 \pm 0.15 \%$$

$$\mu = 484.3 \pm 0.7$$

$$\sigma = 40.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.70 \pm 0.15 \%$$

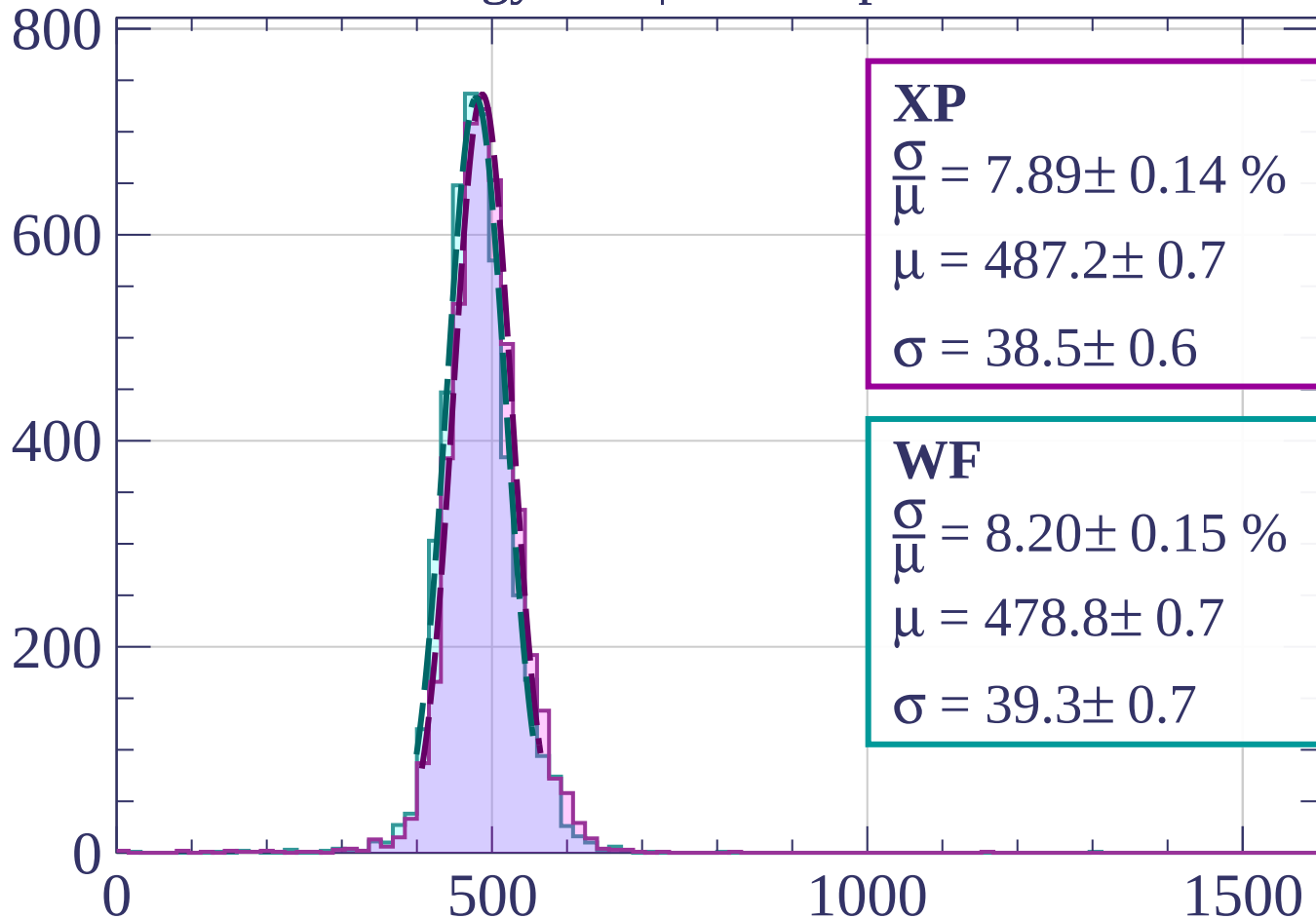
$$\mu = 477.0 \pm 0.7$$

$$\sigma = 41.5 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $1520 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 7.89 \pm 0.14 \%$$

$$\mu = 487.2 \pm 0.7$$

$$\sigma = 38.5 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.20 \pm 0.15 \%$$

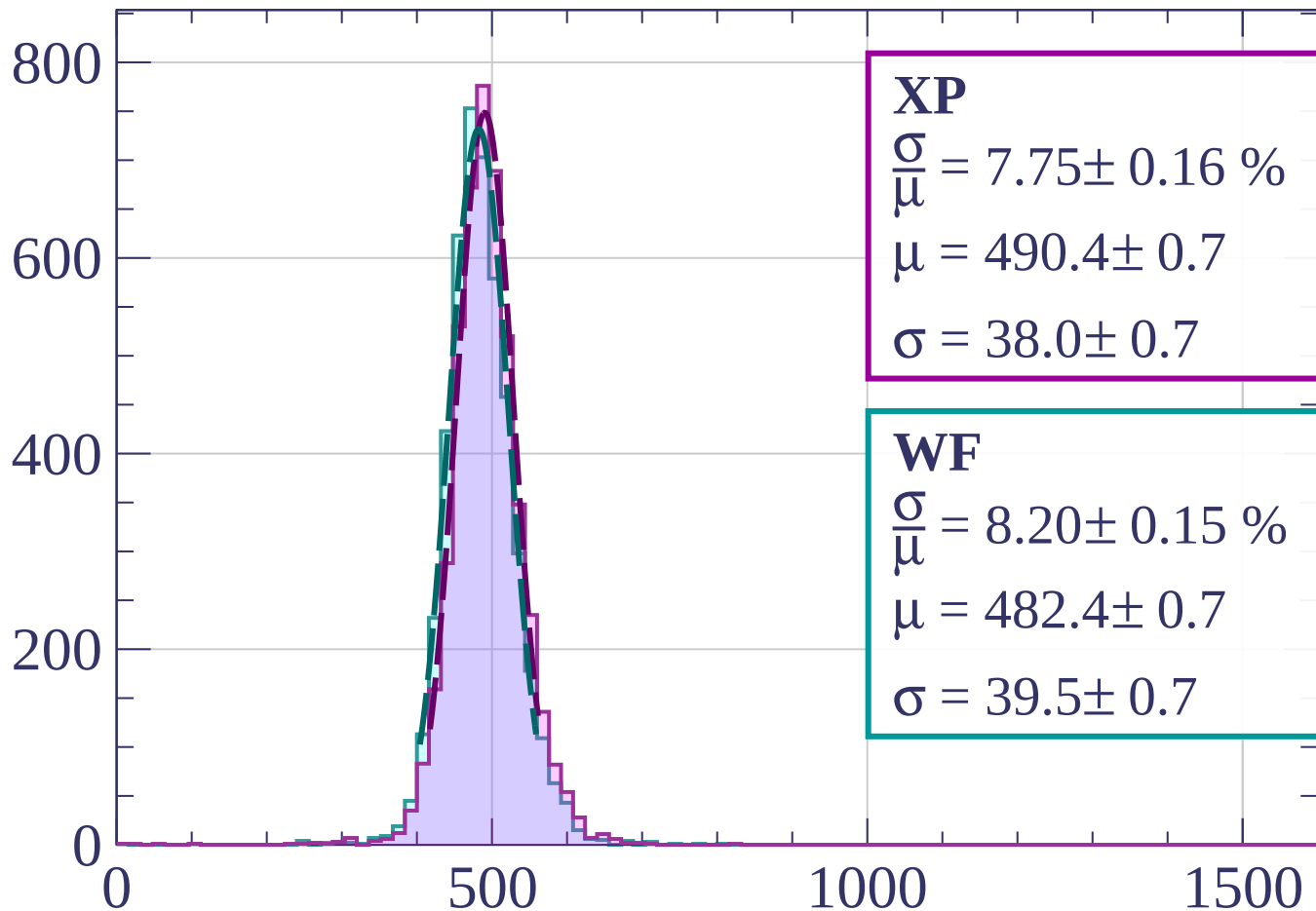
$$\mu = 478.8 \pm 0.7$$

$$\sigma = 39.3 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $1600 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 7.75 \pm 0.16 \%$$

$$\mu = 490.4 \pm 0.7$$

$$\sigma = 38.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.20 \pm 0.15 \%$$

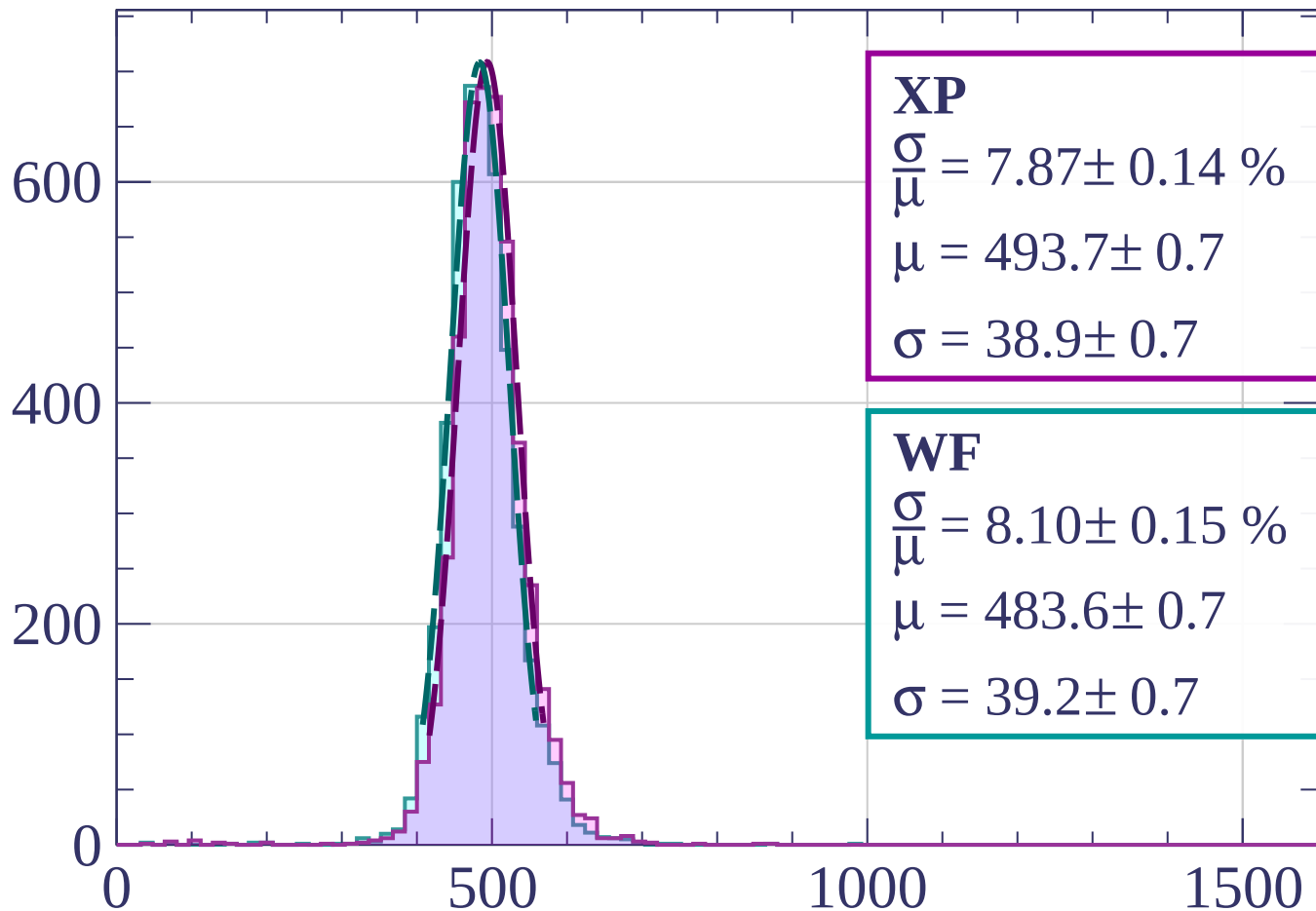
$$\mu = 482.4 \pm 0.7$$

$$\sigma = 39.5 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $1680 < p < 1760$

Count



XP

$$\frac{\sigma}{\mu} = 7.87 \pm 0.14 \%$$

$$\mu = 493.7 \pm 0.7$$

$$\sigma = 38.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.10 \pm 0.15 \%$$

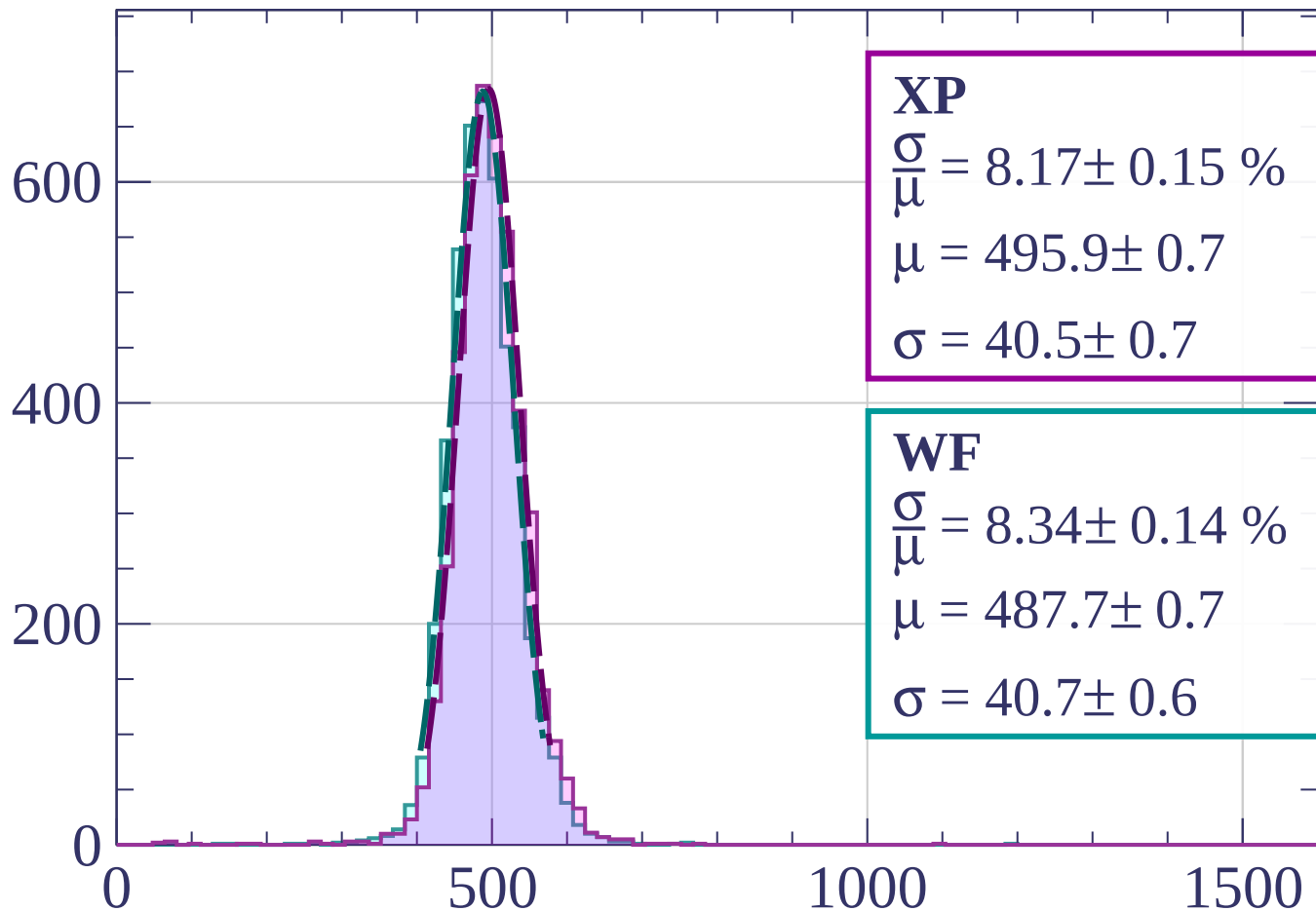
$$\mu = 483.6 \pm 0.7$$

$$\sigma = 39.2 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $1760 < p < 1840$

Count



XP

$$\frac{\sigma}{\mu} = 8.17 \pm 0.15 \%$$

$$\mu = 495.9 \pm 0.7$$

$$\sigma = 40.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.34 \pm 0.14 \%$$

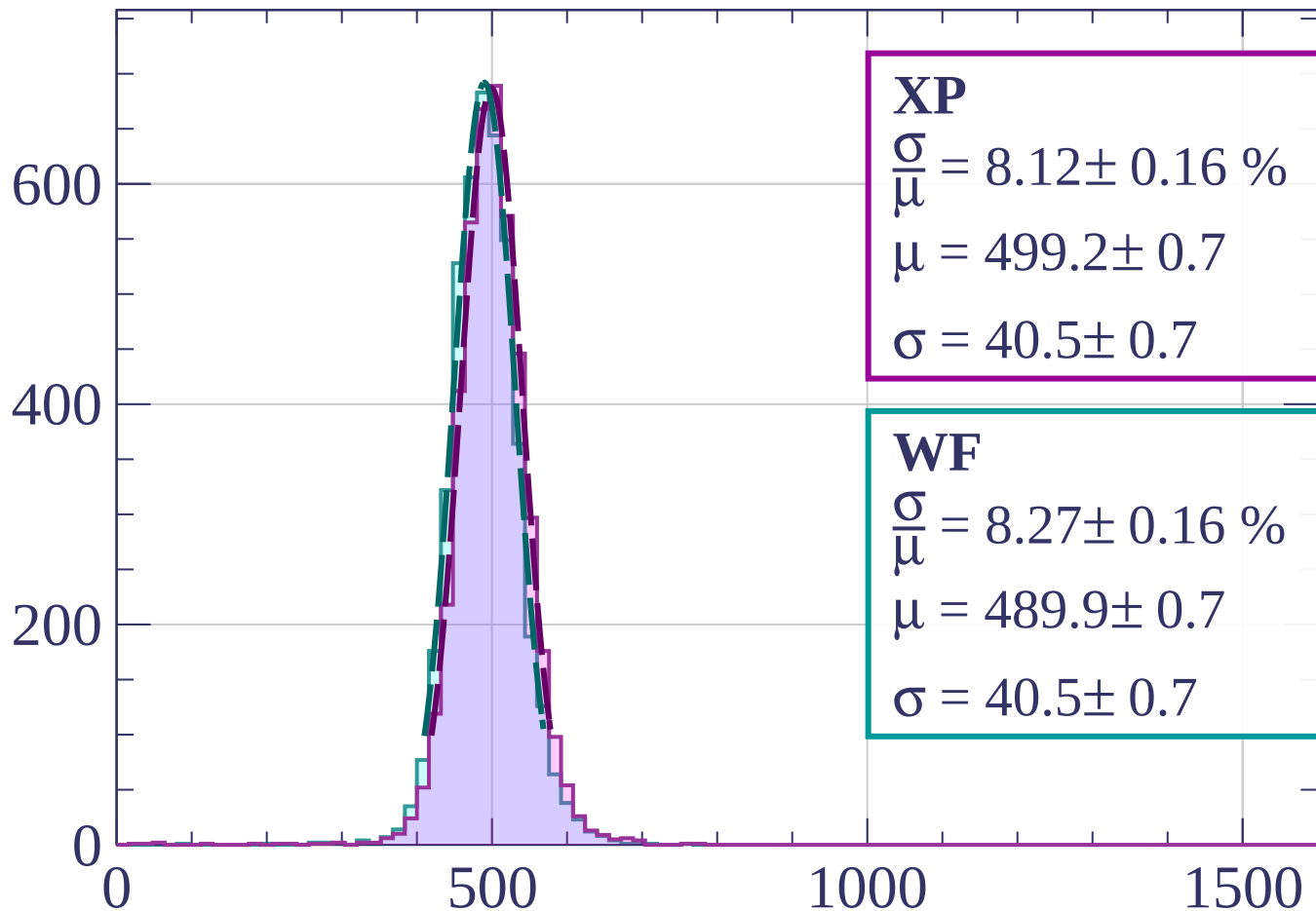
$$\mu = 487.7 \pm 0.7$$

$$\sigma = 40.7 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $1840 < p < 1920$

Count



XP

$$\frac{\sigma}{\mu} = 8.12 \pm 0.16 \%$$

$$\mu = 499.2 \pm 0.7$$

$$\sigma = 40.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.27 \pm 0.16 \%$$

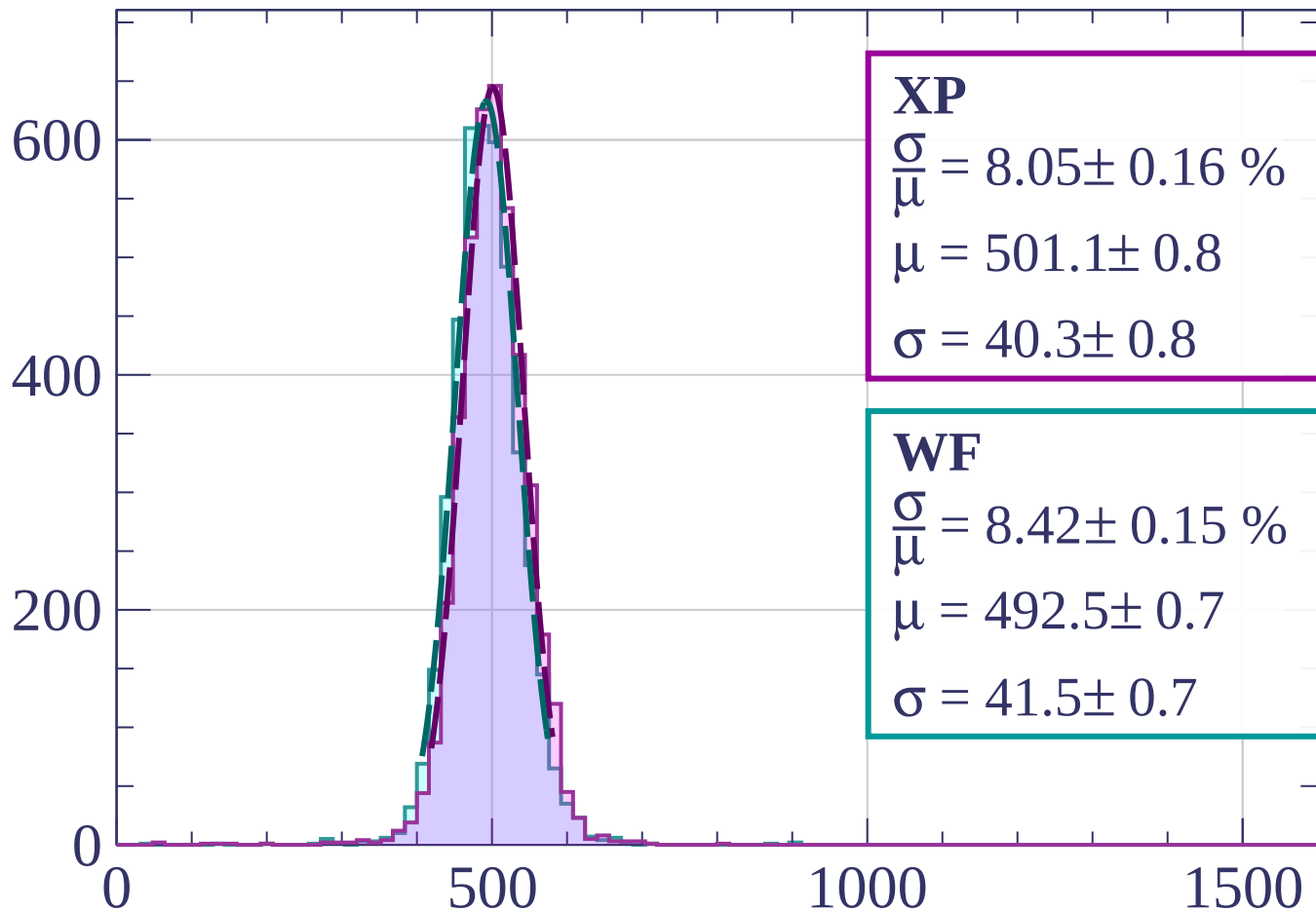
$$\mu = 489.9 \pm 0.7$$

$$\sigma = 40.5 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $1920 < p < 2000$

Count



XP

$$\frac{\sigma}{\mu} = 8.05 \pm 0.16 \%$$

$$\mu = 501.1 \pm 0.8$$

$$\sigma = 40.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.42 \pm 0.15 \%$$

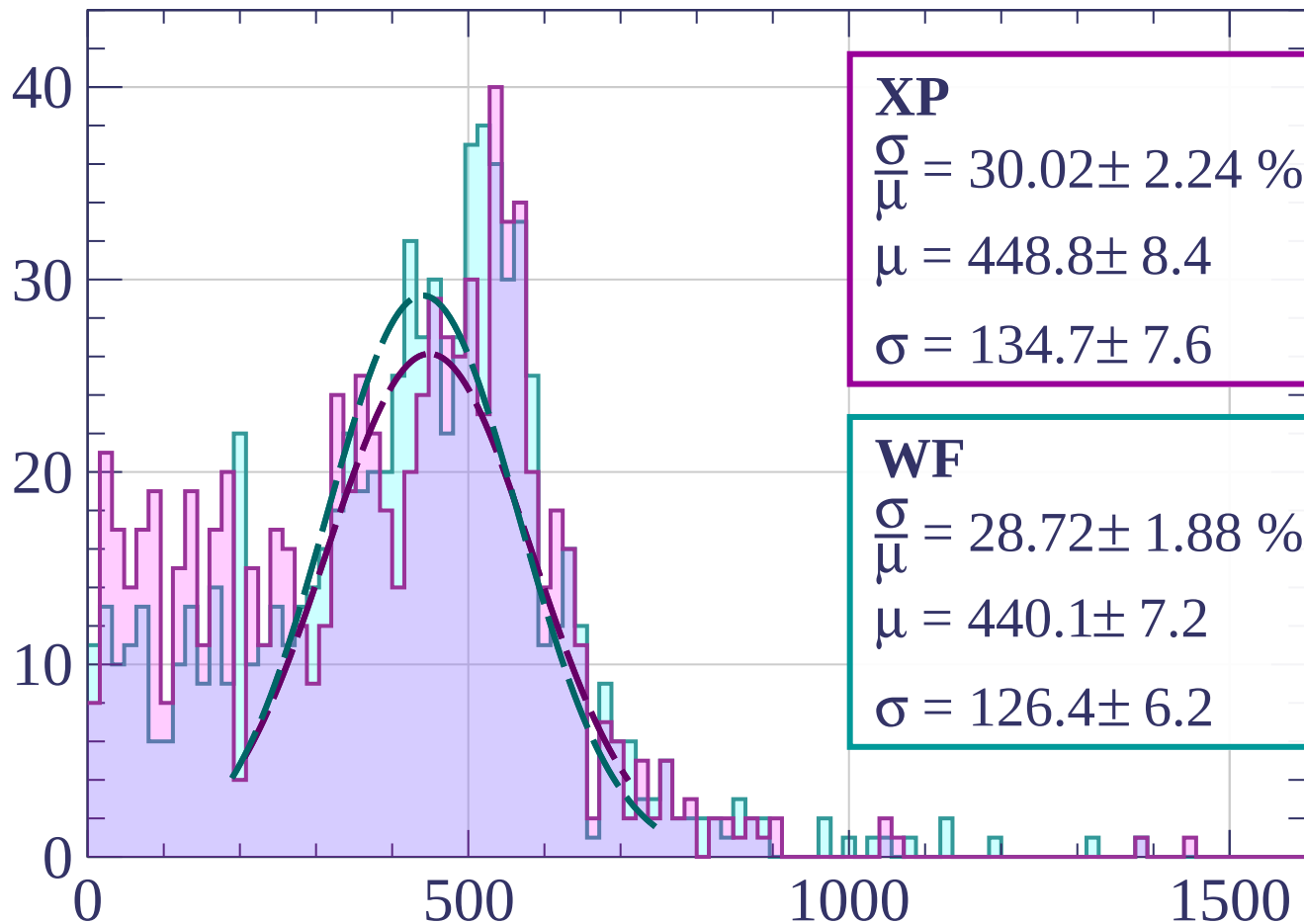
$$\mu = 492.5 \pm 0.7$$

$$\sigma = 41.5 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $0 < \phi < 3$

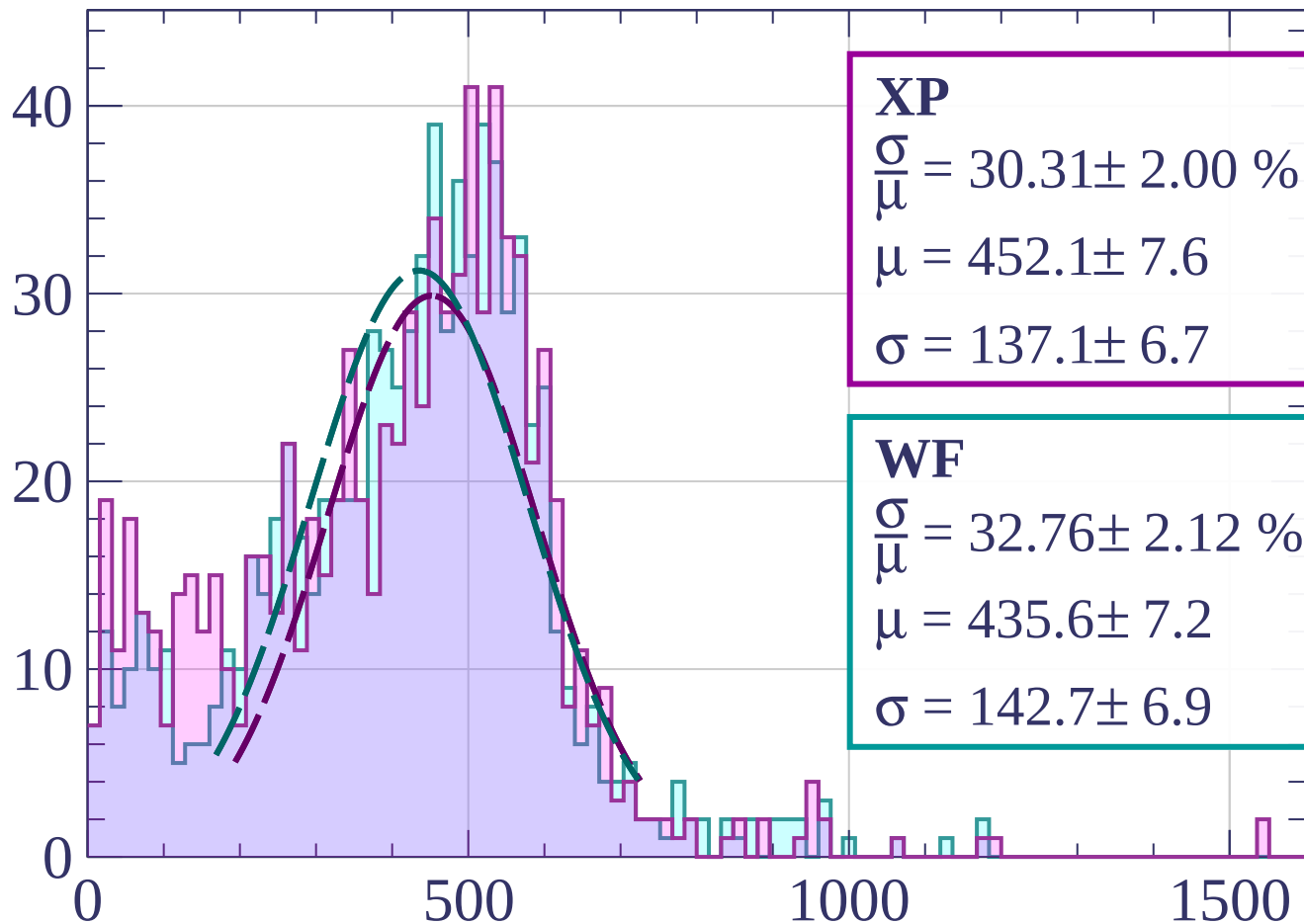
Count



dE/dx [ADC counts/cm]

Energy loss | $3 < \phi < 6$

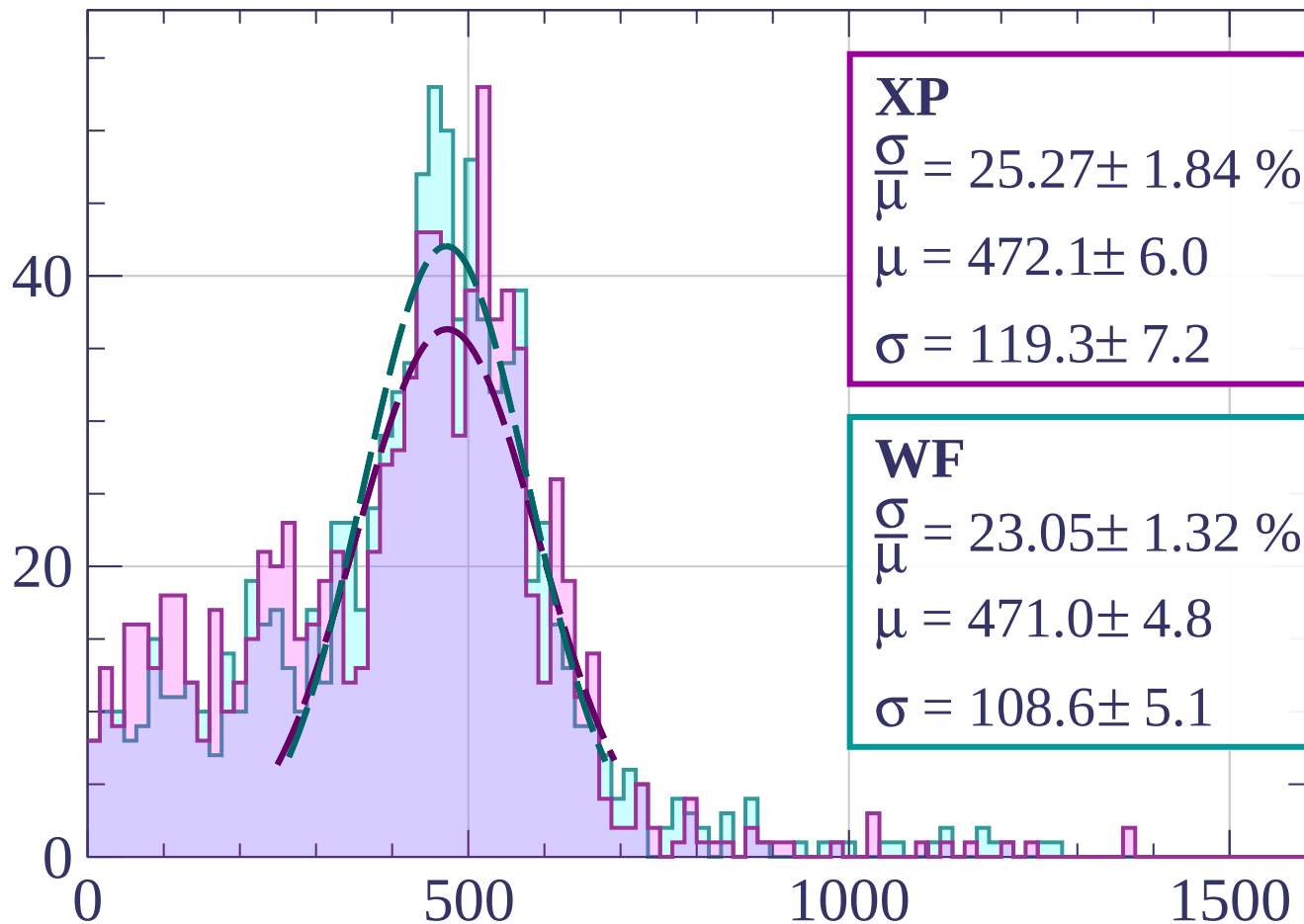
Count



dE/dx [ADC counts/cm]

Energy loss | $6 < \phi < 9$

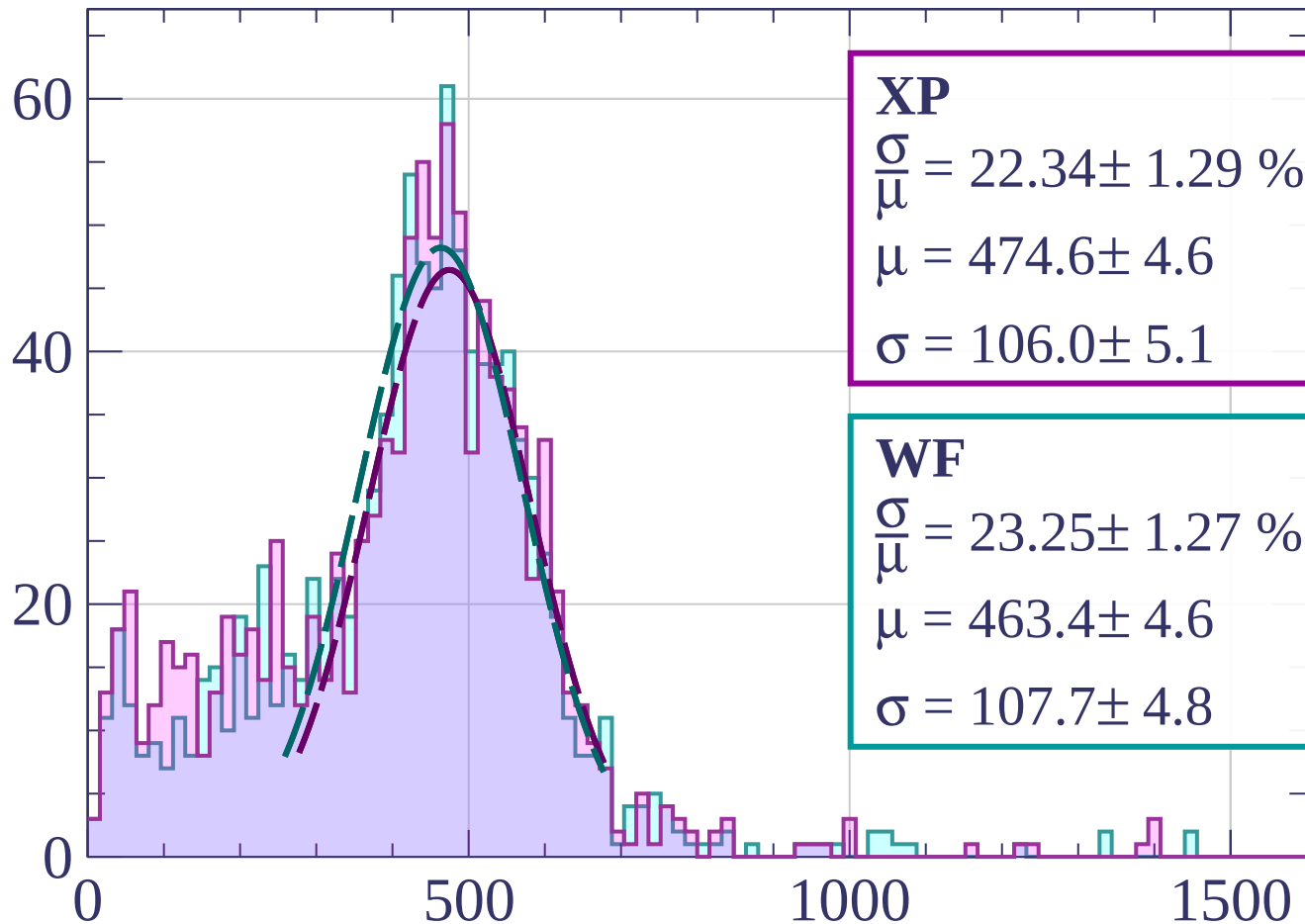
Count



dE/dx [ADC counts/cm]

Energy loss | $9 < \phi < 12$

Count



XP

$$\frac{\sigma}{\mu} = 22.34 \pm 1.29 \%$$

$$\mu = 474.6 \pm 4.6$$

$$\sigma = 106.0 \pm 5.1$$

WF

$$\frac{\sigma}{\mu} = 23.25 \pm 1.27 \%$$

$$\mu = 463.4 \pm 4.6$$

$$\sigma = 107.7 \pm 4.8$$

dE/dx [ADC counts/cm]

Count

Energy loss | $12 < \phi < 15$

XP

$$\frac{\sigma}{\mu} = 18.02 \pm 0.78 \%$$

$$\mu = 487.1 \pm 3.1$$

$$\sigma = 87.8 \pm 3.3$$

WF

$$\frac{\sigma}{\mu} = 18.43 \pm 0.72 \%$$

$$\mu = 474.6 \pm 3.0$$

$$\sigma = 87.5 \pm 2.9$$

0

50

100

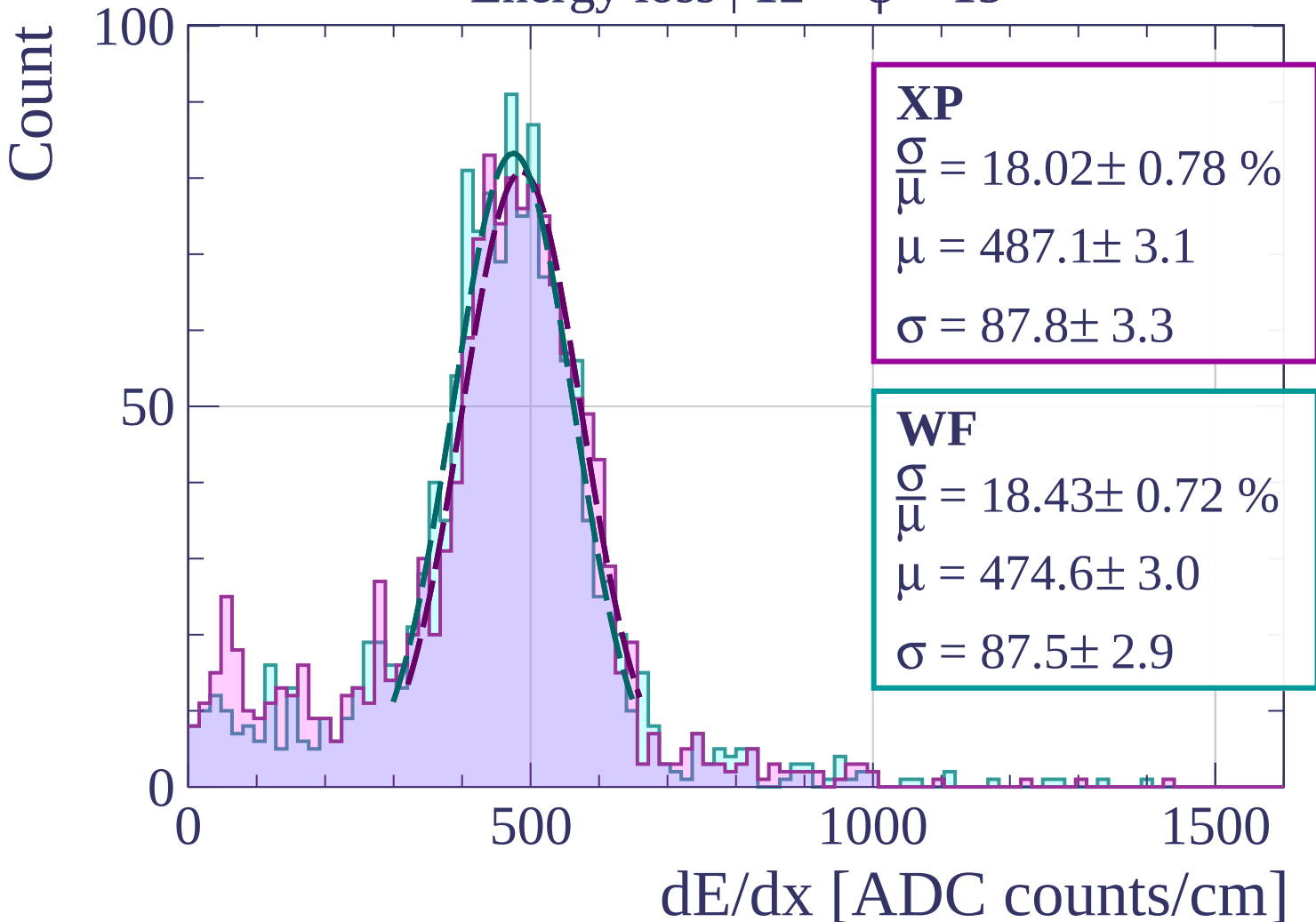
0

500

1000

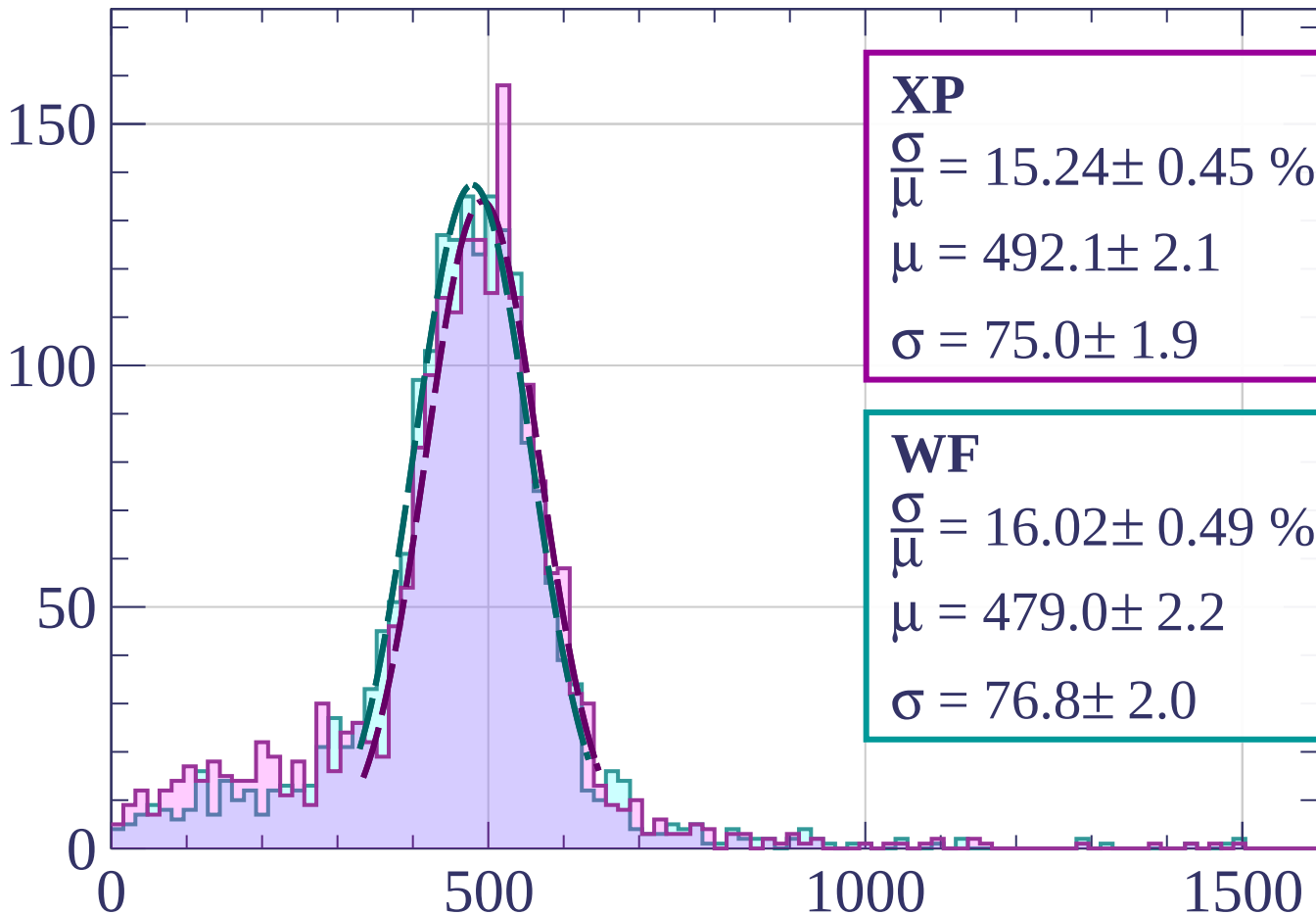
1500

dE/dx [ADC counts/cm]



Energy loss | $15 < \phi < 18$

Count



XP

$$\frac{\sigma}{\mu} = 15.24 \pm 0.45 \%$$

$$\mu = 492.1 \pm 2.1$$

$$\sigma = 75.0 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 16.02 \pm 0.49 \%$$

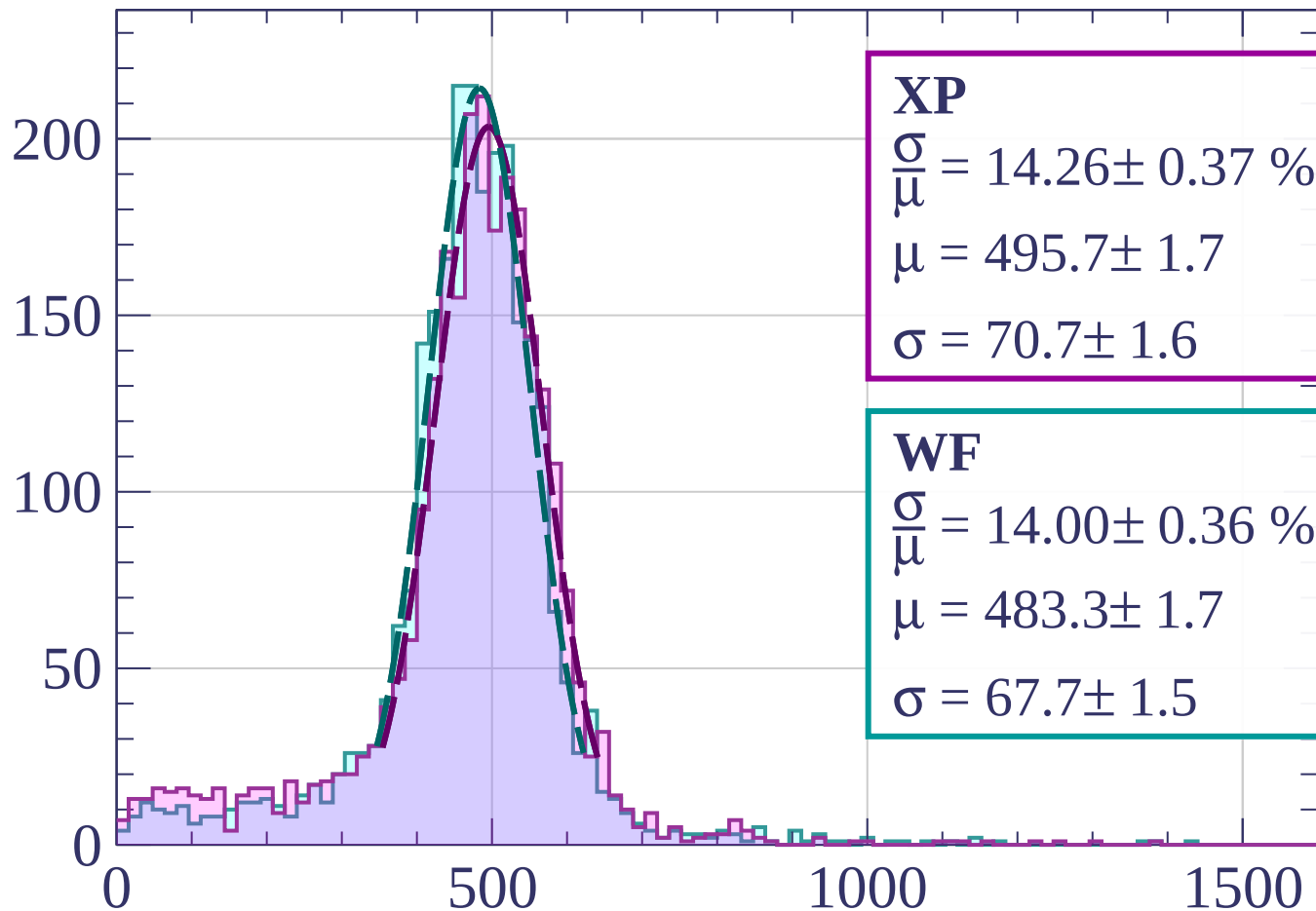
$$\mu = 479.0 \pm 2.2$$

$$\sigma = 76.8 \pm 2.0$$

dE/dx [ADC counts/cm]

Energy loss | $18 < \phi < 21$

Count



XP

$$\frac{\sigma}{\mu} = 14.26 \pm 0.37 \%$$

$$\mu = 495.7 \pm 1.7$$

$$\sigma = 70.7 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 14.00 \pm 0.36 \%$$

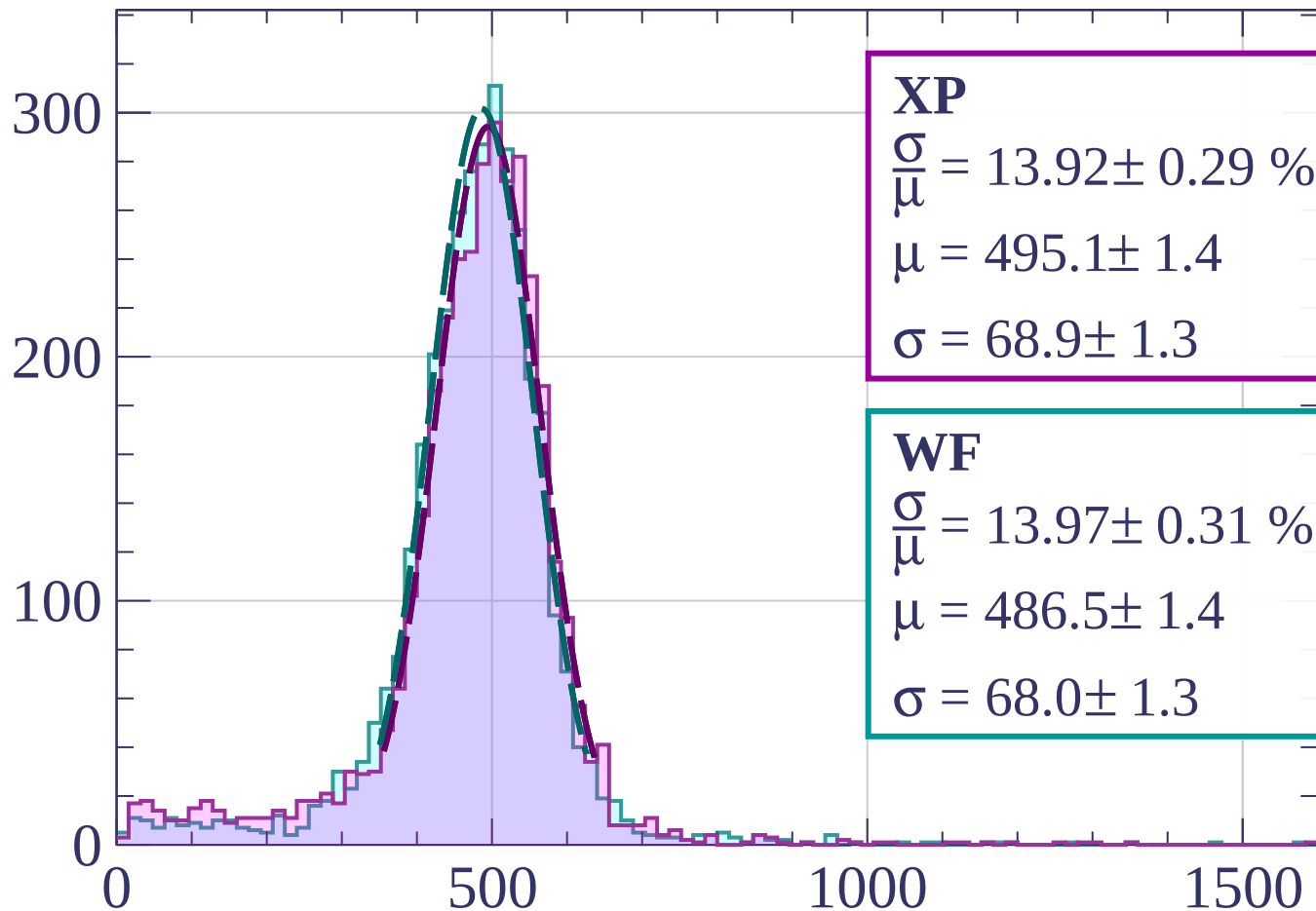
$$\mu = 483.3 \pm 1.7$$

$$\sigma = 67.7 \pm 1.5$$

dE/dx [ADC counts/cm]

Energy loss | $21 < \phi < 24$

Count



XP

$$\frac{\sigma}{\mu} = 13.92 \pm 0.29 \%$$

$$\mu = 495.1 \pm 1.4$$

$$\sigma = 68.9 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 13.97 \pm 0.31 \%$$

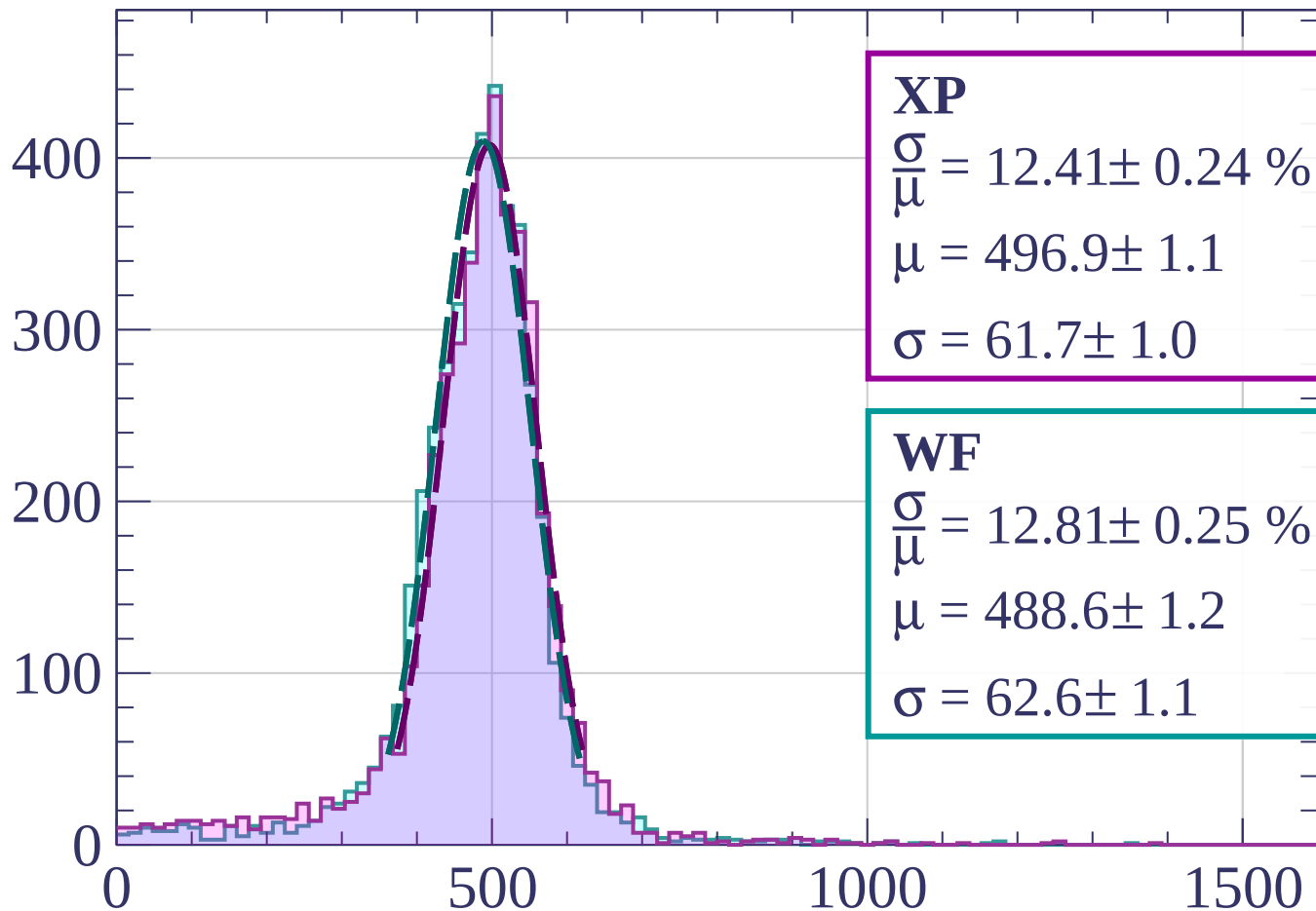
$$\mu = 486.5 \pm 1.4$$

$$\sigma = 68.0 \pm 1.3$$

dE/dx [ADC counts/cm]

Energy loss | $24 < \phi < 27$

Count



XP

$$\frac{\sigma}{\mu} = 12.41 \pm 0.24 \%$$

$$\mu = 496.9 \pm 1.1$$

$$\sigma = 61.7 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 12.81 \pm 0.25 \%$$

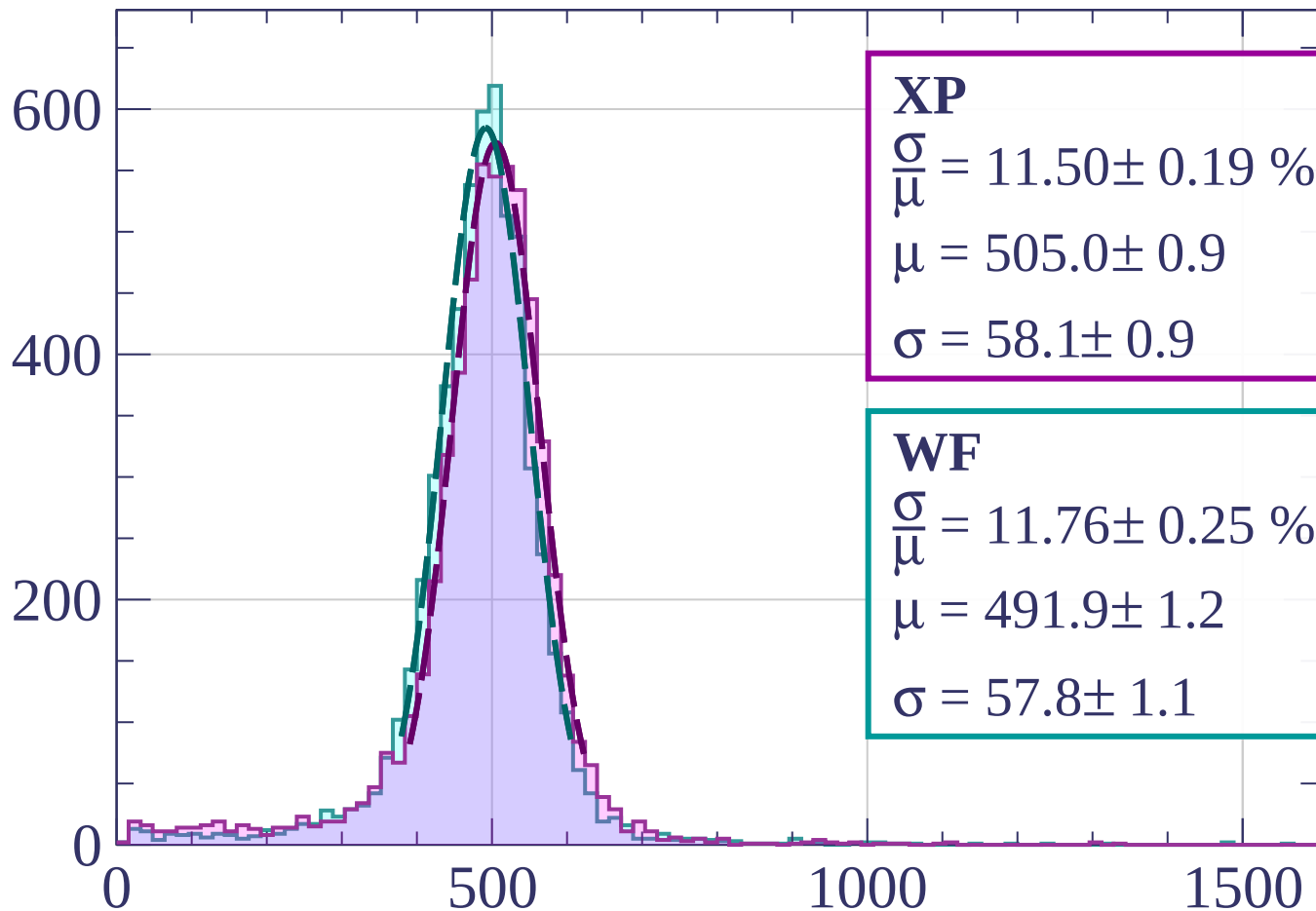
$$\mu = 488.6 \pm 1.2$$

$$\sigma = 62.6 \pm 1.1$$

dE/dx [ADC counts/cm]

Energy loss | $27 < \phi < 30$

Count



XP

$$\frac{\sigma}{\mu} = 11.50 \pm 0.19 \%$$

$$\mu = 505.0 \pm 0.9$$

$$\sigma = 58.1 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 11.76 \pm 0.25 \%$$

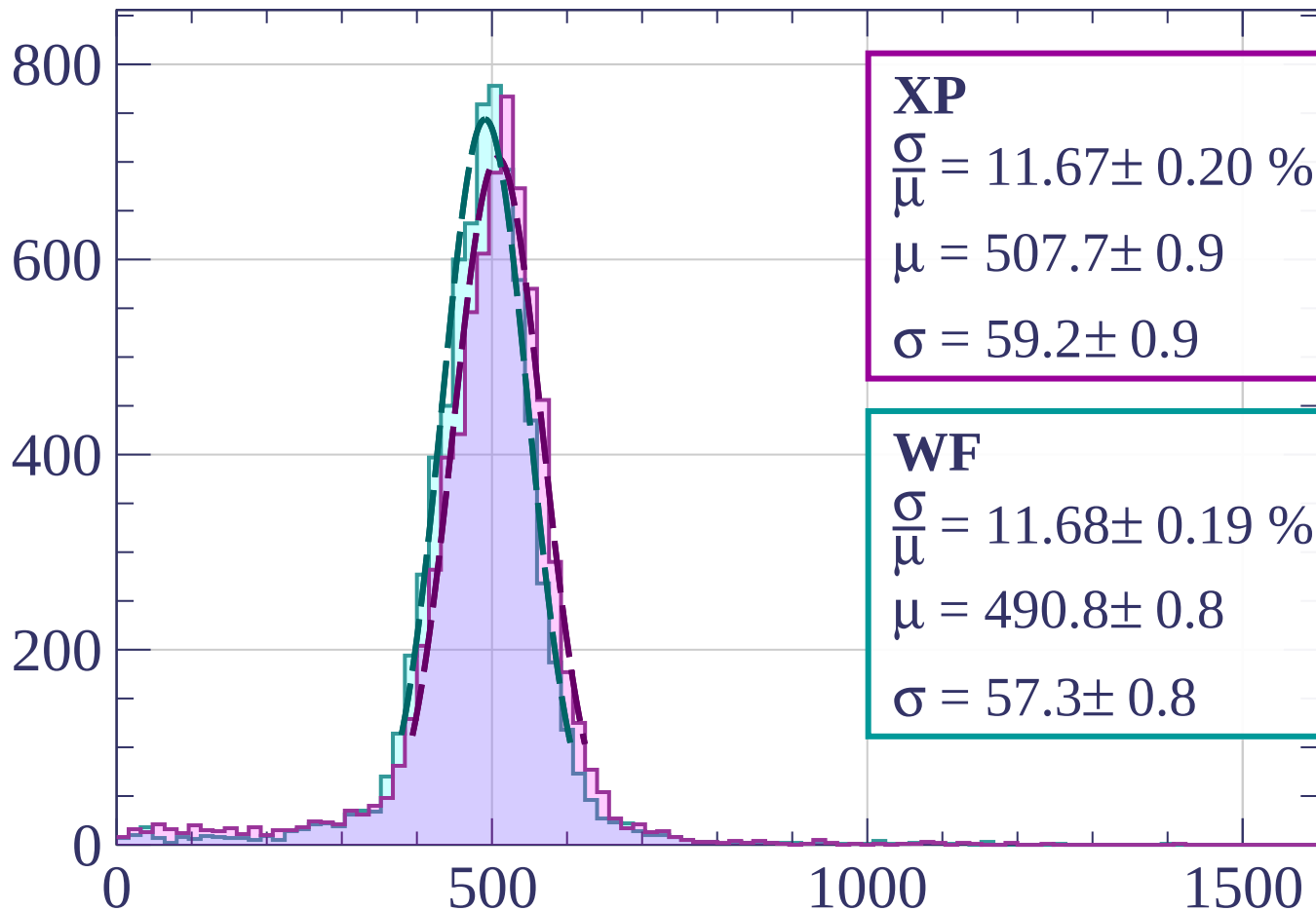
$$\mu = 491.9 \pm 1.2$$

$$\sigma = 57.8 \pm 1.1$$

dE/dx [ADC counts/cm]

Energy loss | $30 < \phi < 33$

Count



XP

$$\frac{\sigma}{\mu} = 11.67 \pm 0.20 \%$$

$$\mu = 507.7 \pm 0.9$$

$$\sigma = 59.2 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 11.68 \pm 0.19 \%$$

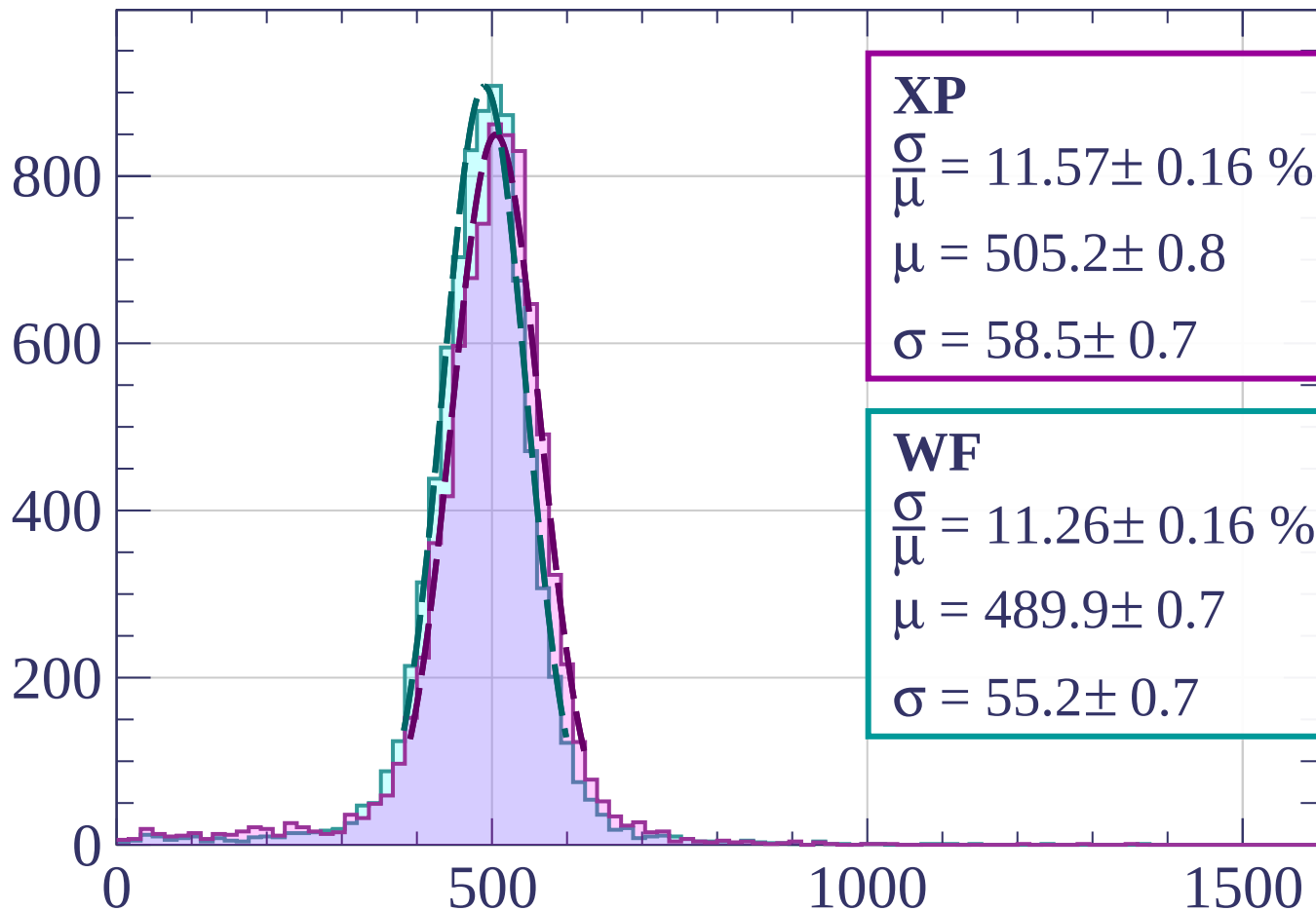
$$\mu = 490.8 \pm 0.8$$

$$\sigma = 57.3 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $33 < \phi < 36$

Count



XP

$$\frac{\sigma}{\mu} = 11.57 \pm 0.16 \%$$

$$\mu = 505.2 \pm 0.8$$

$$\sigma = 58.5 \pm 0.7$$

WF

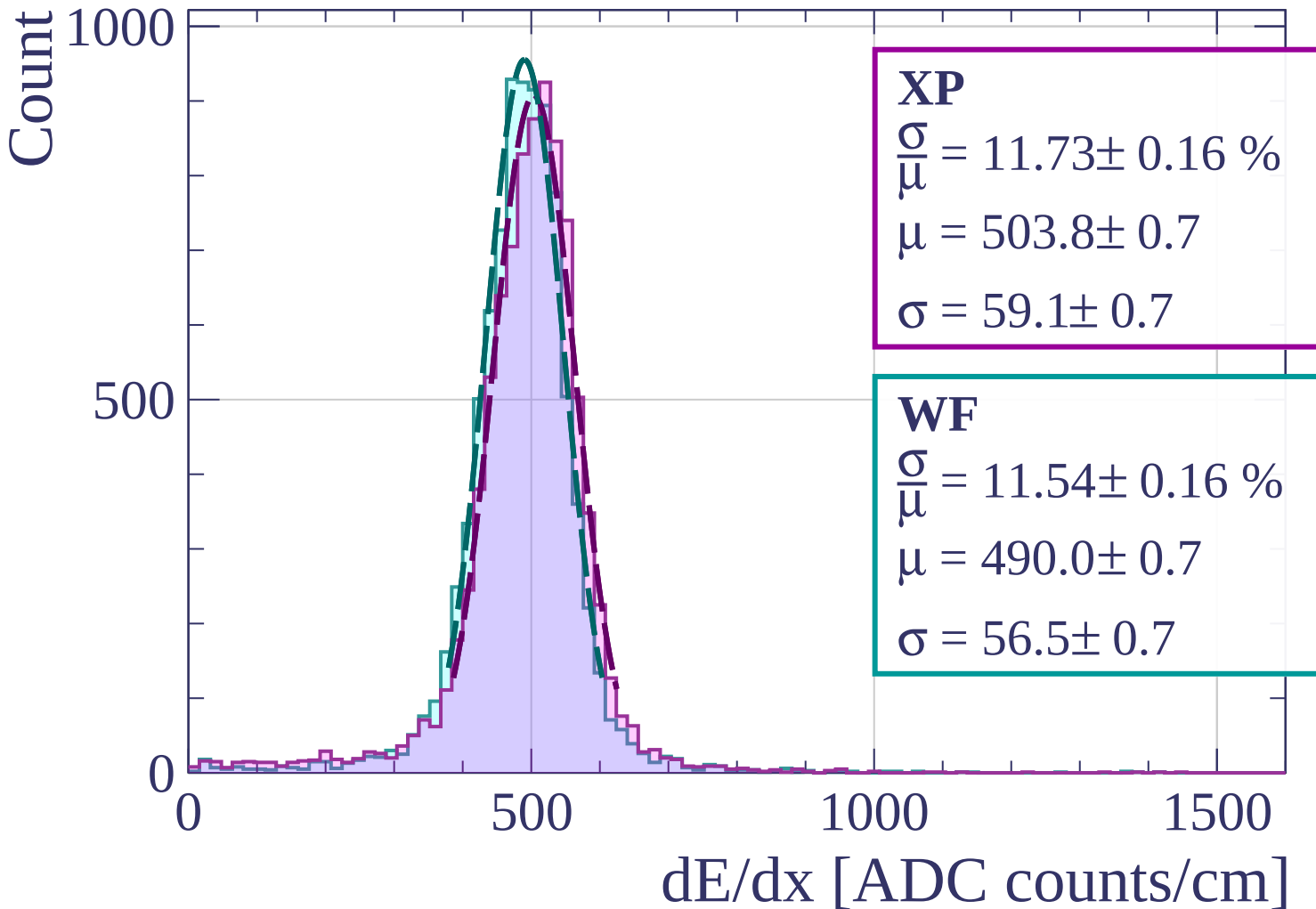
$$\frac{\sigma}{\mu} = 11.26 \pm 0.16 \%$$

$$\mu = 489.9 \pm 0.7$$

$$\sigma = 55.2 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $36 < \phi < 39$



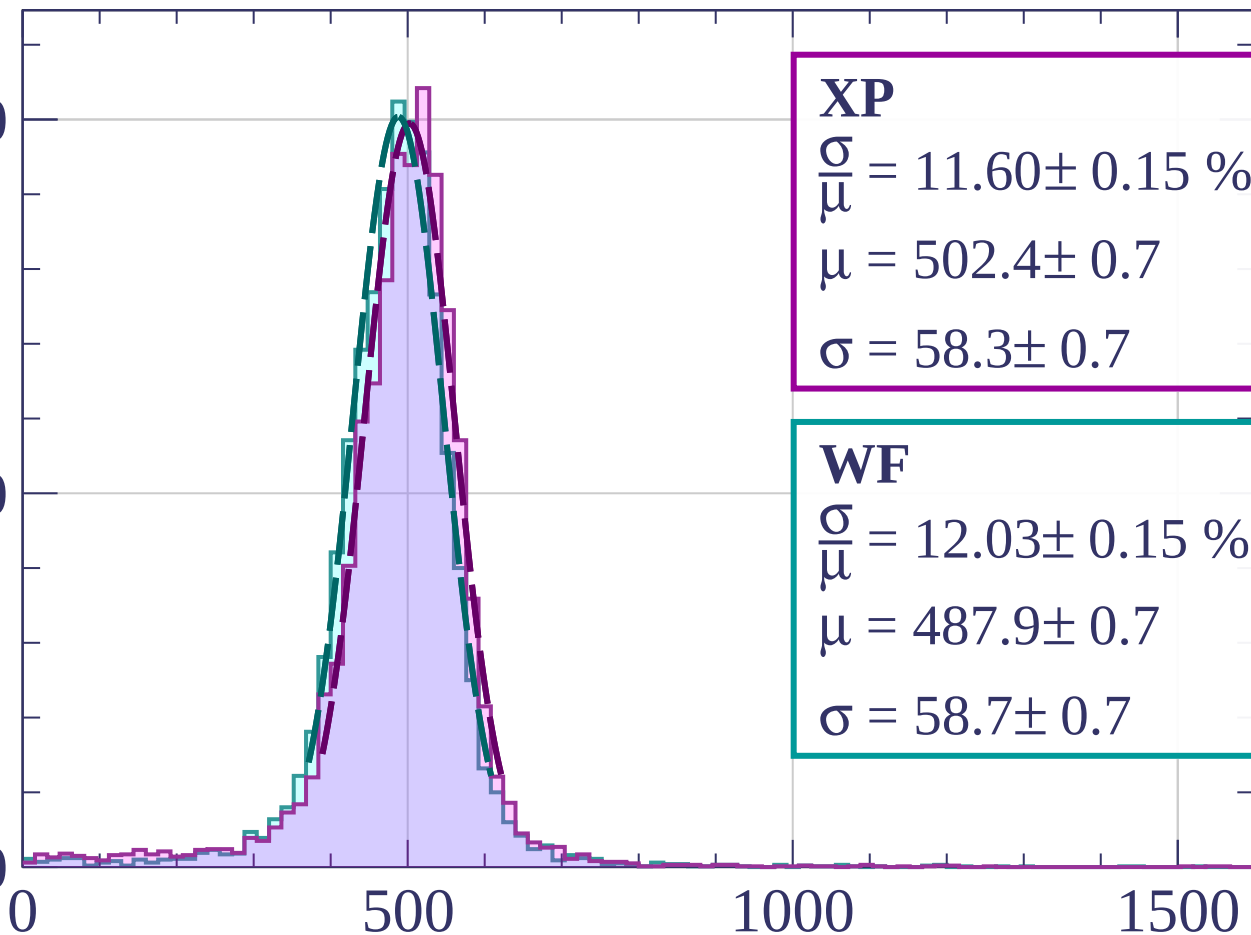
Energy loss | $39 < \phi < 42$

Count

1000

500

0



XP

$$\frac{\sigma}{\mu} = 11.60 \pm 0.15 \%$$

$$\mu = 502.4 \pm 0.7$$

$$\sigma = 58.3 \pm 0.7$$

WF

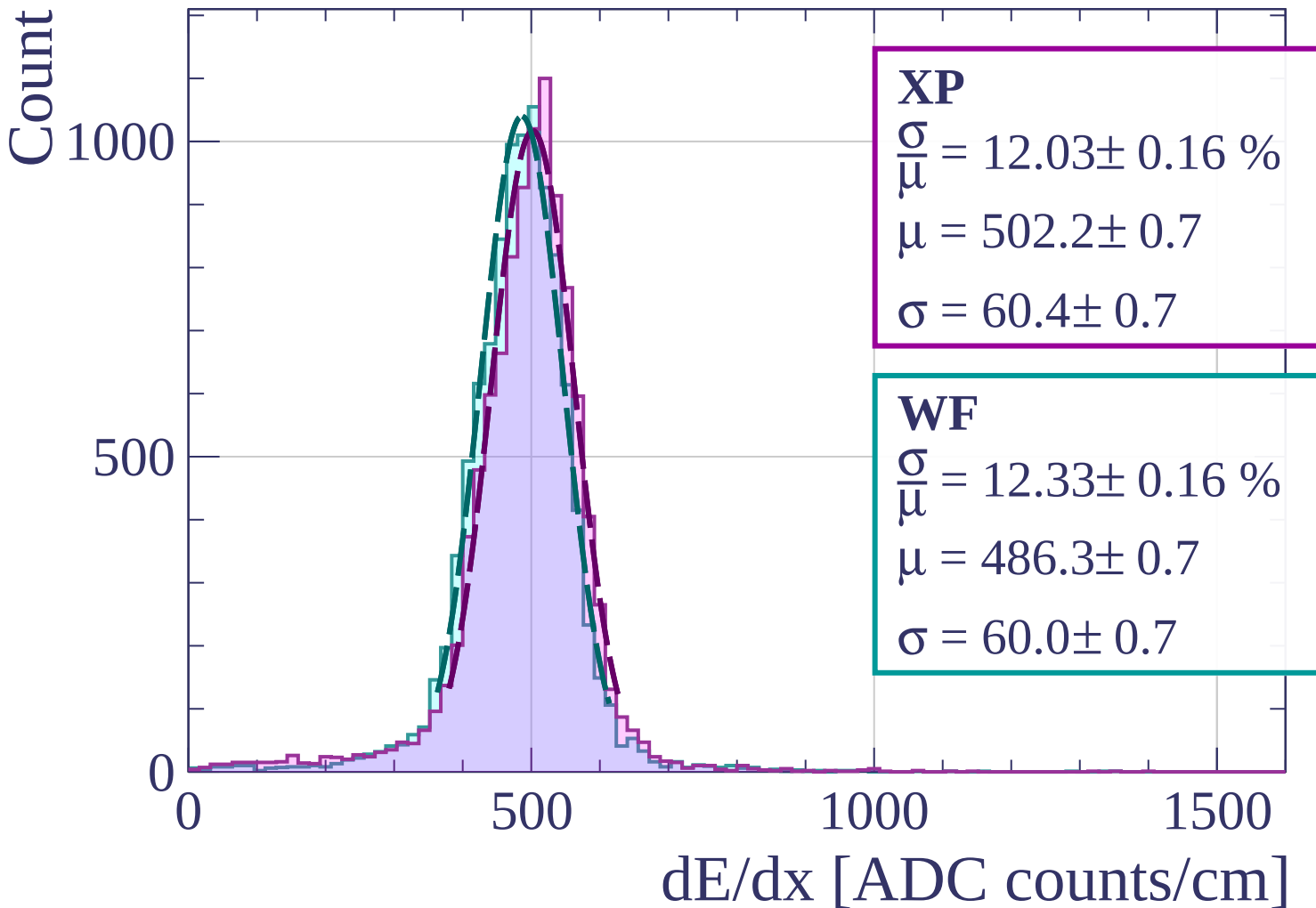
$$\frac{\sigma}{\mu} = 12.03 \pm 0.15 \%$$

$$\mu = 487.9 \pm 0.7$$

$$\sigma = 58.7 \pm 0.7$$

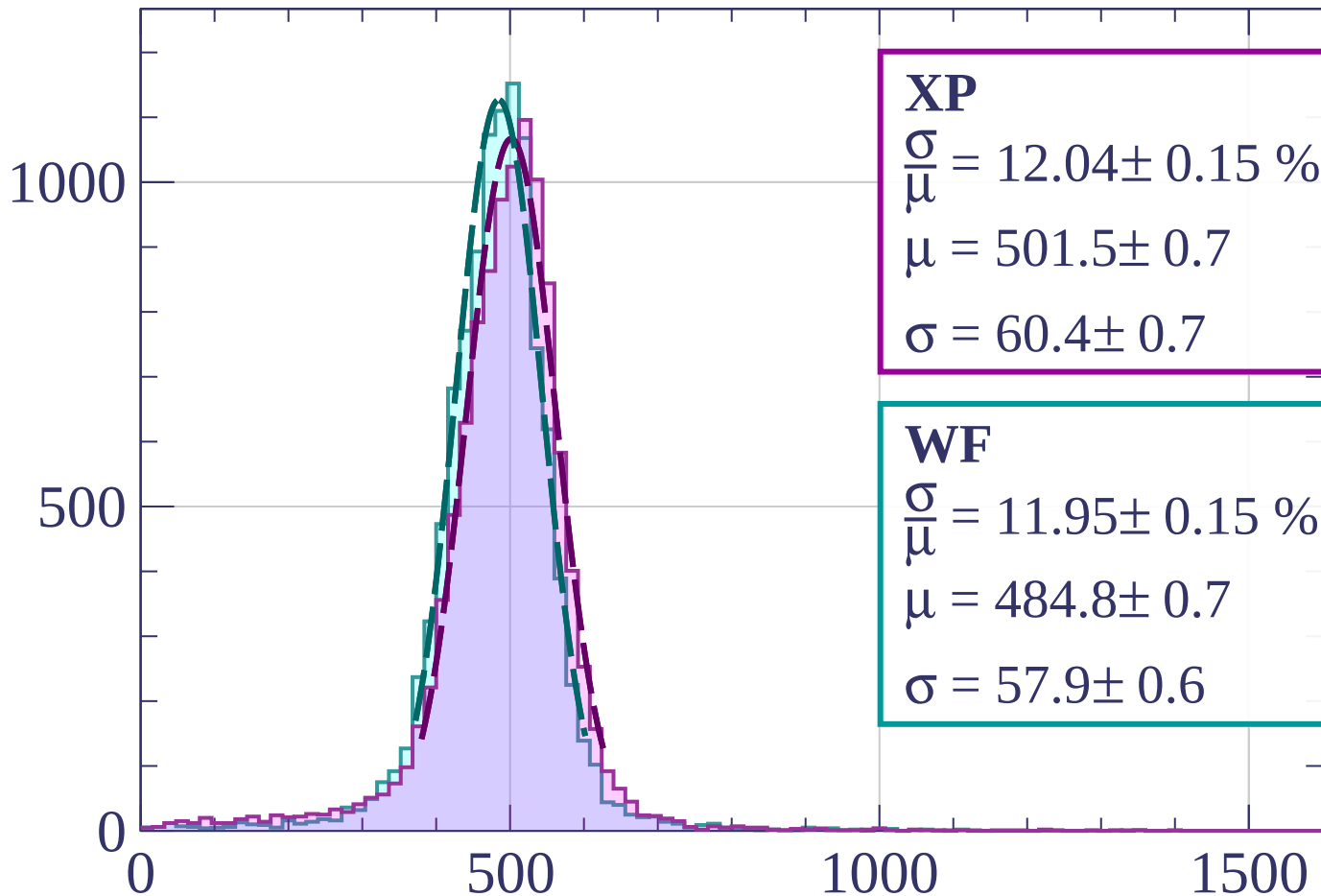
dE/dx [ADC counts/cm]

Energy loss | $42 < \phi < 45$



Energy loss | $45 < \phi < 48$

Count



XP

$$\frac{\sigma}{\mu} = 12.04 \pm 0.15 \%$$

$$\mu = 501.5 \pm 0.7$$

$$\sigma = 60.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 11.95 \pm 0.15 \%$$

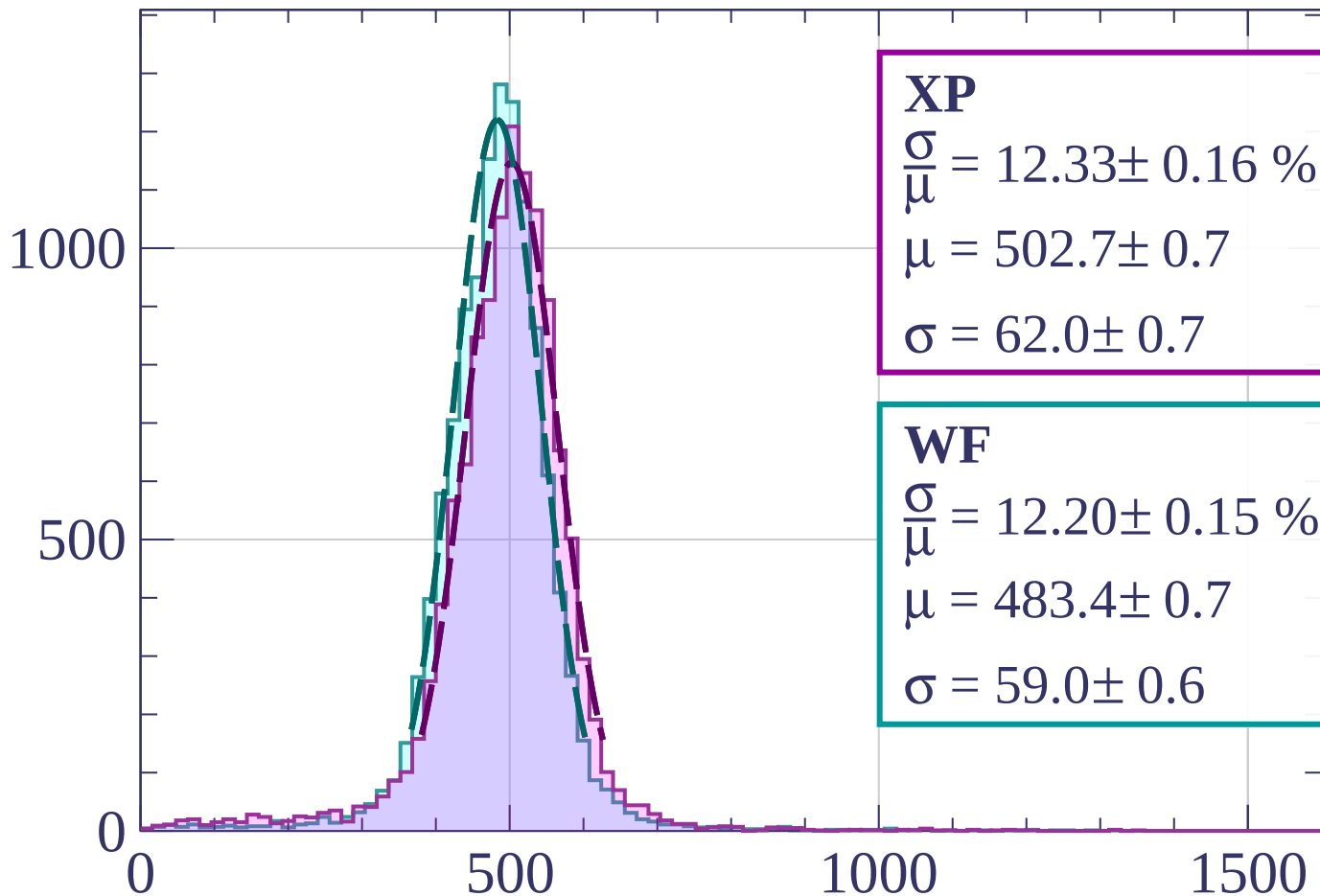
$$\mu = 484.8 \pm 0.7$$

$$\sigma = 57.9 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $48 < \phi < 51$

Count



XP

$$\frac{\sigma}{\mu} = 12.33 \pm 0.16 \%$$

$$\mu = 502.7 \pm 0.7$$

$$\sigma = 62.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 12.20 \pm 0.15 \%$$

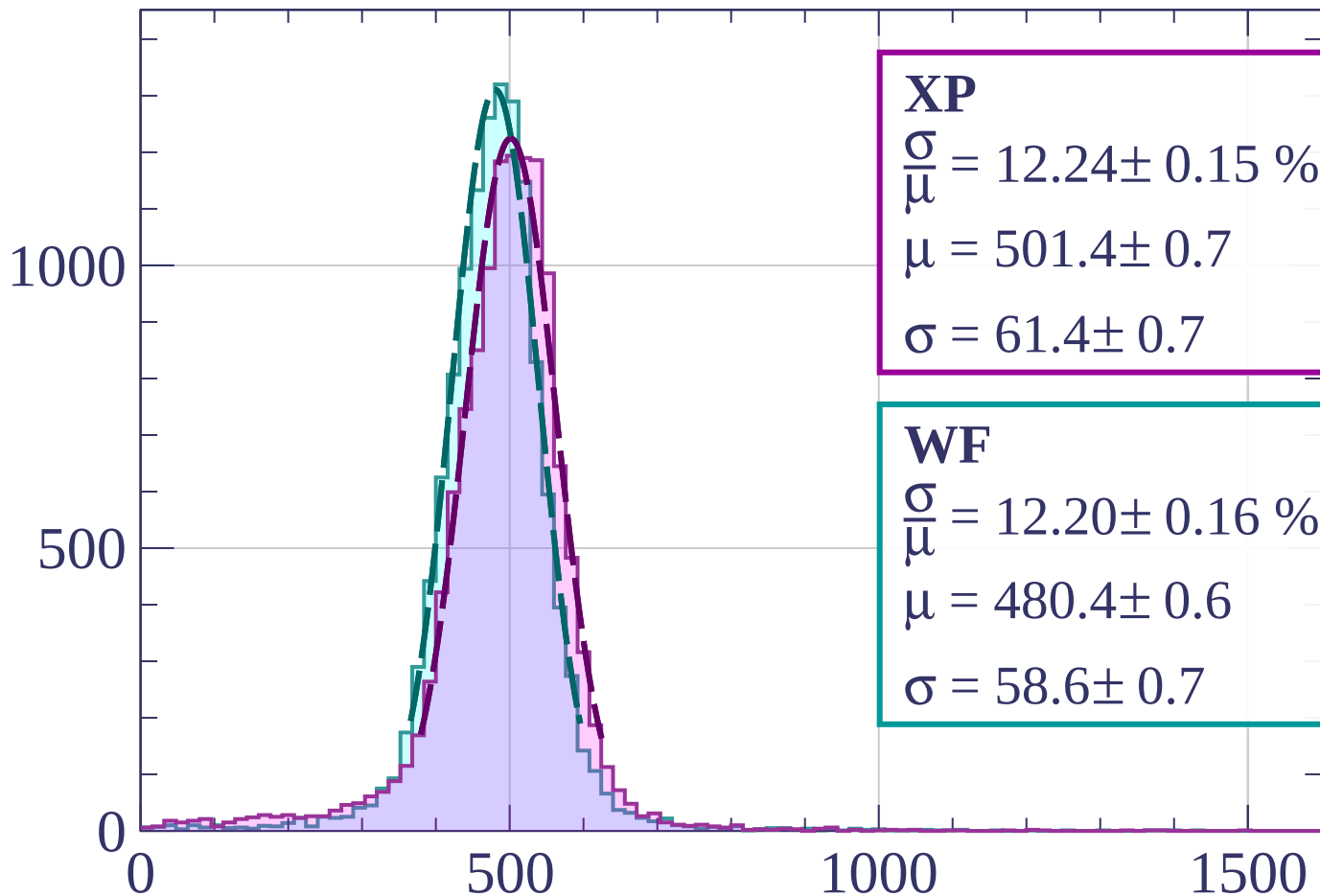
$$\mu = 483.4 \pm 0.7$$

$$\sigma = 59.0 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $51 < \phi < 54$

Count



XP

$$\frac{\sigma}{\mu} = 12.24 \pm 0.15 \%$$

$$\mu = 501.4 \pm 0.7$$

$$\sigma = 61.4 \pm 0.7$$

WF

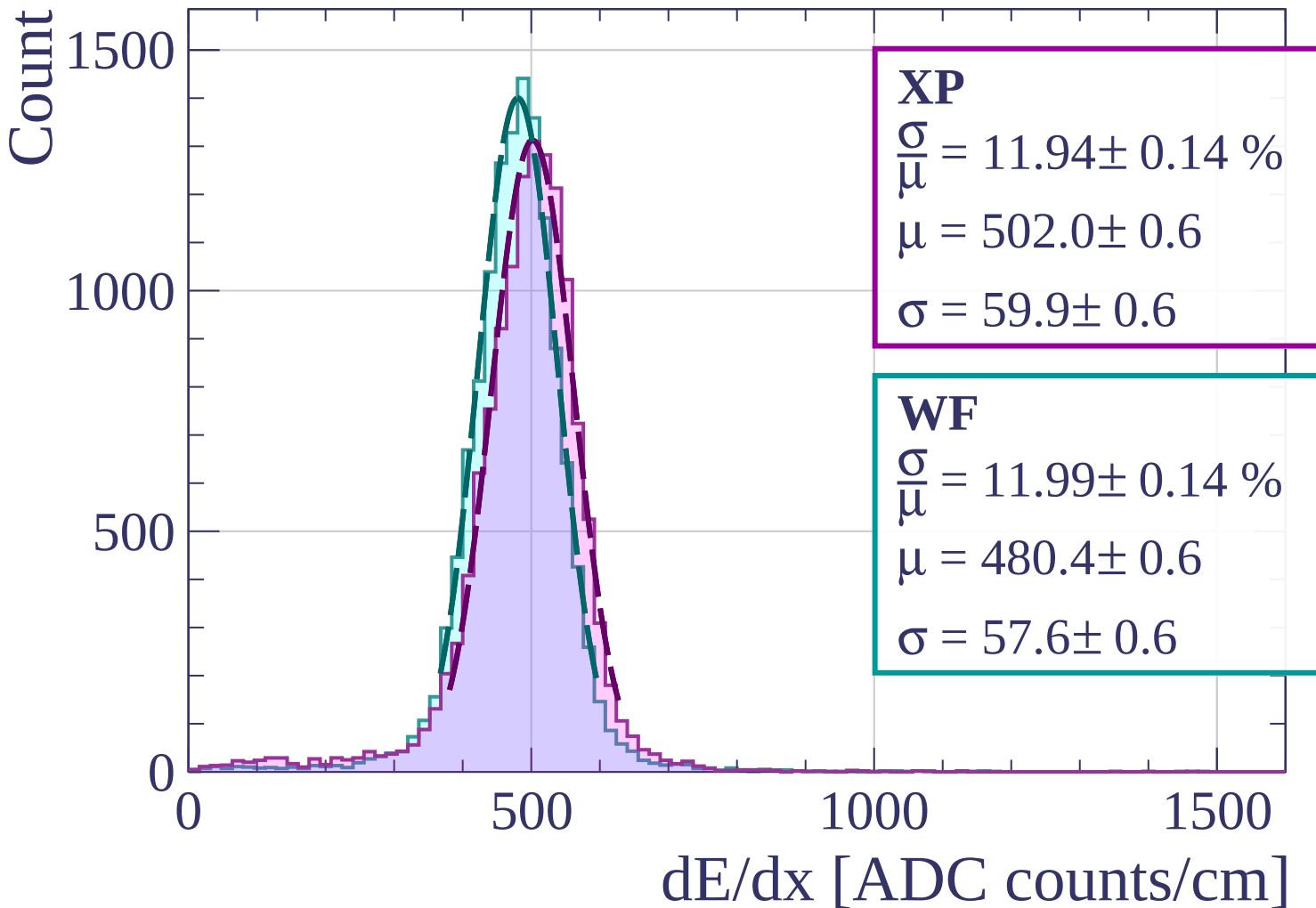
$$\frac{\sigma}{\mu} = 12.20 \pm 0.16 \%$$

$$\mu = 480.4 \pm 0.6$$

$$\sigma = 58.6 \pm 0.7$$

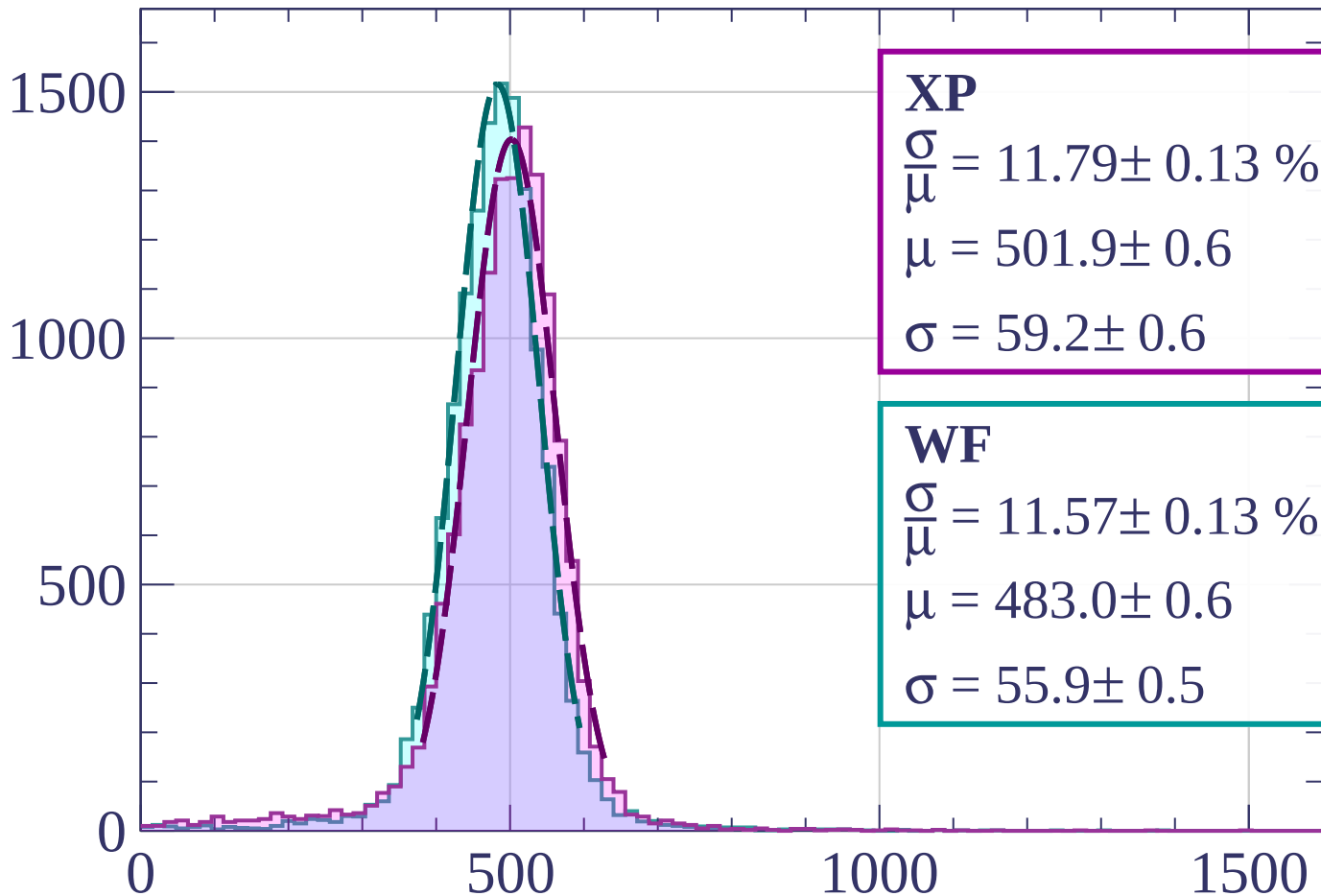
dE/dx [ADC counts/cm]

Energy loss | $54 < \phi < 57$



Energy loss | $57 < \phi < 60$

Count



XP

$$\frac{\sigma}{\mu} = 11.79 \pm 0.13 \%$$

$$\mu = 501.9 \pm 0.6$$

$$\sigma = 59.2 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 11.57 \pm 0.13 \%$$

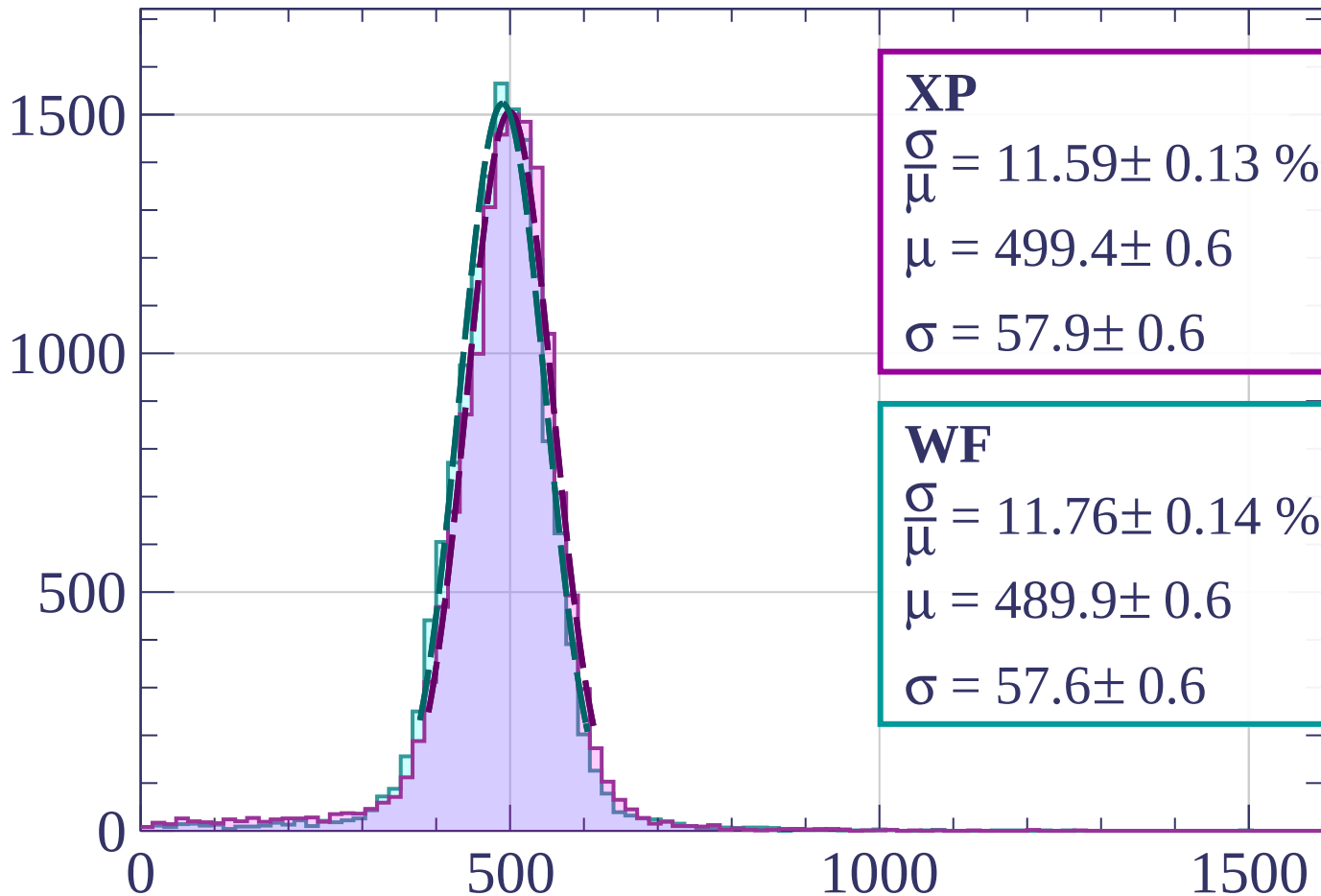
$$\mu = 483.0 \pm 0.6$$

$$\sigma = 55.9 \pm 0.5$$

dE/dx [ADC counts/cm]

Energy loss | $60 < \phi < 63$

Count



XP

$$\frac{\sigma}{\mu} = 11.59 \pm 0.13 \%$$

$$\mu = 499.4 \pm 0.6$$

$$\sigma = 57.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 11.76 \pm 0.14 \%$$

$$\mu = 489.9 \pm 0.6$$

$$\sigma = 57.6 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $63 < \phi < 66$

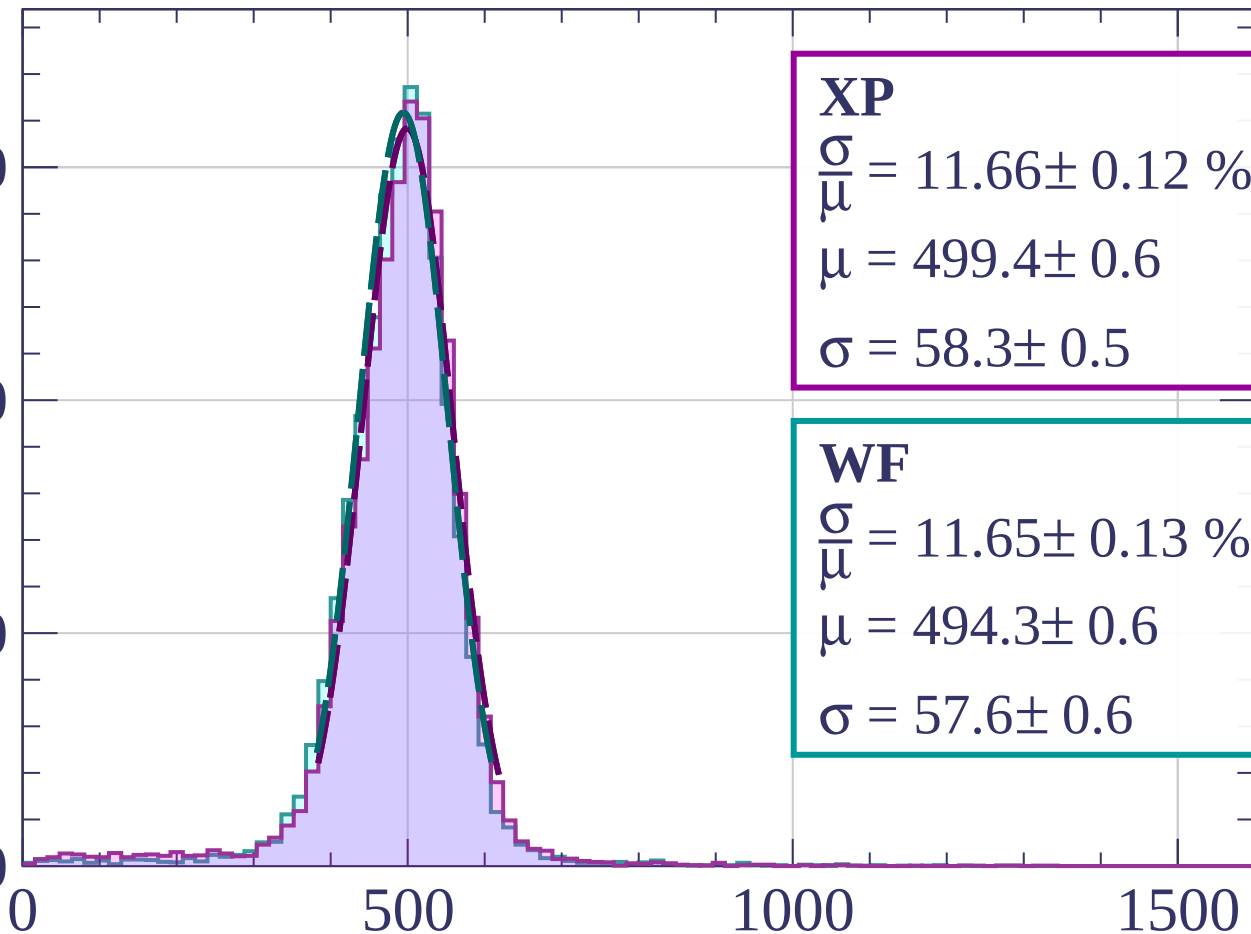
Count

1500

1000

500

0



XP

$$\frac{\sigma}{\mu} = 11.66 \pm 0.12 \%$$

$$\mu = 499.4 \pm 0.6$$

$$\sigma = 58.3 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 11.65 \pm 0.13 \%$$

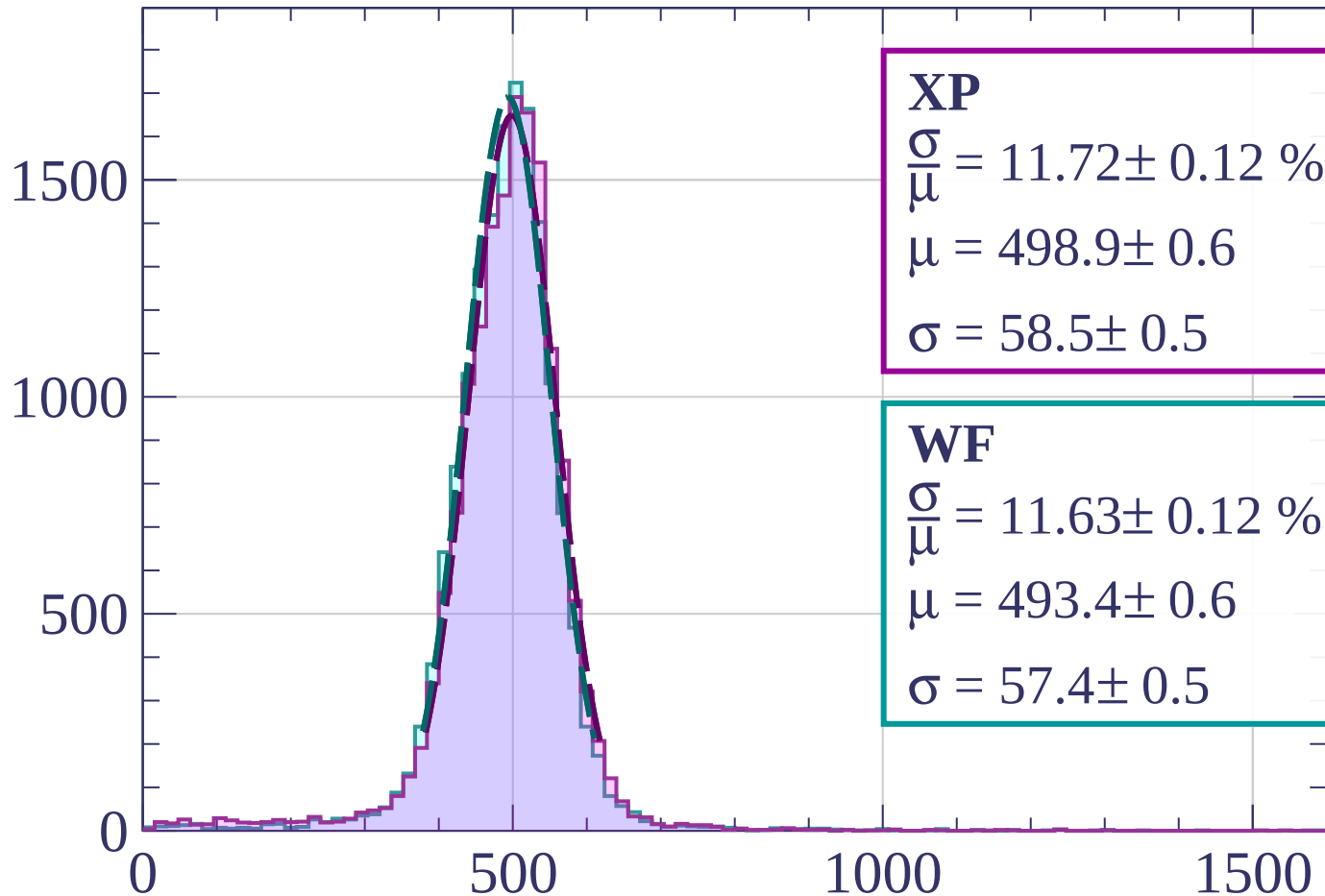
$$\mu = 494.3 \pm 0.6$$

$$\sigma = 57.6 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $66 < \phi < 69$

Count



XP

$$\frac{\sigma}{\mu} = 11.72 \pm 0.12 \%$$

$$\mu = 498.9 \pm 0.6$$

$$\sigma = 58.5 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 11.63 \pm 0.12 \%$$

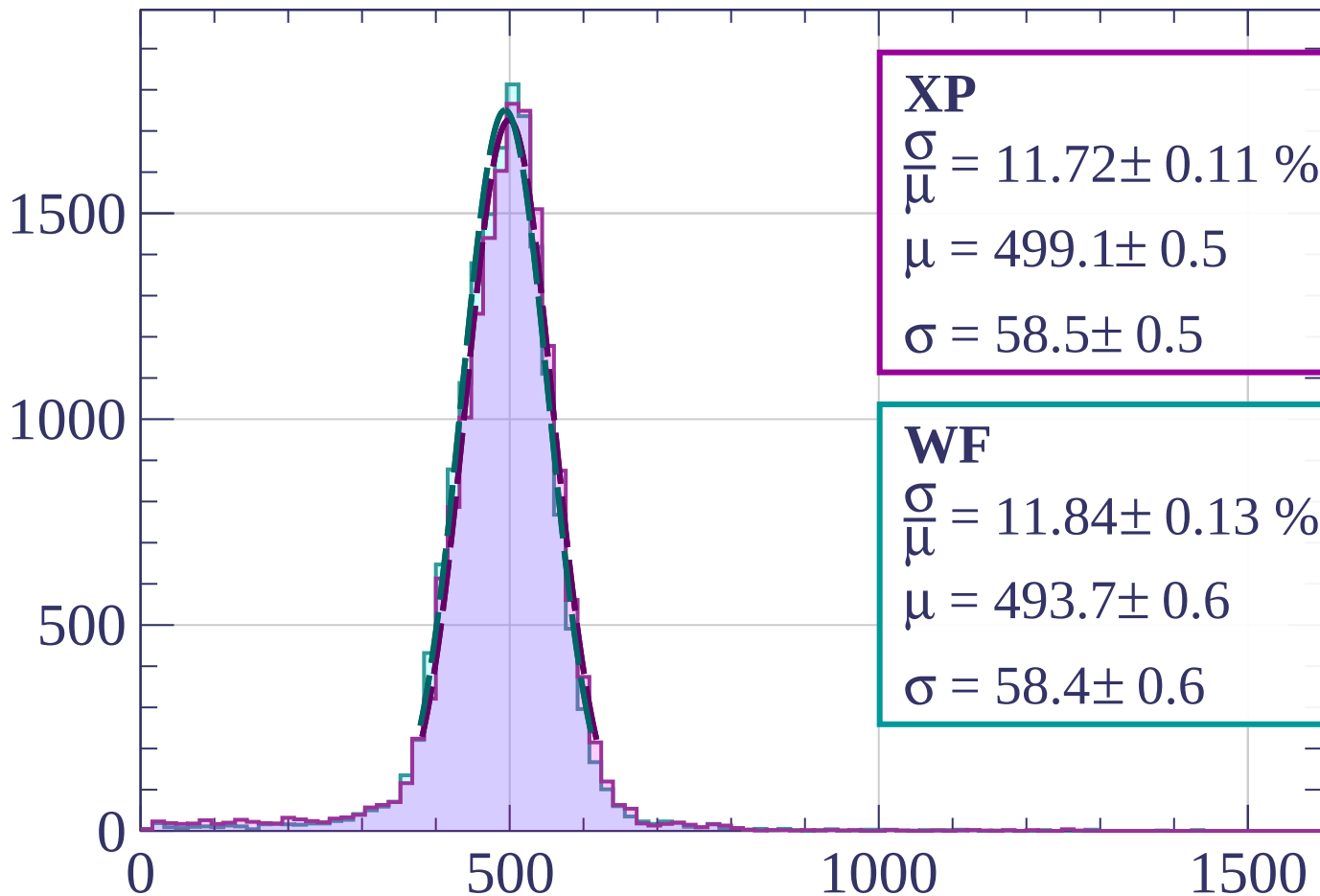
$$\mu = 493.4 \pm 0.6$$

$$\sigma = 57.4 \pm 0.5$$

dE/dx [ADC counts/cm]

Energy loss | $69 < \phi < 72$

Count



XP

$$\frac{\sigma}{\mu} = 11.72 \pm 0.11 \%$$

$$\mu = 499.1 \pm 0.5$$

$$\sigma = 58.5 \pm 0.5$$

WF

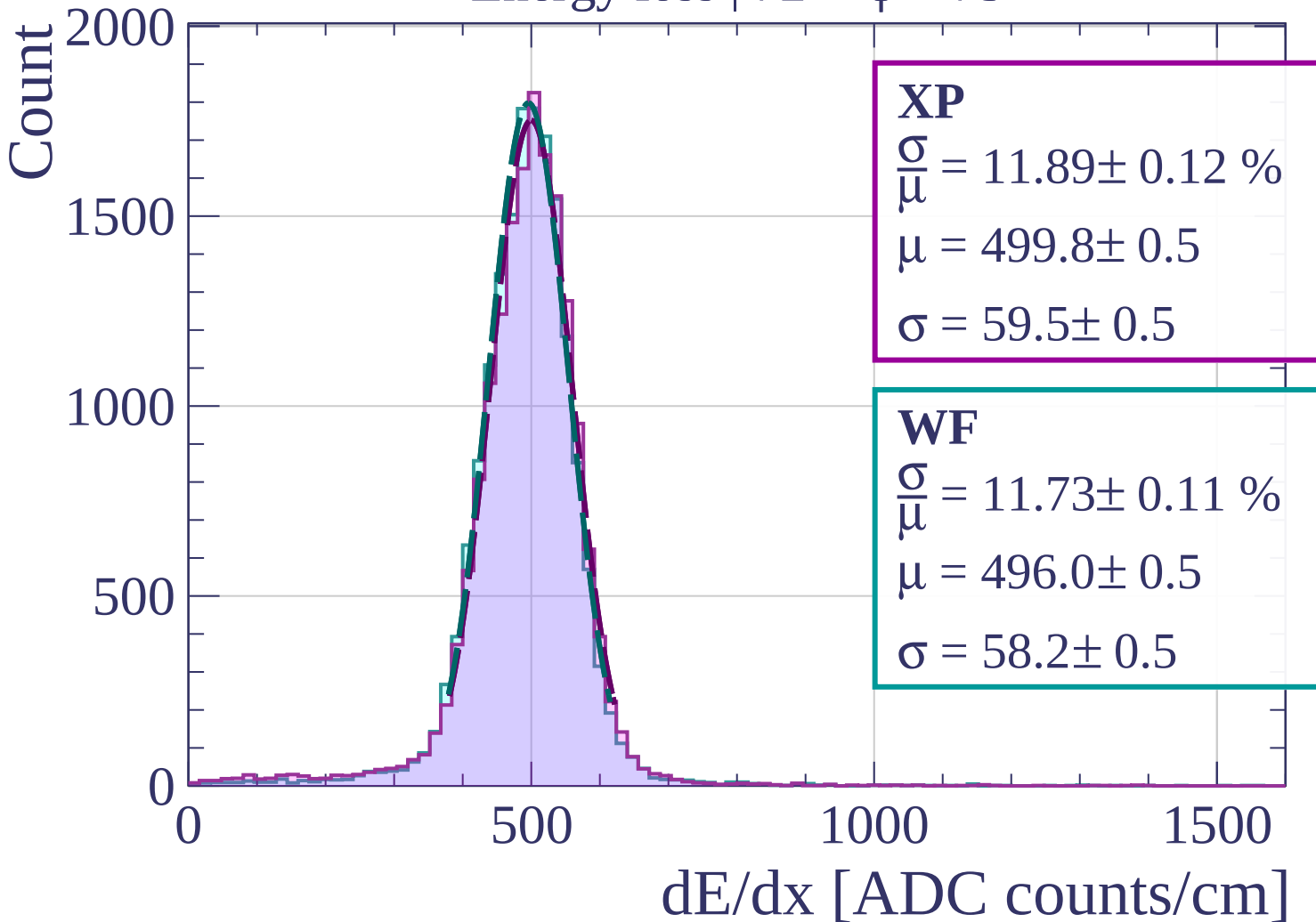
$$\frac{\sigma}{\mu} = 11.84 \pm 0.13 \%$$

$$\mu = 493.7 \pm 0.6$$

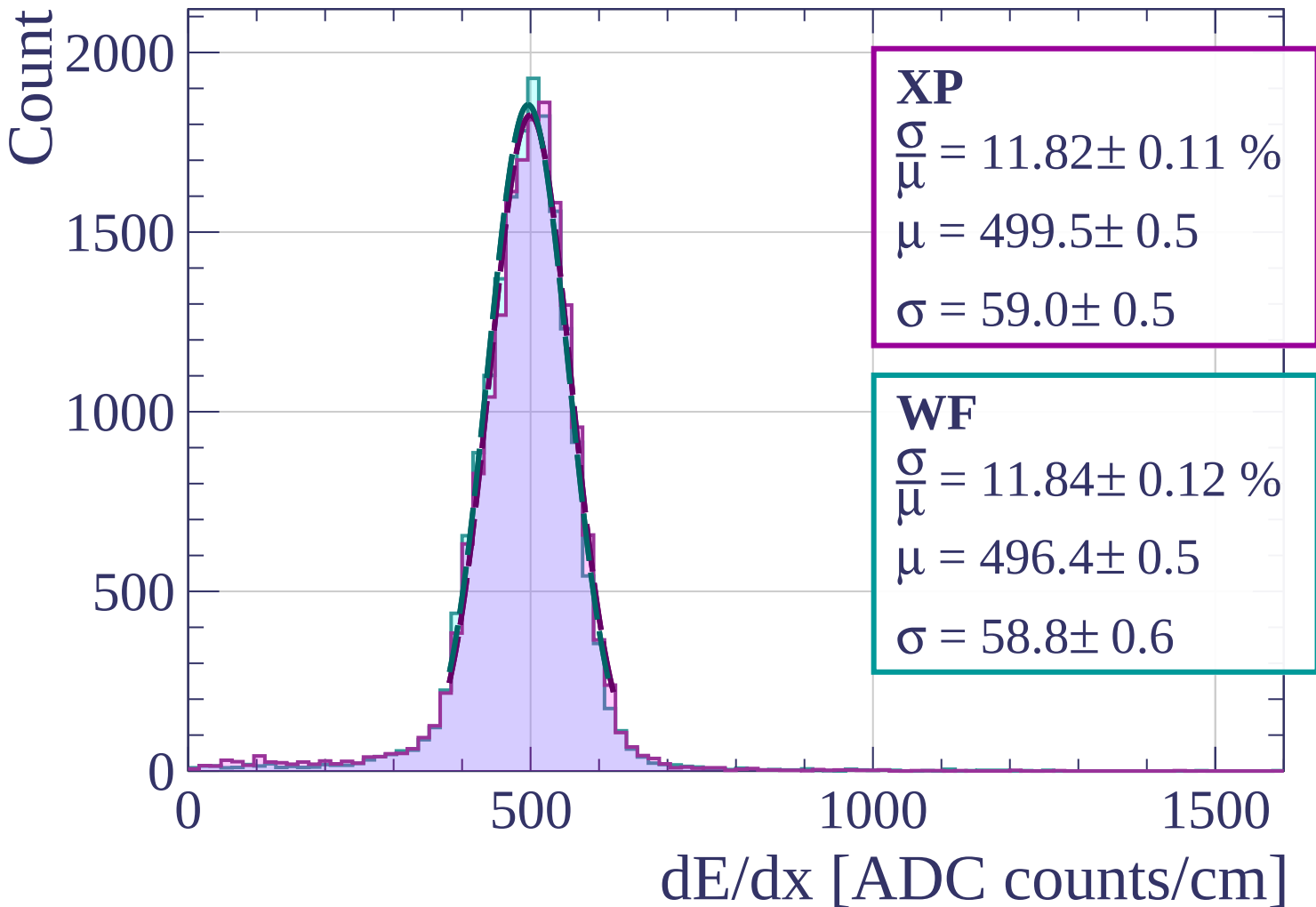
$$\sigma = 58.4 \pm 0.6$$

dE/dx [ADC counts/cm]

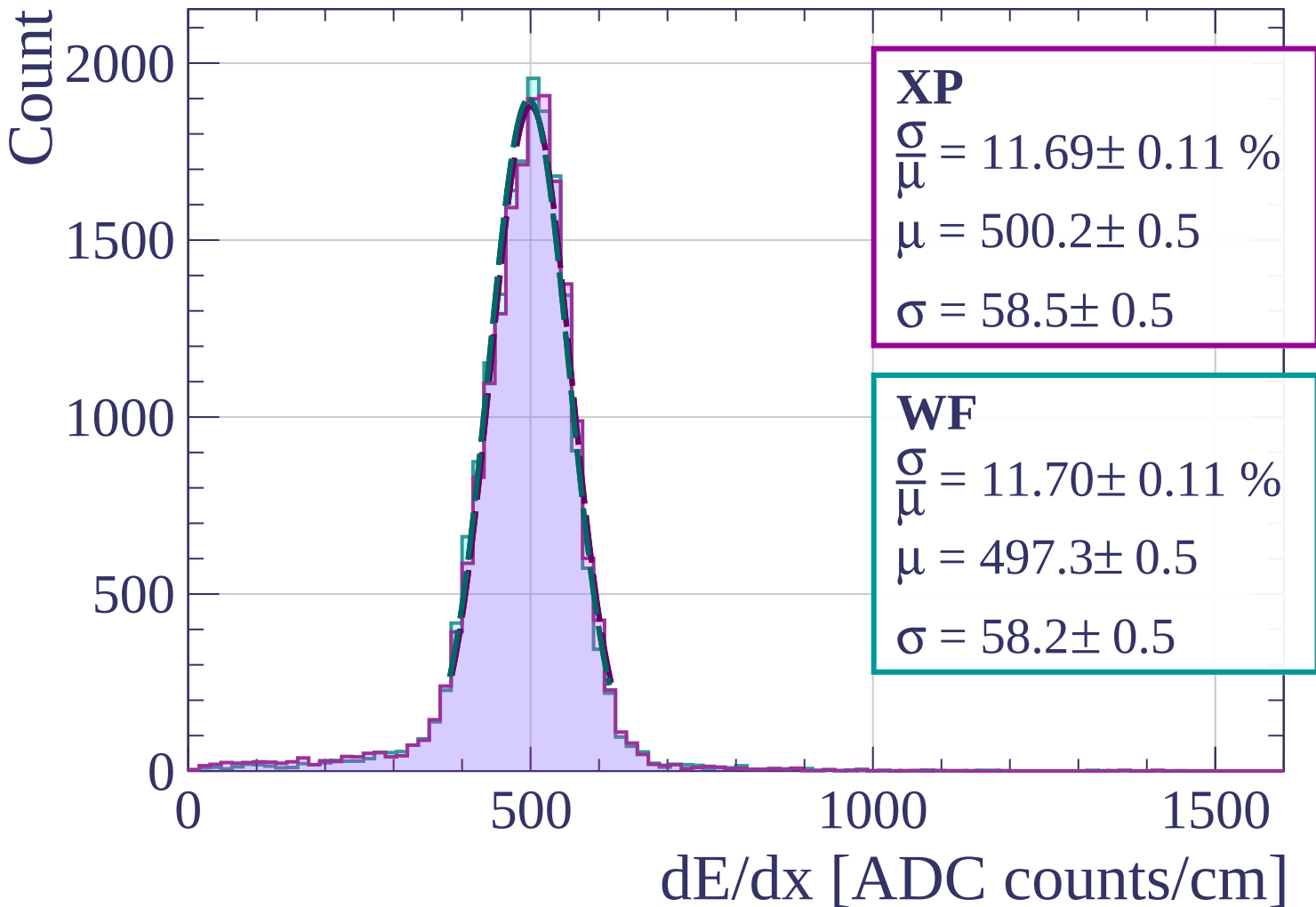
Energy loss | $72 < \phi < 75$



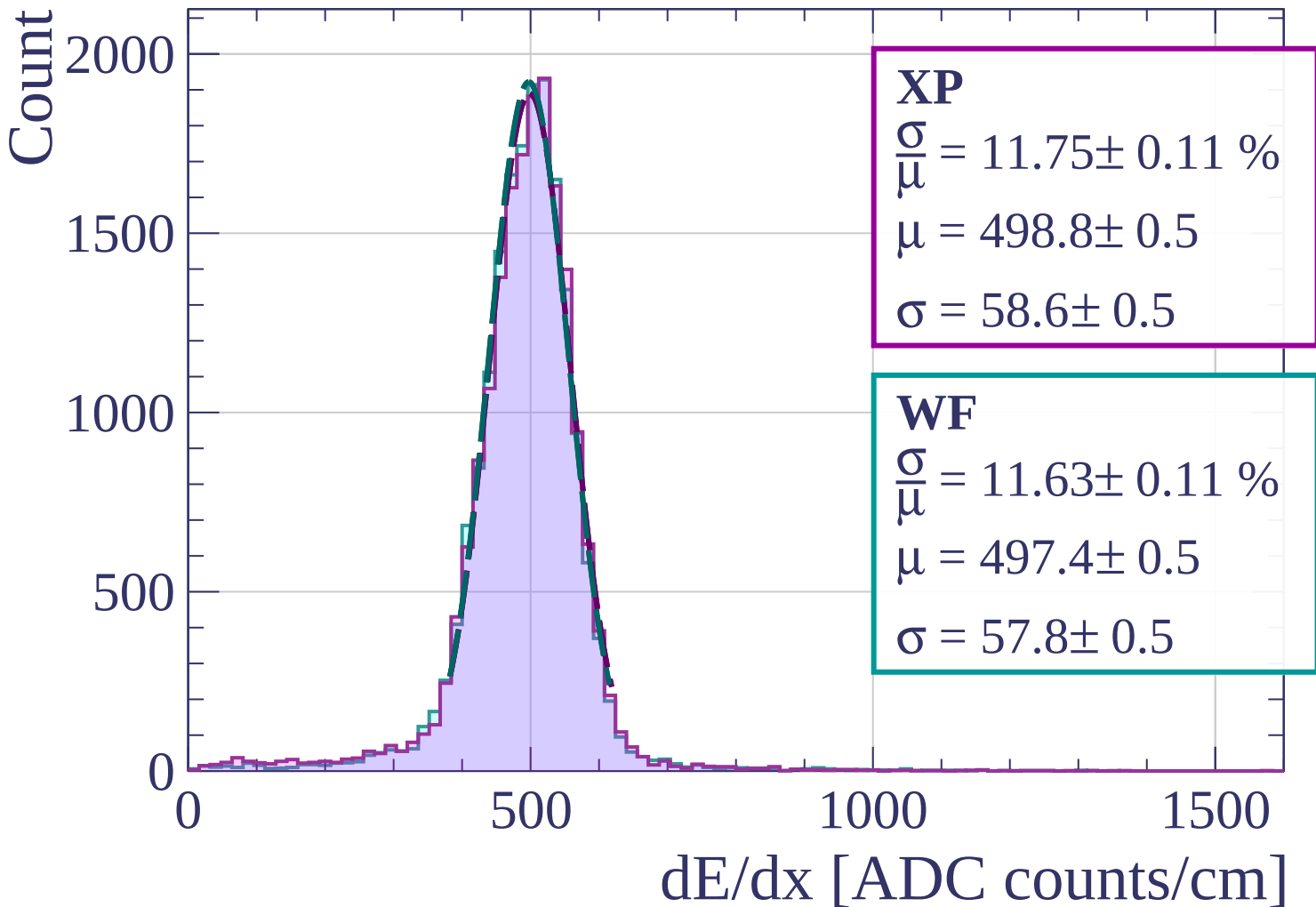
Energy loss | $75 < \phi < 78$



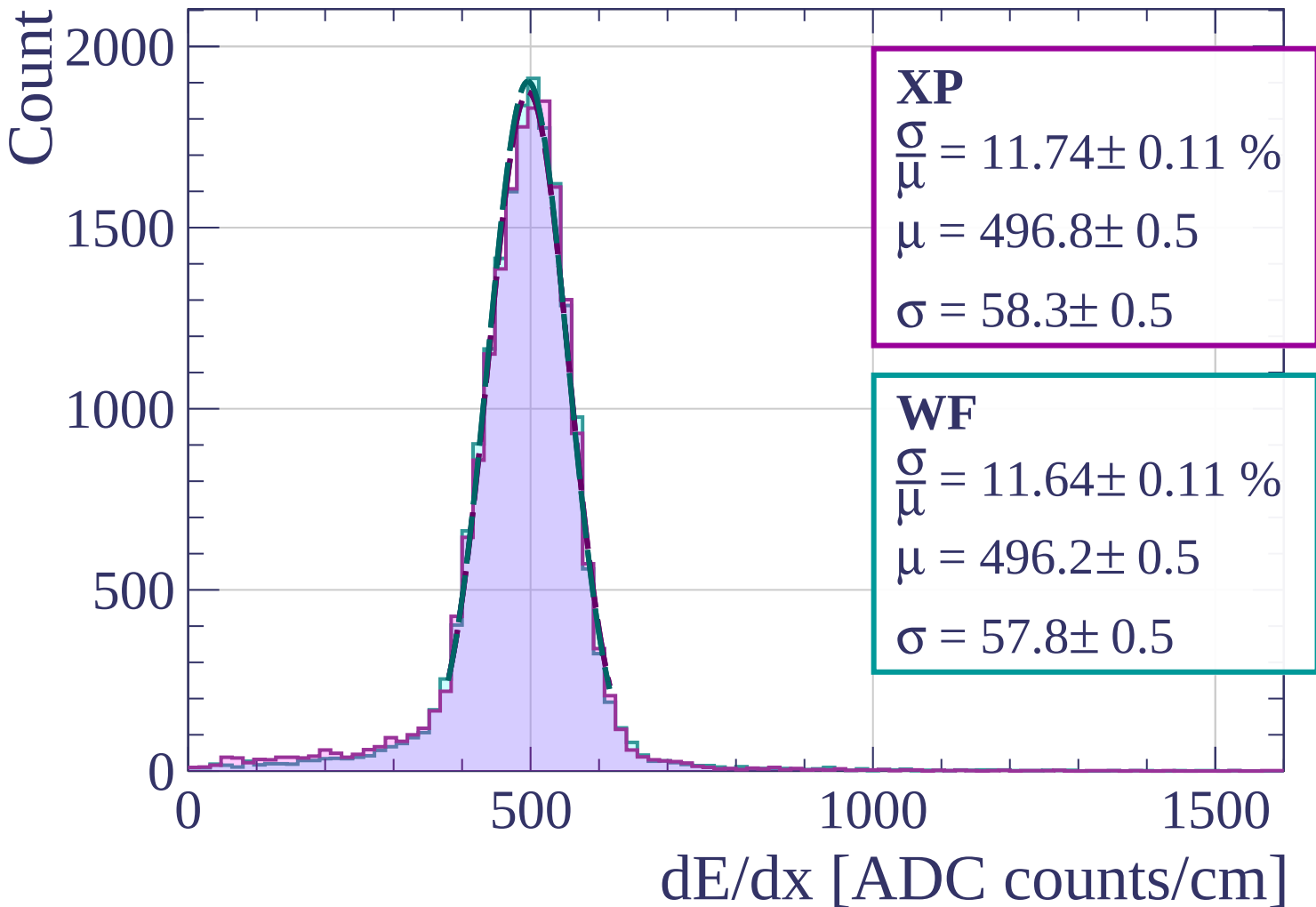
Energy loss | $78 < \phi < 81$



Energy loss | $81 < \phi < 84$



Energy loss | $84 < \phi < 87$



Energy loss | $87 < \phi < 90$

Count

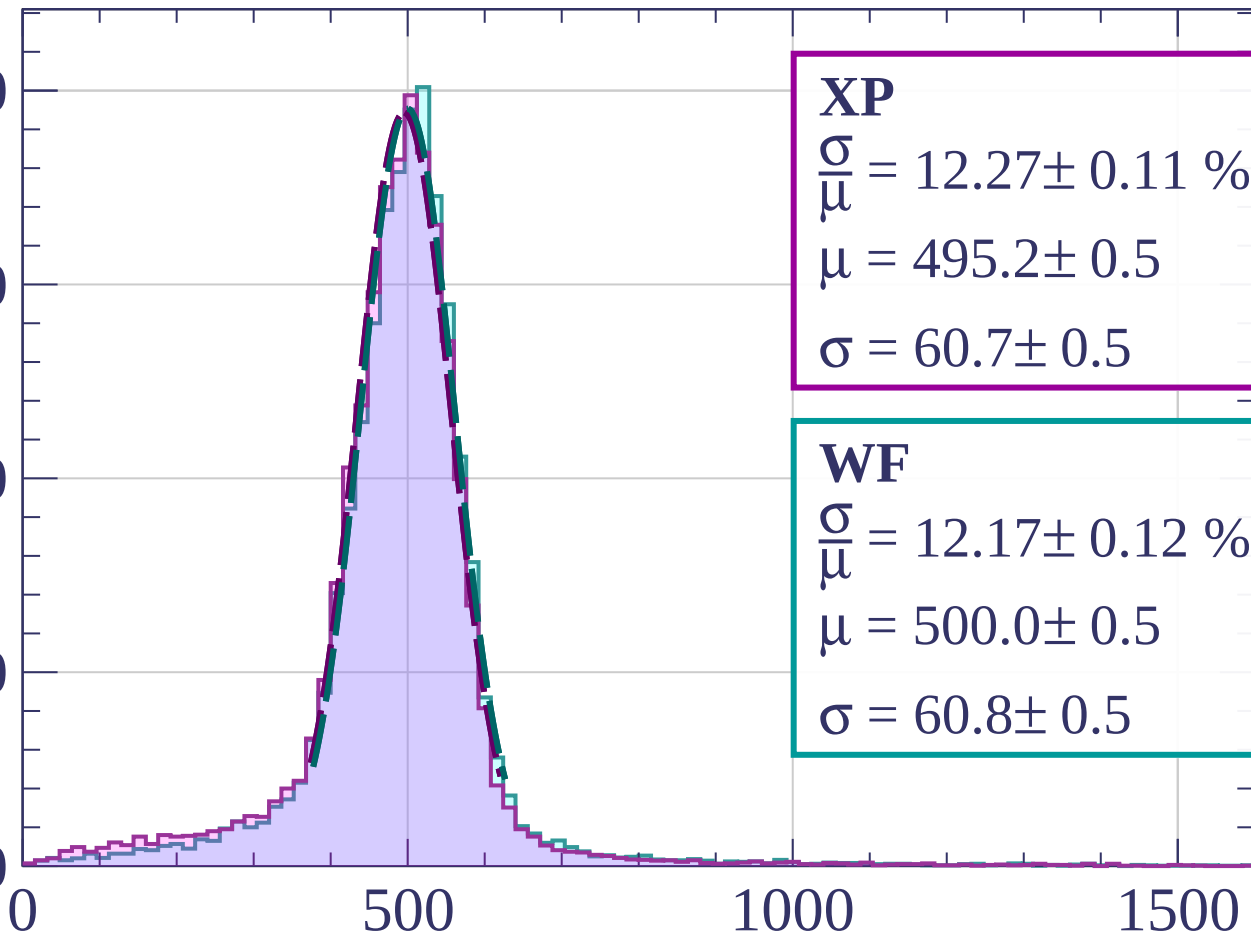
2000

1500

1000

500

0



XP

$$\frac{\sigma}{\mu} = 12.27 \pm 0.11 \%$$

$$\mu = 495.2 \pm 0.5$$

$$\sigma = 60.7 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 12.17 \pm 0.12 \%$$

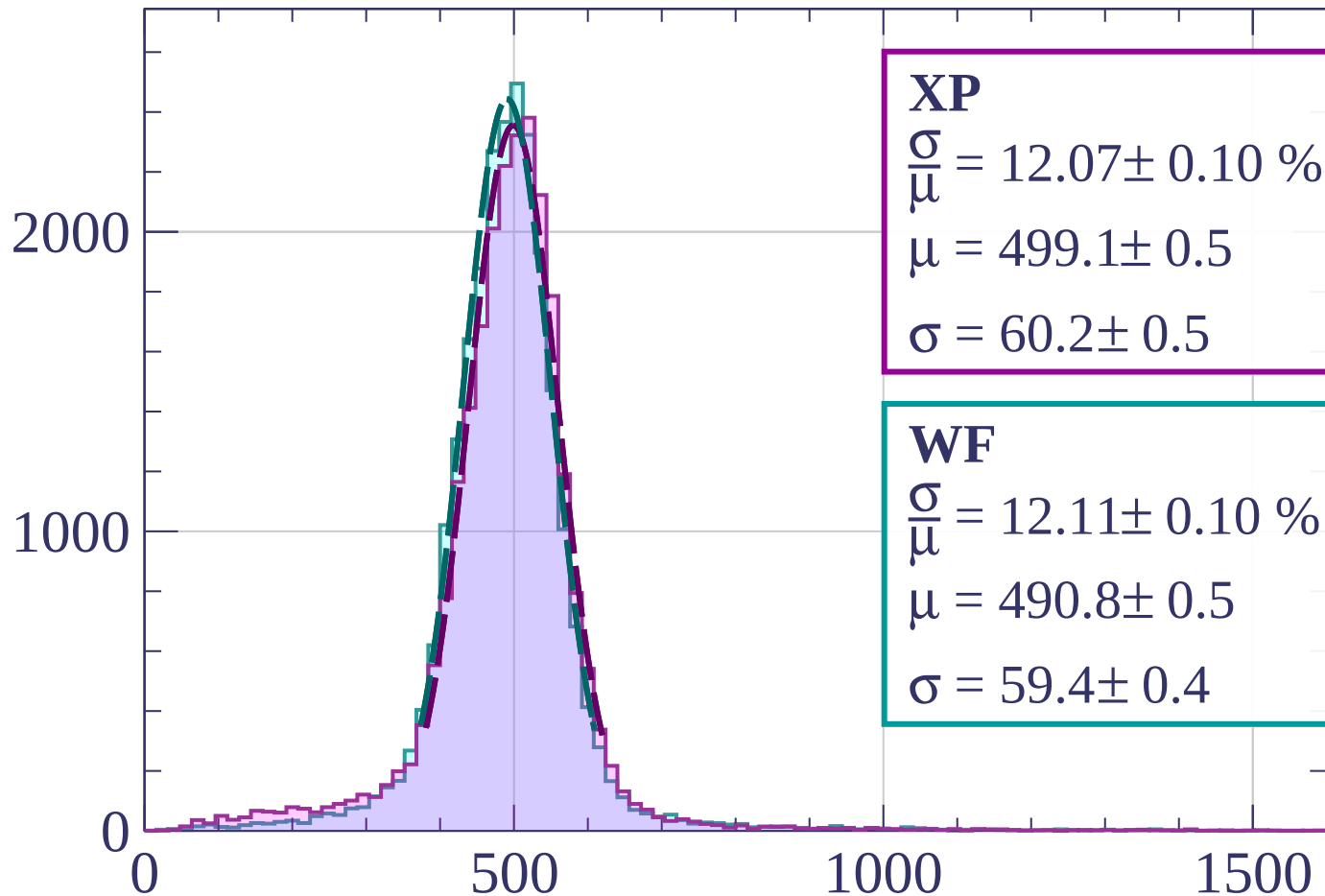
$$\mu = 500.0 \pm 0.5$$

$$\sigma = 60.8 \pm 0.5$$

dE/dx [ADC counts/cm]

Energy loss | $0 < \theta < 3$

Count



XP

$$\frac{\sigma}{\mu} = 12.07 \pm 0.10 \%$$

$$\mu = 499.1 \pm 0.5$$

$$\sigma = 60.2 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 12.11 \pm 0.10 \%$$

$$\mu = 490.8 \pm 0.5$$

$$\sigma = 59.4 \pm 0.4$$

dE/dx [ADC counts/cm]

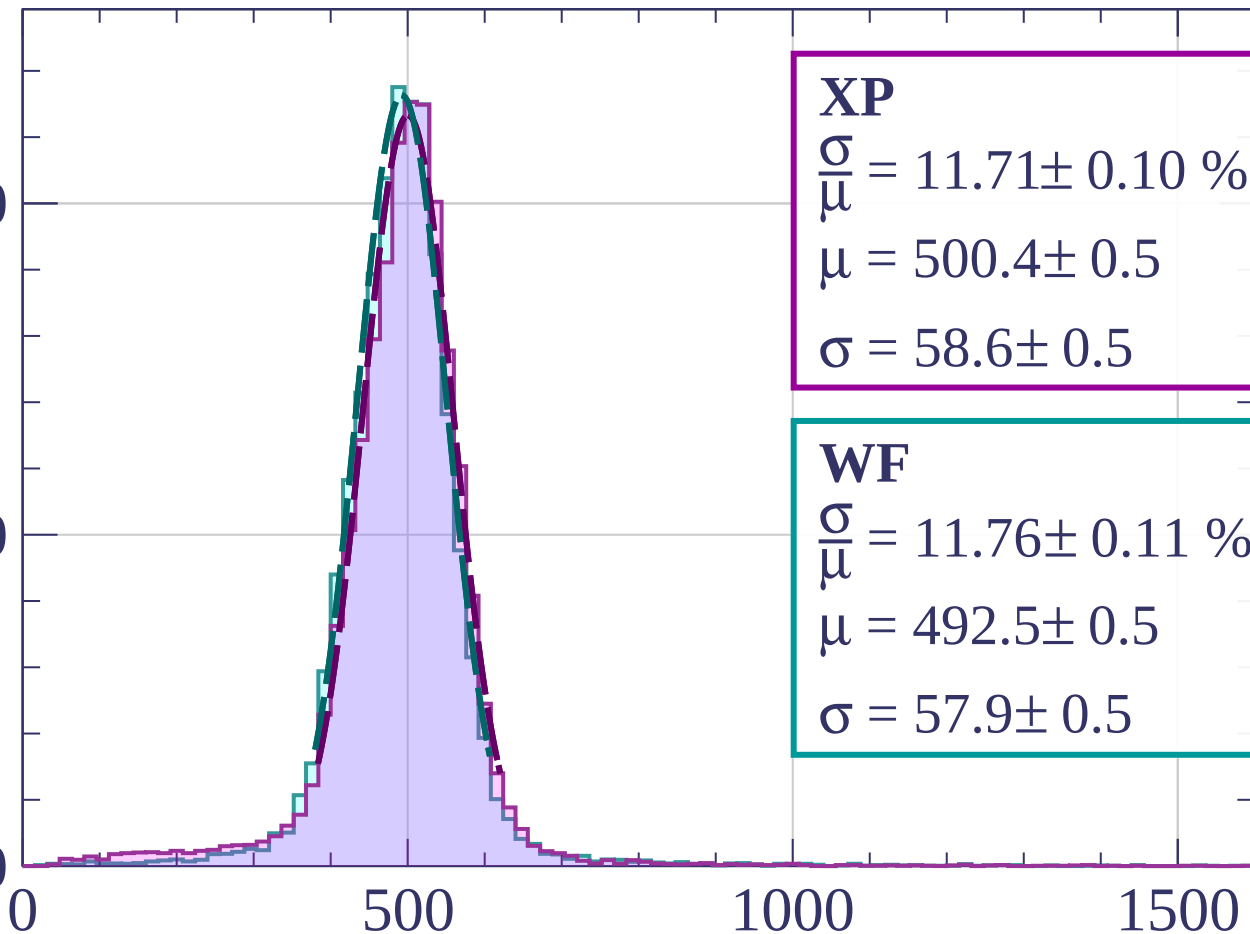
Energy loss | $3 < \theta < 6$

Count

2000

1000

0



XP

$$\frac{\sigma}{\mu} = 11.71 \pm 0.10 \%$$

$$\mu = 500.4 \pm 0.5$$

$$\sigma = 58.6 \pm 0.5$$

WF

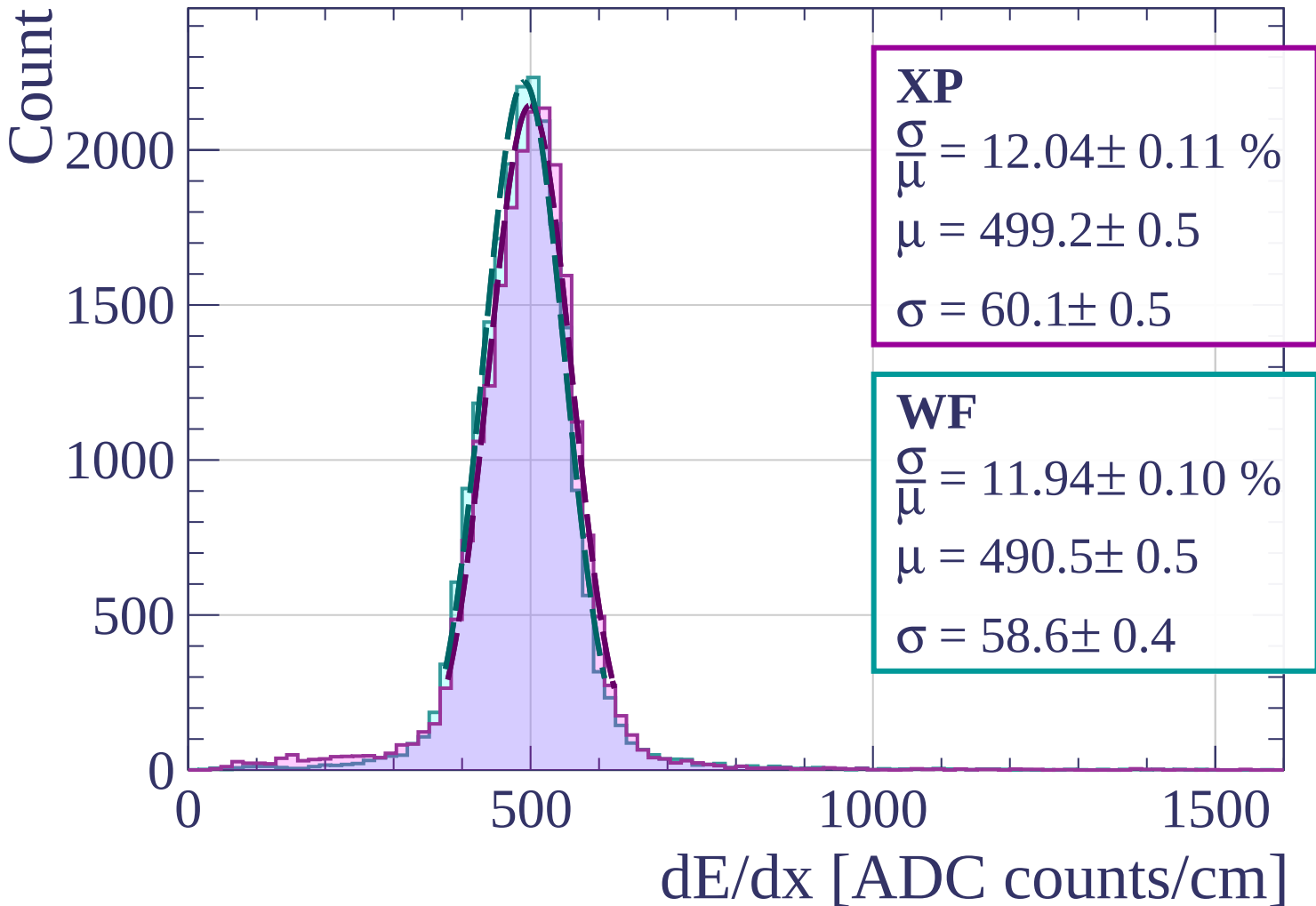
$$\frac{\sigma}{\mu} = 11.76 \pm 0.11 \%$$

$$\mu = 492.5 \pm 0.5$$

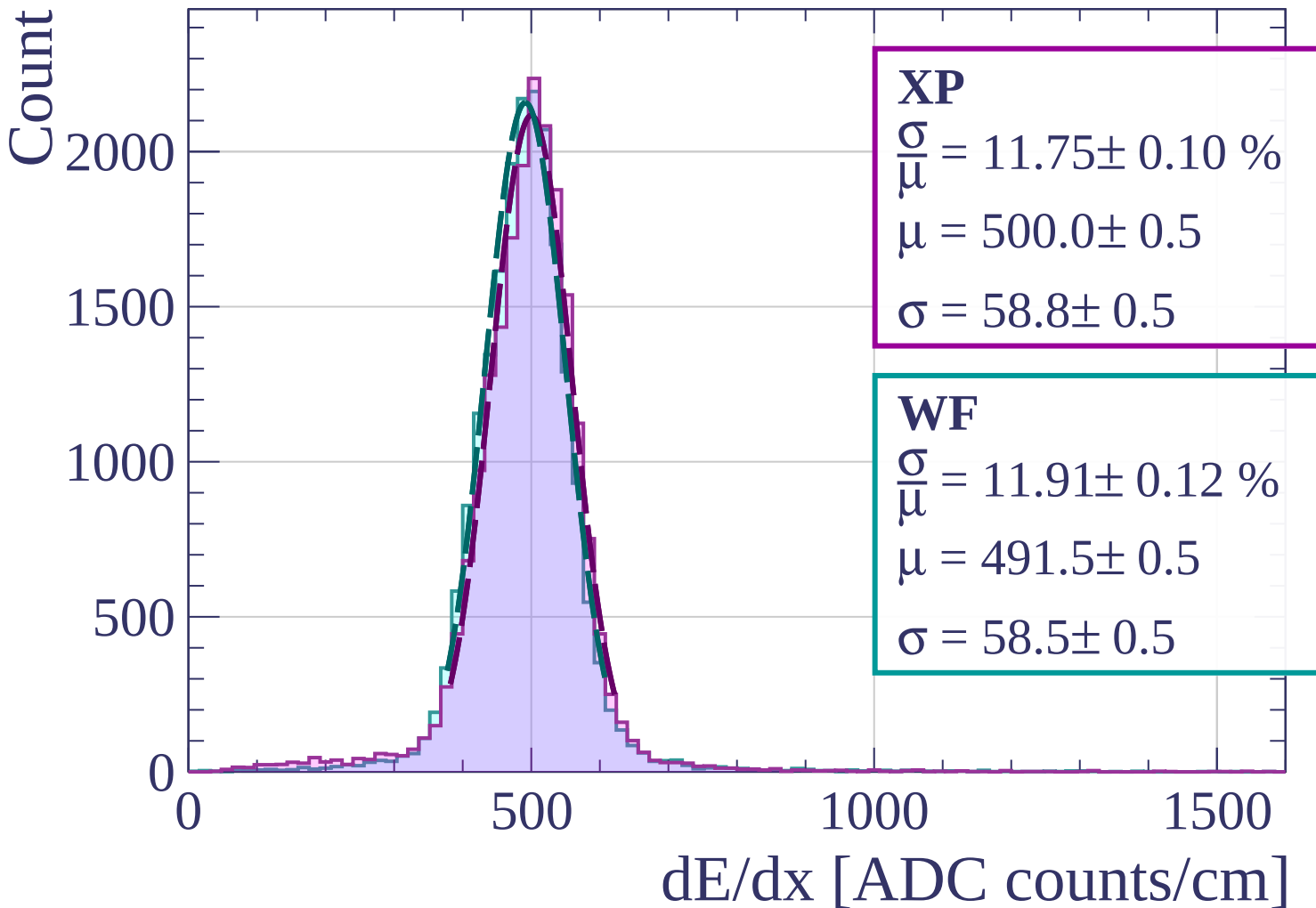
$$\sigma = 57.9 \pm 0.5$$

dE/dx [ADC counts/cm]

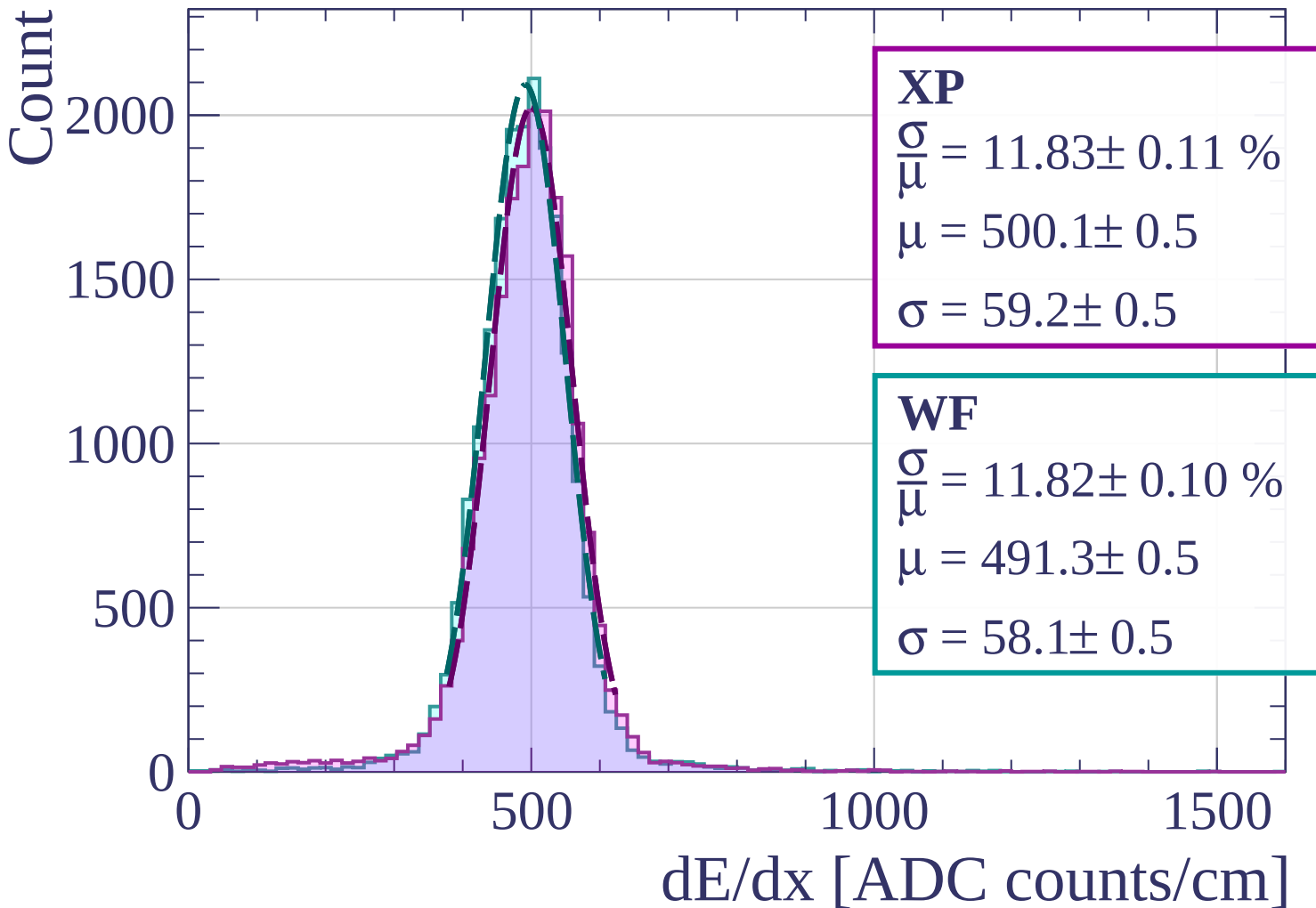
Energy loss | $6 < \theta < 9$



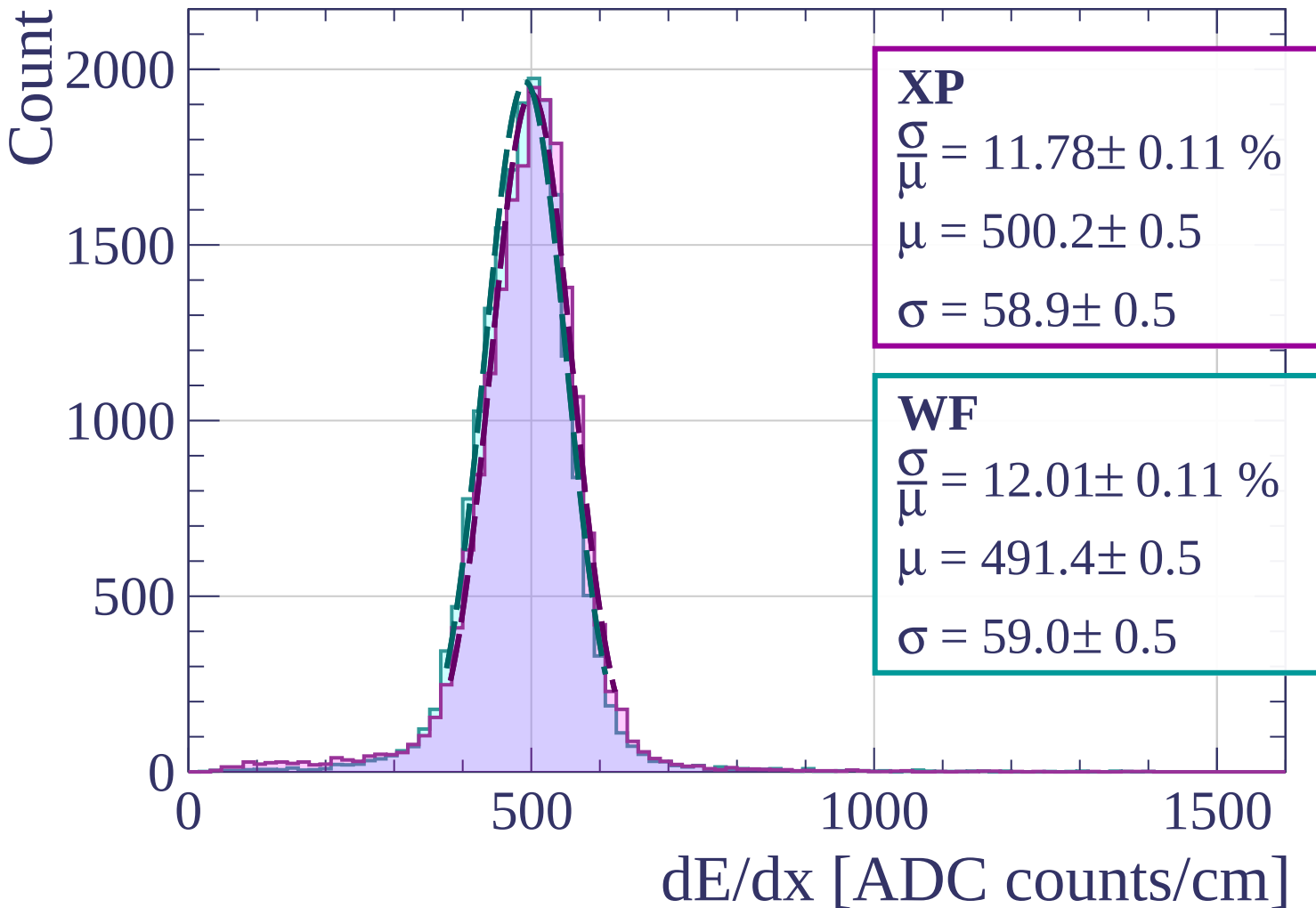
Energy loss | $9 < \theta < 12$



Energy loss | $12 < \theta < 15$



Energy loss | $15 < \theta < 18$



Energy loss | $18 < \theta < 21$

Count

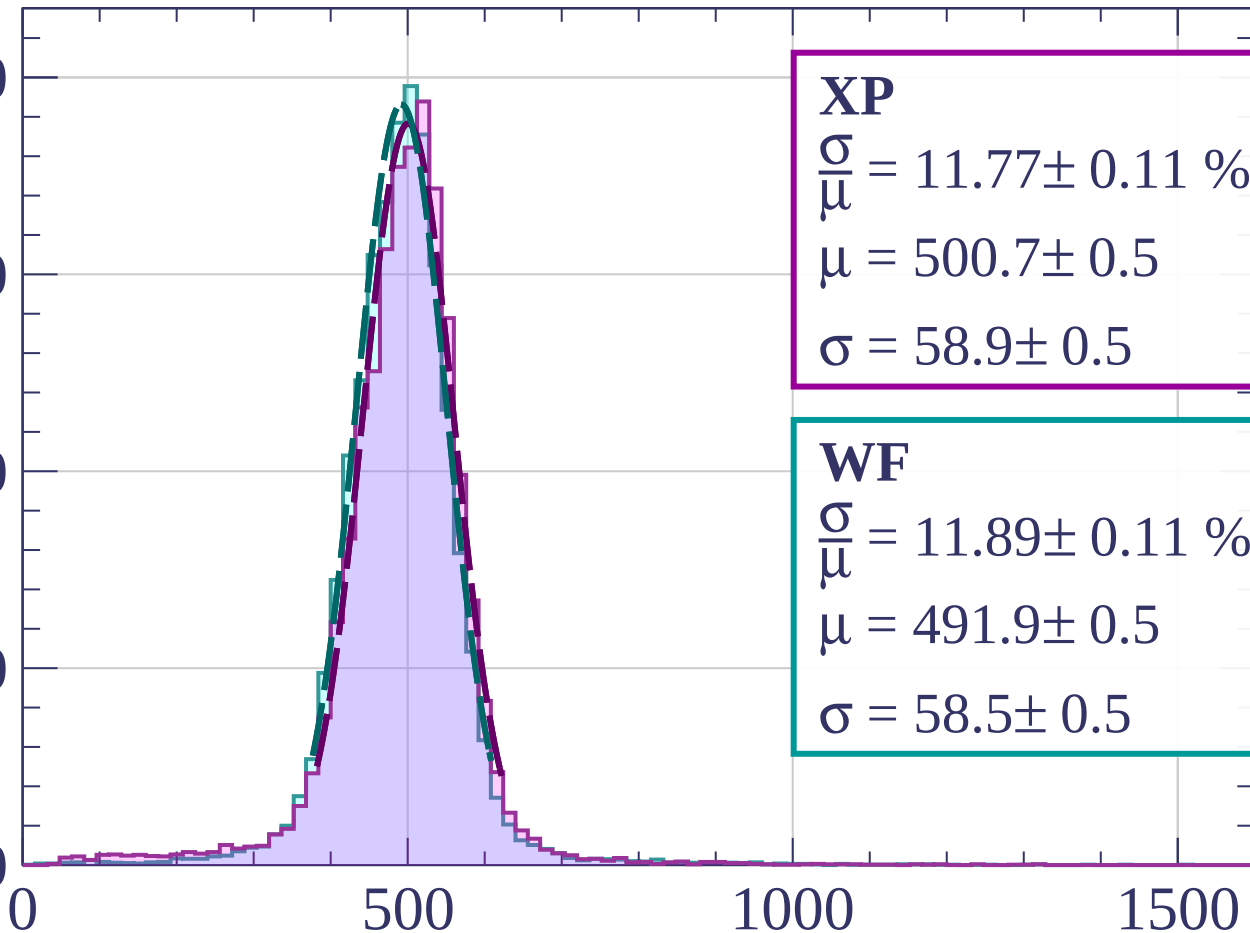
2000

1500

1000

500

0



XP

$$\frac{\sigma}{\mu} = 11.77 \pm 0.11 \%$$

$$\mu = 500.7 \pm 0.5$$

$$\sigma = 58.9 \pm 0.5$$

WF

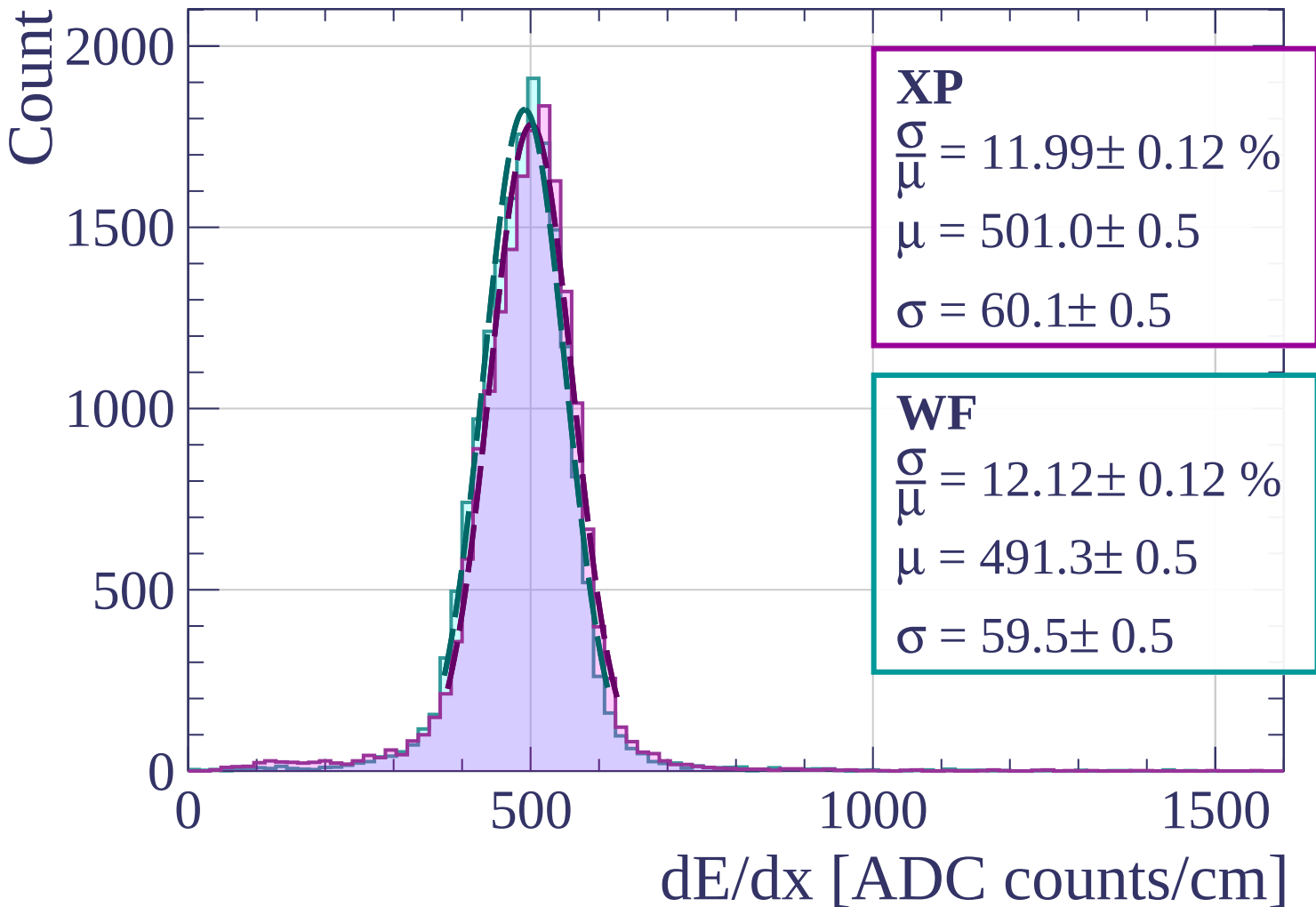
$$\frac{\sigma}{\mu} = 11.89 \pm 0.11 \%$$

$$\mu = 491.9 \pm 0.5$$

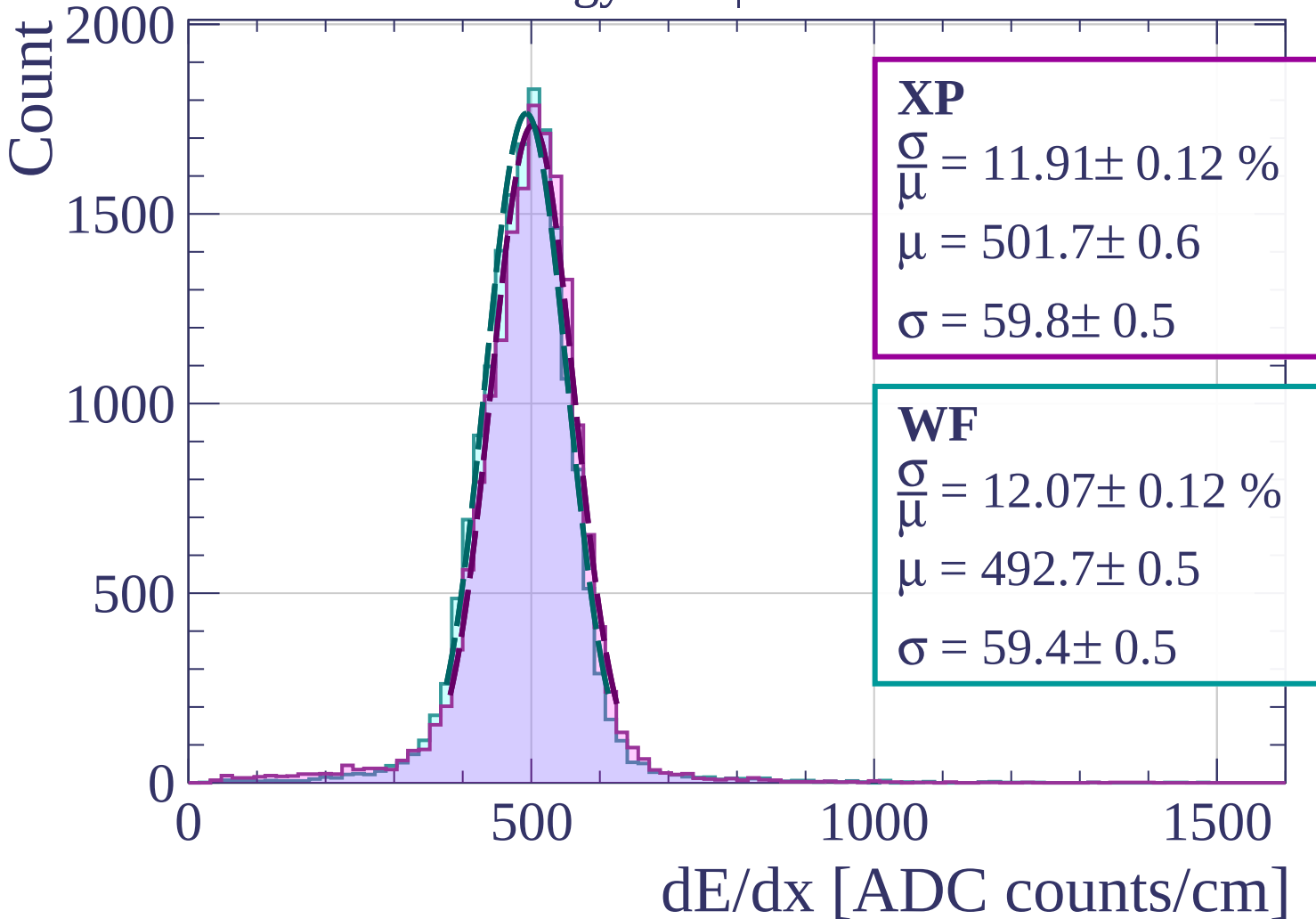
$$\sigma = 58.5 \pm 0.5$$

dE/dx [ADC counts/cm]

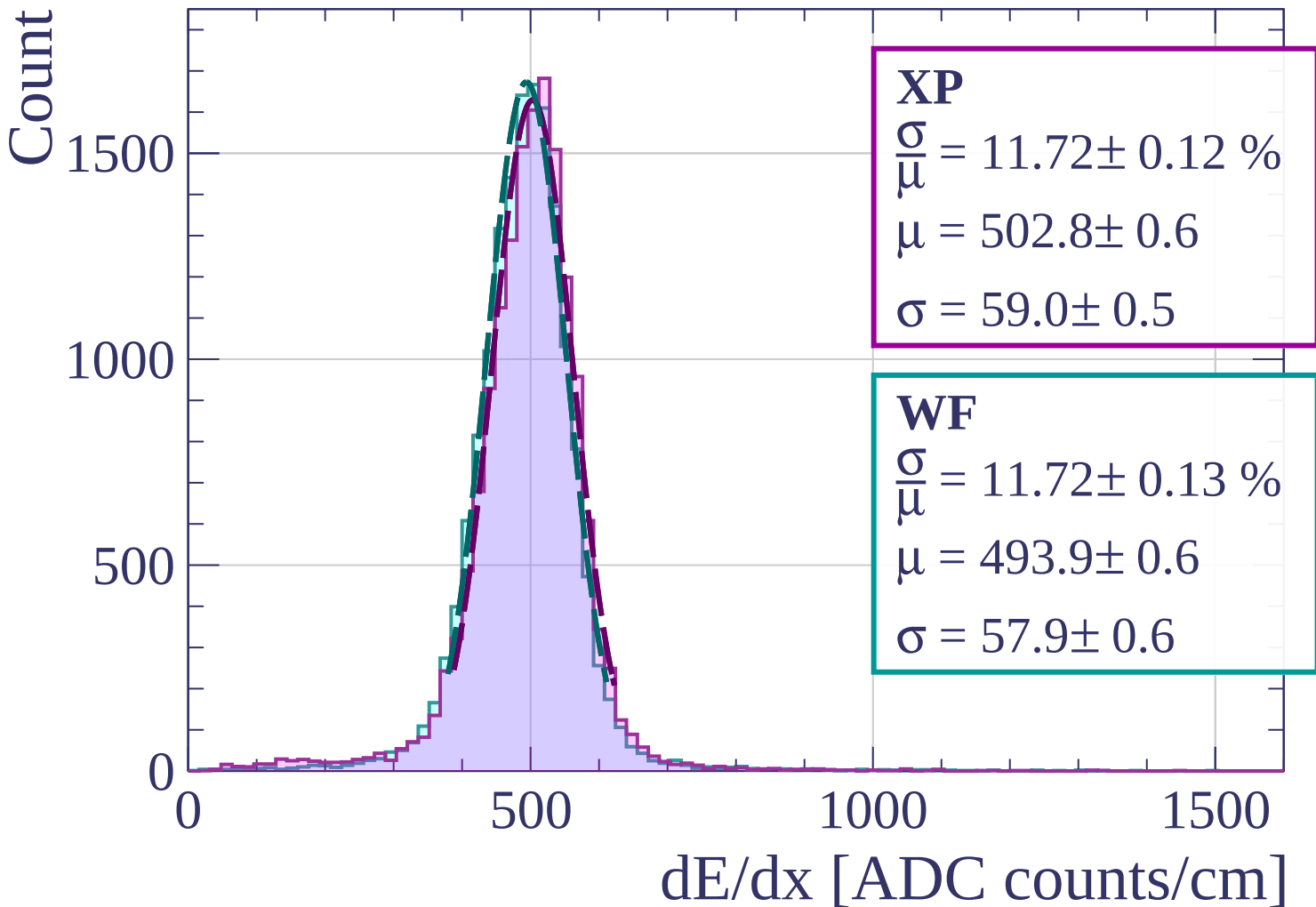
Energy loss | $21 < \theta < 24$



Energy loss | $24 < \theta < 27$

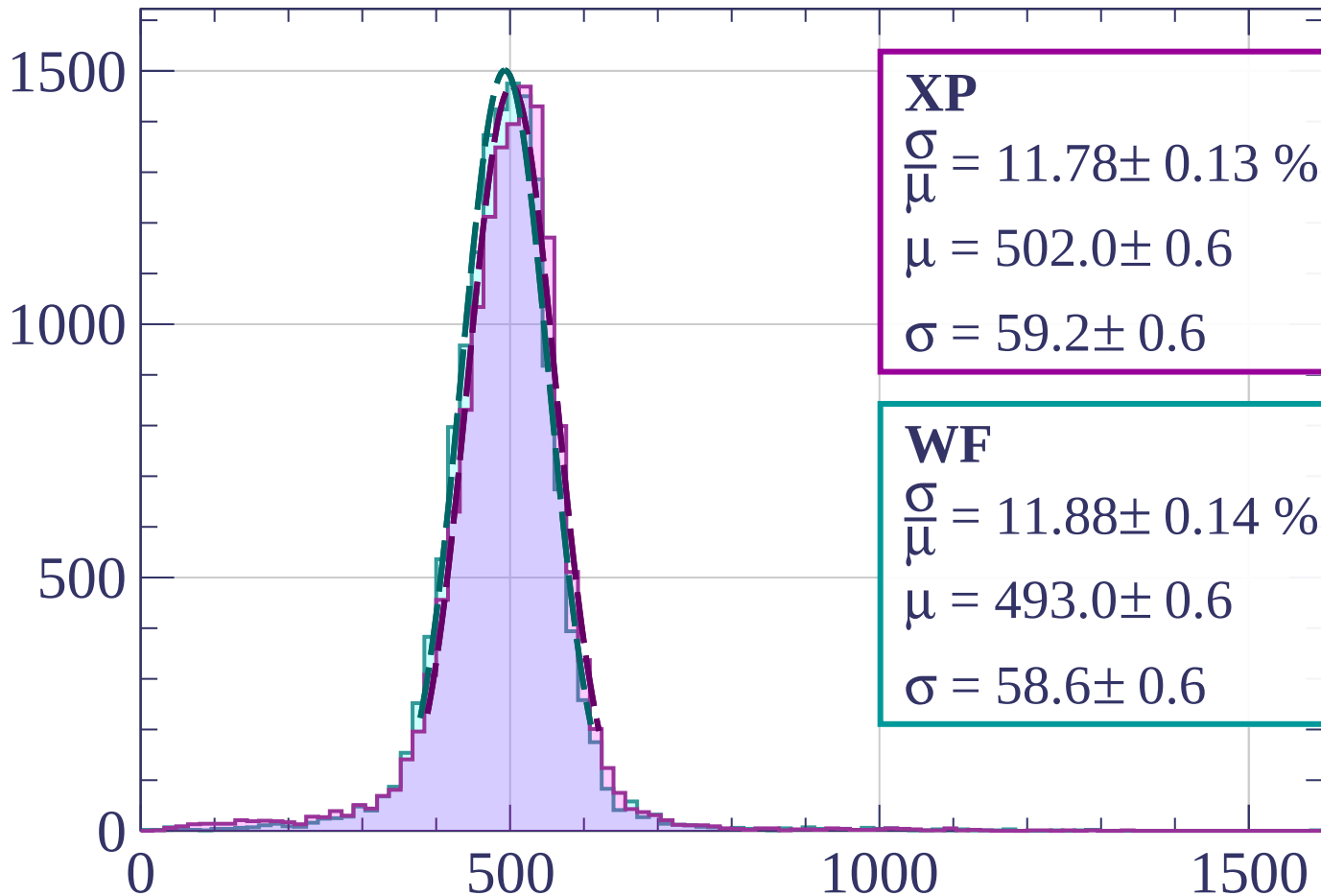


Energy loss | $27 < \theta < 30$



Energy loss | $30 < \theta < 33$

Count



XP

$$\frac{\sigma}{\mu} = 11.78 \pm 0.13 \%$$

$$\mu = 502.0 \pm 0.6$$

$$\sigma = 59.2 \pm 0.6$$

WF

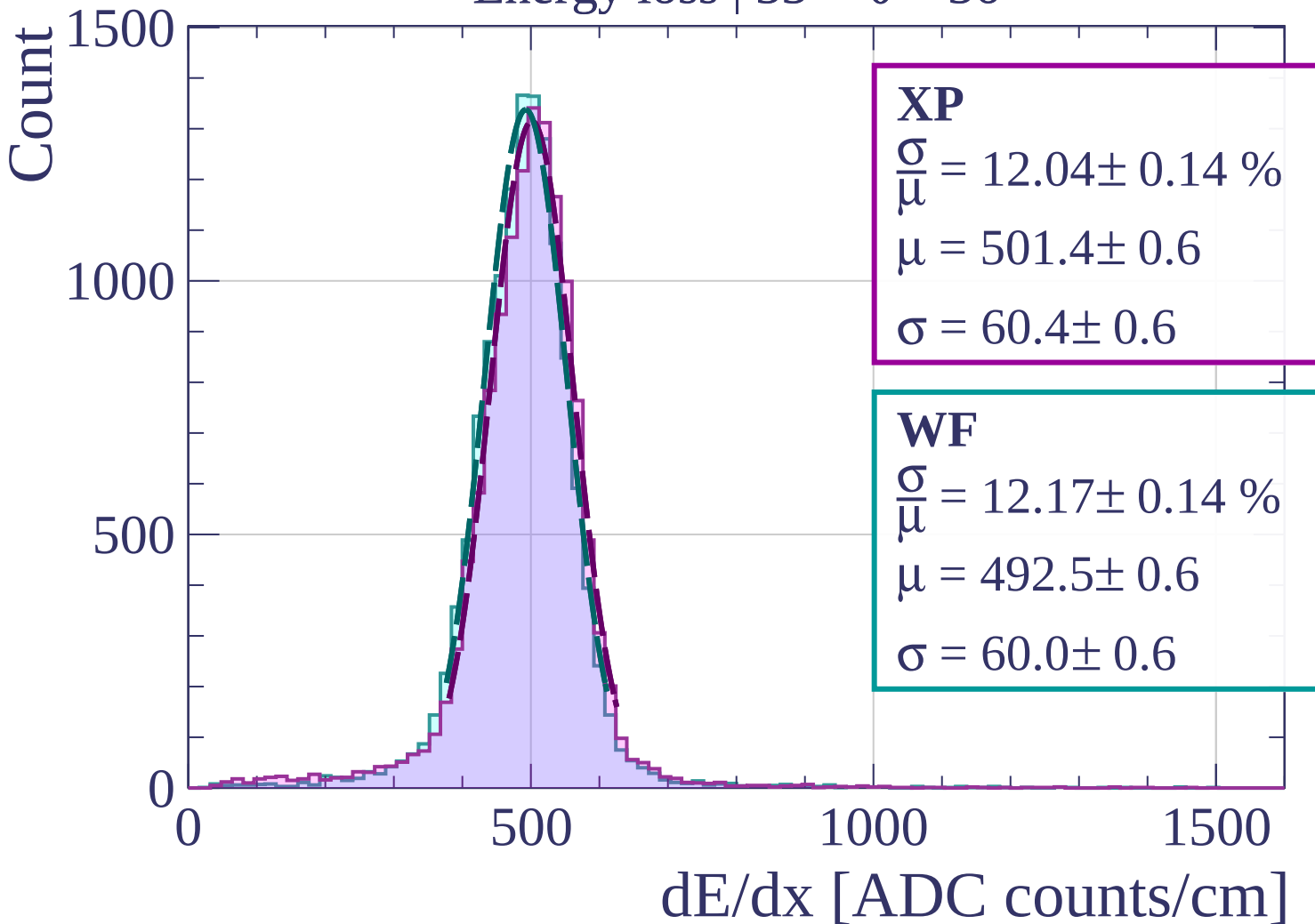
$$\frac{\sigma}{\mu} = 11.88 \pm 0.14 \%$$

$$\mu = 493.0 \pm 0.6$$

$$\sigma = 58.6 \pm 0.6$$

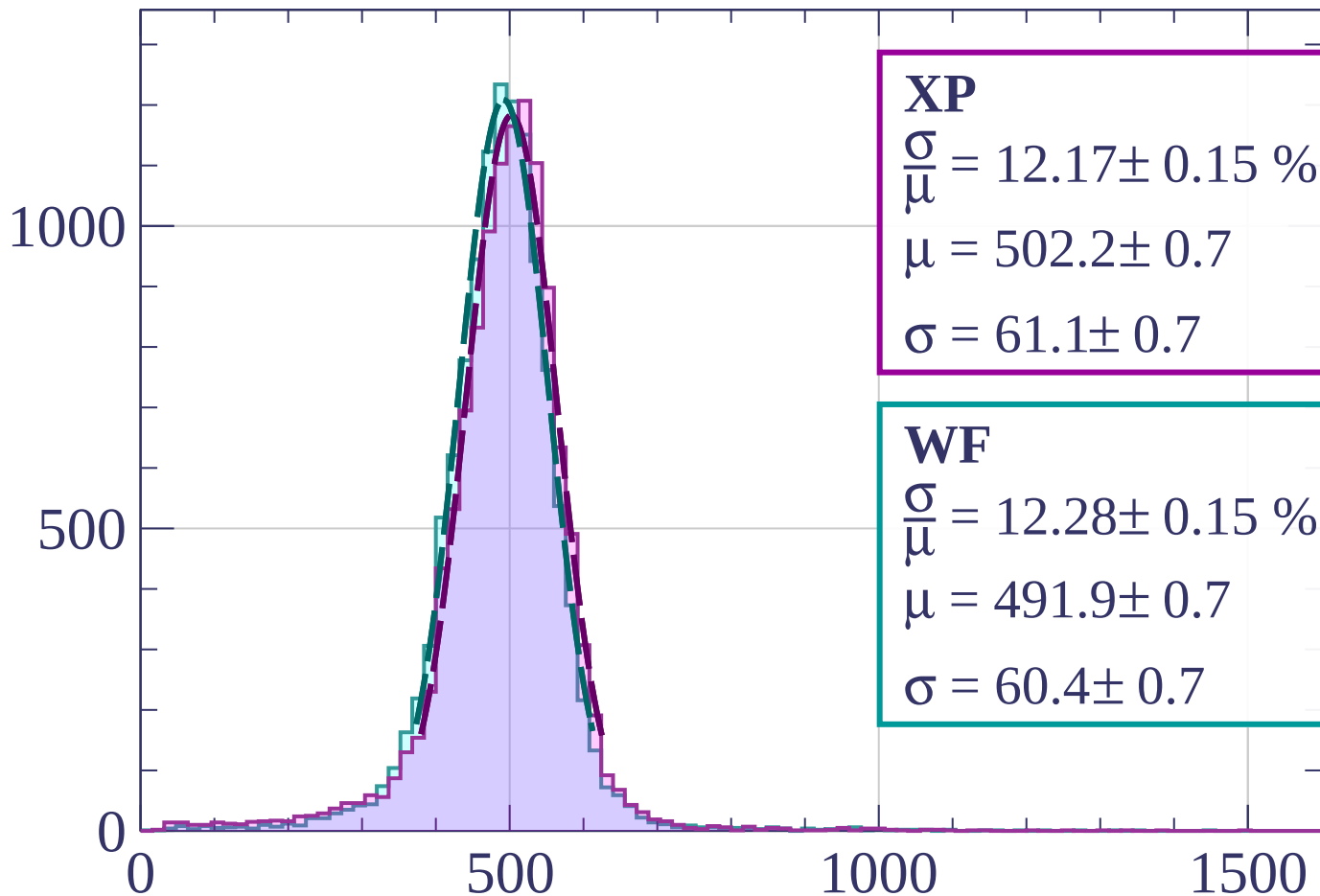
dE/dx [ADC counts/cm]

Energy loss | $33 < \theta < 36$



Energy loss | $36 < \theta < 39$

Count



XP

$$\frac{\sigma}{\mu} = 12.17 \pm 0.15 \%$$

$$\mu = 502.2 \pm 0.7$$

$$\sigma = 61.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 12.28 \pm 0.15 \%$$

$$\mu = 491.9 \pm 0.7$$

$$\sigma = 60.4 \pm 0.7$$

dE/dx [ADC counts/cm]

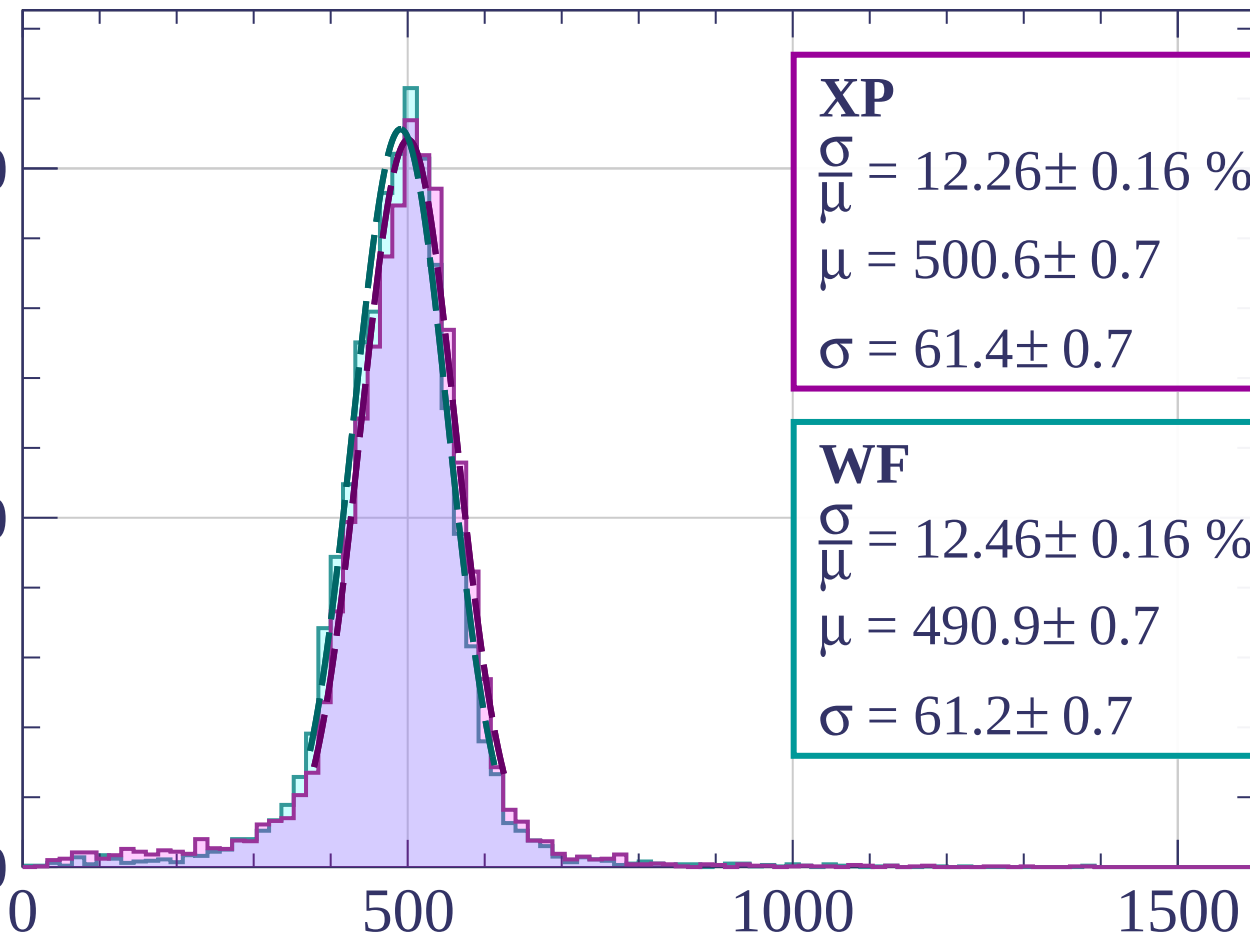
Energy loss | $39 < \theta < 42$

Count

1000

500

0



XP

$$\frac{\sigma}{\mu} = 12.26 \pm 0.16 \%$$

$$\mu = 500.6 \pm 0.7$$

$$\sigma = 61.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 12.46 \pm 0.16 \%$$

$$\mu = 490.9 \pm 0.7$$

$$\sigma = 61.2 \pm 0.7$$

dE/dx [ADC counts/cm]

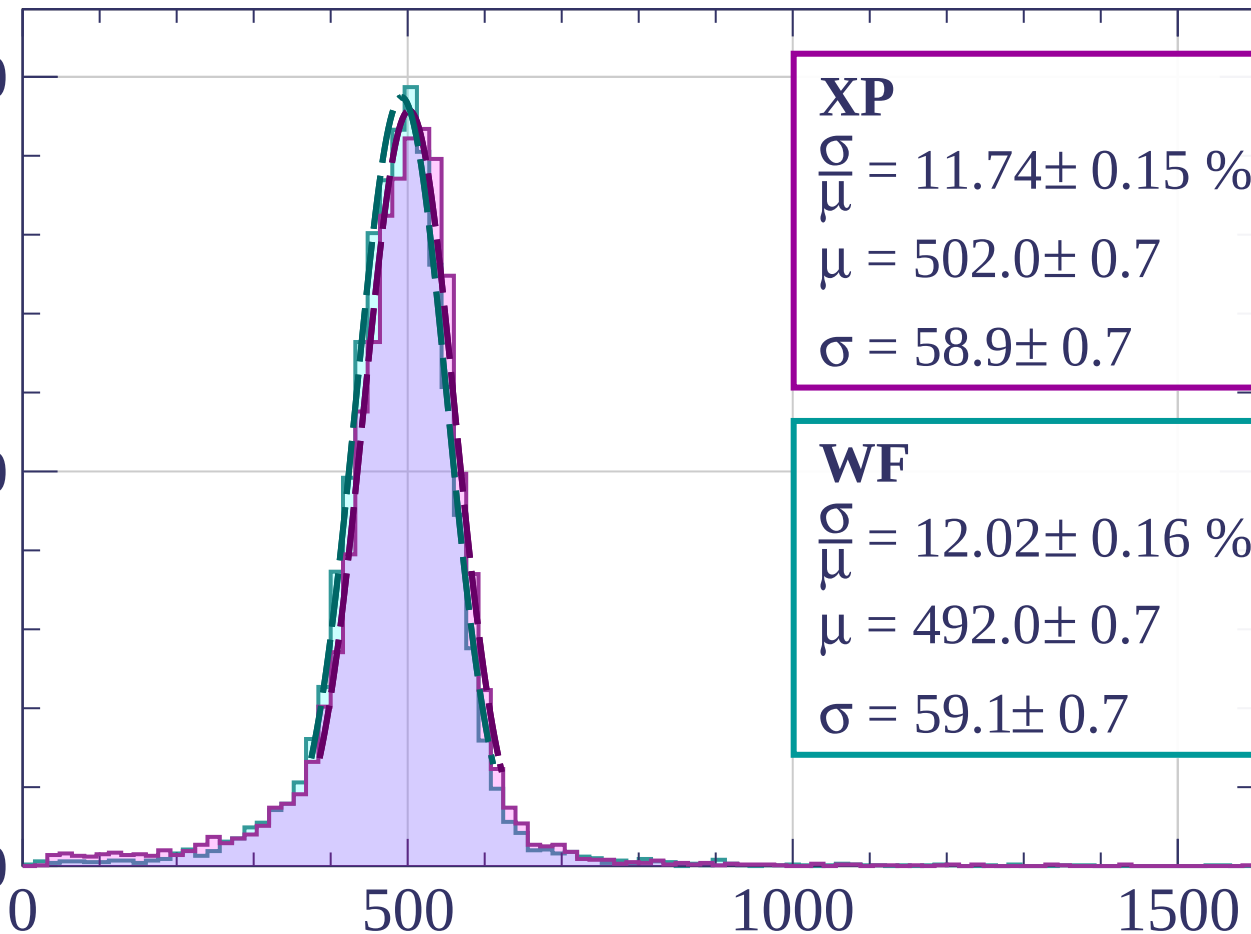
Energy loss | $42 < \theta < 45$

Count

1000

500

0



XP

$$\frac{\sigma}{\mu} = 11.74 \pm 0.15 \%$$

$$\mu = 502.0 \pm 0.7$$

$$\sigma = 58.9 \pm 0.7$$

WF

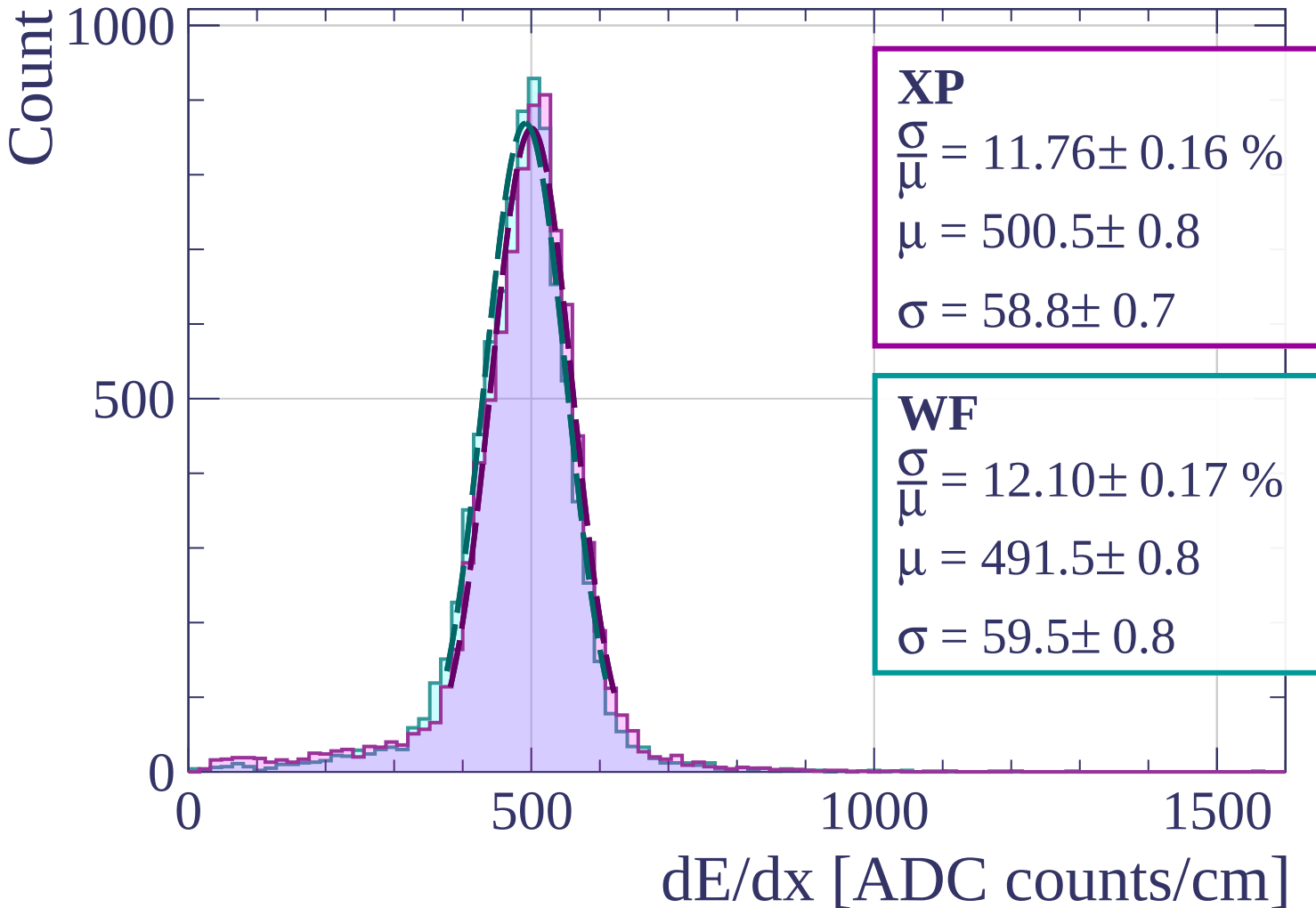
$$\frac{\sigma}{\mu} = 12.02 \pm 0.16 \%$$

$$\mu = 492.0 \pm 0.7$$

$$\sigma = 59.1 \pm 0.7$$

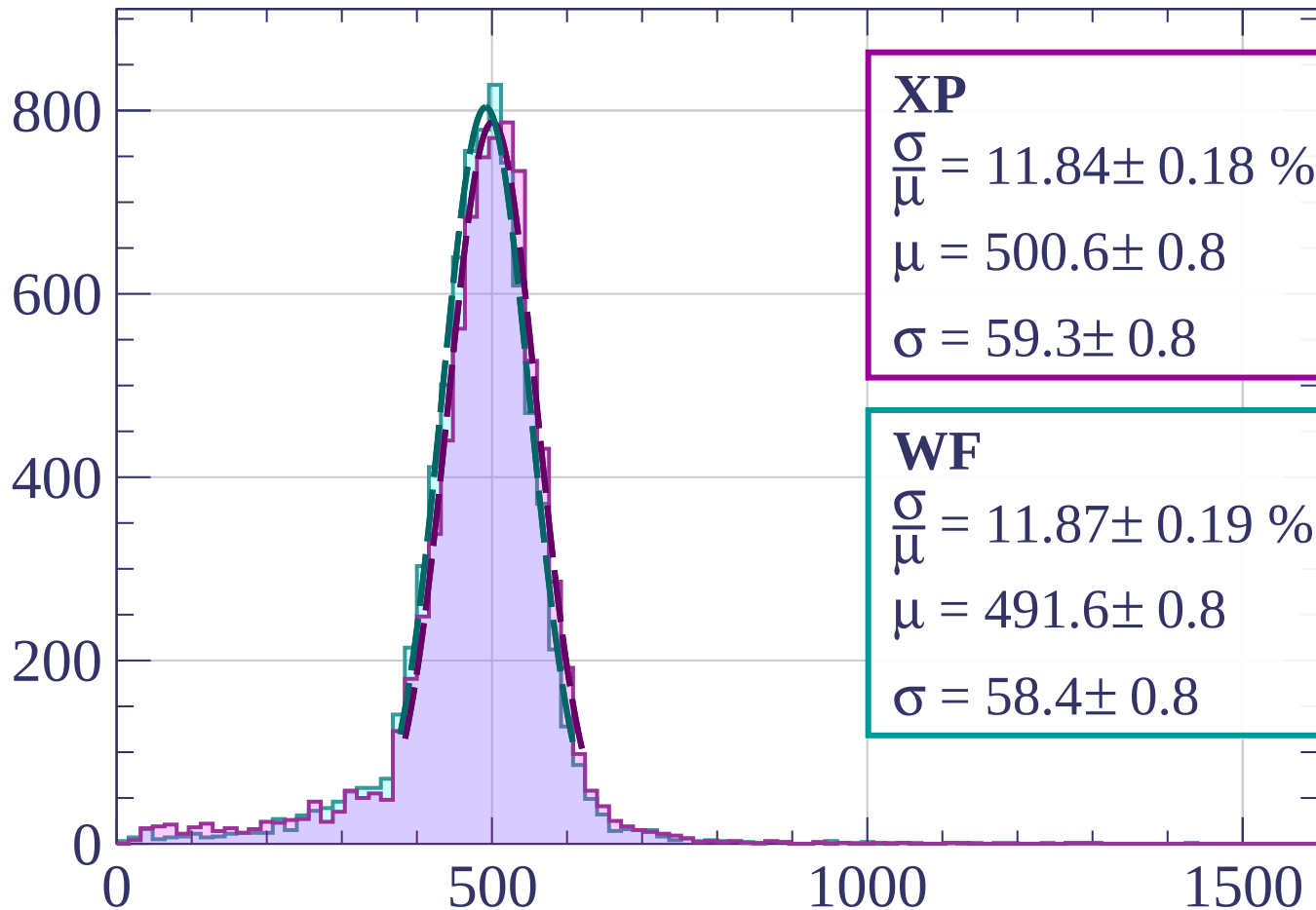
dE/dx [ADC counts/cm]

Energy loss | $45 < \theta < 48$



Energy loss | $48 < \theta < 51$

Count



XP

$$\frac{\sigma}{\mu} = 11.84 \pm 0.18 \%$$

$$\mu = 500.6 \pm 0.8$$

$$\sigma = 59.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 11.87 \pm 0.19 \%$$

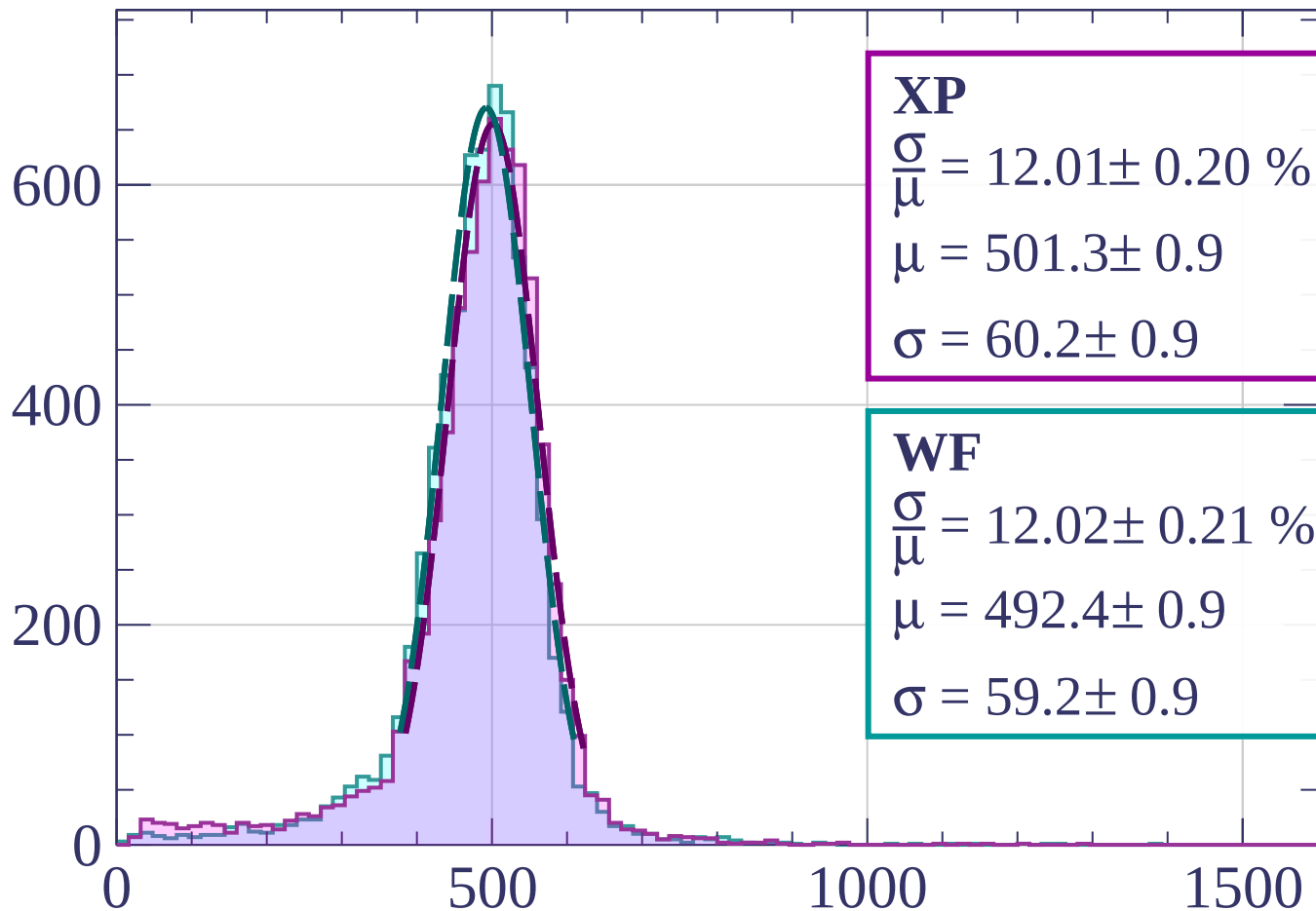
$$\mu = 491.6 \pm 0.8$$

$$\sigma = 58.4 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $51 < \theta < 54$

Count



XP

$$\frac{\sigma}{\mu} = 12.01 \pm 0.20 \%$$

$$\mu = 501.3 \pm 0.9$$

$$\sigma = 60.2 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 12.02 \pm 0.21 \%$$

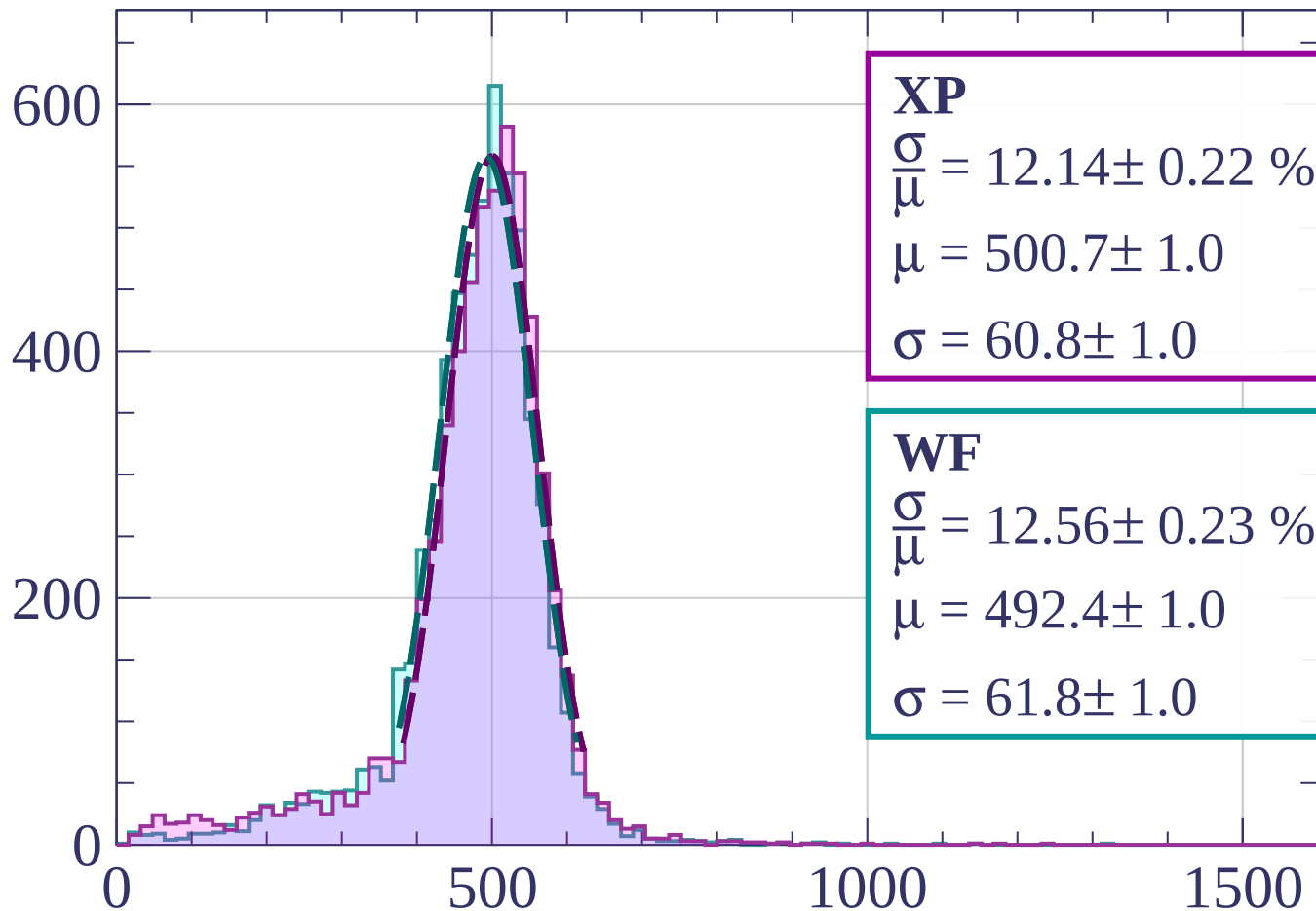
$$\mu = 492.4 \pm 0.9$$

$$\sigma = 59.2 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $54 < \theta < 57$

Count



XP

$$\frac{\sigma}{\mu} = 12.14 \pm 0.22 \%$$

$$\mu = 500.7 \pm 1.0$$

$$\sigma = 60.8 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 12.56 \pm 0.23 \%$$

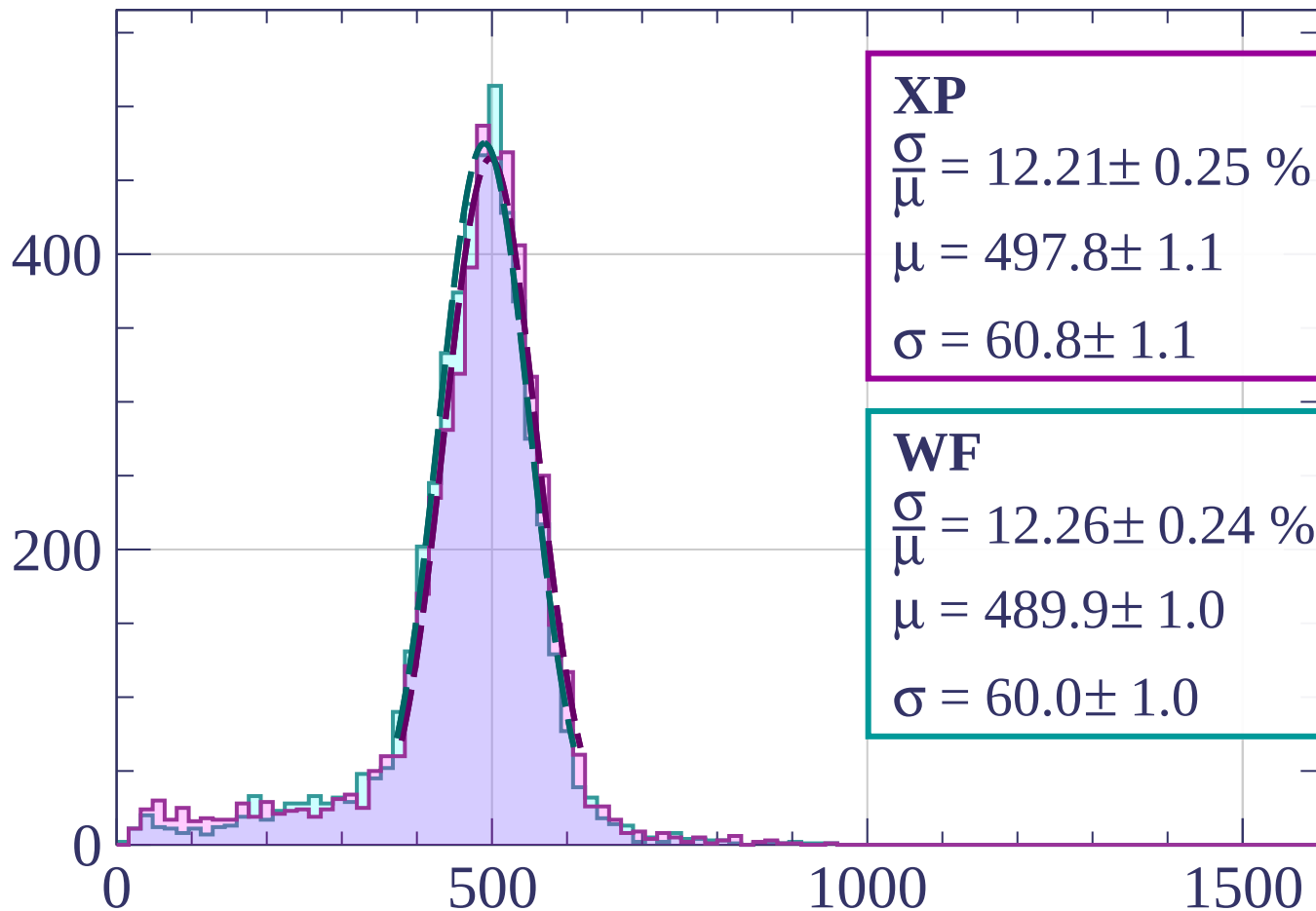
$$\mu = 492.4 \pm 1.0$$

$$\sigma = 61.8 \pm 1.0$$

dE/dx [ADC counts/cm]

Energy loss | $57 < \theta < 60$

Count



XP

$$\frac{\sigma}{\mu} = 12.21 \pm 0.25 \%$$

$$\mu = 497.8 \pm 1.1$$

$$\sigma = 60.8 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 12.26 \pm 0.24 \%$$

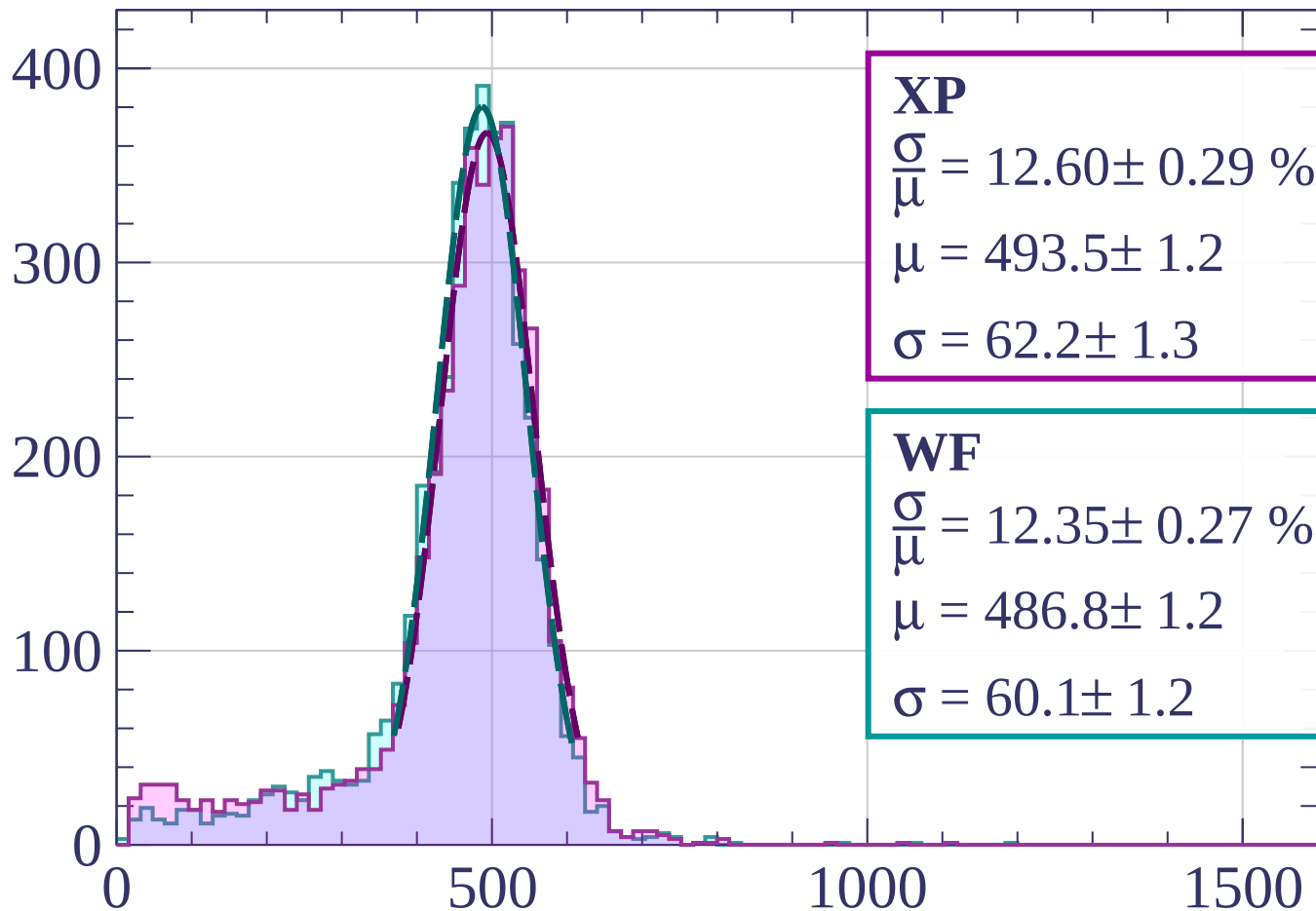
$$\mu = 489.9 \pm 1.0$$

$$\sigma = 60.0 \pm 1.0$$

dE/dx [ADC counts/cm]

Energy loss | $60 < \theta < 63$

Count



XP

$$\frac{\sigma}{\mu} = 12.60 \pm 0.29 \%$$

$$\mu = 493.5 \pm 1.2$$

$$\sigma = 62.2 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 12.35 \pm 0.27 \%$$

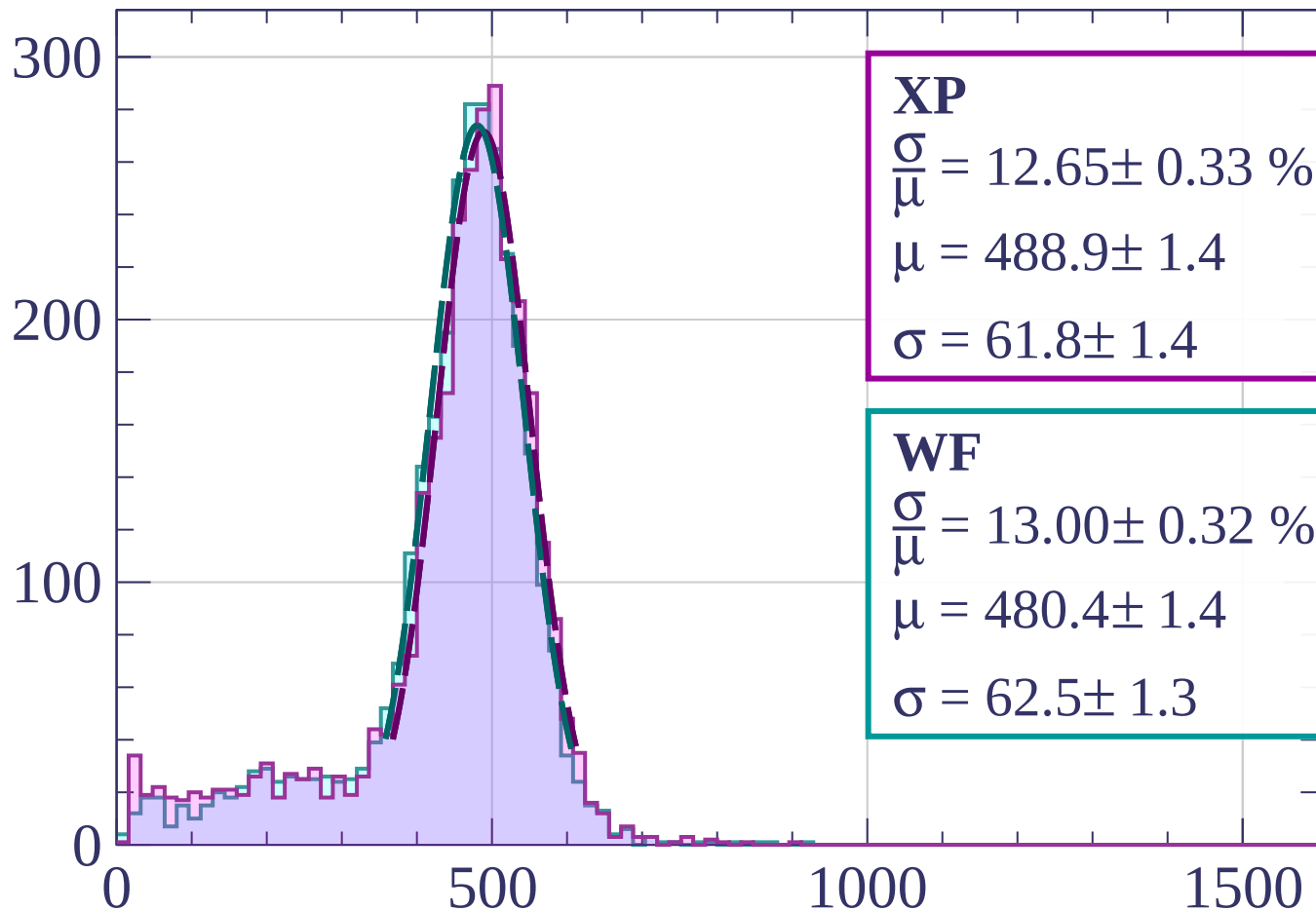
$$\mu = 486.8 \pm 1.2$$

$$\sigma = 60.1 \pm 1.2$$

dE/dx [ADC counts/cm]

Energy loss | $63 < \theta < 66$

Count



XP

$$\frac{\sigma}{\mu} = 12.65 \pm 0.33 \%$$

$$\mu = 488.9 \pm 1.4$$

$$\sigma = 61.8 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 13.00 \pm 0.32 \%$$

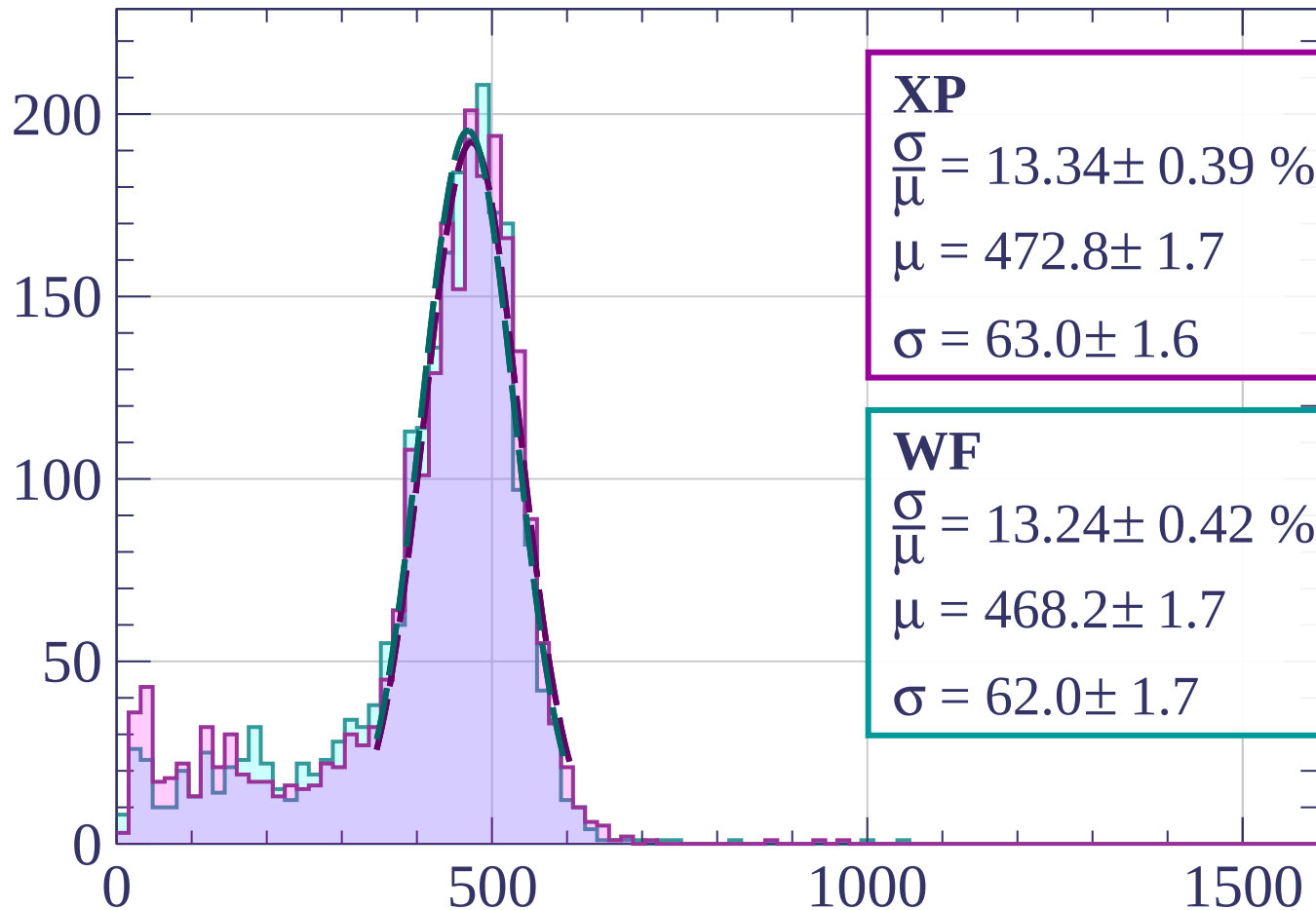
$$\mu = 480.4 \pm 1.4$$

$$\sigma = 62.5 \pm 1.3$$

dE/dx [ADC counts/cm]

Energy loss | $66 < \theta < 69$

Count



XP

$$\frac{\sigma}{\mu} = 13.34 \pm 0.39 \%$$

$$\mu = 472.8 \pm 1.7$$

$$\sigma = 63.0 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 13.24 \pm 0.42 \%$$

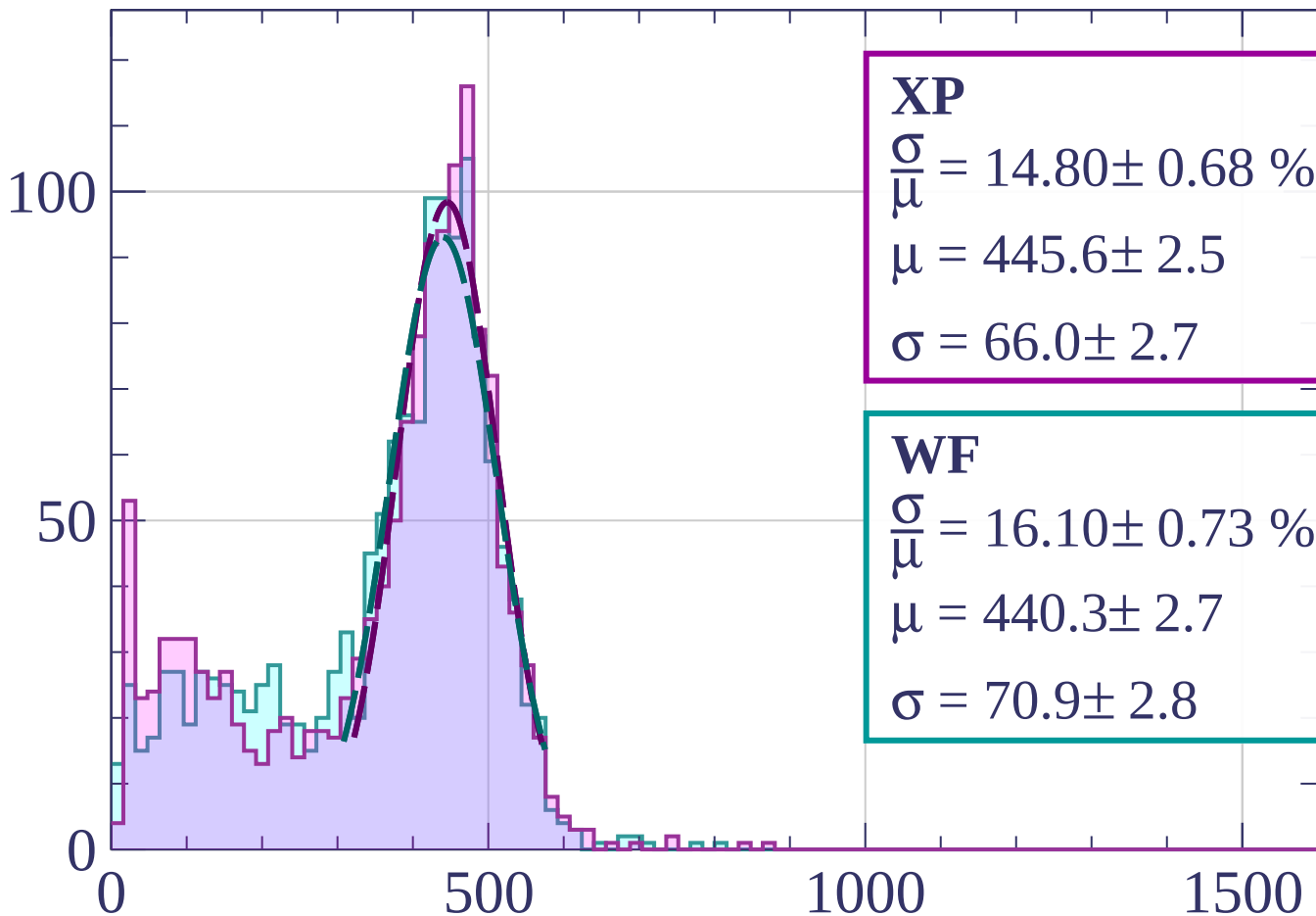
$$\mu = 468.2 \pm 1.7$$

$$\sigma = 62.0 \pm 1.7$$

dE/dx [ADC counts/cm]

Energy loss | $69 < \theta < 72$

Count



XP

$$\frac{\sigma}{\mu} = 14.80 \pm 0.68 \%$$

$$\mu = 445.6 \pm 2.5$$

$$\sigma = 66.0 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 16.10 \pm 0.73 \%$$

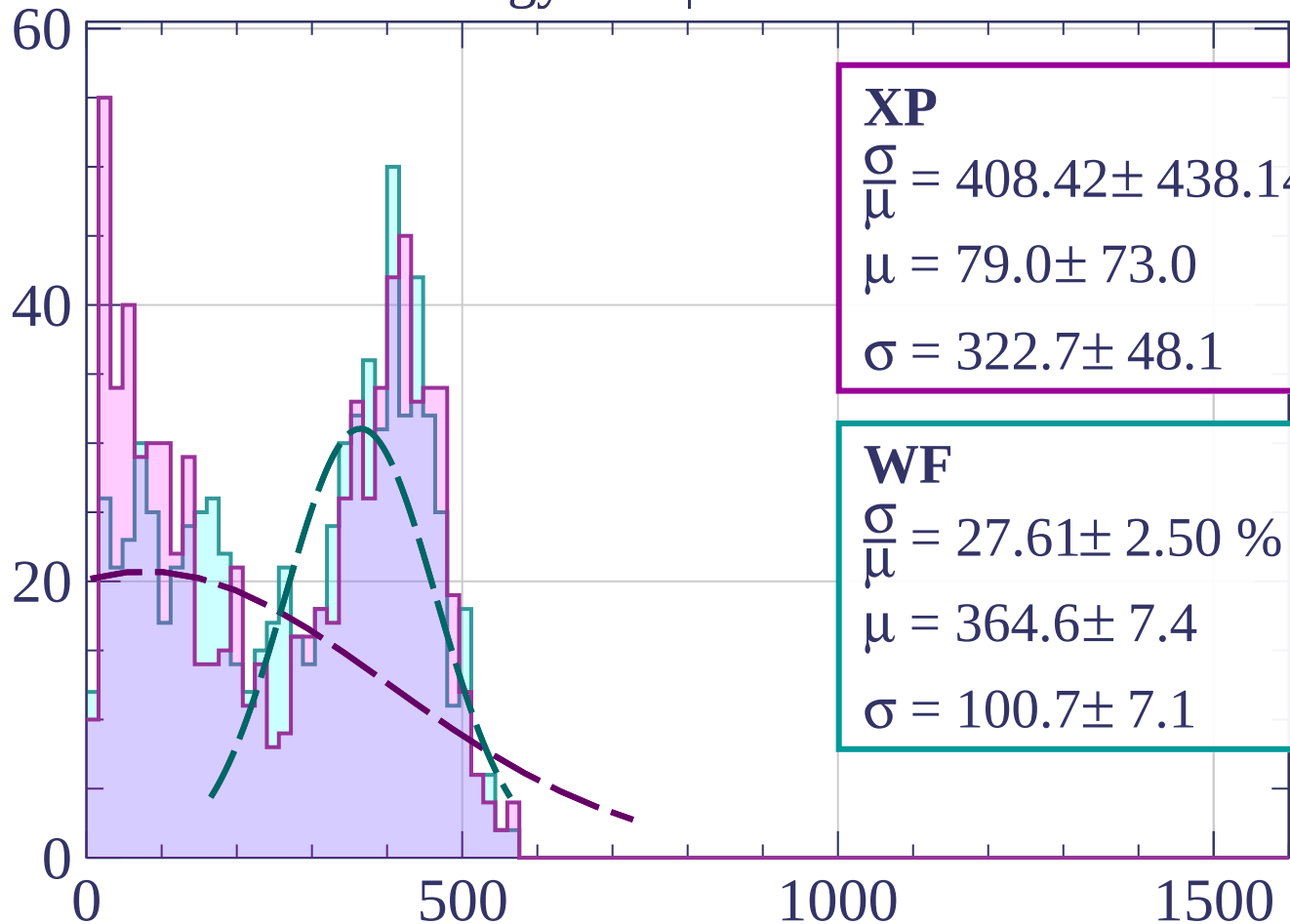
$$\mu = 440.3 \pm 2.7$$

$$\sigma = 70.9 \pm 2.8$$

dE/dx [ADC counts/cm]

Energy loss | $72 < \theta < 75$

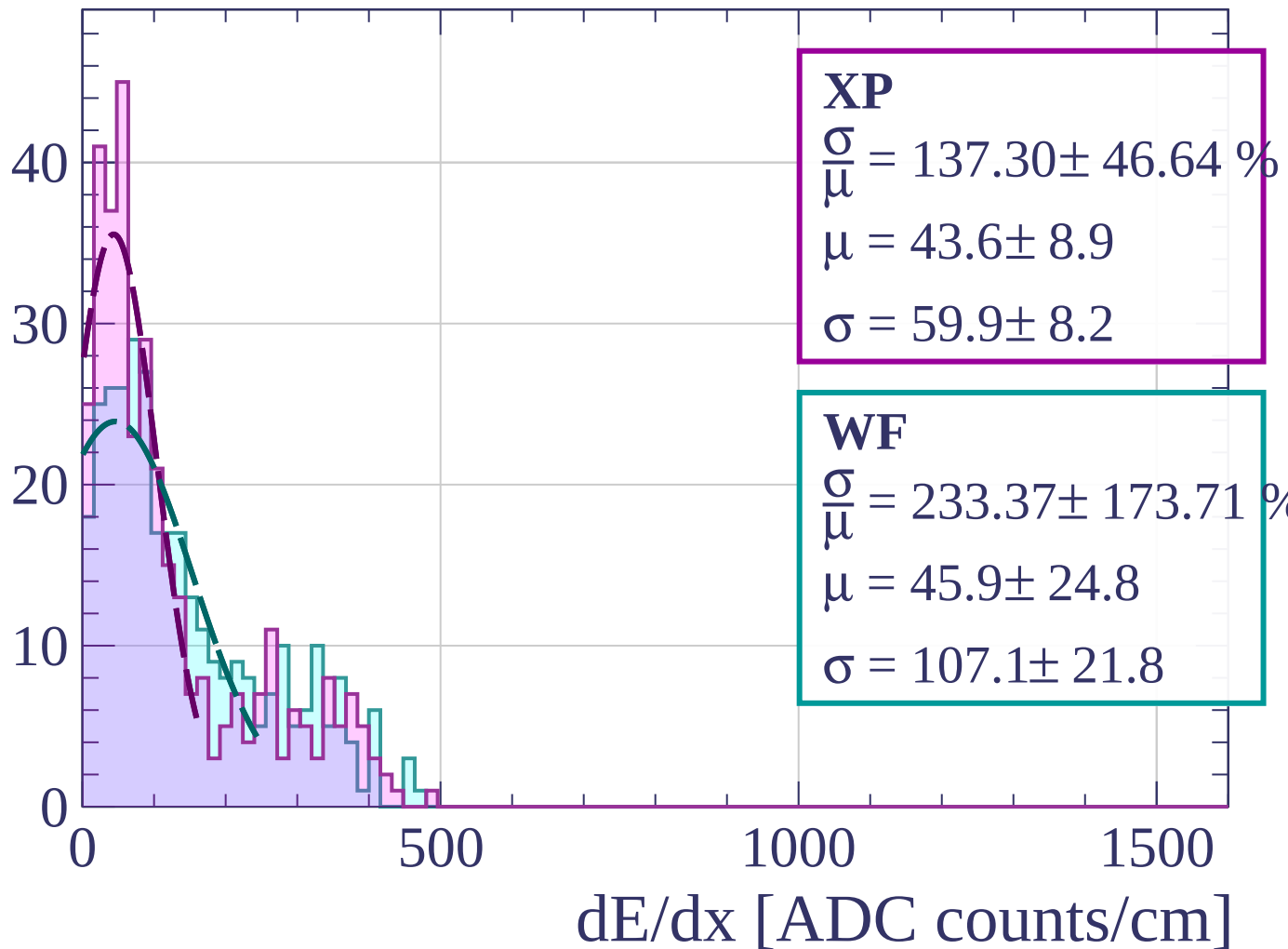
Count



dE/dx [ADC counts/cm]

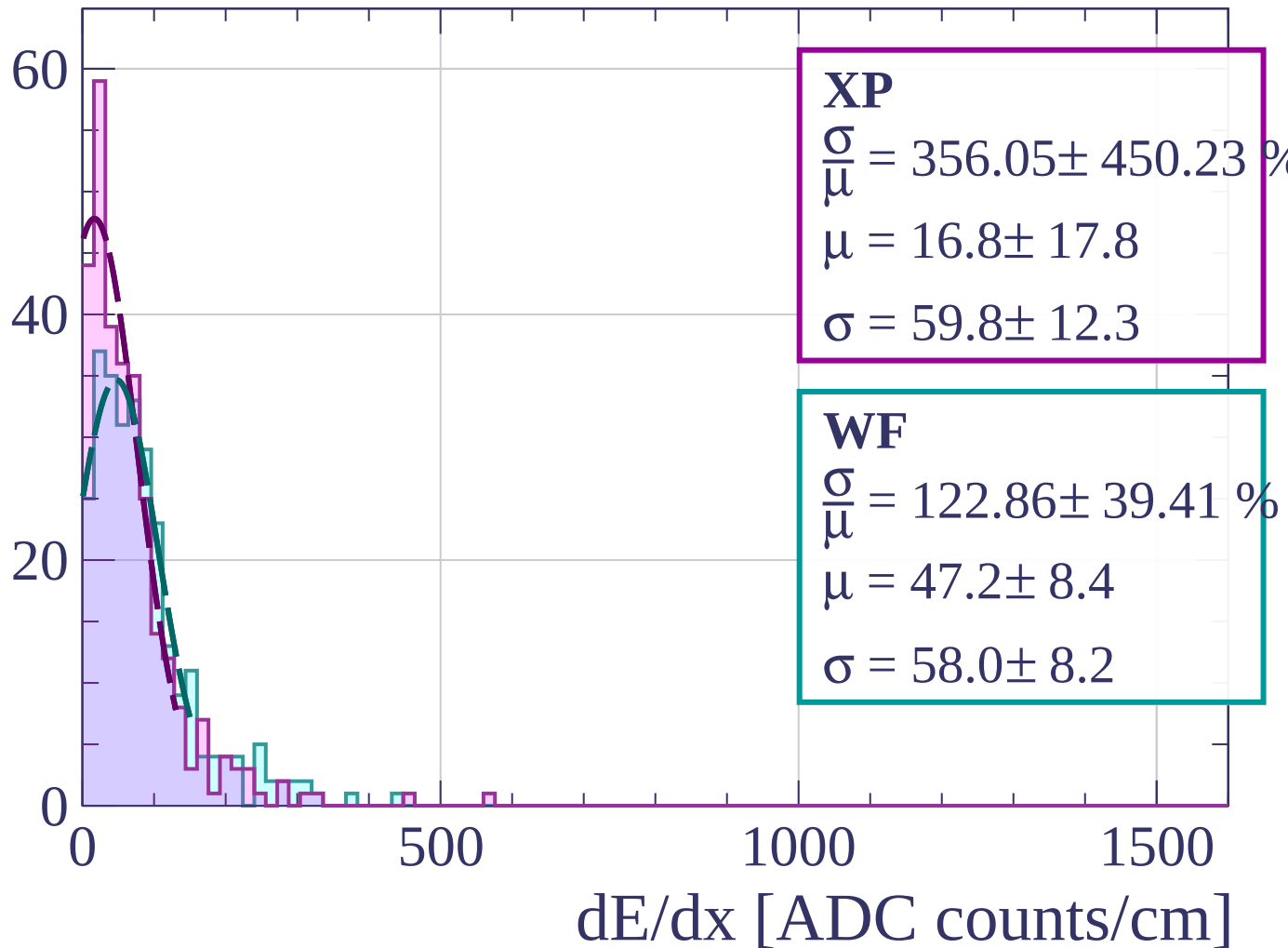
Energy loss | $75 < \theta < 78$

Count



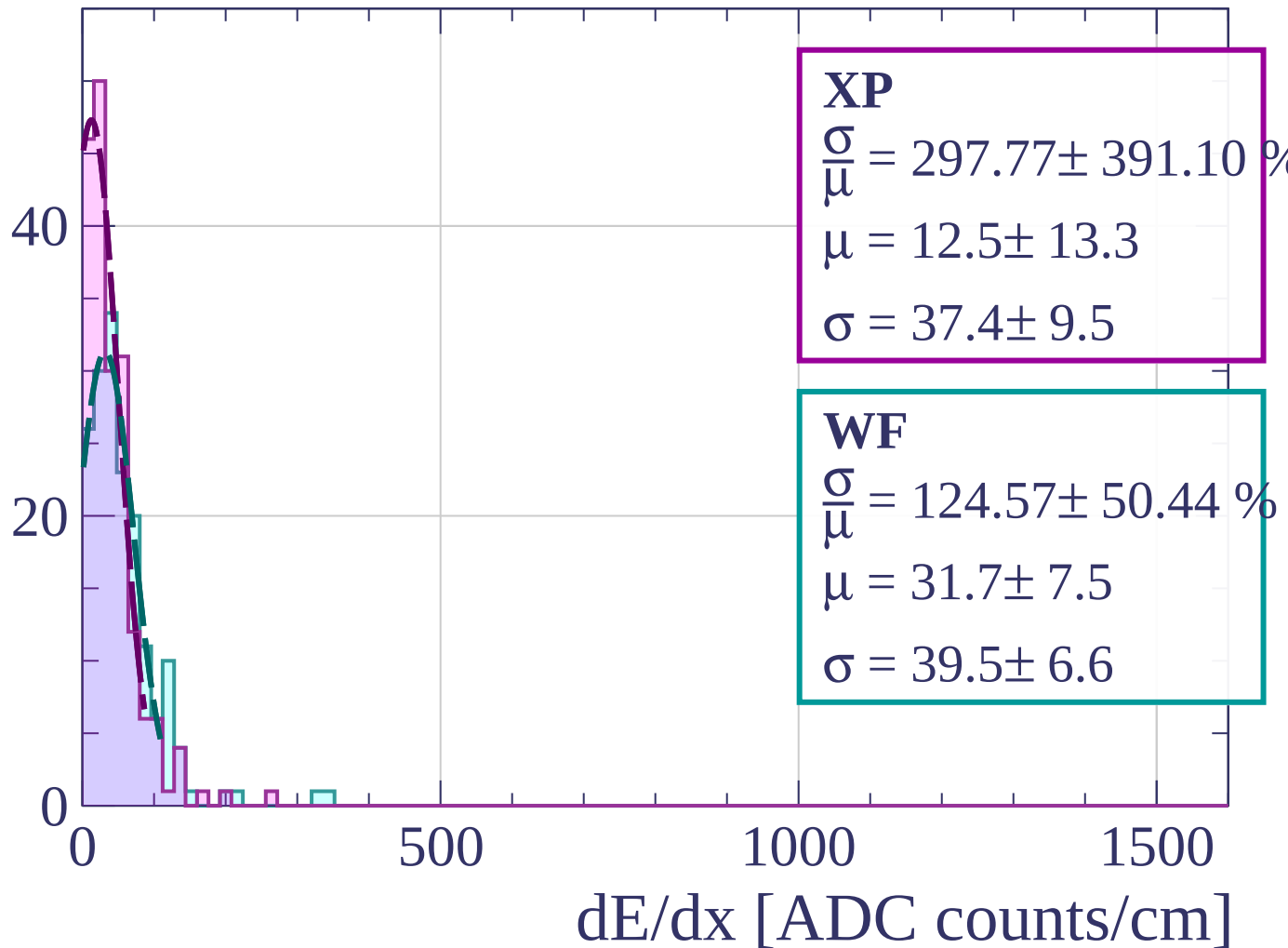
Energy loss | $78 < \theta < 81$

Count



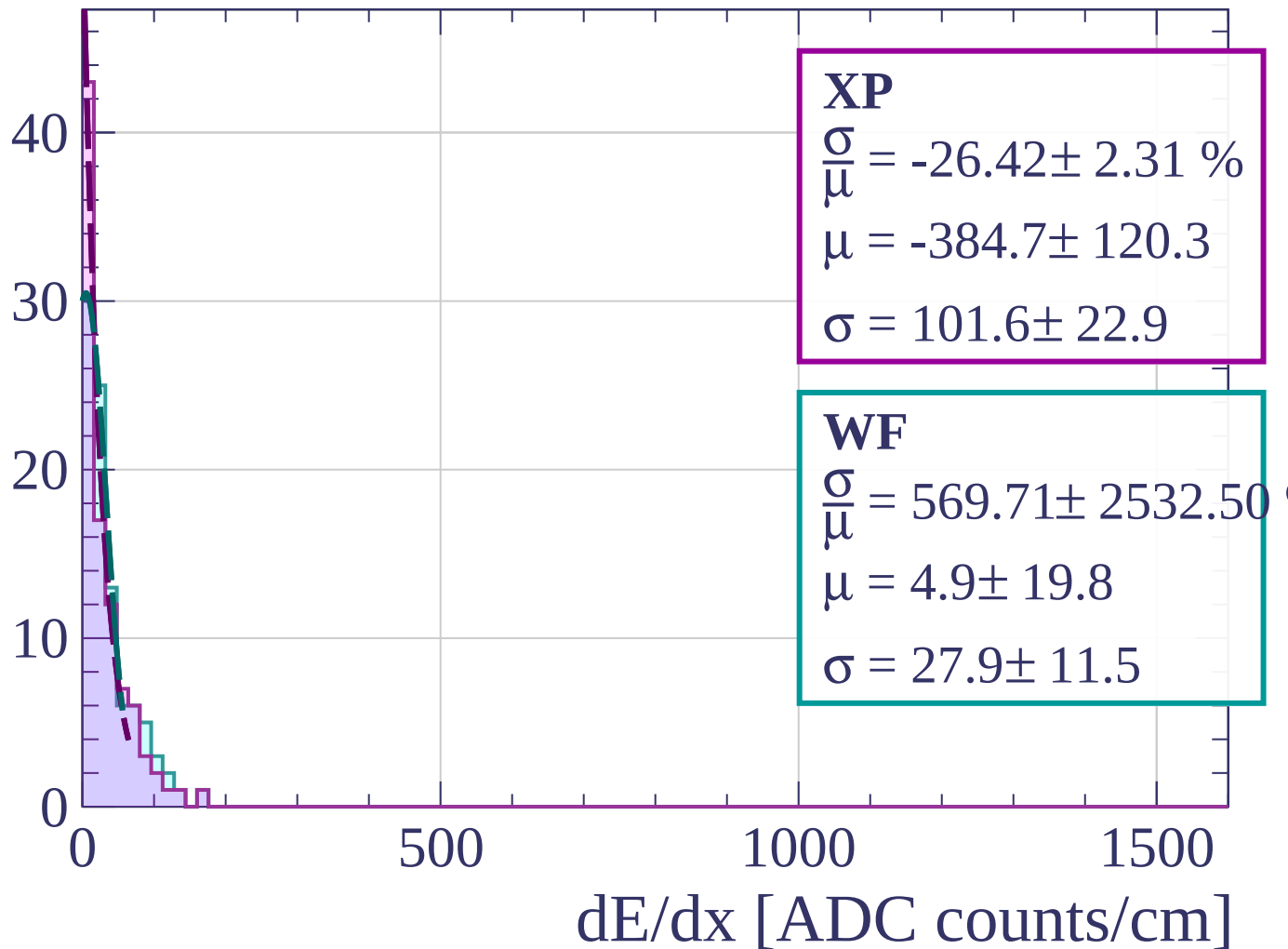
Energy loss | $81 < \theta < 84$

Count



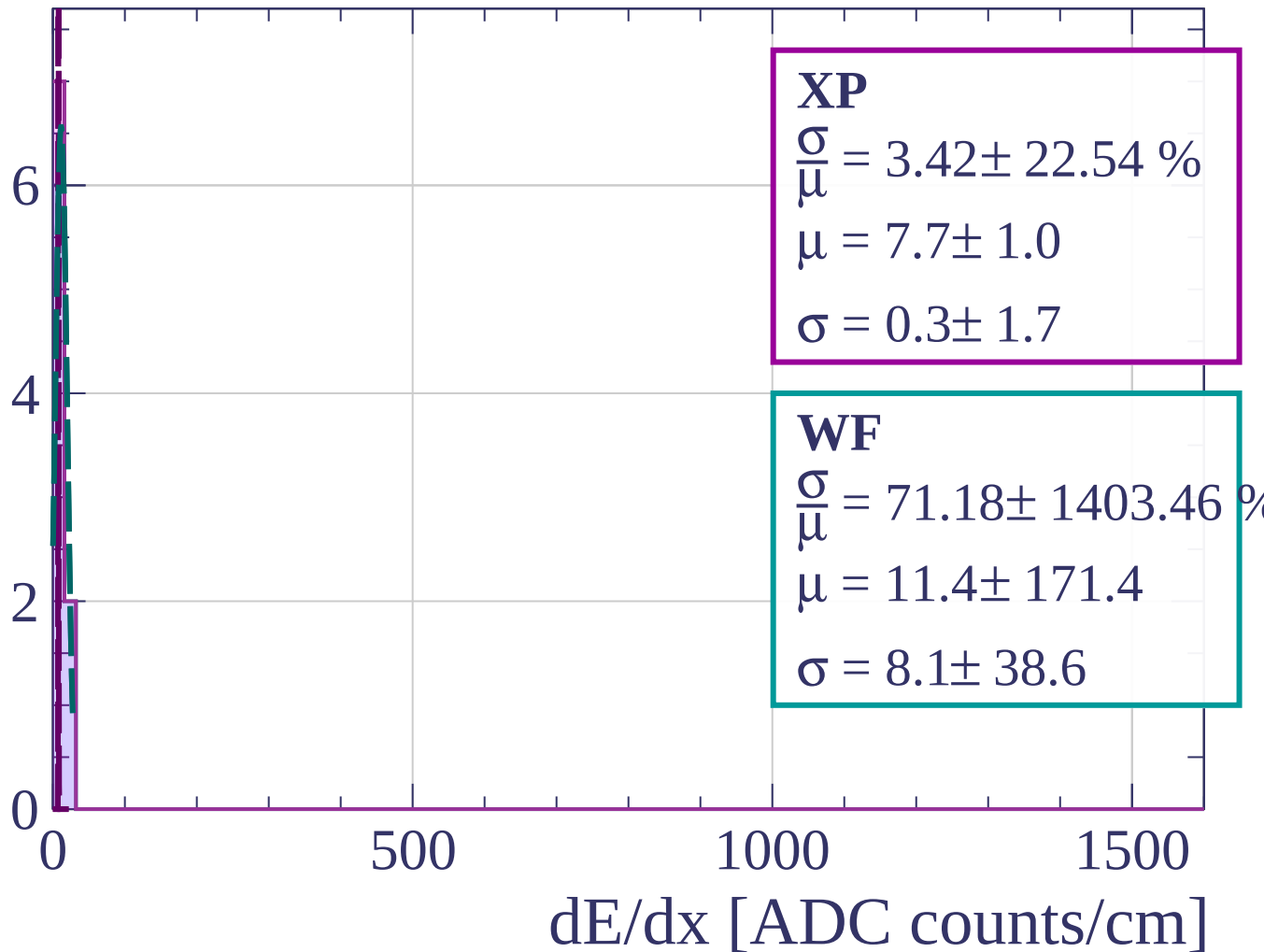
Energy loss | $84 < \theta < 87$

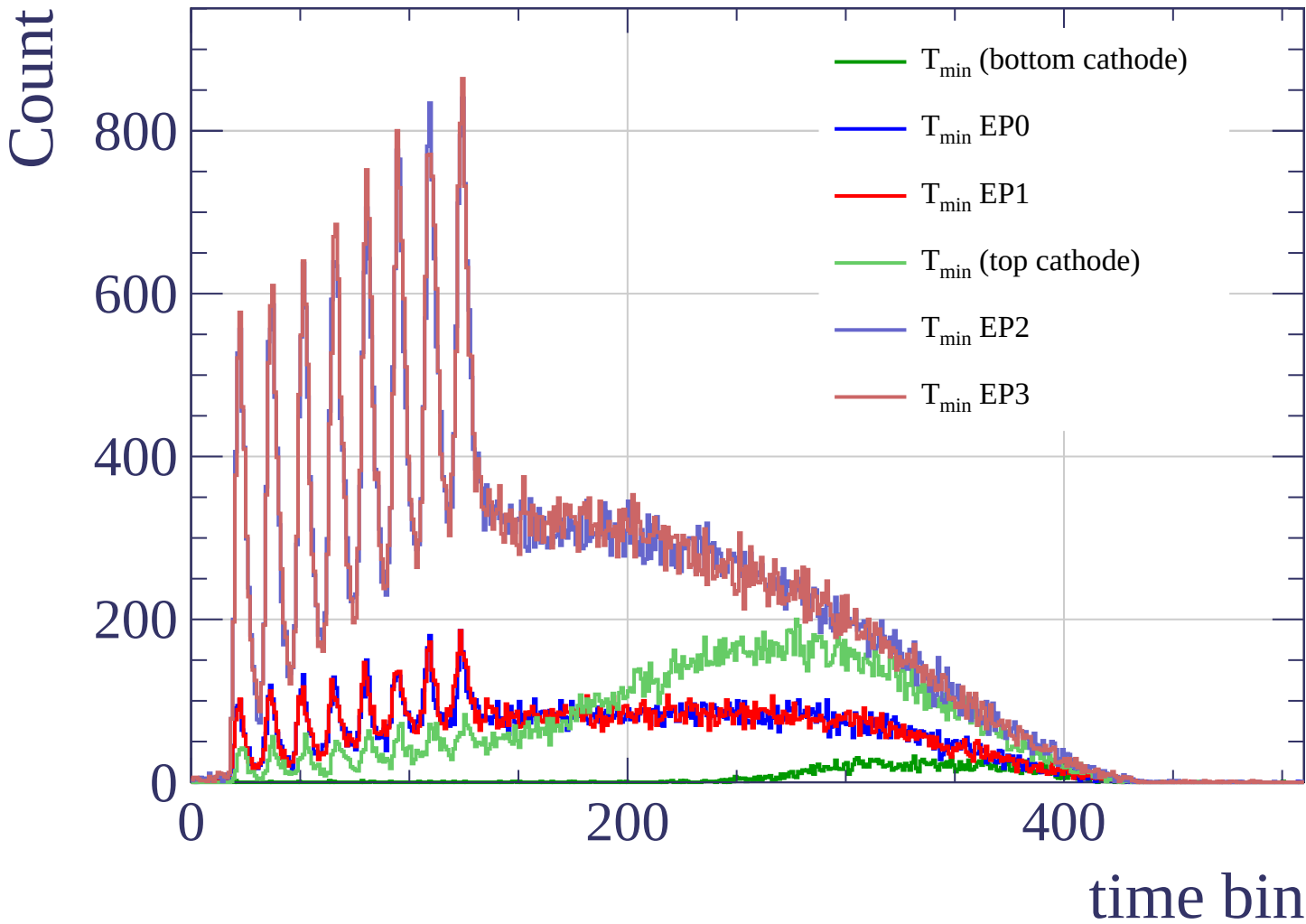
Count



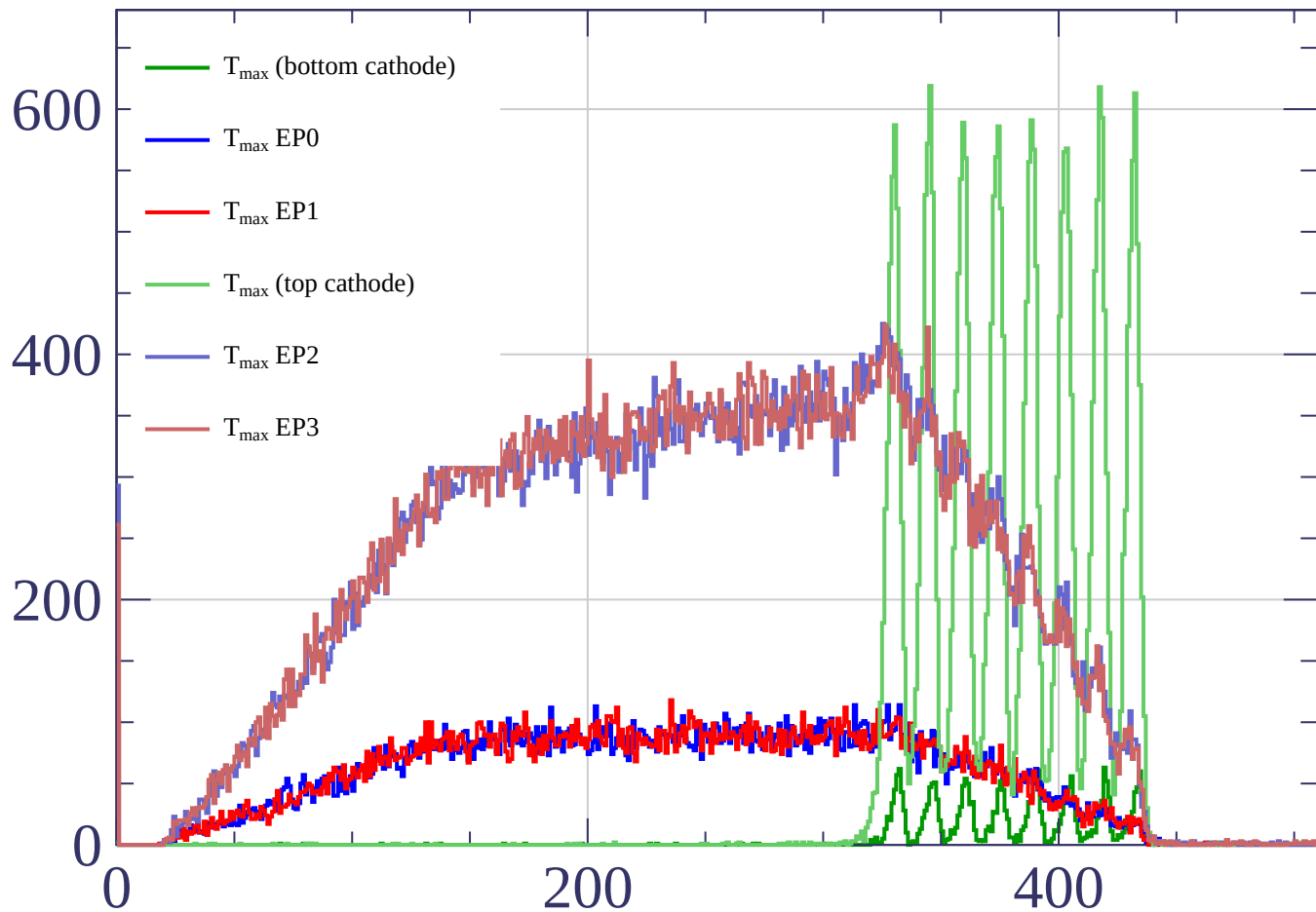
Energy loss | $87 < \theta < 90$

Count





Count



time bin